

Topological data analysis for time series

Alexandre Bois

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Topological data analysis for time series

Analyse topologique de données pour les séries temporelles

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et le co-encadrement de **Brian TERVIL**, Chercheur

Thèse soutenue à Paris-Saclay, le 26 novembre 2024, par

Alexandre BOIS

Composition du jury

Membres du jury avec voix délibérative

Mathilde MOUGEOT Professeure des Universités, ENS Paris-Saclay	Présidente
Elizabeth MUNCH Associate Professor, Michigan State University	Rapporteure & Examinatrice
Julien TIERNY Directeur de recherche, Sorbonne Université	Rapporteur & Examinateur
Frederic Chazal Directeur de recherche, INRIA Saclay	Examinateur

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Titre : Analyse topologique de données pour les séries temporelles
Mots clés : analyse topologique de données, séries temporelles, homologie persistante

Résumé : Dans cette thèse, nous présentons de nouvelles méthodes d'analyse topologique de données pour les séries temporelles. Ce travail est motivé par l'étude de signaux physiologiques. En effet, ces signaux possèdent généralement une certaine structure qui peut être étudiée à l'aide d'outils comme l'homologie persistante de manière non-supervisée et interprétable. Nous passons en revue de l'état de l'art sur l'utilisation de l'homologie persistante pour l'étude de séries temporelles, et proposons trois contributions.
La première contribution est une méthode non paramétrique d'analyse de signaux de marche mesurés par des centrales inertielles placées sur les pieds, que nous avons appliquée à l'étude de la sclérose en plaques. Cette méthode peut donner des informations sur la sévérité de la maladie et son évolution dans le temps chez un patient.
La seconde contribution est un algorithme de clustering basé sur une nouvelle filtration qui intègre une information sur la densité de chaque point et permet donc d'exclure les points isolés. Nous montrons que notre filtration possède une propriété de stabilité et que l'algorithme peut être performant sur différents types de données.
La troisième contribution est un algorithme non-supervisé de détection d'anomalies pour les séries temporelles univariées. Cet méthode est construite à partir d'un modèle de séries temporelles composées d'une succession de motifs, qui nous permet de bien définir le problème de détection d'anomalies et d'étudier théoriquement notre méthode. Nous montrons qu'elle est compétitive par rapport à l'état de l'art et étudions l'influence des paramètres et du bruit.

Title : Topological data analysis for time series
Keywords : topological data analysis, time series, persistent homology

Abstract : In this thesis, we develop new methods for time series analysis based on topological data analysis. This work is motivated by the study of physiological signals. These signals usually have a certain structure that can be studied using tools such as persistent homology in an unsupervised and interpretable way. We review the state of the art on the use of persistent homology for time series, and propose three contributions.
The first contribution is a non-parametric method for analyzing gait signals measured by inertial units placed on the feet, which we have applied to the study of healthy subjects and patients with multiple sclerosis. This method can provide information on the severity of the disease and its evolution over time in a patient.
The second contribution is a clustering algorithm based on a new filtration that incorporates information about the density of each point and thus makes it possible to exclude isolated points. We show that our filtration has a stability property and that the algorithm can perform well on various types of data.
The third contribution is an unsupervised anomaly detection algorithm for univariate time series. This method is built from a model of time series composed of a succession of patterns, which allows us to define the anomaly detection problem and to study our method theoretically. We show that it is competitive with the state-of-the art and study the influence of parameters and noise.

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Contents

Contents **6**

Résumé (en français) **9**

Abstract **11**

Introduction (en français) **13**
0.1 Contexte et motivations . . . 14
0.1.1 Contexte . . . 14
0.1.2 Motivations . . . 15
0.2 Contributions et plan du manuscrit . . . 16
0.2.1 Chapitre II : Contexte mathématique . . . 16
0.2.2 Chapitre III : Revue de la littérature sur l'homologie persistante pour les séries temporelles . . . 17
0.2.3 Chapitre IV : Une méthode pour l'analyse de signaux de marche avec une application à l'étude de la sclérose en plaques . . . 17
0.2.4 Chapitre V : Clustering basé sur la persistance avec une filtration retirant les valeurs aberrantes . . . 20
0.2.5 Chapitre VI : Détection non supervisée d'anomalies dans les séries temporelles . . . 20
0.3 Liste des articles . . . 21

1 Introduction **22**
1.1 Context and motivations . . . 23
1.1.1 Context . . . 23
1.1.2 Motivations . . . 24
1.2 Contributions and outline . . . 25
1.2.1 Chapter II: Mathematical background . . . 25
1.2.2 Chapter III: Literature review on persistent homology for time series 25
1.2.3 Chapter IV: A method for gait signals analysis with an application to the study of multiple sclerosis . . . 26
1.2.4 Chapter V: Persistence-based clustering with outlier-removing filtration . . . 28
1.2.5 Chapter VI: Unsupervised anomaly detection in time series . . . 28
1.3 List of papers . . . 29

2 Mathematical background **30**
2.1 Signals, time series and delay embeddings . . . 31
2.1.1 Signals and time series . . . 31
2.1.2 Delay embeddings . . . 31

2.2 Algebraic topology background: simplicial and singular homology 32
2.2.1 Simplicial complexes 32
2.2.2 Simplicial homology 33
2.2.3 Singular homology 35
2.3 Persistence 36
2.3.1 Filtrations and persistence modules 36
2.3.2 Examples of filtrations of simplicial complexes 37
2.3.3 Persistent homology 39
2.3.4 Distances between persistence diagrams and stability 42

Appendix A: Topology background **44**

3 Literature review: persistent homology for time series **45**
3.1 Introduction 46
3.2 Application fields and tasks 46
3.3 Methods 48
3.3.1 General framework 48
3.3.2 Sublevel sets methods 48
3.3.3 Delay embedding methods 53
3.3.4 Multivariate time series 61
3.4 Discussion and perspectives 66

4 A method for gait signals analysis with an application to the study of multiple sclerosis **68**
4.1 Introduction 69
4.2 Persistent homology of periodic functions 70
4.3 Protocol and data 74
4.4 Method description 75
4.4.1 Overview of our method and guidelines for the clinician 75
4.4.2 Mathematical description our of method 80
4.4.3 Feature extraction 84
4.5 Results 86
4.5.1 HS/MS experiment 87
4.5.2 EDSS experiments 87
4.5.3 Longitudinal experiment 88
4.6 Analysis of the results and discussion 89
4.6.1 Analysis of the results 89
4.7 Discussion 92
4.8 Conclusion and perspectives 93

Appendix B: Analysis of gait signals with standard features **94**

5 Persistence-based clustering with outlier-removing filtration **98**
5.1 Introduction 99
5.2 Nearest Neighbor Vietoris-Rips (NNVR) filtration and stability 99
5.3 Outlier-removing clustering 102
5.3.1 Algorithm 102
5.3.2 Choice of thresholds 102
5.4 Experiments 103
5.4.1 Karate club graph 104

5.4.2 Point clouds . . . 104
5.4.3 Synthetic graphs . . . 105
5.5 Application to gait signals . . . 108
5.6 Conclusion . . . 111

6 Unsupervised anomaly detection in time series 112
6.1 Introduction . . . 113
6.2 Model and problem formulation . . . 114
6.3 Method . . . 117
6.3.1 Motivation . . . 117
6.3.2 Algorithm description . . . 118
6.4 Theoretical study . . . 122
6.4.1 Overview of existing stability theorems for DTM filtrations . . . 122
6.4.2 Upper bound on the interleaving distance between the real and ideal filtrations . . . 123
6.5 Experimental setup . . . 129
6.5.1 Synthetic dataset . . . 129
6.5.2 Real-world datasets and methods from TSB-UAD . . . 130
6.6 Results . . . 131
6.6.1 Benchmark on real-world data . . . 131
6.6.2 Influence of the embedding dimension and delay . . . 133
6.6.3 Influence of the number of neighbors and points on the diagram . . . 134
6.6.4 Computation time and influence of the subsampling parameter . . . 134
6.6.5 Robustness to noise and benchmark on synthetic data . . . 135
6.7 Application to gait signals . . . 136
6.8 Conclusion . . . 139

7 Conclusion and perspectives 140

Bibliography 142

Résumé (en français)

Dans cette thèse, nous présentons le développement de nouvelles méthodes d'analyse topologique de données pour les séries temporelles. Ce travail est motivé par l'étude de signaux physiologiques en neurologie comportementale, et en particulier par l'étude de la marche. En effet, ces signaux possèdent généralement une certaine structure (périodicité, répétitions de motifs) qui peut être étudiée à l'aide d'outils comme l'homologie persistante de manière non-supervisée et interprétable.

Après avoir introduit les principaux concepts mathématiques liés aux séries temporelles et à l'analyse topologique de données utilisés dans les chapitres suivants, nous menons une revue de l'état de l'art sur l'utilisation de l'homologie persistante pour l'étude de séries temporelles. Nous proposons ensuite trois contributions.

La première contribution est une méthode non paramétrique d'analyse de signaux de marche mesurés par des centrales inertielles placées sur les pieds, que nous avons appliquée à l'étude de sujets sains et de patients atteints de sclérose en plaques. Cette méthode, basée sur les ensembles de sous-niveaux de fonctions et la distance de bottleneck entre codes-barre de persistance, permet de visualiser des différences entre signaux et d'en déduire des informations, notamment sur la sévérité de la maladie ou son évolution dans le temps chez un même patient. Plus précisément, nous calculons la filtration par sous-niveaux de chaque signal et le code-barre de persistance associé auquel on retire les plus grandes barres. Nous calculons ensuite la matrice des distances de Bottleneck entre chaque paire de codes barres, visualisons ces distances à l'aide de l'algorithme UMAP et proposons différents moyens d'interpréter le résultat. Dans ce chapitre, nous étudions aussi une propriété théorique de l'homologie persistante des sous-niveaux de fonctions périodiques, qui motive la méthode car elle relie le nombre de périodes observées à la multiplicité des barres du code-barres de persistance.

La seconde contribution est un algorithme de clustering pour les points d'espaces métriques quelconques, basé sur une nouvelle filtration appelée NNVR (Nearest Neighbor Vietoris-Rips). Cette filtration est proche de la filtration de Vietoris-Rips mais y ajoute une information sur la proximité du plus proche voisin de chaque point. La lecture du diagramme de persistance permet alors d'exclure des points isolés puis de choisir un seuil adapté pour le clustering hiérarchique. Nous montrons que notre filtration possède la même propriété de stabilité que la filtration de Vietoris-Rips par rapport à la distance de Gromov-Hausdorff, décrivons l'algorithme et montrons qu'il peut être performant sur différents types de données même dans sa version non-paramétrique. Enfin, nous illustrons son utilité dans un algorithme de détection de motifs appliqué à des signaux de marche.

La troisième contribution est un algorithme non-supervisé de détection d'anomalies pour les séries temporelles univariées. Il utilise une transformation des séries temporelles en nuage de points, une filtration basée sur la notion de distance à une mesure, l'extraction de cycles représentant des classes d'homologie persistante en dimension 1, et une lecture du diagramme de persistance permettant de définir des cycles normaux. Un score

d’anomalie est alors défini pour chaque point du nuage de point comme sa distance aux cycles normaux, puis on en déduit un score pour chaque point de la série temporelle. Cet méthode est construite à partir d’un modèle de séries temporelles composées d’une succession de motifs, que nous décrivons formellement en début de chapitre. Il nous permet de proposer une définition rigoureuse d’un problème de détection d’anomalies et d’étudier théoriquement notre méthode. Nous montrons que notre méthode est compétitive par rapport à l’état de l’art et étudions l’influence des paramètres et du bruit. Enfin, nous illustrons son utilisation sur des signaux de marche.

Abstract

In this thesis, we present the development of new methods for time series analysis based on topological data analysis (TDA). This work is motivated by the study of physiological signals in behavioral neurology, and in particular by the study of gait. Indeed, these signals usually have a certain structure (periodicity, repeating patterns) that can be analyzed using tools such as persistent homology in an unsupervised and interpretable way.

After introducing the main mathematical concepts related to time series and topological data analysis which are used in the following chapters, we review the state of the art on the use of persistent homology for time series analysis. We then propose three contributions.

The first contribution is a non-parametric method for analyzing gait signals measured by inertial measurement units (IMUs) placed on the feet, which we have applied to the study of healthy subjects and multiple sclerosis patients. This method, based on sublevel sets of functions and on the bottleneck distance between persistence barcodes, makes it possible to visualize differences between each pair of signals and to deduce information, notably on the severity of the disease or its evolution over time for a given patient. More precisely, we calculate the sublevel filtration of each signal and the corresponding persistence barcode from which the largest bars are removed. We then calculate the bottleneck distance matrix between each pair of barcodes, visualize these distances using the UMAP algorithm, and propose several ways of interpreting the result. In this chapter, we also study a theoretical property of persistent homology of sublevel sets of periodic functions, which motivates the method because it makes the link between the number of observed periods and the multiplicity of bars in the persistence barcode.

The second contribution is a clustering algorithm for points in any metric space, based on a new filtration called NNVR (Nearest Neighbor Vietoris-Rips). This filtration is close to the Vietoris-Rips filtration, but integrates information about the proximity of each point's nearest neighbor. By reading the persistence diagram, we can then exclude isolated points and choose a suitable threshold for hierarchical clustering. We show that our filtration has the same stability property as the Vietoris-Rips filtration with respect to the Gromov-Hausdorff distance, describe the algorithm and show that it can perform well on various types of data, even in its non-parametric version. Finally, we illustrate its usefulness in a pattern detection algorithm applied to gait signals.

The third contribution is an unsupervised anomaly detection algorithm for univariate time series. It uses a delay embedding technique (transformation of time series into point clouds), a filtration based on the notion of distance to a measure (DTM), the extraction of cycles representing persistent homology classes in dimension 1, and a reading of the persistence diagram to define normal cycles. An anomaly score is then defined for each point in the point cloud as its distance from normal cycles, and a score is deduced for each point in the time series. This method is based on a time series model composed of a succession of patterns, which we describe formally at the beginning of this chapter.

This model enables us to propose a rigorous definition of an anomaly detection problem and to study our method theoretically. We show that our method is competitive with state-of-the-art anomaly detection methods on various datasets and study the influence of parameters and noise. Finally, we illustrate its use on gait signals.

Introduction (en français)

Contents

0.1 Contexte et motivations **14**
0.1.1 Contexte 14
0.1.2 Motivations 15
0.2 Contributions et plan du manuscrit **16**
0.2.1 Chapitre II : Contexte mathématique 16
0.2.2 Chapitre III : Revue de la littérature sur l'homologie persistante pour les séries temporelles 17
0.2.3 Chapitre IV : Une méthode pour l'analyse de signaux de marche avec une application à l'étude de la sclérose en plaques 17
0.2.4 Chapitre V : Clustering basé sur la persistance avec une filtration retirant les valeurs aberrantes 20
0.2.5 Chapitre VI : Détection non supervisée d'anomalies dans les séries temporelles 20
0.3 Liste des articles **21**

0.1 Contexte et motivations

L'objectif général de cette thèse est d'introduire de nouvelles méthodes d'analyse de séries temporelles basées sur l'analyse topologiques de données.

0.1.1 Contexte

Les recherches présentées dans cette thèse ont été menées au Centre Borelli [1], une unité mixte de recherche affiliée à l'École Normale Supérieure Paris Saclay, à l'Université Paris Saclay, à l'Université Paris Cité, au CNRS, à la SSA et à l'INSERM. Une des spécificités du Centre Borelli est de réunir des mathématiciens, des médecins, des cliniciens et des biologistes pour développer des connaissances fondamentales en neurosciences et en mathématiques ainsi que de nouvelles méthodes d'analyse de données biomédicales. Le Centre Borelli est également impliqué dans des recherches ayant des applications dans l'industrie, les sciences physiques ou les sciences sociales.

L'un des principaux axes de recherche en neurosciences au Centre Borelli est l'étude quantitative du comportement humain et animal dans des situations normales, altérées ou pathologiques. L'objectif est d'utiliser des capteurs non intrusifs et peu coûteux pour enregistrer des signaux physiologiques et de développer de nouveaux algorithmes pour les analyser, ce qui peut permettre aux cliniciens d'accéder à des informations quantitatives sur les patients qui peuvent les aider lors des examens cliniques. Par exemple, des chercheurs du Centre Borelli ont utilisé des signaux d'accélération ou de vitesse enregistrés par des centrales inertielles (IMUs, *inertial measurement units*) pour étudier les signaux de marche de sujets sains et pathologiques [VJOM$^+$19, VJOM$^+$20, BTM$^+$22, JdlO$^+$23], les trajectoires du centre de pression chez les personnes âgées pour étudier le contrôle postural et prédire les chutes [BAV$^+$18, QNC$^+$21], des signaux de position 3D provenant de marqueurs CODA (*Cartesian Optoelectronic Dynamic Anthropometer*) placés sur les membres supérieurs de sujets lors d'un examen de routine [CBM$^+$24, CMO24] ou des signaux de pléthysmographie pour étudier les cycles respiratoires chez les souris [GTOK22]. La figure 0.1 montre un exemple de protocole utilisant des IMUs pour enregistrer les signaux de marche.

Deux grands types d'approche peuvent être identifiés :

- **Analyse individuelle des signaux**: on étudie chaque signal (c'est-à-dire chaque réalisation d'un protocole médical) individuellement pour obtenir des informations sur le sujet, par exemple en détectant des événements récurrents ou anormaux [GTO24b, BTO24b], en produisant une représentation symbolique qui peut être facilement interprétée par un clinicien [QNC$^+$21, GTOK22, CMO24], ou en faisant une prédiction [BAV$^+$18].

- **Comparaison de signaux**: on constitue des groupes de de sujets similaires en comparant des ensembles de signaux appropriés. Les comparaisons peuvent être inter-individuelles, afin de distinguer les sujets sains des sujets pathologiques, les sujets souffrant de pathologies différentes ou d'une même pathologie à des degrés de gravité différents [VJOM$^+$19, VJOM$^+$20, BTM$^+$22]; ou alors intra-individuelles, auquel cas l'objectif principal est d'effectuer un suivi longitudinal, c'est-à-dire de quantifier l'évolution d'un sujet dans le temps. [BTM$^+$22].

Comme dans l'exemple de la figure 0.1, les signaux physiologiques enregistrés au cours de protocoles médicaux présentent généralement une forme d'onde spécifique où

[1] `https://centreborelli.ens-paris-saclay.fr/fr`

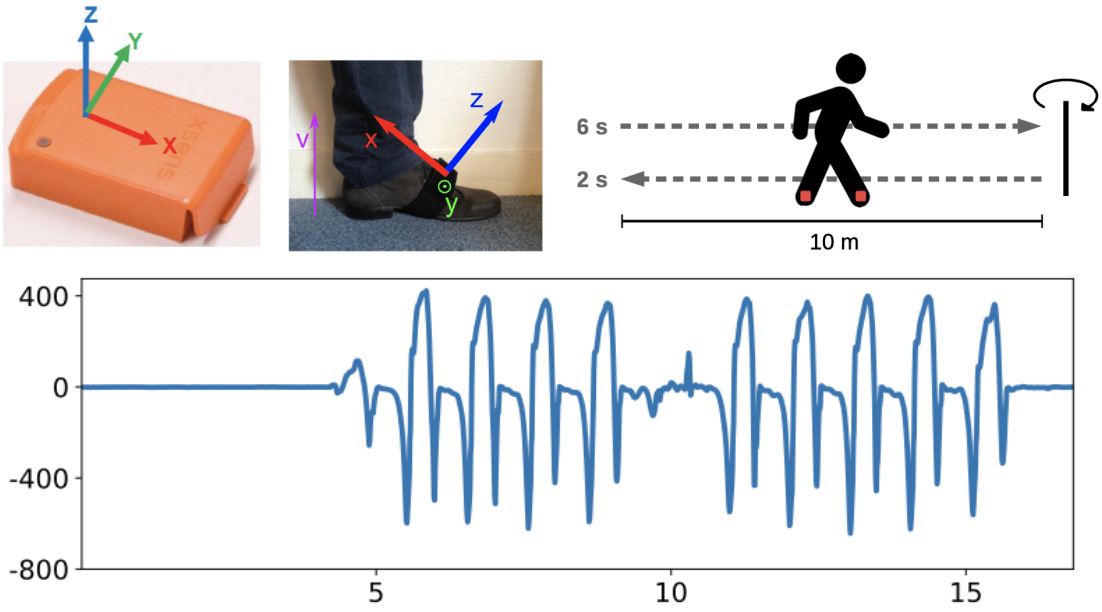


Figure 0.1: Protocole d'enregistrement des signaux de marche à l'aide d'IMUs placés sur les pieds. **En haut à gauche**: position des capteurs [BTM$^+$22]. **En haut à droite**: illustration du protocole [TBMM$^+$19]. **En bas**: Séries temporelles du jeu de données de Truong et al. [TBMM$^+$19] montrant la vitesse angulaire autour de l'axe Y pendant le protocole.

l'on peut distinguer des motifs récurrents, qui correspondent à l'occurrence d'un événement récurrent dans le corps humain (pas, battements de cœur, mouvements des bras...). Il est important que les cliniciens soient en mesure d'interpréter les résultats fournis par les algorithmes en termes médicaux afin de pouvoir les utiliser avec leurs patients. Une source de difficulté est que cette régularité peut être perturbée par du bruit et des événements qui ne correspondent pas aux schémas récurrents, tels que le demi-tour dans le protocole montré en figure 0.1 ou des comportements induits par une pathologie. Enfin, les séries temporelles étudiées sont souvent multivariées (plusieurs capteurs sont utilisés en même temps, par exemple un IMU par pied). Les séries temporelles multivariées peuvent être étudiées en considérant chaque canal comme une série temporelle univariée [BTM$^+$22],mais l'étude des corrélations entre les canaux peut être cruciale dans certains cas, comme dans le cas des électroencéphalogrammes (EEG), afin de tenir compte de la nature hautement interconnectée des régions du cerveau.

0.1.2 Motivations

L'analyse topologique des données (TDA), et plus particulièrement l'homologie persistante, fournit des outils pertinents pour étudier les séries temporelles médicales. L'idée est d'utiliser les séries temporelles pour construire une séquence de *complexes simpliciaux* (une *filtration*) et d'étudier sa structure topologique à différentes échelles. Le *diagramme de persistance* résume l'évolution de cette structure à travers toutes les échelles. Plusieurs études [EGK14, PH15, RMC23] ont déjà montré que les diagrammes de persistance permettent de détecter la périodicité dans les séries temporelles. De plus, les diagrammes de persistance peuvent être comparés à l'aide de distances qui bénéficient de *théorèmes de stabilité* [CDSO14] qui garantissent une certaine robustesse au bruit. Cela rend ces distances pertinentes pour comparer des séries temporelles. Ainsi, notre intuition ini-

tiale pour cette thèse était que **lorsque l'on utilise une filtration appropriée pour un problème donné, le diagramme de persistance reflétera des caractéristiques structurelles intéressantes de la série temporelle**, qui peuvent être facilement identifiées visuellement ou algorithmiquement pour effectuer des comparaisons interindividuelles ou un suivi longitudinal.

La spécificité de la recherche menée dans le cadre de cette thèse est que nous avons utilisé l'homologie persistante pour définir des représentations de séries temporelles qui peuvent être interprétées seules plutôt qu'utilisées comme des caractéristiques supplémentaires dans un algorithme d'apprentissage automatique. L'interprétation est basée sur des propriétés théoriques (additivité pour les diagrammes de persistance, stabilité) et/ou sur des modèles (fonctions périodiques [BTM+22], apprentissage de dictionnaires convolutifs [BTO24b]) des séries temporelles étudiées.

La première motivation de notre travail était de **construire des ponts entre les communautés de l'analyse topologique des données (TDA) et du traitement du signal en développant de nouvelles méthodes topologiques pour l'analyse des séries temporelles**. Ces méthodes doivent être basées sur des idées ou des modèles mathématiques interprétables et s'accompagner de garanties théoriques. L'idée est que la TDA peut être utilisée pour construire des représentations qui reflètent des événements intéressants ou des propriétés des séries temporelles. Dans cette thèse, nous avons passé en revue tous les articles que nous avons pu trouver sur l'application de l'homologie persistante aux séries temporelles et nous avons identifié plusieurs types de méthodes pour différents types de problèmes et de données. Ces méthodes (y compris les trois développées dans cette thèse) peuvent être utilisées pour d'autres applications que celles pour lesquelles elles ont été conçues, ou au moins fournir des caractéristiques pertinentes aux algorithmes d'apprentissage automatique.

La deuxième motivation était l'**analyse des séries temporelles médicales, et en particulier des signaux de marche** pour l'étude des maladies neurodégénératives telles que la sclérose en plaques. En effet, la marche est considérée comme la principale source de handicap dans la sclérose en plaques [HBR+08]. En outre, bien que les signaux de marche tels que celui de la figure 0.1 puissent être plus complexes que les signaux purement périodiques, la marche reste une activité structurée (une suite d'étapes). Nous voulons utiliser l'homologie persistante pour détecter cette structure avec des méthodes interprétables, robustes et non supervisées, avec le moins de paramètres possible.

0.2 Contributions et plan du manuscrit

Le manuscrit est organisé comme suit.

0.2.1 Chapitre II : Contexte mathématique

Dans le chapitre II - Contexte mathématique, nous présentons les objets et les résultats mathématiques utilisés dans le reste de la thèse. Nous commençons par donner les notations et les définitions des séries temporelles et des outils liés tels que les plongements de Takens. Ensuite, nous présentons la construction de l'homologie simpliciale/singulière et de l'homologie persistante ainsi que des objets et théorèmes importants.

Les quatre chapitres suivants contiennent nos contributions et utilisent les notions introduites dans ce chapitre.

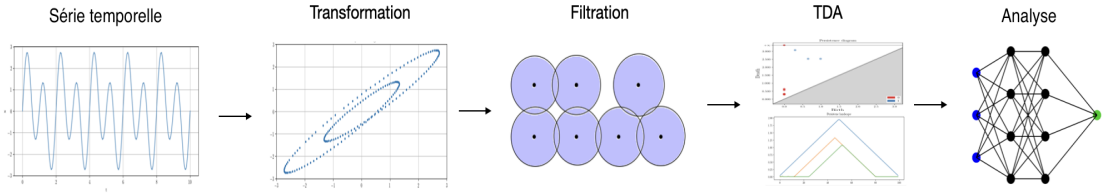


Figure 0.2: Illustration d'une méthode décomposée dans notre cadre.

0.2.2 Chapitre III : Revue de la littérature sur l'homologie persistante pour les séries temporelles

Dans le chapitre III - Revue de la littérature sur l'homologie persistante pour les séries temporelles, nous effectuons une revue de toutes les méthodes de la littérature qui utilisent l'homologie persistante pour l'analyse des séries temporelles (univariées ou multivariées). Au cours des dix dernières années, l'homologie persistante a été de plus en plus utilisée pour analyser la structure et la forme de divers types de données, y compris les séries temporelles. Nous passons en revue 71 articles publiés entre 2014 et 2024 dans plusieurs domaines d'application tels que la biomédecine, l'industrie et l'économie. Nous donnons un aperçu des champs d'application et des tâches et proposons un cadre général pour décrire les principales caractéristiques de toutes les méthodes. Le cadre consiste à décrire cinq étapes clés de chaque méthode : le type de série temporelle, la transformation de la série temporelle, la filtration construite sur les données transformées, les outils de TDA utilisés et la méthode d'analyse, comme illustré sur la figure 0.2 (dans le cas d'une méthode basée sur un plongement de Takens).

0.2.3 Chapitre IV : Une méthode pour l'analyse de signaux de marche avec une application à l'étude de la sclérose en plaques

Dans le chapitre IV - Une méthode d'analyse de signaux de marche avec une application à l'étude de la sclérose en plaques, nous décrivons une méthode basée sur la TDA pour comparer des paires de signaux de marche, visualiser les résultats et les interpréter, qui a été appliquée à l'étude de sujets sains et de patients souffrant de sclérose en plaques [BTM$^+$22]. La sclérose en plaques est une maladie démyélinisante du système nerveux central dont les symptômes comprennent une déficience des membres inférieurs, raison pour laquelle les cliniciens utilisent les exercices de marche pour le suivi des patients. Pour ce faire, les caractéristiques standard de la marche peuvent être utilisées, mais elles ne permettent pas de détecter des changements subtils dans plusieurs pathologies, ce qui motive le développement de nouvelles approches.

La méthode est non paramétrique et ne nécessite pas d'autres données que les signaux de marche acquis à l'aide de centrales inertielles. Les outils de la TDA sont introduits de manière pratique pour les rendre aussi interprétables que possible afin d'encourager leur utilisation par les cliniciens. La méthode peut aider à estimer la gravité de la maladie et peut également être utilisée pour le suivi longitudinal afin de détecter son évolution ou d'autres phénomènes tels que l'asymétrie ou les valeurs aberrantes.

Nous avons également étudié une propriété théorique de l'homologie persistante des sous-niveaux de fonctions périodiques, qui relie le nombre de périodes observées à la multiplicité des barres du code-barres de persistance : nous avons montré que le diagramme de persistance obtenu à partir de la filtration des sous-niveaux d'une fonction T-périodique sur $[0, kT]$ (où k est un entier supérieur à 1) peut être déduit du diagramme

correspondant à la fonction sur $[0, 2T]$, ce qui implique en particulier qu'il existe un groupe de $k-1$ ou k barres identiques pour chaque minimum local de la fonction sur $[0, T]$. Ceci justifie une étape importante de la méthode, où un lien est fait entre le nombre de pas et le nombre de longues barres dans le code-barres.

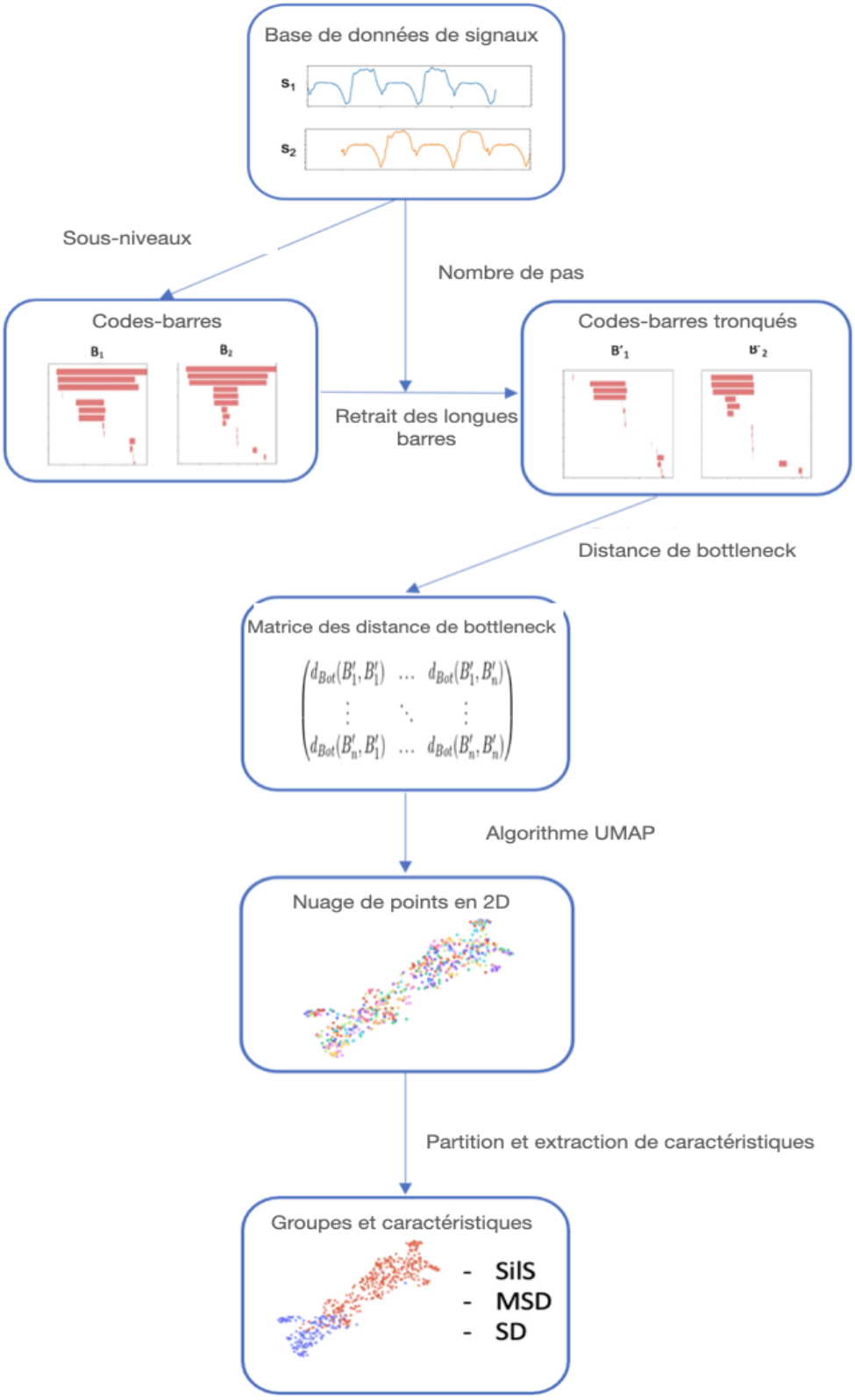


Figure 0.3: Illustration de notre méthode d'analyse de signaux de marche.

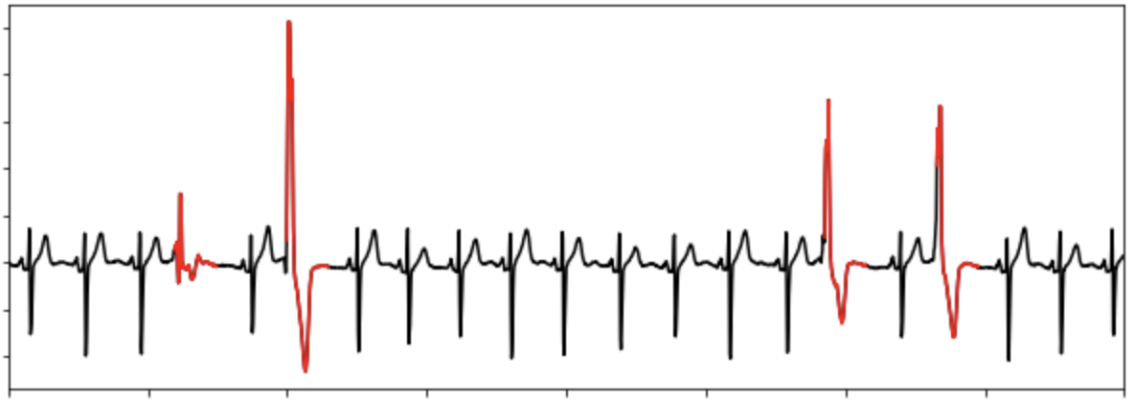


Figure 0.4: Illustration d'une série temporelle d'électrocardiogramme avec des anomalies (en rouge) dues à des contractions ventriculaires prématurées.

0.2.4 Chapitre V : Clustering basé sur la persistance avec une filtration retirant les valeurs aberrantes

Dans le chapitre V - Clustering basé sur la persistance avec une filtration retirant les valeurs aberrantes, nous décrivons un algorithme de clustering hiérarchique non paramétrique avec une étape d'élimination des valeurs aberrantes, basé sur l'analyse topologique de données. Nous utilisons une nouvelle filtration d'espaces métriques appelée NNVR (*Nearest Neighbor Vietoris-Rips*), qui est une variante de la filtration de Vietoris-Rips qui ajoute des informations sur le plus proche voisin des points au diagramme de persistance. En lisant le diagramme de persistance, nous pouvons alors exclure les points isolés et choisir un seuil approprié pour le clustering hiérarchique. Nous prouvons que notre filtration a la même propriété de stabilité que la filtration de Vietoris-Rips par rapport à la distance de Gromov-Hausdorff, nous décrivons l'algorithme et nous montrons qu'il peut donner de bons résultats sur des nuages de points et des graphes. Enfin, nous illustrons son utilité dans un algorithme de détection de motifs appliqué à des signaux de marche.

0.2.5 Chapitre VI : Détection non supervisée d'anomalies dans les séries temporelles

Dans le chapitre VI - Détection non supervisée d'anomalies dans les séries temporelles, nous proposons un nouvel algorithme basé sur la TDA pour la détection non supervisée d'anomalies dans les séries temporelles univariées. La détection d'anomalies dans les séries temporelles est un problème important en science des données et a des applications dans de nombreux domaines, y compris la biomédecine (voir par exemple la figure 0.4). L'algorithme repose sur le plongement de Takens (transformation des séries temporelles en nuages de points) et sur l'extraction de cycles persistants à partir de l'homologie persistante en dimension 1 construite à partir de la filtration dite de Rips DTM. Cette filtration permet d'identifier des cycles en dimension 1 (c'est-à-dire des boucles) correspondant à des motifs récurrents en tirant parti des informations de densité. Les points constituant ces cycles sont considérés comme normaux, et l'algorithme peut alors attribuer un score d'anomalie à tout point correspondant à sa distance à l'ensemble normal.

Cette méthode est basée sur un modèle de séries temporelles composées d'une succession de motifs, que nous décrivons formellement au début de ce chapitre, à la fois dans le cas continu et dans le cas discret. Ce modèle nous permet de proposer une définition rigoureuse d'un problème de détection d'anomalies et d'étudier théoriquement

notre méthode. Plus précisément, nous considérons le cas idéal où nos données sont une fonction continue sans bruit et sans anomalie et le cas où les données sont une série temporelle discrète et bruitée avec des motifs anormaux. Sous certaines hypothèses de régularité, nous obtenons une borne supérieure pour la distance d'entrelacement entre les modules de persistance associés (et la distance de bottleneck entre les diagrammes de persistance correspondants).

Nous montrons que notre méthode est compétitive par rapport aux méthodes de détection d'anomalies les plus récentes sur différents jeux de données et nous étudions l'influence des paramètres et du bruit. Enfin, nous illustrons son utilisation sur des signaux de marche.

0.3 Liste des articles

Articles de revue:

- [BTM+22]: A. Bois, B. Tervil, A. Moreau, A. Vienne-Jumeau, D. Ricard and L. Oudre. "A topological data analysis-based method for gait signals with an application to the study of multiple sclerosis". In Plos one, 17(5), e0268475, 2022.
- [BTO24a]: A. Bois, B. Tervil and L. Oudre. "Persistence-based clustering with outlier-removing filtration". In Frontiers in Applied Mathematics and Statistics, 10, 1260828, 2024.

Articles en relecture:

- [BTO24b]: A. Bois, B. Tervil and L. Oudre. "A persistent homology- based algorithm for unsupervised anomaly detection in time series". 2024.
- [BTO24c] A. Bois, B. Tervil and L. Oudre. "Persistent homology for time series: a review". 2024

Articles de conférence:

- A. Bois, B. Tervil and L. Oudre. "Approches topologiques pour l'analyse de signaux physiologiques". In Proceedings of the Groupe de Recherche et d'Etudes en Traitement du Signal et des Images (GRETSI), Nancy, France, 2022.
- T. Germain, A. Bois, C. Truong, B. Tervil and L. Oudre. "Détection non supervisée de motifs sur séries temporelles". In Proceedings of the Groupe de Recherche et d'Etudes en Traitement du Signal et des Images (GRETSI), Grenoble, France, 2023.
- A. Bois, B. Tervil and L. Oudre. "Topological data analysis for unsupervised anomaly detection in time series". In Proceedings of the 32nd European Signal Processing Conference, EUSIPCO, Lyon, France, 2024.

Chapter 1

Introduction

Contents

1.1 Context and motivations **23**
1.1.1 Context 23
1.1.2 Motivations 24
1.2 Contributions and outline **25**
1.2.1 Chapter II: Mathematical background 25
1.2.2 Chapter III: Literature review on persistent homology for time series 25
1.2.3 Chapter IV: A method for gait signals analysis with an application to the study of multiple sclerosis 26
1.2.4 Chapter V: Persistence-based clustering with outlier-removing filtration 28
1.2.5 Chapter VI: Unsupervised anomaly detection in time series . . . 28
1.3 List of papers **29**

1.1 Context and motivations

The general objective of this thesis is to introduce new methods for time series analysis based on topological data analysis.

1.1.1 Context

The research presented in this thesis was conducted in Centre Borelli [1], a joint research unit affiliated with Ecole Normale Supérieure Paris Saclay, Université Paris Saclay, Université Paris Cité, CNRS, SSA and INSERM. A specificity of the Centre Borelli is to bring together mathematicians, medical doctors, clinicians and biologists to develop fundamental knowledge in neuroscience and mathematics as well as new methods to analyze biomedical data. The Centre Borelli is also involved in research with applications in the industry, physical sciences or social sciences.

A major neuroscience research axis at Centre Borelli is the quantitative study of human and animal behavior in normal, altered or pathological situations. The goal is to use non-intrusive and inexpensive sensors to record physiological signals and to develop new algorithms to analyze them, which can give clinicians access to quantitative information about patients that can help them during clinical examinations. For example, researchers from Centre Borelli have used acceleration or velocity signals recorded with inertial measurement units (IMUs) to study gait signals from healthy and pathological subjects [VJOM$^+$19, VJOM$^+$20, BTM$^+$22, JdlO$^+$23], center of pressure trajectories in elderly subjects to study postural control and predict falls [BAV$^+$18, QNC$^+$21], 3D position signals from Cartesian Optoelectronic Dynamic Anthropometer (CODA) markers placed on the upper limb of subjects during routine examination [CBM$^+$24, CMO24] or plethysmography signals to study respiratory cycles in mice [GTOK22]. An example of protocol using IMUs to record gait signals is shown on Figure 1.1.

Two main types of approach can be identified:

- **Single signal analysis**: study each signal (i.e. each realization of a medical protocol) individually to get information about the subject, for example by detecting recurring or abnormal events [GTO24b, BTO24b], producing a symbolic representation that can be easily interpreted by a clinician [QNC$^+$21, GTOK22, CMO24], or making a prediction about a future event [BAV$^+$18].

- **Signals comparison**: building similarity groups of subjects by comparing suitable sets of signals. Comparisons can be inter-individual, to distinguish healthy and pathological subjects, subjects suffering from different pathologies or from the same pathology with different degrees of severity [VJOM$^+$19, VJOM$^+$20, BTM$^+$22]; or they can be intra-individual, in which case the main objective is to perform longitudinal follow-up, i.e. to quantify the evolution of a subject over time [BTM$^+$22].

As in the example shown in Figure 1.1, physiological signals recorded during medical protocols usually have a specific waveform where recurring patterns can be distinguished, which correspond to the occurrences of a recurring event in the human body (steps, heartbeats, arm movements...). An important issue is that clinicians should be able to interpret the results given by algorithms in medical terms so that they can use these results with their patients. A source of difficulty is that this regularity can be perturbed by noise and events that do not correspond to the recurring patterns, such as the U-turn in the protocol

[1] `https://centreborelli.ens-paris-saclay.fr/en`

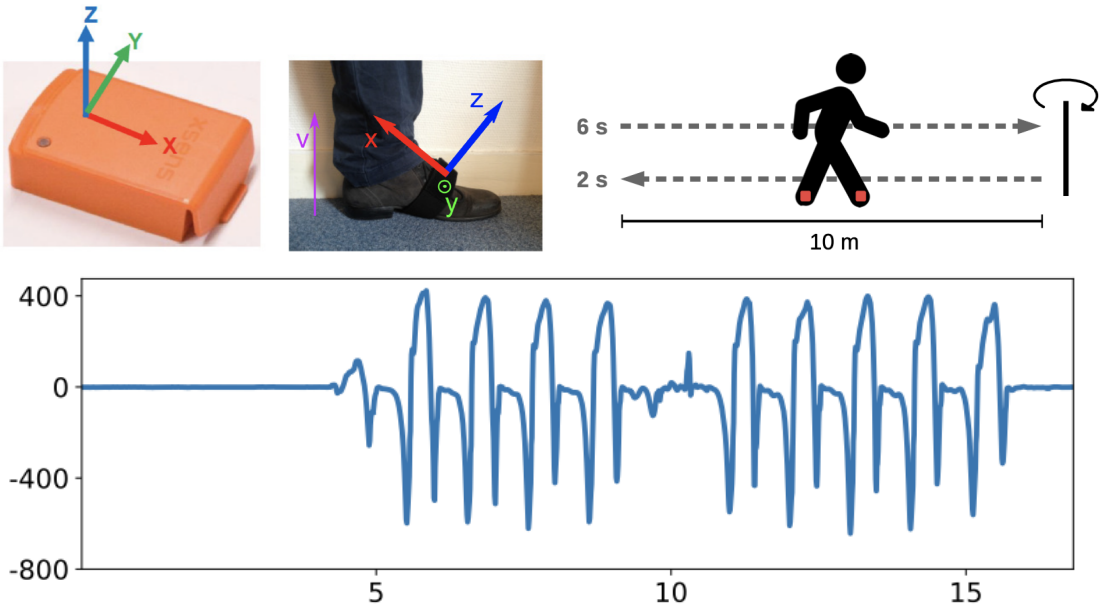


Figure 1.1: Protocol to record gait signals using IMUs on the feet. **Top left**: sensor placement [BTM+22]. **Top right**: illustration of the protocol [TBMM+19]. **Bottom**: Time series from Truong et al. 's dataset [TBMM+19] showing the angular velocity around the Y-axis during the protocol.

from Figure 1.1 or behaviors induced by a pathology. Finally, the studied time series are often multivariate (several sensors are used at the same time, for example one IMU per foot). Multivariate time series can be studied by considering each channel as a univariate time series [BTM+22], but studying the correlations between channels can be crucial in some cases as in the case of electroencephalograms (EEGs), in order to take into account the highly interconnected nature of brain regions.

1.1.2 Motivations

Topological data analysis (TDA), and more specifically persistent homology, provides relevant tools to study medical time series. The idea is to use the time series to build a sequence of *simplicial complexes* (a *filtration*) and to study its topological structure at various scales. The *persistence diagram* sums up the evolution of this structure through all scales. Several studies [EGK14, PH15, RMC23] have already shown that persistence diagrams can detect periodicity in time series. Moreover, persistence diagrams can be compared using distances that benefit from *stability theorems* [CDSO14] which guarantee a certain robustness to noise. This makes these distances relevant to compare time series. Thus, our initial intuition for this thesis was that **when using a suitable filtration for a given problem, the persistence diagram will reflect interesting structural features of the time series**, which can be easily identified visually or algorithmically to perform inter-individual comparisons or longitudinal follow-up.

The specificity of the research carried out for this thesis is that we used persistent homology to define representations of time series that can be interpreted on their own rather than used as additional features in a machine learning algorithm. The interpretation is based on theoretical properties (additivity for persistence diagrams, stability) and/or on models (periodic functions [BTM+22], convolutional dictionary learning [BTO24b])

of the studied times series.

The first motivation of our work was to **build bridges between the topological data analysis (TDA) and signal processing communities by developing new topological methods for time series analysis**. These methods should be based on interpretable mathematical insights or models, and come with theoretical guarantees. The idea is that TDA can be used to build representations that reflect interesting events or properties of the time series. In this thesis, we reviewed all the articles we could find that applied persistent homology to time series and identified various types of methods for different types of problems and data. These methods (including the three ones developed in this thesis) can be used for other applications than the one they were designed for, or at least provide relevant features to machine learning algorithms.

The second motivation was the **analysis of medical time series, and in particular gait signals** for the study of neurodegenerative diseases such as multiple sclerosis. Indeed, gait is considered the most important source of disability in multiple sclerosis [HBR+08]. Moreover, although gait signals such as the one from Figure 1.1 can be more complex than purely periodic signals, walking is still a structured activity (a sequence of steps). We want to use persistent homology to detect this structure with interpretable, robust and unsupervised methods, with as few parameters as possible.

1.2 Contributions and outline

The manuscript is organized as follows.

1.2.1 Chapter II: Mathematical background

In Chapter II - Mathematical background, we introduce the mathematical objects and results used in the rest of the thesis. We start by giving notations and definitions of times series and related objects such as delay embeddings. Then, we present the construction of simplicial/singular homology and persistent homology as well as important objects and theorems.

The following four chapters contain our contributions and use the notions introduced in this chapter.

1.2.2 Chapter III: Literature review on persistent homology for time series

In Chapter III - Literature review on persistent homology for time series, we conduct a review of all methods in the literature that use persistent homology for (univariate of multivariate) time series analysis. Over the past ten years persistent homology has been increasingly used to analyze the structure and shape of various types of data including time series. We review 71 articles published between 2014 and 2024 in several domains of application such as biomedicine, industry and economics. We give an overview of the applications fields and tasks and propose a general framework to describe the main characteristics of all the methods. The framework consists in describing five key steps of each methods: the type time series, transformation of time series, filtration built on the transformed data, TDA tools used and analysis method, as illustrated on Figure 1.2 (in the case of a delay embedding-based method).

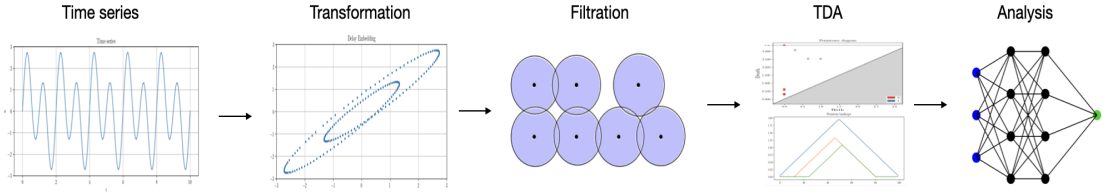


Figure 1.2: Illustration of a method decomposed into our framework.

1.2.3 Chapter IV: A method for gait signals analysis with an application to the study of multiple sclerosis

In Chapter IV - A method for gait signals analysis with an application to the study of multiple sclerosis, we describe a TDA-based method to compare pairs of gait signals, visualize the results and interpret them, which was applied to the study of healthy subjects and patients suffering from multiple sclerosis [BTM$^+$22]. Multiple sclerosis is a demyelinating disease of the central nervous system whose symptoms include lower limb impairment, which is why gait trials are used by clinicians for patient follow-up. To do so, standard gait features can be used but they fail to detect subtle changes in several pathologies, which motivates the development of new approaches.

The method is non-parametric and requires no data other than gait signals acquired with inertial measurement units. Tools from TDA are introduced in a practical way to make it as interpretable as possible in order to encourage its use by clinicians. The method can help estimate the severity of the disease and also be used for longitudinal follow-up to detect its evolution or other phenomena such as asymmetry or outliers.

We also studied a theoretical property of persistent homology of sublevel sets of periodic functions, which links the number of observed periods to the multiplicity of bars in the persistence barcode: we showed that the persistence diagram obtained from the sublevel set filtration of a T-periodic function on $[0, kT]$ (where k is an integer greater than 1) can be deduced from the diagram corresponding to the function on $[0, 2T]$, which implies in particular that there is a group of $k-1$ or k identical bars for each local minimum of the function on $[0, T]$. This justifies an important step of the method, where a link is made between number of steps and number of long bars in the barcode.

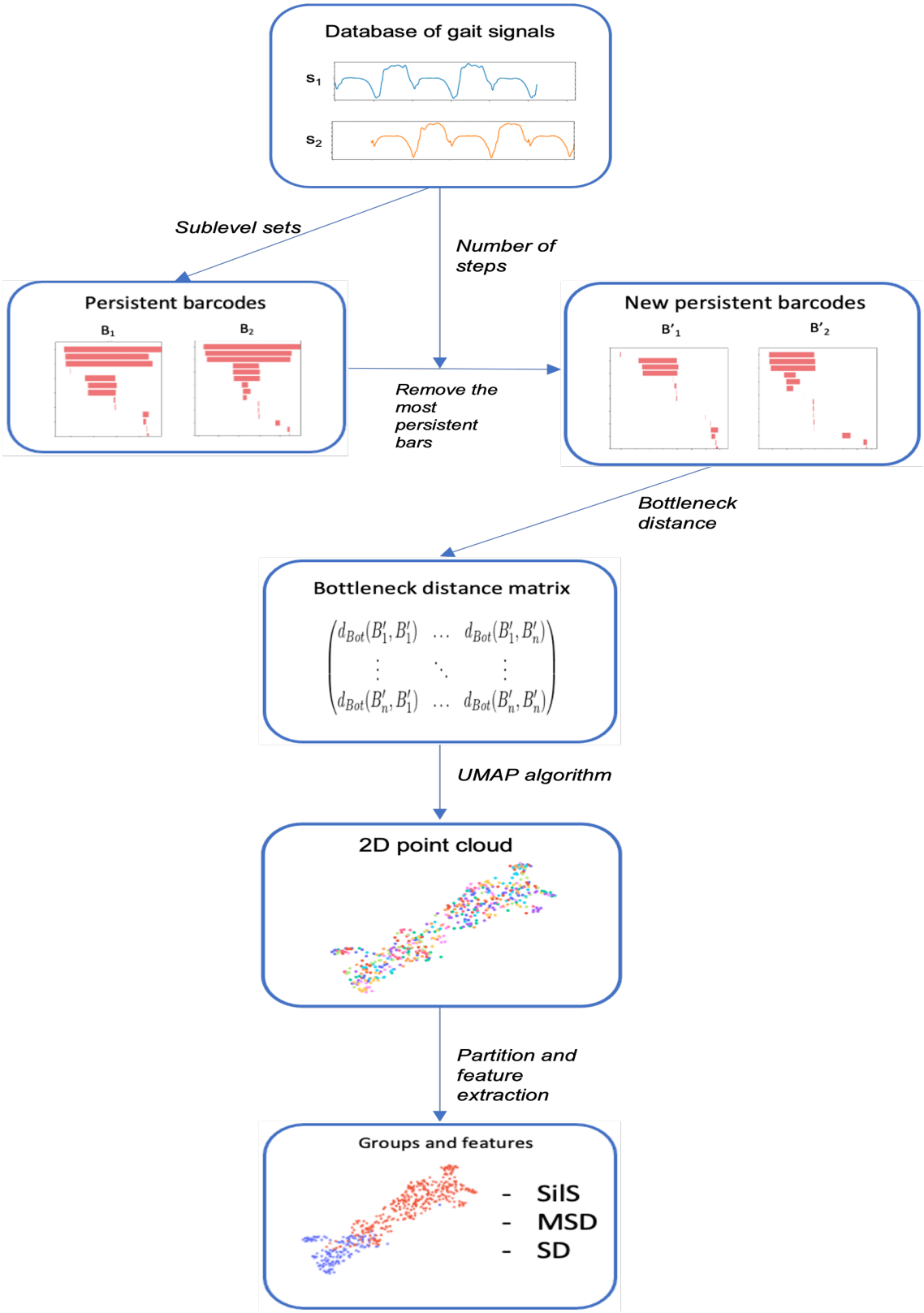


Figure 1.3: Illustration of our method for gait signals analysis.

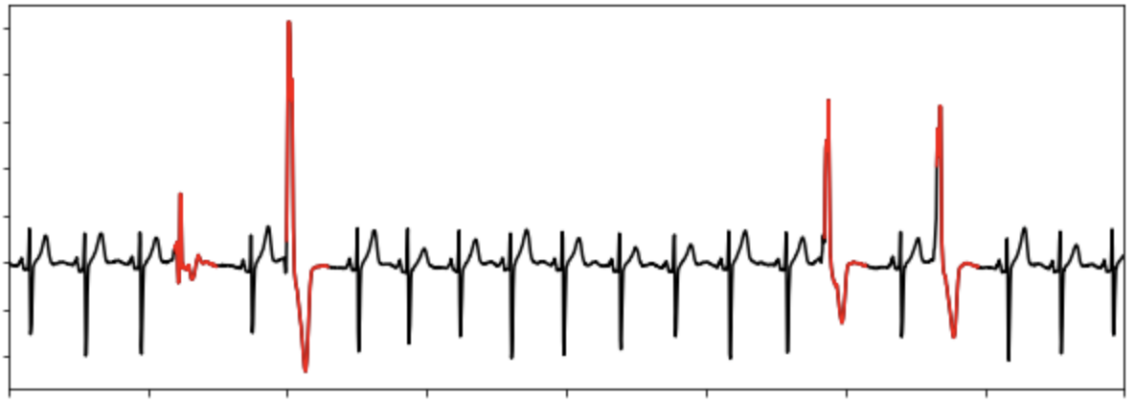


Figure 1.4: Illustration of an electrocardiogram time series with anomalies (in red) due to ventricular premature contractions.

1.2.4 Chapter V: Persistence-based clustering with outlier-removing filtration

In Chapter V - Persistence-based clustering with outlier-removing filtration, we describe a non-parametric hierarchical clustering algorithm with an outlier removal step, based on topological data analysis. We use a new filtration on metric spaces called NNVR (Nearest Neighbor Vietoris-Rips), which is a variant of the Vietoris-Rips filtration that adds information about the points' nearest neighbor to the persistence diagram. By reading the persistence diagram, we can then exclude isolated points and choose a suitable threshold for hierarchical clustering. We prove that our filtration has the same stability property as the Vietoris-Rips filtration with respect to the Gromov-Hausdorff distance, describe the algorithm and show that it can perform well on point cloud and graph datasets. Finally, we illustrate its usefulness in a pattern detection algorithm applied to gait signals.

1.2.5 Chapter VI: Unsupervised anomaly detection in time series

In Chapter VI - Unsupervised anomaly detection in time series, we propose a new TDA-based algorithm for unsupervised anomaly detection in univariate time series. Anomaly detection in time series is an important problem in data science and has application in many fields including biomedicine (see Figure 1.4 for an example). The algorithm relies on delay embeddings (transformations of time series into point clouds) and on the extraction of persistent cycles from the 1-dimensional persistent homology constructed from the distance to measure Rips filtration. This filtration makes it possible to identify 1-cycles (i.e. loops) corresponding to recurrent patterns by leveraging density information. Points in those cycles are considered as normal, and the algorithm can then assign an anomaly score to any point which is its distance to the normal set.

This method is based on a time series model composed of a succession of patterns, which we describe formally at the beginning of this chapter, both in the continuous and discrete case. This model enables us to propose a rigorous definition of an anomaly detection problem and to study our method theoretically. More precisely, we consider the ideal case where our data is a continuous function with no noise and no anomalies and the case where the data is a discrete, noisy time series with abnormal patterns. Under some regularity assumptions, we derive an upper bound on the interleaving distance between the associated persistence modules (and bottleneck distance between the corresponding persistence diagrams).

We show that our method is competitive with state-of-the-art anomaly detection methods on various datasets and study the influence of parameters and noise. Finally, we illustrate its use on gait signals.

1.3 List of papers

Journal papers:

- [BTM+22]: A. Bois, B. Tervil, A. Moreau, A. Vienne-Jumeau, D. Ricard and L. Oudre. "A topological data analysis-based method for gait signals with an application to the study of multiple sclerosis". In Plos one, 17(5), e0268475, 2022.
- [BTO24a]: A. Bois, B. Tervil and L. Oudre. "Persistence-based clustering with outlier-removing filtration". In Frontiers in Applied Mathematics and Statistics, 10, 1260828, 2024.

Papers under review:

- [BTO24b]: A. Bois, B. Tervil and L. Oudre. "A persistent homology- based algorithm for unsupervised anomaly detection in time series". 2024.
- [BTO24c] A. Bois, B. Tervil and L. Oudre. "Persistent homology for time series: a review". 2024

Conference papers:

- A. Bois, B. Tervil and L. Oudre. "Approches topologiques pour l'analyse de signaux physiologiques". In Proceedings of the Groupe de Recherche et d'Etudes en Traitement du Signal et des Images (GRETSI), Nancy, France, 2022.
- T. Germain, A. Bois, C. Truong, B. Tervil and L. Oudre. "Détection non supervisée de motifs sur séries temporelles". In Proceedings of the Groupe de Recherche et d'Etudes en Traitement du Signal et des Images (GRETSI), Grenoble, France, 2023.
- A. Bois, B. Tervil and L. Oudre. "Topological data analysis for unsupervised anomaly detection in time series". In Proceedings of the 32nd European Signal Processing Conference, EUSIPCO, Lyon, France, 2024.

Chapter 2

Mathematical background

Contents

2.1 Signals, time series and delay embeddings **31**
2.1.1 Signals and time series 31
2.1.2 Delay embeddings 31
2.2 Algebraic topology background: simplicial and singular homology **32**
2.2.1 Simplicial complexes 32
2.2.2 Simplicial homology 33
2.2.3 Singular homology 35
2.3 Persistence **36**
2.3.1 Filtrations and persistence modules 36
2.3.2 Examples of filtrations of simplicial complexes 37
2.3.3 Persistent homology 39
2.3.4 Distances between persistence diagrams and stability 42

2.1 Signals, time series and delay embeddings

2.1.1 Signals and time series

A **univariate signal** is a continuous real-valued function $\mathbf{x} = (\mathbf{x}[t])_{t\in I}$, where $I \subset \mathbb{R}$ is an interval. Let $n > 1 \in \mathbb{N}$. A **multivariate signal** with m channels is a continuous function with values in $\mathbb{R}^m$: $\mathbf{x} = (\mathbf{x}[t])_{t\in I} = ((\mathbf{x}_1, \dots, \mathbf{x}_m)[t])_{t\in I}$. In what follows, unless specified otherwise, we will always consider that $I = [0, 1]$ and time series will always be univariate (except in one section of Chapter 3 dedicated to multivariate time series).

A **univariate time series** $\hat{\mathbf{x}} = (\hat{\mathbf{x}}[i])_{1\leq i\leq n}$ is a finite subsampling of size n of a given signal $\mathbf{x}$, where n is called the length of the time series. From now on, unless precised otherwise, sampling is assumed to be uniform (the time between two consecutive indices is always the same) so $\hat{\mathbf{x}} = (\hat{\mathbf{x}}[i])_{1\leq i\leq n}$, where $\hat{\mathbf{x}}[i] = \mathbf{x}\left[\frac{i-1}{n-1}\right]$. A subwindow of length $l < n$ of a time series $\hat{\mathbf{x}}$ is a sequence of the form $\hat{\mathbf{x}}[i : i + l - 1] = (\hat{\mathbf{x}}[i], \hat{\mathbf{x}}[i + 1], \dots, \hat{\mathbf{x}}[i + l - 1])$. This definition naturally extends to multivariate time series.

Remark 1. *We will only make the distinction between the term signal (for continuous functions) and time series (for finite subsamplings of signals) in Chapter 6 where it is important. Otherwise, we use the two terms equivalently.*

The **Pearson correlation coefficient** between two univariate time series $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ is

$$\rho_{\hat{\mathbf{x}}\hat{\mathbf{y}}} = \frac{\frac{1}{n}\sum_{i=1}^{n}(\hat{\mathbf{x}}[i] - \mathbb{E}(\hat{\mathbf{x}}))(\hat{\mathbf{y}}[i] - \mathbb{E}(\hat{\mathbf{y}}))}{\sigma_{\hat{\mathbf{x}}}\sigma_{\hat{\mathbf{y}}}}$$

where $\mathbb{E}(\hat{\mathbf{x}}) = \frac{1}{n}\sum_{i=1}^{n}\hat{\mathbf{x}}[i]$ is the mean of $\hat{\mathbf{x}}$ and $\sigma_{\hat{\mathbf{x}}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{\mathbf{x}}[i] - \mathbb{E}(\hat{\mathbf{x}}))^2}$ is its standard deviation.

2.1.2 Delay embeddings

A **delay embedding** is a way of transforming a signal (resp. a time series) into a curve (resp. a point cloud) of chosen dimension d. Delay embeddings come from the field of dynamical systems, with strong theoretical results [Tak81, SYC91] which guarantee that, when studying a dynamical system whose evolution is described by a function $f : M \mapsto M$ where M is a D-dimensional manifold, a delay embedding of dimension $d \geq 2D + 1$ can be used to reconstruct a manifold with the same topological properties as M. More generally, delay embeddings are a useful tool to study signals as each point represents variations of the signal on a given subwindow, so it gives information about the local shape of the signal.

Definition 1. *Using the above notations, we define the delay embedding of a signal $\mathbf{x}$ with dimension $d \geq 2$ and delay $\tau \in \mathbb{R}$ as the following curve in $\mathbb{R}^d$:*

$$X_{d,\tau} = (X_{d,\tau}[t])_{t\in[0,1-(d-1)\tau]} = ((\mathbf{x}[t], \mathbf{x}[t + \tau], \dots, \mathbf{x}[t + (d - 1)\tau]))_{t\in[0,1-(d-1)\tau]}.$$

The delay embedding of the time series $\hat{\mathbf{x}}$ with dimension $d \geq 2$ and delay $\hat{\tau} \in \mathbb{N}$ is the following point cloud in $\mathbb{R}^d$:

$$\begin{aligned}\hat{X}_{d,\hat{\tau}} &= (\hat{X}_{d,\hat{\tau}}[i])_{1\leq i\leq n-(d-1)\hat{\tau}} \\ &= ((\hat{\mathbf{x}}[i], \hat{\mathbf{x}}[i + \hat{\tau}], \dots, \hat{\mathbf{x}}[i + (d - 1)\hat{\tau}]))_{1\leq i\leq n-(d-1)\hat{\tau}} \\ &= \left(\left(\mathbf{x}\left[\frac{i}{n-1}\right], \mathbf{x}\left[\frac{i+\hat{\tau}}{n-1}\right], \dots, \mathbf{x}\left[\frac{i}{n-1} + (d-1)\frac{\hat{\tau}}{n-1}\right]\right)\right)_{0\leq i\leq n-1-(d-1)\hat{\tau}}.\end{aligned}$$

An example of delay embedding is shown on Figure 2.1.

Note that if $\tau = \frac{\hat{\tau}}{n-1}$ then $\hat{X}_{d,\hat{\tau}} \subset X_{d,\tau}$. In the rest of the thesis, we fix d and $\hat{\tau}$ for time series and use d and $\tau = \frac{\hat{\tau}}{n-1}$ for signals. When there is no ambiguity, we use X and $\hat{X}$ instead of $X_{d,\tau}$ and $\hat{X}_{d,\hat{\tau}}$.

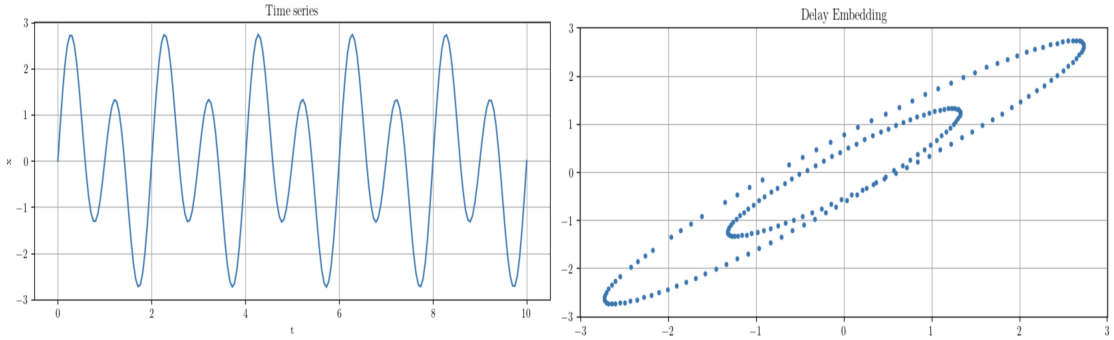


Figure 2.1: **Left:** A univariate 2-periodic time series: $\mathbf{x}[t] = \sin(\pi t) + 2\sin(2\pi t)$. **Right:** the corresponding delay embedding $\hat{X}_{d,\hat{\tau}}$ with $d = 2$ and $\hat{\tau} = 5$. The delay embedding is made of two concentric ellipses: the small one corresponds to the smallest oscillations of the signal (roughly between -1 and 1) and the large one corresponds to the largest oscillations (roughly between -3 and 3).

2.2 Algebraic topology background: simplicial and singular homology

Algebraic topology is a branch of mathematics, developed by H. Poincaré in the late 19th century in a series of papers about what he called *analysis situs* [Poi99]. The main idea behind algebraic topology is to associate to a given topological space so-called **topological invariants**, that is algebraic structures (for instance groups) that remain unchanged when the topological space is transformed into a topologically equivalent one (see Appendix A for precise definitions). Those invariants are compressed representations of the spaces: they summarize their global structure but they loose information since two topologically equivalent spaces have the same invariants, and therefore they cannot be distinguished (even though they might carry different additional information). Moreover two non-equivalent topological spaces might have the same topological invariants. In the following sections, we will define two important examples of invariants: homology groups and Betti numbers, which are the invariants we will use for TDA. We refer to [Hat00] for a complete introduction to algebraic topology.

2.2.1 Simplicial complexes

The main idea behind TDA is that data are a finite subset of samples of an underlying topological space. The question is now: can the invariants of this space be recovered from the data only ? Studying the finite point cloud data as a topological space actually gives no information except for the number of points as the only non-trivial topology on it is the discrete one, which only treats points individually. Thus, we need to impose some structure to the data for it to have a different topology. To do so, we can draw lines

between some pairs of points, then fill the surface of some triangles we may have created, fill the volume of tetrahedrons we may have created with the triangles, and generalize this in higher dimensions by adding elementary shapes called **simplices**. This is called creating a **simplicial complex** (formal definitions below). Simplicial complexes have a more complex topology than the discrete topology and their homology groups can be computed efficiently.

Remark 2. *In what follows, (X, d_X) will denote any metric space. In several applications, X will be a delay embedding $X_{d,\tau}$ which is why we use the same notation. In this case, unless specified otherwise, X is endowed with the Euclidean distance of $\mathbb{R}^d$.*

Definition 2 (Simplicial complexes)**.** *A k-simplex on a set $X \subset \mathbb{R}^d$ is an unordered tuple $\sigma = [x_0, ..., x_k]$ of $k+1$ distinct elements of X. The elements $x_0, ..., x_k$ are called the vertices of σ. If each vertex of a simplex ρ is also a vertex of σ, then ρ is called a face of σ. A simplicial complex K is a set of simplices such that any face of a simplex of K is a simplex of K.*

Two well-known complexes are the **Čech and Vietoris-Rips (or Rips, or VR) complexes**. In what follows, we let $X = \{x_1, \dots, x_n\}$ be a point cloud and $B_\alpha(x)$ be the open ball of radius α and centered at x.

Definition 3. *The Čech complex* $\mathsf{Cech}(X, \alpha)$ *is defined by:*

$$[x_{i_1}, \dots, x_{i_k}] \in \mathsf{Cech}(X, \alpha) \iff \bigcap_{j=1}^{k} B_\alpha(x_{i_j}) \neq \emptyset.$$

The nerve theorem, a result in algebraic topology, implies that if the balls $B_\alpha(x)$ have empty or contractible (i.e. homotopy equivalent to a point, see Appendix A for a definition) intersections, then $\mathsf{Cech}(X, \alpha)$ is homotopy equivalent to $\bigcup_x B_\alpha(x)$. This means that the simplicial complex carries the topology of the cover of X by open balls. The nerve theorem is also used to prove the theorem from [NSW08] which, as mentioned above, guarantees that there exists a scale such that $\mathsf{Cech}(X, \alpha)$ is homotopy equivalent to the topological space of which X is a sampling.

Definition 4. *The Rips complex* $\mathsf{Rips}(X, \alpha)$ *is defined by:*

$$[x_{i_1}, \dots, x_{i_k}] \in \mathsf{Rips}(X, \alpha) \iff \forall j, l,\ d(x_{i_j}, x_{i_l}) \leq \alpha$$

Figure 2.2 shows a cover of a point cloud by a union of balls with the same radius, and the corresponding Čech and Rips complexes.

2.2.2 Simplicial homology

Here, we introduce simplicial homology, which is a topological invariant that represents the number of "holes" of a simplicial complex in each dimension from 0 to $d-1$. For example, a 0-dimensional hole is a connected component, a 1-dimensional hole is a loop (this correspond to the intuitive definition) and a 2-dimensional hole is a cavity.

Let K be a simplicial complex with maximal simplex dimension d, $\mathbb{G}$ be an abelian group and $0 \leq k \leq d$.

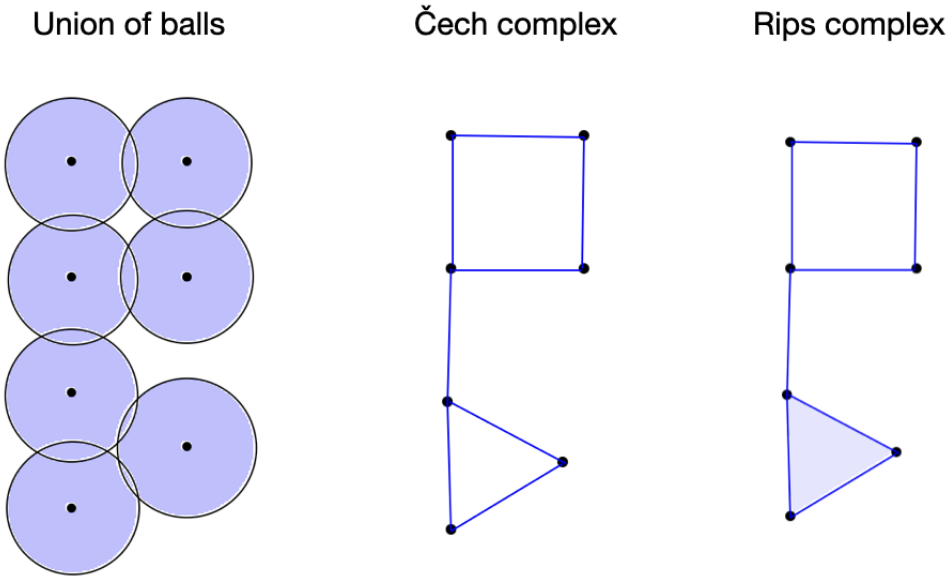


Figure 2.2: An example of cover of a point cloud by a union of balls, and the corresponding Čech and Rips complexes. The three lower balls have an empty intersection but each pair of balls intersect, so the triangle in the Rips complex is not present in the Čech complex.

Definition 5. *The space $C_k(K;\mathbb{G})$ of k-chains is defined as the set of formal sums of k-simplices of K with coefficients in $\mathbb{G}$, that is to say, if all the k-simplices of K are $\sigma_1,\dots,\sigma_{n_k}$, all the elements of the form:*

$$c = \sum_{i=1}^{n_k} a_i\sigma_i, \quad a_i \in \mathbb{G}.$$

The set $C_k(K)$ is abelian group whose addition is naturally defined: if $c = \sum_{i=1}^{n_k} a_i\sigma_i$ and $c' = \sum_{i=1}^{n_k} a'_i\sigma_i$ then:

$$c + c' = \sum_{i=1}^{n_k}(a_i + a'_i)\sigma_i.$$

Using a group $\mathbb{G}$ gives the more general definition, but from now on, we will only consider $\mathbb{G} = \mathbb{Z}/2\mathbb{Z}$, so the coefficients are modulo 2, which allows to avoid orientation considerations. In this case, $C_k(K)$ is a vector space with scalar multiplication defined as

$$\lambda c = \sum_{i=1}^{n_k}(\lambda a_i)\sigma_i \quad \text{for} \quad \lambda \in \mathbb{Z}/2\mathbb{Z}.$$

Definition 6. *Let $\sigma = [v_1,\dots,v_k]$ be a k-simplex with vertices $v_1,\dots,v_k$, and $[v_1,\dots,\hat{v_i},\dots,v_k]$ be the $(k-1)$-simplex spanned by those points minus v_i. The boundary operator ∂ is defined as:*

$$\partial : \begin{cases} C_k(K) & \longrightarrow & C_{k-1}(K) \\ \sigma & \longmapsto & \partial\sigma = \sum_{i=1}^{k}(-1)^i[v_1,\dots,\hat{v_i},\dots,v_k]. \end{cases}$$

We have the following sequence of linear maps:

$$C_d(K) \xrightarrow{\partial} C_{d-1}(K) \xrightarrow{\partial} \dots \xrightarrow{\partial} C_1(K) \xrightarrow{\partial} C_0(K) \xrightarrow{\partial} \{0\}.$$

They satisfy $\partial \circ \partial = 0$: we call such a sequence of maps a chain complex. This constitutes the setup for homology. We can now define cycles and boundaries, homology groups and Betti numbers.

Definition 7. *We define the set $Z_k(K)$ of k-cycles of K as :*

$$Z_k(K) = \mathsf{Ker}(\partial : C_k(K) \to C_{k-1}(K))$$

and the set $B_k(K)$ of k-boundaries of K as :

$$B_k(K) = \mathsf{Im}(\partial : C_{k+1}(K) \to C_k(K)).$$

We have $B_k(K) \subset Z_k(K) \subset C_k(K)$ [BCY18] so we can define the kth homology group as:

$$H_k(K) = Z_k(K)/B_k(K)$$

and the kth Betti number:

$$\beta_k(K) = \mathsf{dim}(H_k(K)).$$

A k-cycle is a sum of k-simplices. A k-boundary is a k-cycle made of all the faces of some $k+1$-simplices in K. A k-dimensional hole is an element of the homology group $H_k(K)$, so it is the homology class of a cycle that is not a boundary, β_k represents the number of k-dimensional holes. For example, β_0 is the number of connected components of K, β_1 is the number of loops, β_2 is the number of voids. On the examples from Figure 2.2, the four highest edges and the three lowest ones form 1-cycles of the Čech and Rips complex. In the Rips complex, the lowest one is the boundary of a triangle (notice that it is not the case in the Čech complex). Thus, the Betti numbers of the Čech complex are $\beta_0 = 1$ and $\beta_1 = 2$, and for the Rips complex $\beta_0 = 1$ and $\beta_1 = 1$.

2.2.3 Singular homology

Singular homology extends simplicial homology to a larger class of topological spaces by considering the continuous maps from simplices to any topological space. In practice, data are a discrete space so we use simplicial homology, but in Chapter 6 we model time series as a finite sampling of signals, which implies that delay embeddings of time series are discrete samplings of curves in $\mathbb{R}^d$. Singular homology enables us to study these curves and compare them to point clouds studied in practice.

Definition 8 (Standard simplex)**.** *Let $k \in \mathbb{N}$ and let $(e_0, \dots, e_k)$ be the canonical basis of $\mathbb{R}^{k+1}$. The standard k-simplex Δ^k is the convex hull of $e_0, \dots, e_k$. We use the following notations:*

$$\Delta^k = [e_0, \dots, e_k] = \left\{ \sum_{i=0}^{k} \alpha_i e_i \ : \ \sum_{i=0}^{k} \alpha_i = 1, \ \alpha_i \geq 0 \right\}.$$

Let X be any topological space. A **singular k-simplex** is a continuous map $\sigma : \Delta^k \to X$. For example, a singular 0-simplex is a map $\{1\} \to X$ so there are as many 0-simplices as there are points in X, and a singular 1-simplex is a map from the line segment Δ^1 to X, so there are as many 1-simplices as there are continuous paths in X. Note that in many cases, the set of singular simplices is uncountable so singular homology is not computable by an algorithm.

The space of singular k-chains $C_k(X, \mathbb{G})$ is defined the same way as simplicial k-chains and for $k \geq 1$, the boundary operator ∂ is defined on any singular k-simplex σ by:

$$\partial\sigma = \sum_{i=0}^{k} (-1)^i \sigma|_{[e_0,\ldots,\hat{e_i},\ldots e_k]}$$

where $\sigma|_{[e_0,\ldots,\hat{e_i},\ldots e_k]}$ is the restriction of σ to the face $[e_0, \ldots, \hat{e_i}, \ldots e_k]$ of Δ^k (the convex hull of all the e_j except e_i), and where $[e_0, \ldots, \hat{e_i}, \ldots e_k]$ is canonically identified to Δ^{k-1} by preserving the order of vertices so that $\partial\sigma$ is well defined on Δ^{k-1} [Hat00]. Cycles, boundaries, homology groups and Betti number are then defined the same way as for simplicial homology. For example, singular 1-cycles are closed curves in X.

An important result in algebraic topology [Hat00] states that if two topological spaces are homotopy equivalent (in particular, it is the case if they are homeomorphic, see Appendix A for definitions) then their homology groups are isomorphic.

2.3 Persistence

Now that we have defined simplicial complexes, the question is: is there a way to construct a simplicial complex on our data whose invariants are the same as those of the underlying space ? The answer is yes, if we choose the right scale: theoretical results from algebraic topology [NSW08] guarantee that if the sampling is dense enough, there exist scales α for which the Čech complex $\text{Cech}(X, \alpha)$ is homotopy equivalent to the underlying topological space and thus they have the same invariants. However, there is no way to determine the right scale from the data only. This is where the central notion of **persistence** (first introduced in [ELZ02] and [CZCG05]) comes into play: we should look at all the possible scales, and look at the evolution of the invariants when α increases from 0 to $+\infty$ (the obtained nested family of topological spaces or simplicial complexes is called a **filtration**). Features that are present for long intervals of scales brings information about the topology of the underlying topological space.

2.3.1 Filtrations and persistence modules

Definition 9 (Filtrations and interleavings)**.** *A filtration is a family $V = (V_\alpha)_{\alpha\in\mathbb{R}^+}$ of topological spaces or simplicial complexes such that $\alpha \leq \beta \Rightarrow V_\alpha \subset V_\beta$. For $\epsilon \geq 0$, two filtrations V and W are called ϵ-interleaved if for all $\alpha \geq 0$, $V_\alpha \subset W_{\alpha+\epsilon}$ and $W_\alpha \subset V_{\alpha+\epsilon}$. The interleaving pseudo-distance is then defined as*

$$d_i(V, W) = \inf\{\epsilon \geq 0 | \textit{V and W are } \epsilon\textit{-interleaved}\}.$$

In the case of a filtration of simplicial complexes over a finite set, only a finite number of values of α correspond to a strict inclusion in the filtration. We call **filtration value** of a simplex σ the lowest α such that $\sigma \in V_\alpha$. See Figure 2.3 (a-d) for an example of filtration.

Definition 10 (Persistence modules)**.** *Let K be a field. A persistence module is a family $\mathbb{V} = (\mathbb{V}_b, (v_a^b)_{0\leq a\leq b})$ where the $\mathbb{V}_b$ are K-vector spaces and the v_a^b are linear maps $\mathbb{V}_a \to \mathbb{V}_b$ such that for all real numbers $a \leq b \leq c$, $v_a^a = Id$ and $v_a^b \circ v_b^c = v_a^c$.*

Definition 11 (Morphisms and interleavings of persistence modules)**.** *Let $\epsilon \geq 0$. An ϵ-morphism between two persistence modules $\mathbb{V}$ and $\mathbb{W}$ is a family of linear maps $\phi = (\phi_\alpha : \mathbb{V}_\alpha \to \mathbb{W}_{\alpha+\epsilon})_{\alpha\in\mathbb{R}^+}$ such that the following diagram commutes for all $a \leq b$:*

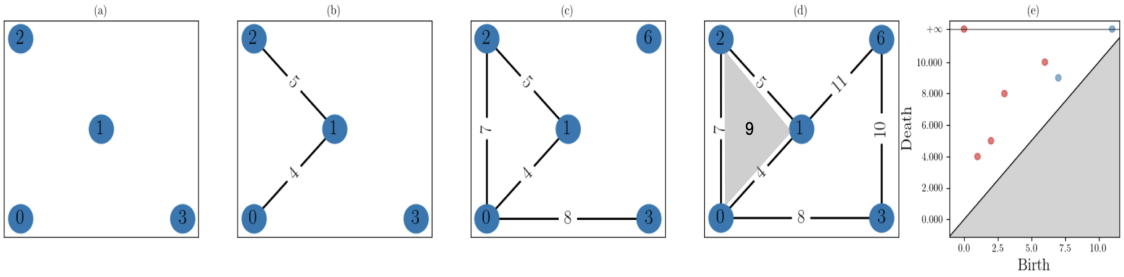


Figure 2.3: A filtration and the corresponding persistence diagram. Vertices are added at steps 0, 1, 2, 3 and 6, edges are added at steps 4, 5, 7, 8 10 and 11, and a triangle is added at step 9. **(a)**: steps 0 to 3. **(b)**: steps 4 and 5. **(c)**: steps 6, 7, 8. **(d)**: steps 9, 10, 11. **(e)**: Persistence diagram in dimension 0 (red) and 1 (blue).

$$\begin{array}{ccc} \mathbb{V}_a & \xrightarrow{v_a^b} & \mathbb{V}_b \\ \downarrow{\scriptstyle \phi_a} & & \downarrow{\scriptstyle \phi_b} \\ \mathbb{W}_{a+\epsilon} & \xrightarrow{v_{a+\epsilon}^{b+\epsilon}} & \mathbb{W}_{b+\epsilon} \end{array} .$$

If $\epsilon = 0$ and if all the ϕ_α are isomorphisms, ϕ is called an isomorphism of persistence modules.

An ϵ-interleaving between two persistence modules $\mathbb{V}$ and $\mathbb{W}$ is a pair (ϕ, ψ) of morphisms such that the following diagrams commute for all $a \in \mathbb{R}$:

$$\begin{array}{ccc} \mathbb{V}_a & \xrightarrow{v_a^{a+2\epsilon}} & \mathbb{V}_{a+2\epsilon} \\ & {\scriptstyle \phi_a} \searrow \quad \nearrow {\scriptstyle \psi_{a+\epsilon}} & \\ & \mathbb{W}_{a+\epsilon} & \end{array} \qquad \begin{array}{ccc} & \mathbb{V}_{a+\epsilon} & \\ {\scriptstyle \psi_a} \nearrow & & \searrow {\scriptstyle \phi_{a+\epsilon}} \\ \mathbb{W}_a & \xrightarrow{w_a^{a+2\epsilon}} & \mathbb{W}_{a+2\epsilon} \end{array} .$$

The interleaving pseudo-distance is then defined as

$$d_i(\mathbb{V}, \mathbb{W}) = \inf\{\epsilon \geq 0 \quad | \quad \mathbb{V} \text{ and } \mathbb{W} \text{ are } \epsilon\text{-interleaved}\}.$$

2.3.2 Examples of filtrations of simplicial complexes

In this section, we introduce filtrations of simplicial complexes that are important in the next chapters.

Let $f : X \to \mathbb{R}$ be a real-valued function defined on some set $X \subset \mathbb{R}^d$. Inspired by Morse theory [Mil63], a natural way of constructing a filtration on X is to consider the sublevel sets $f^{-1}((-\infty, \alpha])$ for increasing values of $\alpha \in \mathbb{R}$. The d-dimensional simplicial complex SL_α is then defined by:

$$\sigma \in SL_\alpha \iff \forall v \in \sigma,\ v \in f^{-1}((-\infty, \alpha]).$$

And the **sublevel set filtration** is $(SL_\alpha)_{\alpha \geq 0}$.

The definitions of **Rips and Čech filtrations** follow from the above definitions of the Rips and Čech complexes: $\text{Rips}(X) = \text{Rips}(X, \alpha)_{\alpha \geq 0}$ and $\text{Cech}(X) = \text{Cech}(X, \alpha)_{\alpha \geq 0}$. The Rips filtration is easier to compute than the Čech filtration because only distances are

needed instead of intersections of balls.

Let us now define **DTM filtrations**, which are generalizations of Rips and Čech filtrations introduced by the authors of [ACG+20] that play a central role in Chapter 6. Let $X \subset \mathbb{R}^d$, $f : X \to \mathbb{R}^+$ and $p \geq 1$ be a real number. For $x \in X$ and $\alpha \in \mathbb{R}^+$, we define $r_x(\alpha)$ as:

$$r_x(\alpha) = \begin{cases} -\infty & \text{if } \alpha < f(x) \\ (\alpha^p - f(x)^p)^{\frac{1}{p}} & \text{otherwise} \end{cases}$$

or $r_x(\alpha) = \alpha$ for $p = \infty$. Now, let us denote by $\bar{B}_f(x, \alpha)$ the closed ball $\bar{B}(x, r_x(\alpha))$ (by convention, a ball is empty if its radius is $-\infty$). We can now define the **weighted Čech filtration** as a union of growing balls.

Definition 12. *With the above notations, the weighted Čech filtration with parameters* (X, f, p)*,* $\mathsf{Cech}[X, f, p]$*, is defined by:*

$$\mathsf{Cech}[X, f, p]_\alpha = \bigcup_{x \in X} \bar{B}_f(x, \alpha).$$

Let $\mathcal{N}(\mathsf{Cech}[X, f, p]) = (\mathcal{N}(\mathsf{Cech}[X, f, p]_\alpha))_{\alpha \geq 0}$ denote the nerve of the above cover of X by closed Euclidean balls, i.e. the filtration of simplicial complexes whose $(i-1)$-simplices (with $i - 1 \leq d$) are defined by

$$[x_1, \ldots, x_i] \in \mathcal{N}(\mathsf{Cech}[X, f, p]_\alpha) \iff \bigcap_{j=1}^{i} \bar{B}_f(x_j, \alpha) \neq \emptyset.$$

The persistent nerve theorem ([CO08], Lemma 3.4) states that the persistence (singular) homology module associated to the weighted Čech filtration is isomorphic to the persistence (simplicial) homology module associated to its nerve. The latter is computable in practice [EH10, BCY18].

Definition 13. *With the above notations, the weighted Vietoris-Rips filtration with parameters* (X, f, p)*,* $\mathsf{Rips}[X, f, p]$ *is the flag complex of* $\mathcal{N}(\mathsf{Cech}[X, f, p])$*, i.e. the filtered simplicial complex such that each vertex (or 0-simplex)* $x \in X$ *has filtration value* $f(x)$*, and such that for* $2 \leq i \leq d + 1$*:*

$$[x_1, \ldots, x_i] \in \mathsf{Rips}[X, f, p]_\alpha \iff \forall (j, k),\ \bar{B}_f(x_j, \alpha) \cap \bar{B}_f(x_k, \alpha) \neq \emptyset.$$

The Vietoris-Rips complex is easier to compute than the Čech complex in practice because one only needs to know the filtration values of the 0 and 1 simplices to characterize the whole filtration. The following proposition gives this value in the case $p = 1$, which we will use in this paper (see [ACG+20] for values when $p = 2$ or $p = \infty$).

Proposition 1 ([ACG+20])**.** *Let* $x, y \in X$*. The filtration value of* $[x]$ *in* $\mathsf{Rips}[X, f, 1]$ *is* $f(x)$*, and the filtration value of* $[x, y]$ *in* $\mathsf{Rips}[X, f, 1]$ *is:*

$$\begin{cases} \max(f(x), f(y)) & \text{if } ||x - y|| < |f(x) - f(y)| \\ \frac{||x-y|| + f(x) + f(y)}{2} & \text{otherwise.} \end{cases}$$

In the rest of this paper, the function f will always be a **distance to measure function**. Those functions were introduced in [ACG+20] to make weighted filtrations robust to outliers.

Definition 14 (DTM). *Let μ be a probability measure on $\mathbb{R}^d$ and let $m \in [0, 1[$. Let $\delta_{\mu,m}$ be the function defined on $\mathbb{R}^d$ by $\delta_{\mu,m}(x) = \inf\left\{r > 0 \mid \mu(\bar{B}(x, r)) > m\right\}$. The distance to measure (DTM) μ with parameter m is the function defined on $\mathbb{R}^d$ by:*

$$d_{\mu,m}(x) = \sqrt{\frac{1}{m}\int_0^m \delta_{\mu,u}^2(x)du}.$$

If $\hat{X}$ is a finite subset of $\mathbb{R}^d$, we denote by $\mu_{\hat{X}}$ the empirical measure on $\hat{X}$. If $m = \frac{q}{\mathsf{Card}(\hat{X})}$ with $q \in \mathbb{N}$, then:

$$\forall x \in \mathbb{R}^d,\ d_{\mu_{\hat{X}},m}(x) = \sqrt{\frac{1}{q}\sum_{i=1}^{q} ||x - NN^{(i)}(x)||^2}$$

where $NN^{(i)}(x)$ is the ith nearest neighbor to x. The **DTM Čech and Vietoris-Rips filtrations** for a measure μ and a parameter m are defined as the filtrations $\mathsf{Cech}[X, d_{\mu,m}, p]$ and $\mathsf{Rips}[X, d_{\mu,m}, p]$.

See [BCY18, DSC04] for other examples of filtrations.

2.3.3 Persistent homology

Persistent homology consists in studying the persistence module obtained when considering the simplicial (or singular) homology groups of a filtered simplicial complex (or topological space). An essential property of persistent homology is that, under generic assumptions, the persistence module can be decomposed into a sum of elementary modules that describe the appearance and disappearance of homology classes in each dimension.

Formally, applying the (singular or simplicial) homology functor to a filtration and its inclusion maps gives a persistence module of homology groups $\mathbb{V} = (H_i(V_\alpha))_{\alpha\in\mathbb{R}^+}$ for each dimension $i \in [0, d]$. If for all $a \leq b$, v_a^b is of finite rank, then $\mathbb{V}$ is called **q-tame** and $\mathbb{V}$ can be decomposed into a direct sum of **interval modules** $\bigoplus_j \mathbb{I}_{[\alpha_{l(j)},\alpha_j)}$, where the $[\alpha_{l(j)}, \alpha_j)$ are left closed, right open intervals and each $\mathbb{I}_{[\alpha_{l(j)},\alpha_j)}$ is a basic persistence module that represent a feature that appears at index $\alpha_{l(j)}$ and disappears at index α_j (see [CDSGO12] for more details). In this case, the **persistence diagram** of $\mathbb{V}$ can be defined. The persistence diagram $\mathsf{Diag}(\mathbb{V})$ of a i-dimensional persistent homology module $\mathbb{V}$ is a multiset $\{(\alpha_{l(j)}, \alpha_j)\}$, such that a new homology class that appears in $H_i(V_{\alpha_{l(j)}})$ disappears in $H_i(V_{\alpha_j})$ ($\alpha_{l(j)}$ is called its birth date and α_j its death date, if the component never dies, $\alpha_j = \infty$). We call $\alpha_j - \alpha_{l(j)}$ the **persistence** of the point. By convention and to be able to define the **bottleneck distance**, we include every points (α, α) to the diagram, with infinite multiplicity.

Let us now describe persistent homology in the concrete case of filtrations of finite simplicial complexes. Let $(K^{\alpha_i})_{0\leq i\leq m}$ be a filtered finite simplicial complex such that for each index i, we go from K^{α_i} to $K^{\alpha_{i+1}}$ by adding a simplex σ_{i+1} to K^{α_i}.

We call $C_k^i, Z_k^i, B_k^i, H_k^i, \beta_k^i$ the respective spaces of k-chains, k-cycles, k-boundaries, k^{th} homology group and k^{th} Betti number of K^i. The goal is to follow the evolution of H_k^i as i increases. It can be shown ([BCY18]) that when a k-simplex σ_{i+1} ($k > 0$) is added, it either creates a new homology class in H_k^{i+1} (i.e. a new k-cycle that is independent of those of H_k^i) or it fills a $k-1$-dimensional hole of H_{k-1}^{i-1}, so H_{k-1}^i has one less homology class

than H_{k-1}^{i-1}, in that case we say that σ_{i+1} killed a homology class (by convention, we always consider that when two classes merge, the younger class gets killed). If $k = 0$, each new vertex creates a homology class in H_0.

The final result of persistent homology is the set of all so-called **persistent pairs** $(\sigma_{l(j)}, \sigma_j)$ such that for each j, $\sigma_{l(j)}$ creates a component and σ_j kills it. We say that the persistence (or lifetime) of such a pair is $j - l(j) - 1$. We refer the reader to Boissonnat et al. [BCY18] for the detailed description of the algorithm to compute persistent pairs. The k-dimensional **persistence diagram (PD)** is the set of points of coordinates $(\alpha_{l(j)}, \alpha_j)$ such that $\sigma_{l(j)}$ is a k-simplex (counted with multiplicity). The points of the diagonal $y = x$ are added with infinite multiplicity (it is useful to define distances). See Figure 2.3 (e) for an example of persistence diagram. A connected component is created each time a vertex is added, and edges 4, 5, 8 and 10 kill connected components, which gives the five red points on the 0D-persistence diagram. Edges 7 creates a 1-cycle that is killed by triangle 9 an edge 11 creates a 1-cycle that is never killed, which gives the two blue points on the 1D-persistence diagram.

Figure 2.4 shows the persistence diagram corresponding to the Rips filtration of the point cloud from Figure 2.1.

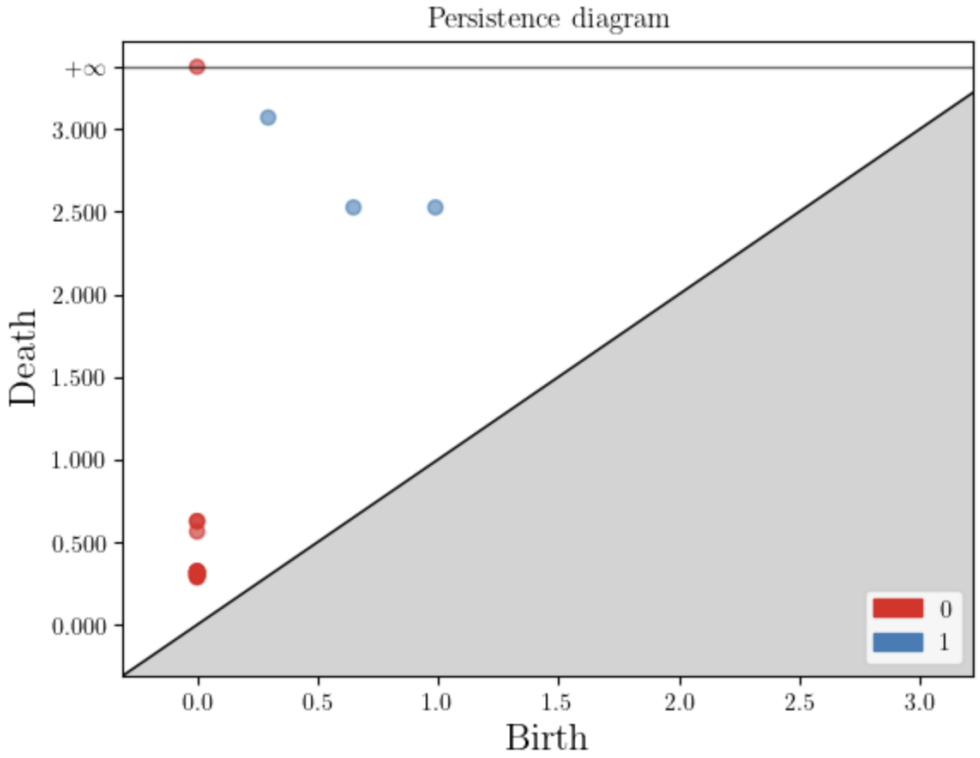


Figure 2.4: Persistence diagram (0D and 1D) of the Rips filtration of the point cloud from Figure 2.1.

The **persistence barcode** is a representation equivalent to the persistence diagram, where each pair $(\sigma_{l(j)}, \sigma_j)$ is represented as a line that starts at $l(j)$ and ends at (j). See Figure 2.5 for an example of persistence barcode.

From a persistence diagram of given dimension, one can obtain a representation called **persistence landscapes (PL)** [Bub15] by transforming the diagram into a set of

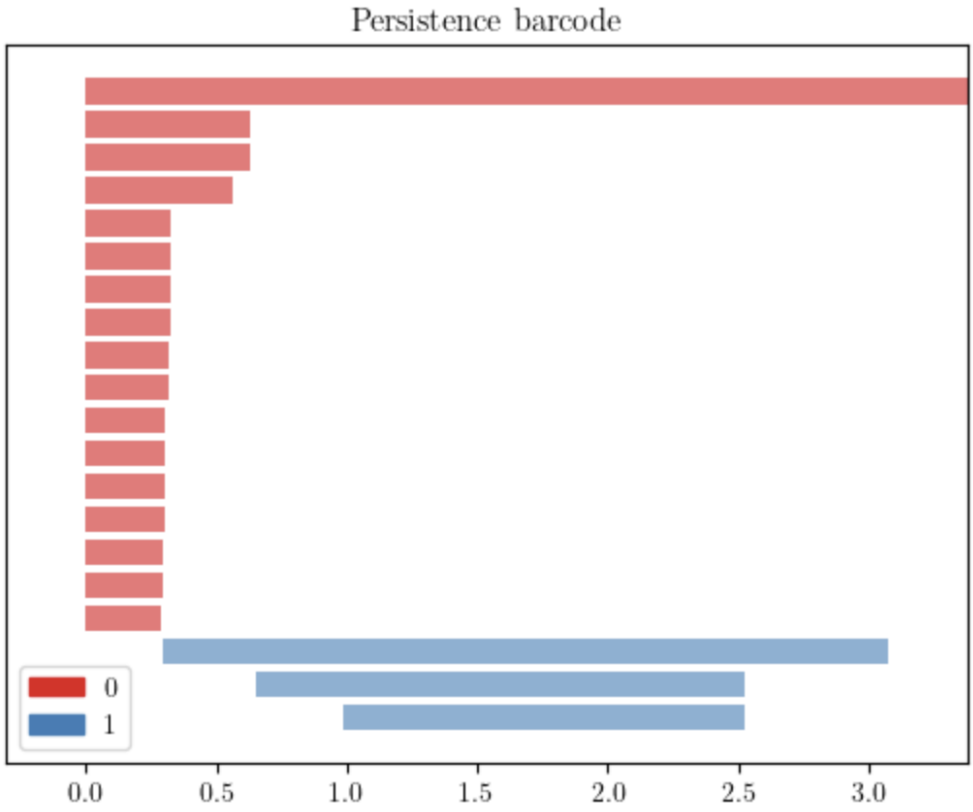


Figure 2.5: Persistence barcode (0D and 1D) of the Rips filtration of the point cloud from Figure 2.1.

piecewise affine functions (each one is called a landscape).

Definition 15. *For each point (b, d) in a persistence diagram D, we define the function $f_{b,d} : \mathbb{R} \longrightarrow [0, \infty]$ as:*

$$f_{b,d}(x) = \begin{cases} 0 & \textit{if} \quad x \notin [b, d] \\ x - b & \textit{if} \quad x \in [b, \frac{b+d}{2}] \\ d - x & \textit{if} \quad x \in [\frac{b+d}{2}, d] \end{cases} .$$

The k-th persistence landscape of D is the function:

$$\lambda_k : \begin{cases} \mathbb{R} & \longrightarrow & [0, \infty] \\ x - b & \longmapsto & \mathsf{kmax}(\{f_{b,d}(x)\}_{(b,d)\in D}) \end{cases}$$

where $\mathsf{kmax}(S)$ *is the k-th largest value of the set S.*

Landscapes represent persistence in a vector space of function, so L^p norms and the induced distances can be defined to compare landscapes, and statistics can be computed on landscapes (e.g. it is possible to average landscapes). Figure 2.6 shows an example of persistence landscape.

A representation of a persistence diagram in a Hilbert space is called a **vectorization** of the diagram. Several other vectorizations have been defined such as Betti curves, persistence images, silhouettes, paths, or persistence kernels. These representation are more suitable than persistence diagrams for analysis with usual machine learning methods. The **k-th Betti curve** of a filtered simplicial complex, introduced by Umeda [Ume17],

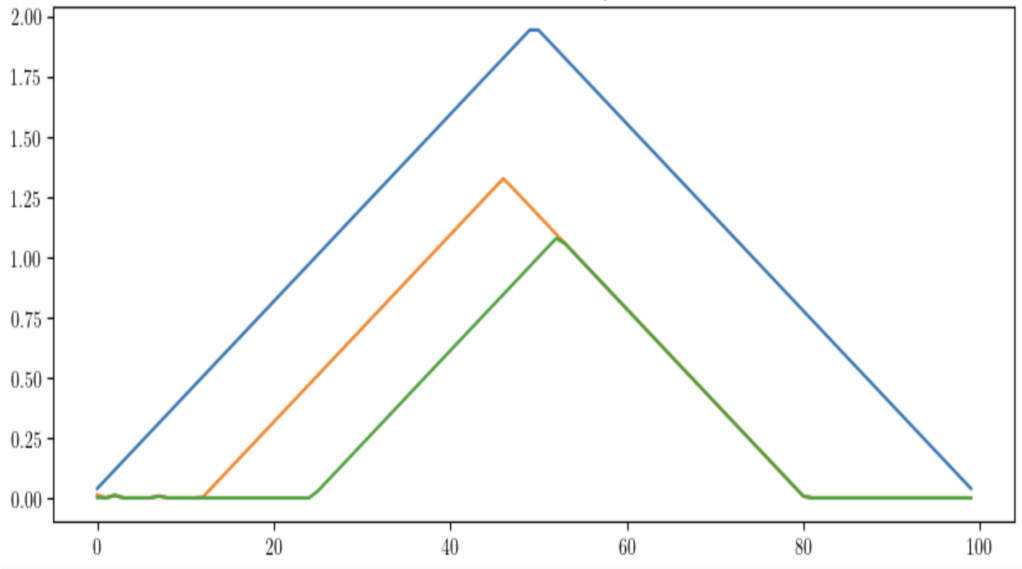


Figure 2.6: Fist three 1D persistence landscapes of the persistence diagram from Figure 2.4.

is the function $\alpha \mapsto \beta_k(K_\alpha)$. See [AAJ+23] for a survey on vectorization methods where the above vectorizations are defined, and [HMR21] for a survey on TDA for machine learning.

2.3.4 Distances between persistence diagrams and stability

A useful property of persistence diagrams is the fact that there exist distances that make it possible to compare them. This means that the topological features of two datasets can be compared through the distance between their persistence diagrams. Moreover, stability theorems [CSEH07, CSO08, CDSGO16, CDSO14] prove that under generic assumptions, close data have close diagrams. In other words, a small change in the data (some noise, for example) cannot lead to a major change in its persistent homology.

Definition 16. *The bottleneck distance between two persistence diagrams D and D' is defined as:*

$$d_b(D, D') = \inf_{\gamma \in \Gamma(D,D')} \sup_{(\alpha,\beta)\in D} ||(\alpha, \beta) - \gamma((\alpha, \beta))||_\infty$$

where $\Gamma(D, D')$ is the set of bijections from D to D', and where $||(\alpha, \beta) - (\alpha', \beta')||_\infty = |\alpha - \alpha'|$ when $\beta = \beta' = +\infty$.

It is necessary to include the diagonal in persistence diagrams so there exists bijections between diagrams that do not have the same number of persistent pairs.

Remark 3. *Note that if two filtrations are ϵ-interleaved, then the corresponding persistence modules are also ϵ-interleaved. Moreover, the isometry theorem [CDSGO12] states that $d_i(\mathbb{V}, \mathbb{W}) = d_b(\text{Diag}(\mathbb{V}), \text{Diag}(\mathbb{W}))$ for q-tame modules $\mathbb{V}$ and $\mathbb{W}$. We will implicitly use this in Chapter 6, as a bound on the interleaving distance between filtrations induces a bound on the bottleneck distance between persistence diagrams, which are the objects studied by our algorithm.*

Definition 17. *For a given norm $||.||_l$, the q-Wasserstein distance is defined as:*

$$d_{W_{q,l}}(D, D') = \inf_{\gamma \in \Gamma(D, D')} \left(\sum_{p \in D} ||p - \gamma(p)||_l^q \right)^{\frac{1}{q}}.$$

We conclude this section by giving three stability results for the bottleneck distance, related to Chapters 4, 5 and 6. See [ST20] for a study of stability of the Wasserstein distance between persistence diagrams.

The first theorem states the stability of the sublevel set filtration with respect to the infinity norm.

Theorem 1 ([BCY18])**.** *let D_f, D_g be the persistence diagrams associated to the sublevel sets of continuous real-valued functions f and g, then*

$$d_b(D_f, D_g) \leq ||f - g||_\infty.$$

The second theorem states the stability of the Rips and Čech filtrations with respect to the **Gromov-Hausdorff distance** d_{GH} [BBI22] (this distance will be defined and used in Chapter 5).

Theorem 2 ([CDSO14])**.** *Let (X, d_X) and (Y, d_Y) be totally bounded metric spaces. Let D_X and D_Y be the persistence diagrams of* Rips(X) *and* Rips(Y) *or* Cech(X) *and* Cech(Y)*. Then:*

$$d_b(D_X, D_Y) \leq 2 d_{GH}(X, Y).$$

The third result states the stability of DTM filtrations with respect to the **Hausdorff distance** between sets, defined as

$$d_H(X, Y) = \max \left(\sup_{x \in X} d(x, Y), \sup_{y \in Y} d(X, y) \right).$$

and the **Wasserstein distance with quadratic cost** between probability measures μ and ν , defined as

$$W_2(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \sqrt{\int_{\mathbb{R}^d \times \mathbb{R}^d} ||u - v||^2 d\pi(u, v)}$$

where $\Pi(\mu, \nu)$ is the set of transport plans between μ and ν (see [San15] for more details).

Proposition 2 ([ACG$^+$20], Prop. 4.3)**.** *Let $V[X, d_{\mu,m}, p]$ denote either a DTM Čech or DTM Vietoris-Rips filtration, and let μ_X and μ_Y be two probability measures on $\mathbb{R}^d$ with compact supports $X, Y \subset \mathbb{R}^d$. Then*

$$d_i(V[X, d_{\mu_X,m}, p], V[Y, d_{\mu_Y,m}, p]) \leq m^{-\frac{1}{2}} W_2(\mu_X, \mu_Y) + 2^{\frac{1}{p}} d_H(X, Y).$$

Note that a particular case of this result is the stability of the Rips and Čech filtrations with respect to the Hausdorff distance.

Appendix A
Topology background

Definition 18 (**Topological space**). *A topology on a set X is a family $\mathcal{T}$ of subsets of X such that:*

- $\emptyset \in \mathcal{T}$ *and* $X \in \mathcal{T}$
- *Any union of elements of $\mathcal{T}$ is an element of $\mathcal{T}$*
- *Any finite intersection of elements of $\mathcal{T}$ is an element of $\mathcal{T}$*

$(X, \mathcal{T})$ is called a topological spaces, the elements of $\mathcal{T}$ are called open sets, their complements in X are called closed sets.

Let $(X, \mathcal{T})$ and $(Y, \mathcal{T}')$ be topological spaces.

Definition 19 (**Continuous map**). *A map $f : X \to Y$ is continuous if and only if the pre-image of any open set of Y is an open set of X, or equivalently if the pre-image of any closed set of Y is a closed set of X.*

Definition 20 (**Homeomorpism**). *A continuous map $f : X \to Y$ is called a homeomorphism is it is bijective and if its inverse f^{-1} is also continuous. Two topological spaces are called homeomorphic if there exists a homeomorphism between them.*

Definition 21 (**Homotopy equivalence**). *Two maps $f, g : X \to Y$ are homotopic if there exists a continuous map $H : [0,1] \times X \to Y$ such that $H(0, \cdot) = f$ and $H(1, \cdot) = g$. Two topological spaces X and Y have the same homotopy type (or are homotopy equivalent) if there exist two continuous maps $f : X \to Y$ and $h : Y \to X$ such that $f \circ h$ is homotopic to the identity map in Y and $h \circ f$ is homotopic to the identity map in X.*

Algebraic topology studies topological spaces up to homeomorphism or up to homotopy equivalence by associating algebraic objects to topological spaces. A class of objects is called a topological (resp. homotopy) invariant if for all pairs of homeomorphic (resp. homotopy equivalent) spaces, their associated objects are identical, up to isomorphism in the case of algebraic structures. Note that a homotopy invariant is always a topological invariant.

Chapter 3

Literature review: persistent homology for time series

Over the last ten years persistent homology has been increasingly used to analyze the structure and shape of various types of data including time series. This chapter is a review of persistent homology applied to (univariate of multivariate) time series data. We review 71 articles which apply methods involving persistent homology to time series data, published between 2014 and 2024 in several domains of application such as biomedicine, industry and economics. We give an overview of the applications fields and tasks and propose a general framework to describe the main characteristics of all the methods.

Contents

3.1 Introduction **46**
3.2 Application fields and tasks **46**
3.3 Methods **48**
3.3.1 General framework 48
3.3.2 Sublevel sets methods 48
3.3.3 Delay embedding methods 53
3.3.4 Multivariate time series 61
3.4 Discussion and perspectives **66**

3.1 Introduction

Topological data analysis (TDA) is a research area that uses tools from algebraic topology to study the shape of data. Persistent homology [EH10, BCY18] is a popular tool of TDA which has had many applications in the last decades[CM21, SL22]. Recently, an increasing number of studies used persistent homology to analyze time series, in several domains of application such as biomedicine, industry and economics. Existing reviews on persistent homology for time series [SDB16, GZ18, RC19, Per19, XDR21, HLMMG24] either focus on one application (cardiology, neuroscience), do not mention certain important articles and methods or do not describe a general framework to identify different types of methods. Table 3.1 sums up the characteristics of existing reviews and ours. The "Type of methods" column indicates if a review deals with univariate time series only or univariate and multivariate, and if the reviewed methods focus on a specific type of data transformation (delay embedding, graph, other). The "Framework" column indicate if the review defines a general framework to decompose methods in several steps. The "Number of references" column refers to the number of cited articles that directly apply TDA to time series: articles introducing the fundamentals of TDA or applying other methods to time series are not counted.

Review	Year	Type of data	Type of methods	Framework	Number of references
Seversky et al. [SDB16]	2016	All	Delay embedding (univariate and multivariate data)	Yes	8
Gholizadeh and Zadrozny [GZ18]	2018	All	Delay embedding and graph (univariate and multivariate data)	No	22
Ravishanker and Chen[RC19]	2019	All	All types for univariate data	Yes	12
Perea [Per19]	2019	All	Delay embedding (univariate and multivariate data)	No	15
Xu et al. [XDR21]	2021	EEG	All	Yes	12
Hernández-Lemus et al. [HLMMG24]	2024	Cardiovascular	All	Yes	12
Ours	2024	All	All	Yes	71

Table 3.1: Comparison of existing literature reviews

This chapter is a review of all the articles we could find that apply persistent homology to time series analysis. Our aim is both to present the main trends in the field and to provide readers with an exhaustive list of possible ideas and methods. Note that topological methods other than persistent homology are also labeled as TDA, such as Mapper [SMC+07], UMAP [MHM18], simplicial complex signal processing [BS20], or sheaves-based approaches [Rob14]. The scope of this review is limited to persistent homology.

The chapter is organized as follows. Section 3.2 gives an overview of the application fields of persistent homology for time series. In Section 3.3, we define a general framework to describe the main characteristics of all the methods and review the 71 articles. In Section 3.4, we discuss the reviewed methods and identify some research perspectives.

3.2 Application fields and tasks

Persistent homology has been applied to time series from many fields. Figure 3.1 shows the proportion of articles belonging to each of the 10 categories we identified in the liter-

ature. Note that some have several applications and are counted multiple times in Figure 3.1. The vast majority has biomedical applications (53%). In particular, cardiology and neurology are by far the most represented applications, and neurophysiology (the study of neurological diseases through physiological time series such as motion sensor data) also represents 7% of the articles. This can be explained by the fact that biomedical signals are often very structured (periodicity for ECGs and motion signals, correlations for EEGs) which is suitable for TDA-based methods (more concrete details will be provided in section 3.3) and by the fact that biomedicine is an important field of research in data analysis in general. For example, persistent homology was used to improve ECG classification for arrythmia detection [Ume17, LWY23] or atrial fibrillation detection and classification [SG19, JXZZ22], and to define visualization methods based on fMRI data that enabled researcher to better understand changes in brain connectivity during motor learning [SHP17] or after consuming psychedelic mushrooms [PET+14].

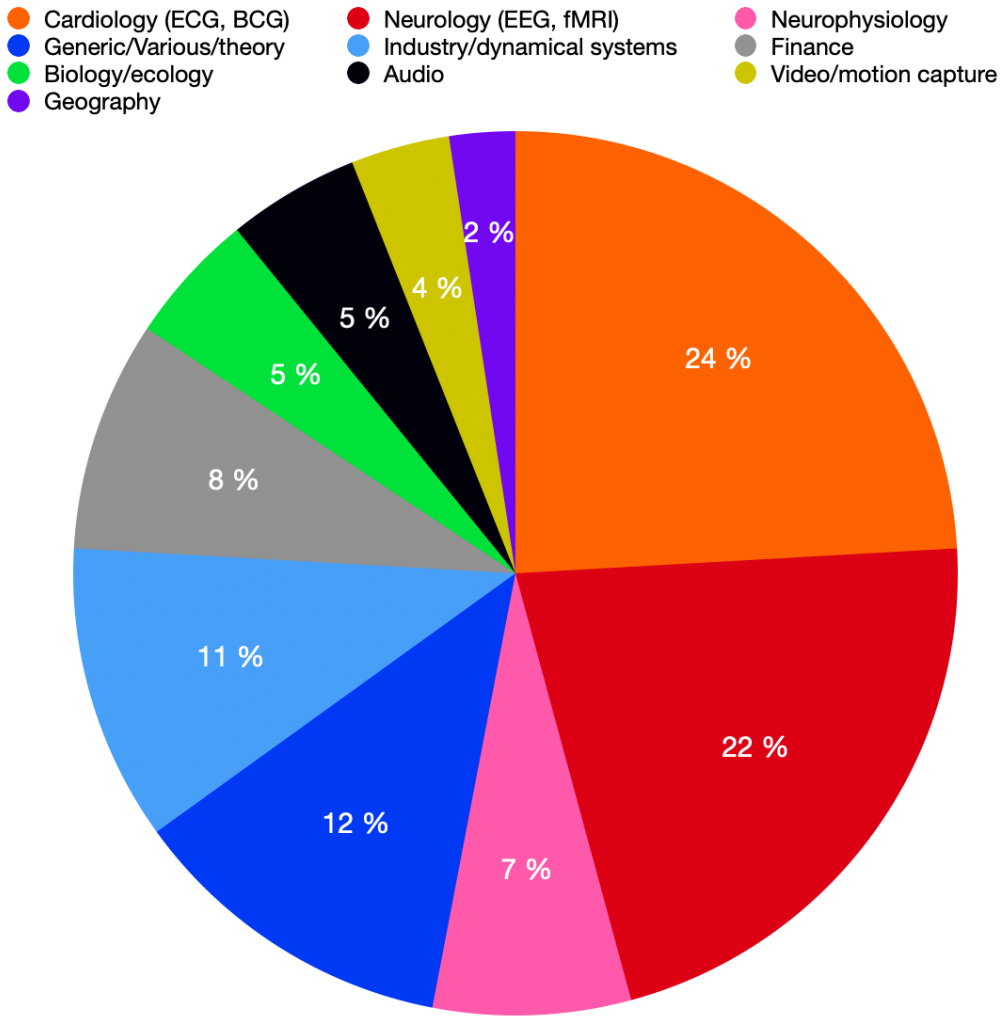


Figure 3.1: Application fields of the 71 reviewed articles.

Another important field is dynamical systems, often with industrial applications such as chatter detection in turning processes [KM16, KMP18, YKO22]. This is mainly due to the fact that data from this field are also very structured, with different regimes such as regular behavior and chaotic behavior. As explained in Section 3.3.3, delay embeddings

are a popular tool to study dynamical systems, and persistent homology can be useful to study their topology.

Persistent homology has also been applied to detect periodicity in video and motion capture data, classify or identify musical signals [SSM+17, RFD+23], detect changes points in financial time series [UIY22, Gid17, GK18, GGK+20], or classify/cluster time series from biology [CJ17], ecology [CR23] and geography [CZRK19]. Persistent homology is also a subject of mathematical study [PH15, Per16, GP24, RMC23].

The field of each article can be found in Tables 3.2, 3.3, 3.4.

Persistent homology was used to create new algorithms for generic tasks in time series analysis such as change point detection [UIY22, FBMG23], anomaly detection [BTO24b, CRL24], motif detection [GTO24b], time series forecasting [ZGHK21], periodicity quantification [DQR19] or zero-crossing detection [TKM23]. In concrete applications, other common machine learning or data analysis tasks are performed on features from persistent homology, such as time series classification using machine learning classifiers or deep learning, statistical analysis or data visualization. Note that most methods are generic and can be applied to other fields than the one they were introduced in.

3.3 Methods

This section is a review of the articles using persistent homology for time series analysis. We first introduce the typical pipeline that is common to all methods, then review methods for univariate and multivariate data. Two subsections are dedicated to univariate time series, corresponding to the two main categories of methods we distinguished: those using sublevel set filtrations and 0D persistent homology, and those that transform time series into a point cloud via a delay embedding before studying persistent homology in higher dimensions. A small number of methods do not exactly fit in those two categories, we describe them along with the most related methods. The last subsection is about multivariate time series.

3.3.1 General framework

Each method of our review can be decomposed using the following framework: time series $\rightarrow$ transformation $\rightarrow$ filtration $\rightarrow$ TDA tools $\rightarrow$ analysis. See Figure 3.2 for an illustration.

The input data is always a time series (univariate or multivariate). It can be transformed into another object on which a filtration is defined. Then, persistent homology can be studied with various TDA tools (persistence diagrams, vectorizations, homology representatives...) and then a specific analysis (statistics, visualization, machine learning...) is performed. Tables 3.2, 3.3, 3.4 list all the 71 methods reviewed here and decompose them in our framework. We also specify when articles include theoretical contributions such as new tools or mathematical results.

3.3.2 Sublevel sets methods

Sublevel set filtrations (defined in Chapter 2) are a way of directly applying persistent homology to graphs of functions. Figure 3.2 illustrates a typical framework using sublevel sets.

In the case of univariate time series $\mathbf{x} = (\mathbf{x}[i])_{1\leq i\leq n}$ (seen as a real-valued function on $\mathbb{N}$), $[i]$ is a vertex of SL_α if and only if $\mathbf{x}[i] \leq \alpha$ and $[i,j]$ is an edge of SL_α if and only if $\max(\mathbf{x}[i],\mathbf{x}[j]) \leq \alpha$. Concretely, the persistence diagram will be made of pairs

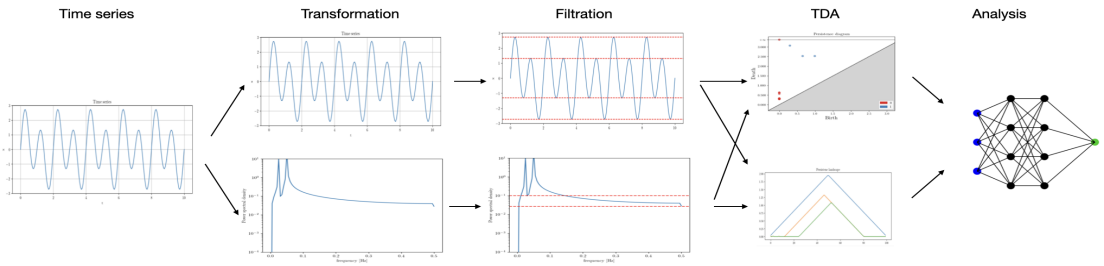


Figure 3.2: Typical framework of persistence homology for time series using sublevel sets.

(i,j) such that $\mathbf{x}[i]$ is a local minimum and $\mathbf{x}[j]$ is a local maximum. This filtration can be computed in $\mathcal{O}(n)$. Its persistent homology module has a stability property: let D_f, D_g be the persistence diagrams associated to the sublevel sets of f and g, then $d_b(D_f, D_g) \leq ||f - g||_\infty$ [BCY18]. It is also invariant under reparametrisation: the persistence module is the same for a sampling of $f : \mathbb{R} \to \mathbb{R}$ of $f \circ \gamma$, where $\gamma : \mathbb{R} \to \mathbb{R}$ is an increasing bijection.

Those properties were used in [BCMR22, RMC23] to define methods to study noisy reparameterized samplings of periodic functions. These articles also highlight the additivity property of the sublevel set filtration of periodic functions: the persistence diagram corresponding to the sublevel filtration of N periods is related to N times the persistence diagram corresponding to one period. In [BCMR22], the authors introduce an algorithm to count the number of periods and find an odometric sequence for reparametrised periodic signals, for an industrial application (magnetic odometry). They use the fact that, seeing a periodic function as a function on the circle, the diagram of sublevel sets of N periods is equal to N times the diagram of one period, so they count the multiplicity of points on the diagram. They show the correctness of their estimator under some assumptions. In [RMC23] topological signatures for periodic signals are defined (each time series is assigned a signature time series that should be independent of reparametrisations). They study the additivity property of diagrams of sublevel sets of periodic functions defined on any intervals, and prove convergence and stability results for their signatures. The additivity property is also leveraged in [BTM$^+$22] to study multiple sclerosis through motion sensor time series: barcodes from sublevel sets are truncated to reduce the impact of the number of steps, then the bottleneck distance is used in UMAP [MHM18] to globally analyze a cohort of healthy subjects and patients with different degree of severity of the disease.

Sublevel set filtrations have also been used on **truncated time series** (time series where points under or above a certain value are ignored) to create robust algorithms: [KM18] introduces an algorithm to count pulses in piecewise constant (binary) industrial signals in the presence of noise and "digital ringing" (spurious pulses very close to normal pulses) and with varying time between pulses. They define a model of their signals and use it to derive a theorem stating that with small noise, dense enough sampling and enough visible pulses, their algorithm counts the right number of pulses. In [TKM23], sublevel set filtrations are studied separately for the positive and negative values of the time series to find zero-crossings in noisy time series. They threshold both persistence diagrams and merge the remaining values to define intervals with potential zero crossings. Thresholding the diagrams makes the method robust by avoiding finding multiple crossings around each true one due to noisy oscillations.

Several studies, mostly with biomedical applications, use the sublevel set filtration with vectorizations of persistence diagrams used as inputs for machine learning or statis-

tics. This pipeline has been used on EEG time series to study aphasia [WBJF20, WBJ$^+$20] and epileptic seizures [WOC18, WOC19] using permutation-based statistical tests on persistence landscapes to detect statistically significant differences between time series before and after a seizure or coming from pathological and healthy subjects. More precisely, in [WOC18, WOC19], Fourier series approximations of the EEGs are computed and the test compares persistence landscapes obtained when exchanging Fourier coefficients between time series. The authors empirically show that the test they defined is invariant to translations, scaling and frequency scaling and claim that it detects topological changes. Also note that [WBJ$^+$20] introduces the **gradient filtration**, that generalises the sublevel set filtration by allowing level lines with a non zero slope. The authors of [MAMD20] and [PRTM18] use **persistent entropy** [CGGD$^+$15] as a feature for classifiers to analyze EEGs to respectively detect autism spectrum disorders and epileptic seizures. In [NOBM19], a Bayesian learning approach is developed with Poisson point processes as prior for the distribution of persistence diagrams, with an application to brain state classification from EEGs. Sublevel sets have also been used for heart rate variability detection in segmented ECG signals (i.e. signals where heartbeats are isolated) [GGP$^+$21], using various features describing the persistence diagram in a SVM classifier.

Deep learning architectures have also been designed to leverage properties of sublevel sets filtrations. Betti curves [DUC20] are fed to 1D convolutions in neural networks ([Ume17] claimed that 1D convolutions are suitable for Betti curves) and added to non-TDA features to improve arrhythmia detection and classification in ECG data. The authors of [ZGHK21] aimed to integrate local topological information on time series by using sublevel filtrations of subwindows and learnable barcode vectorizations [CCI$^+$20, HKN19, HKND19] to design a topological attention mechanism [VSP$^+$17] to improve state-of-the-art time series forecasting neural networks.

All the above methods perform a direct analysis of graphs of time series. Another approach consists in transforming the time series into a more relevant form such as converting it to the frequency domain as did the authors of [CZRK19, CR23], who then used the Euclidean distance between persistence landscapes from sublevel set filtrations into a k-means clustering algorithm. In [CZRK19], the **Walsh-Fourier Transform** (a frequency domain transform adapted to categorical time series) is applied to categorical time series (with few possible states) which transforms it into a real-valued time series that is more suitable for signal processing tools such as sublevel set filtrations. In [CR23], the filtration is applied to the smoothed **periodogram** (a Fourier-based transform that estimates the spectral density of a signal) to study environmental time series. A related approach that does not use the sublevel set filtration but the filtered cubical complex [KMM04] (a related filtration used for image processing) is described in [RFD$^+$23]. The authors filter the **spectrogram** (Fourier-based 2D transform of a time series) of subwindows and use the L^1 distance between Betti curves in 0 and 1D followed by a minimum-cost matching approach to perform audio signals identification, with empirical evidence of robustness to several types of signal obfuscations (noise, reverb, filtering, tempo/pitch shift).

Finally, another relevant method to directly filter a time series' graph is introduced in [Dlu22] to detect P,Q,S, or T-waves in ECG signals. The authors study the graph of the ECG (as a subset of $\mathbb{R}^2$) to which they add the x-axis to form loops (1-cycles) that are detected using the Rips filtration and volume-optimal representatives [Oba18]. Each cycle is identified to a kind of ECG wave according to criteria based on medical knowledge.

Table 3.2 sums up the characteristics of all the articles cited in this section.

Table 3.2: Articles from Section 3.3.2 and their characteristics.

Start of Table 3.2

Article	Year	Field	Type of time series	Transformation	Filtration	TDA tools	Method	Theoretical contribution
Khasawneh and Munch [KM18]	2018	Industry	Binary signals with noise	Truncated time series	Sublevel (x-axis)	Barcode	Count bars above a threshold	Model and guarantee of correctness under some assumptions
Wang et al. [WOC18]	2018	Biomedical	EEG	Fourier series approximation	Sublevel	PL	Permutation test	No
Piangerelli et al. [PRTM18]	2018	Biomedical	EEG	Filtering, downsampling	Sublevel (VR for 2nd step)	Barcode and persistent entropy (generators for 2nd step)	Linear classifier (histograms for 2nd step)	No
Wang et al. [WOC19]	2019	Biomedical	EEG	Fourier series approximation	Sublevel	PL	Permutation test	No
Nasrin et al. [NOBM19]	2019	Biomedical	EEG	None	Sublevel	PD	Bayesian learning	No
Chen et al. [CZRK19]	2019	Geographic	Categorical	Walsh-Fourier transform	Sublevel	PL	K-means clustering	No
Dindin et al. [DUC20]	2020	Biomedical	ECG	None	Sublevel	Betti curve	Deep learning	No
Wang et al. [WBJF20]	2020	Biomedical	EEG	Average over trials	Sublevel	PL	Permutation test	No
Wang et al. [WBJ$^+$20]	2020	Biomedical	EEG	Average over trials	Sublevel (gradient filtration)	PL	Permutation test	Introduction of the gradient filtration
Malumber et al. [MAMD20]	2020	Biomedical	EEG	None	Sublevel	Barcode and persistent entropy	SVM classifier	No
Zeng et al. [ZGHK21]	2021	Generic	Univariate	Subwindows	Sublevel	Barcode and learned vectorization	Deep learning	No
Graff et al. [GGP$^+$21]	2021	Biomedical	ECG	Segmentation	Sublevel	Features from PD	SVM classifier	No
Bois et al. [BTM$^+$22]	2022	Biomedical	Motion sensor data	None	Sublevel	Truncated barcode and bottleneck distance	Visualization (UMAP)	No
Dlugas [Dlu22]	2022	Biomedical	ECG	Add x-axis	VR	1-cycle extraction	Feature analysis	No

Continuation of Table 3.2

Article	**Year**	**Field**	**Type of time series**	**Transformation**	**Filtration**	**TDA tools**	**Method**	**Theoretical contribution**
Bonis et al. [BCMR22]	2022	Industry	Magnetic field	Function on circle	Sublevel	PD	Estimator based on counting points	Theoretical properties of the estimator
Reise et al. [RMC23]	2023	Generic	Periodic	None	Sublevel	Topological signatures	None	Introduction of signatures and theoretical properties
Reise et al. [RFD$^+$23]	2023	Audio signal processing	Musical time series	Spectrogram	Filtered cubical complex	Betti curve (L^1 distance)	Minimum-cost matching	No
Tanweer at al. [TKM23]	2023	Generic	Univariate	Truncated time series	Sublevel (x-axis)	Barcode	Intervals from two thresholded PDs	No
Chen and Ravishanker [CR23]	2023	Ecological	Univariate	Periodogram	Sublevel	PL	K-means clustering	No

3.3.3 Delay embedding methods

Delay embeddings are a popular transformation of time series originally used to study dynamical systems, as Takens' theorem [Tak06] states that a smooth attractor (a manifold of dimension D) of a dynamical system can be reconstructed using a delay embedding of an observation function if its dimension is at least $2D + 1$. More generally, points of a delay embedding represent local variations of the time series (in particular, if the delay is $\tau = 1$, each point represents a subsequence of length d) as a point cloud in $\mathbb{R}^d$ that can be studied with different tools included persistent homology in dimension 0 to $d - 1$. Figure 3.3 illustrates a typical framework using a delay embedding.

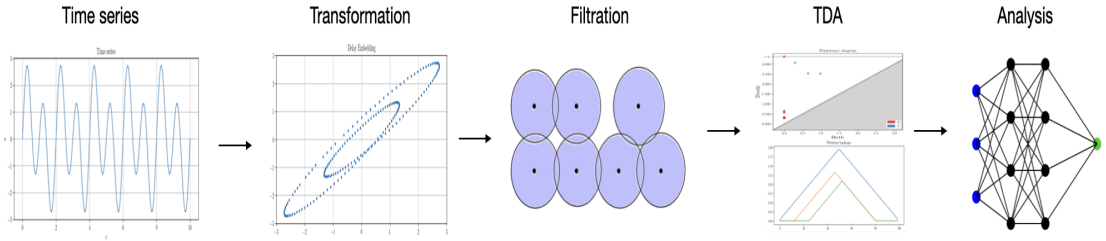


Figure 3.3: Typical framework of persistence homology for time series using a delay embedding.

3.3.3.1 Usual delay embeddings

Many works in the literature study the 1-dimensional persistent homology of a delay embedding of a time series to detect periodic behavior. Perea and Harer [PH15] made a full theoretical study of 1D persistent homology of the Rips filtration of delay embeddings of periodic functions. In particular, they derived a lower bound for maximum persistence of Fourier series approximations (with N frequencies) and their limits as $N \to \infty$, when parameters d and τ satisfy $d\tau = T$ and $d = 2N$ (T being the period of the function). The idea is that the delay embedding of a continuous function is a closed curve so it has at least one persistent feature in its 1D persistent (singular) homology, so the same should happen from time series that are samplings of periodic functions. This work was extended to quasiperiodic functions (sums of complex-valued trigonometric functions whose frequencies are linearly independent over $\mathbb{Q}$) [Per16, GP24]. It was shown that under some hypothesis, the delay embedding of a quasiperiodic function with N frequencies is dense in a N-torus embedded in $\mathbb{C}^{d+1}$. The authors of [GP24] derive lower bounds for the number of persistent features in dimension 1 to N and for their persistence. They show that their work can be used to detect dissonance in music. Following those ideas, maximal persistence in 1D was used to quantify periodicity in several applications such as discovering periodicity in gene expression time series [PDHH15] or wheeze detection in breathing sound signals [EGK14]. The latter introduces a model for wheeze signals and prove that with the right choice of delay embedding parameters, there is at least one persistent feature on the 1D persistence diagram from the Rips filtration, which should not be the case for non-wheeze signals. Khasawneh and Munch [KM16] use maximal 1D persistence of the Rips filtration of delay embeddings for chatter detection in synthetic dynamical systems (with industrial applications). This work was continued in [KMP18] with new features for 0D and 1D homology (maximal persistence and polynomial features from [ACC16] that do not depend on points with zero persistence), used with machine learning (logistic re-

gression classifier). Another method for periodicity detection and period estimation was proposed by Dlotko et al. [DQR19], based on 1D persistent homology. They use the Čech filtration and total p-persistence (pth root of the sum of all persistence values to the power p) to detect periodicity then a symbolization-based method for period estimation (note that the algorithm and choice of parameters not explicitly described).

In recent work [BTO24b], the anomaly detection problem for univariate time series is formalized using a convolutional model: the studied time series are successions of patterns that are normal (i.e. frequent) or abnormal. The above ideas are also relevant for this model. The authors defined an anomaly detection method along with a theoretical analysis. They leverage properties of DTM filtrations [ACG$^+$20] (robustness to outliers, possibility of subsampling a point cloud while integrating density information from the whole point cloud) to identify 1-cycles that correspond to normal patterns, and points far away from them are detected as anomalies.

Umeda [Ume17] introduced Betti curves and showed that Betti curves obtained from Čech filtrations of delay embeddings are sensitive to scale changes in the time series. They claim that 1D convolutional neural networks are suitable to learn from Betti curves and use them in neural networks to improve performance in time series classification on biomedical data. Delay embeddings and Betti curves were used for other biomedical applications. Yamanashi et al. [YKI$^+$21] performed detection of delirium on bispectral EEG using the area under the 1D Betti curve as a score for a statistical test. Yan et al. [YLL$^+$22] used features from Betti curves, persistent landscapes and silhouettes in several classifiers to detect freezing-of-gait episodes in motion sensor data (due to Parkinson's disease).

Delay embeddings can be used to detect changes in time series by studying topological changes in the embeddings of subwindows. Note that a drawback is that this requires many computations of persistence diagrams and bottleneck distances. Ueda et al. [UIY22] proposed a change point detection method (for financial data) based on this idea: they defined statistical models of persistence diagrams and use them to compute a sequence a features from a sequence of persistence diagrams corresponding to successive subwindows. Existing change point detection methods can then be applied to those features. Fernandez et al. [FBMG23] studied the persistent homology of point clouds using Fermat distances. The idea is that in order to recover topological properties of a manifold embedded in $\mathbb{R}^d$, Fermat distances are less dependent on the manifold embedding than the Euclidean distance, and this applies to delay embeddings. The authors propose a method for time series analysis (best suited for change point detection): they study delay embedding of subwindows starting at 0 and of increasing lengths, compute the Rips filtration with Fermat distance which gives a sequence of persistence diagrams. The sequence of bottleneck distance between consecutive diagrams has peaks that correspond to change points. This was only applied to a few examples but the method comes with theoretical guarantees (convergence, robustness).

Computation times for persistent homology of point clouds can be reduced by using dimension reduction [KKR18, ML20], filtrations that have fewer vertices than there are points in the point cloud [FWWS17, SSM$^+$17, YKO22, BTO24b], or **zigzag persistence** [TMK20] (zigzag persistence is an extension of persistent homology to non-increasing sets of simplicial complexes [CDS10]). Kim et al. [KKR18] proposed a generic featurization algorithm for time series, which they applied to financial data. They use delay embeddings and principal component analysis (PCA) for dimension reduction, Rips complexes, and persistent landscapes and silhouettes as inputs for deep learning models. They derived stability theorems with respect to noise, sampling and PCA, under the assumption that,

for l-dimensional PCA, the point cloud lies in a l-dimensional linear subspace, and the PCA matrix $X^\top X$ has at most l positive eigenvalues. They also discuss how PCA can help denoise data while preserving topology (under some assumptions). Majumdar and Laha [ML20] use PCA on delay embeddings of subwindows to save computation time when computing on persistent landscape per subwindow. They studied financial time series using self-organising maps for clustering and random forest for classification. Fraser et al. [FWWS17] proposed an ECG visualization method based on delay embeddings of subwindows and for several delay values. Witness complexes [DSC04] are used to save computation time and memory usage. They are complexes constructed using only a subset of the point as vertices. The output is a 2D plot of the first Betti number of some simplicial complex (not detailed) as a function of time and delay. Sanderson et al. [SSM$^+$17] also use the witness complex to perform binary classification of musical signals using a feature computed from the 1D **persistent rank function** [RT16]. In an update of [KMP18], Yesilli et al. [YKO22] use Bézier curves fitted on the delay embedding to save computation time for the Rips filtration and persistent homology (method from [TA19], about 30 times faster). They experiment with several featurization methods (features from [KMP18], landscapes [BD17], persistent images [AEK$^+$17], a kernel method [RHBK15], and persistence paths signatures [CNO18]) and several classifiers to classify real industrial data. In the work described above, Bois et al. [BTO24b] use a subsampled point cloud but integrate density information from the whole point cloud into their DTM filtration. Tymochko et al. [TMK20] use zigzag persistence [CDS10] to detect bifurcations (qualitative changes in a system's behavior) in dynamical systems. Their data is a sequence of time series (typically, a parameterized dynamical system). They fix a radius and compute all Rips complexes for this radius. Then zigzag persistence is applied to the obtained sequence of simplicial complexes. They identify the appearance and disappearance of bifurcations as the birth and death times of persistent points on the 1D persistence diagrams. One advantage over other approaches is that only one persistence diagram is computed instead of one for each time series, but fixing the radius is a drawback.

Finally, several articles use a similar method to study biomedical time series such as ECG [LWY23, SG19, JXZZ22, LZZJ22, MFVBMRM22, RLX$^+$23, IDE$^+$19, IBM20], motion sensor data [YOX$^+$20, YIMO$^+$20, YLL$^+$22, TZDD21] or financial data [GGK$^+$20]. Applications include classifying diseases, detecting critical events (arrythmia, fibrillation) or finding different regimes. The method consists in computing persistent homology of the Rips or Čech filtration of a delay embedding (maybe after some preprocessing), then using vectorized persistence diagrams or features from persistence diagrams as input for classifiers (SVM, random forest, deep learning) or k-means clustering. See Table 3.3 for more details about the application of each article and specific choices of vectorization and classifiers/clustering method.

3.3.3.2 Other delay embedding-based transformations

A few articles studying univariate time series use transformations that are not exactly delay embeddings but have similarities. Germain et al. [GTO24b] proposed an algorithm for **motif** discovery in time series (a motif is a pattern that appears several times in the time series). They use the **LT-normalized distance** [GTO] (distance invariant to linear trend) to construct a **weighted k-nearest-neighbors graph** (k-nearest non-overlapping subsequences) with additional temporal connections between successive points. Motifs are then reconstructed using a 0D persistence-based clustering algorithm [BTO24a]. Although the set of subsequences can be seen as a delay embedding with delay 1, only the distance to the non-overlapping k nearest neighbors are taken into account for per-

sistent homology (which saves computation time), and time is taken into account with temporal connections, which is significantly different than a Rips filtration on a delay embedding. Temporal connections were also used by Venkataraman et al. [VRT16] for action recognition on motion capture data (note that they have multivariate data but study each dimension independently). They connected successive points of the delay embedding, before computing the Rips filtration and using a nearest-neighbor classifier on persistence diagrams endowed with the 1-Wasserstein distance.

Myers et al. [MMK19, MKM22, MMKM23] use the **ordinal partition network** [MSSI15] to transform time series into unweighted graphs. This transformation takes a delay embedding of parameters d, τ and associates a permutation π of $[1, d]$ to each point $X_t = (\mathbf{x}[t], \mathbf{x}[t+\tau], \ldots, \mathbf{x}[t+(d-1)\tau]) = (X_1, X_2, \ldots, X_d)$, which is the permutation that sorts the d coordinates, i.e. such that $X_{\pi(1)} < X_{\pi(2)} < ... < X_{\pi(d)}$. The set of permutations obtained is the set of vertices of the graph, and an edge goes from i to j if there exists a time t such that permutations i and j respectively correspond to X_t and X_{t+1}. The graph can then be filtered using the Rips filtration from the shortest path distance (weighted or not) or diffusion distance. In [MMK19], the periodicity score from [PDHH15] is extended to unweighted graphs and it is used along with other features from 1D persistent homology such as persistent entropy [CGGD$^+$15] to distinguish periodic behavior from chaotic in dynamical systems. In [MKM22], the bottleneck distance between persistence diagrams is used in multidimensional scaling, then dynamic state detection in dynamical systems is performed with a SVM classifier. The method proposed in [MMKM23] in the case of univariate time series (dynamical systems with intermittency, i.e. irregular transitions from periodic to chaotic dynamics) computes a sequence of ordinal partition networks from subwindows and zigzag persistence. Then they detect intermittency using the coordinates of the most persistent points on the 1D persistence diagram. The multivariate applications of this method is described in Section 3.3.4.

3.3.3.3 Choosing delay embedding parameters

The choice of parameters d and τ can have a major impact on the performance of delay embedding-based methods. When studying continuous periodic functions, theory [PH15] suggests to choose d as high as possible with the window size $d\tau$ equal to the period (the fixed window size is also supported by empirical results [BTO24b]). A high-dimensional embedding with small delay will finely represent the local variations of the function. However, there are several problems in practice. Firstly, working in high dimension can be computationally expensive. Secondly, "irrelevance" and "redundancy" [TAC$^+$23] must be avoided. Irrelevance happens when $d\tau$ is too large and coordinates of each points of the embedding are too far away. In this case, the embedding fails to capture local variations and a point can be influenced by events far apart in time making interpretation difficult. As τ cannot be chosen arbitrarily small in practice, this implies that d should not be too high in practice. This also highlights the importance of signal sampling quality. Redundancy, on the other hand, happens when τ is too small so variations are slow between consecutive points of the time series leading to point clouds that are concentrated on the diagonal $x_1 = x_2 \cdots = x_d$ making it difficult to identify persistent features especially in the presence of noise. We refer to [TAC$^+$23] for a review of methods to choose d and τ. The most used methods are false nearest neighbors methods or methods based on autocorrelation or mutual information to estimate the period of the signal. Note that these methods are heuristics that are not guaranteed to perform well in every application. For example, they can fail in the presence of noise or with non-periodic signals. Moreover, complex

methods can have parameters that are themselves hard to choose. Specific knowledge about the application can help define better and/or simpler heuristics.

Table 3.3: Articles from Section 3.3.3 and their characteristics.

Start of Table 3.3

Article	Year	Field	Type of time series	Transformation	Filtration	TDA tools	Method	Theoretical contribution
Emrani et al. [EGK14]	2014	Biomedical	Breathing sounds	Delay embedding	VR	PD	Max 1D persistence	Model and theoretical study
Perea and Harer [PH15]	2015	Generic	Periodic	Delay embedding	VR	PD	Max 1D persistence	Full theoretical study
Perea et al. [PDHH15]	2015	Biomedical	Gene expression time series	Delay embedding	VR	PD	Max 1D persistence	No
Perea [Per16], Gakhar and Perea [GP24]	2016 and 2024	Audio signal processing	Quasiperiodic	Delay embedding	VR	PD	Max persistence (1D and higher dimension)	Full theoretical study
Khasawneh and Munch [KM16]	2016	Industry	Dynamical systems	Delay embedding	VR	1D max persistence	Visualization	No
Venkataraman et al. [VRT16]	2016	Motion capture	3D position time series	Delay embedding	VR + temporal connections	PD and 1-Wasserstein distance	KNN classifier	No
Umeda [Ume17]	2017	Biomedical	Motion sensor data, ECG, EMG	Delay embedding	Čech	Betti curve	Deep learning	Introduction of Betti curves
Sanderson et al. [SSM+17]	2017	Music	Music time series	Delay embedding	Witness	Persistent rank function	Binary classification	No
Fraser et al. [FWWS17]	2017	Biomedical	ECG	Delay embedding of subwindows	Witness	Betti number	Visualization	No
Khasawneh et al. [KMP18]	2018	Industry	Dynamical systems	Delay embedding	VR	0D and 1D features (max pers. + polynomial)	Machine learning	No
Kim et al. [KKR18]	2018	Generic	Synthetic, financial	Delay embedding then PCA	VR	PL, silhouette	Use PLS or silhouettes as features for DL	Stability with respect to noise, sampling and PCA
Dlotko et al. [DQR19]	2019	Generic	Periodic	Delay embedding	Čech	Total p-persistence	Symbolization	No
Myers et al. [MMK19]	2019	Dynamical systems	Dynamical systems	Delay embedding then unweighted graph (KNN or ordinal partition)	VR (shortest path distance)	Barcode	Features from 1D PH	No
Safarbali and Golpayegani [SG19]	2019	Biomedical	ECG	Delay embedding	Čech	Average 1D persistence	DL	No
Ignacio et al. [IDE+19]	2019	Biomedical	ECG	Delay embedding	VR	Features from barcode	Random forest classifier	No

Continuation of Table 3.3

Article	Year	Field	Type of time series	Transformation	Filtration	TDA tools	Method	Theoretical contribution
Ignacio et al. [IBM20]	2020	Biomedical	ECG	Delay embedding of subwindows	VR	Features from barcode and PL. Persistent entropy	Random forest classifier	No
Gidea et al. [GGK+20]	2020	Financial	Log-return of cryptocurrencies	Delay embedding of subwindows	VR	PLs with L^1 norm	K-means clustering	No
Majumdar and Laha [ML20]	2020	Financial	Stocks	Delay embedding of subwindows and PCA	VR	PL	Self-organizing maps (DL) or random forest classifier	No
Tymochko et al. [TMK20]	2020	Dynamical systems	Sequence of time series	Delay embeddings and VR complexes with fixed radius	Zigzag persistence	PD	Most persistent points	No
Yan et al. [YOX+20]	2020	Biomedical	Motion sensor data	Delay embedding	VR	PL	Random forest classifier	No
Yan et al. [YIMO+20]	2020	Biomedical	Motion sensor data	Delay embedding	VR	PL	Various classifiers	No
Yamanashi et al. [YKI+21]	2021	Biomedical	EEG	Delay embedding	Čech	Betti curve	Area under the 1D Betti curve as a score + statistical test	No
Tong et al. [TZDD21]	2021	Biomedical	Gait features	Delay embedding	Čech	Persistent entropy	SVM classifier	No
Myers et al. [MKM22]	2022	Dynamical systems	Dynamical systems	Graph (ordinal partition)	VR (several graph distances)	PD and bottleneck distance	MDS and SVM	No
Yesilli et al. [YKO22]	2022	Industry	Dynamical systems	Delay embedding and Bézier curves	VR	PL, max pers. + polynomial, or other featurization methods	Machine learning	No
Ueda et al. [UIY22]	2022	Financial	Time series with change points	Delay embedding of subwindows	Čech	PD	Statistical model on PDs, feature, change point detection algorithm.	NML codelength derivation
Yan et al. [YLL+22]	2022	Biomedical	Motion sensor data	Delay embedding	VR	PL, Betti curve, persistent silhouette	Various classifiers	No
Ling et al. [LZZJ22]	2022	Biomedical	ECG	Moving average + delay embedding	VR	Features from PD	Various classifiers	No

Continuation of Table 3.3

Article	Year	Field	Type of time series	Transformation	Filtration	TDA tools	Method	Theoretical contribution
Mjahad et al. [MFVBMRM22]	2022	Biomedical	ECG	Filtering + delay embedding	VR	Features from PD, PL, persistent silhouette	KNN classifier	No
Jiang et al. [JXZZ22]	2022	Biomedical	BCG	Delay embedding	VR	Features from barcode	Various classifiers	No
Liu et al. [LWY23]	2023	Biomedical	ECG	Segmentation and delay embedding	Čech	PL	Random forest classifier	No
Myers et al. [MMKM23]	2023	Various	Dynamical systems with intermittency	Ordinal partition of subwindows	Zigzag persistence	PD	Most persistent points	No
Ren et al. [RLX+23]	2023	Biomedical	ECG	Delay embedding	VR + sublevel of signal	Barcode	DL	No
Fernández et al. [FBMG23]	2023	Generic	Any	Delay embedding	VR (Fermat distance)	PD of subwindows, bottleneck distance	Distance between consecutive PDs	Theoretical analysis
Germain et al. [GTO24b]	2024	Generic	Alterning motifs	KNN Graph (delay embedding with LT-normalized distance)	VR + temporal connections	PD	Persistence-based clustering	No
Bois et al. [BTO24b]	2024	Generic	Normal or abnormal patterns	Delay embedding	DTM Rips	PD and 1-cycles extraction	Distance to normal cycles	Theoretical analysis

3.3.4 Multivariate time series

In this section, we review all the applications of persistent homology to multivariate time series. Figure 3.4 illustrates a typical framework for multivariate time series.

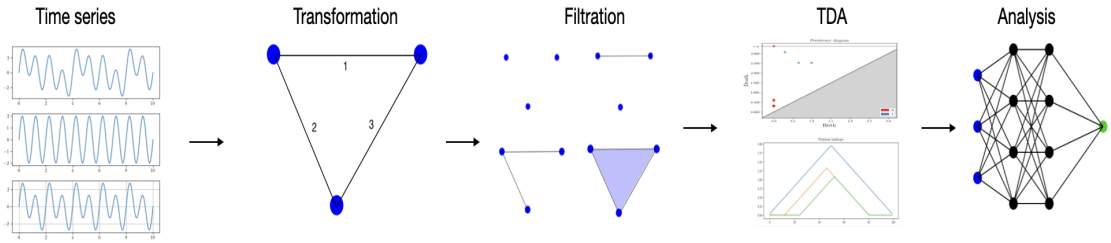


Figure 3.4: Typical framework of persistence homology for multivariate time series.

3.3.4.1 Graph-based methods

The most used method to study multivariate time series is to transform them into an undirected weighted graph whose vertices correspond to each channel of the time series and edge weights represent a distance or similarity between them. The graph is then filtered using a **clique filtration** (ascending for distances, descending for similarity measurements) to study vectorized persistence diagrams. Let $G = (V, E, w)$ be a weighted graph where V is the set of vertices, $E \subset V \times V$ is the set of edges and $w : E \to \mathbb{R}_+$ is a weight function. The ascending clique filtration is defined as follows: all the 0-simplices have a filtration value of 0 and, for $\alpha > 0$ and $k \geq 1$:

$$[v_0, \dots, v_k] \in K_\alpha \iff \forall i \neq j,\ (v_i, v_j) \in E \text{ and } w(v_i, v_j) \leq \alpha.$$

The descending clique filtration is defined the same way, by adding all the vertices at the beginning and working with decreasing values of α. Note that the ascending clique filtration is not exactly a sublevel set filtration because the function is defined on edges instead of vertices. Also note that if V is a metric space and w is its distance then the clique filtration can be seen as a Rips filtration.

Graph-based methods are mainly applied to study brain signals (EEG or fMRI) through a notion of coherence, dependency or correlation, which is motivated by the fact that neurons that are wired together in the brain tend to have correlated activations. Stolz et al. [SHP17, SEN$^+$21] studied multivariate fMRI time series to study motor learning of healthy subjects and schizophrenia. In their first article [SHP17], they use a functional connectivity measure (coherence of the wavelet scale-2 coefficients), a descending clique filtration, and they interpret changes in the persistence landscapes of different days in terms of synchronization. They insist on the fact that low-persistence features can contain important information. In the second article [SEN$^+$21] they use Pearson correlations, a descending clique filtration, and compute persistence landscapes (and average persistence landscapes) and images for clustering and statistical tests. El-Yaagoubi et al. [EYCO23, EYJCO23] studied EEG time series from ADHD patients. They transform their data into graphs using Fourier-based distances (respectively dependency-based distance and frequency-specific distance) and use ascending clique filtrations and statistical tests on persistence landscapes. The test used in [EYCO23] is a permutation test from [RT17] based on the L^2 distance between landscapes to compare EEGs of healthy subjects and patients with ADHD. In the second article [EYJCO23], they introduce an hypothesis test

based on persistence landscapes and derive associated convergence results. Manjunath et al. [MPS23] used the same frequency-specific distance, clique filtration and permutation test on EEGs to study obstructive sleep apnea.

Petri et al. [PET$^+$14] introduced a new object for topological data analysis: the **homological scaffold** (HS), which is a graph whose vertices are the ones from the original graph and whose edge weights encode information about the number or persistence of the homology generators containing each edge. They studied the effect of psychedelic mushrooms on the brain with fMRI data. They built a correlation graph from time series and used a descending clique filtration to compute persistence diagrams and homological scaffolds. They use scaffolds to visualize brain connectivity, and also analyzed statistical features from persistence diagrams and homological scaffolds.

El-Yaagoubi and Ombao [EYO23] also studied EEG signals by transforming them into a directed graph to study epileptic seizures. Their goal is to integrate the directed flow of information between brain regions. First, they use a non-symmetric distance function called the partial directed coherence, which is a Granger causality based dependence measure, to get a distance matrix. A distance is then defined from the anti-symmetric part of the weight matrix which is used to compute the directed graph and its clique filtration and persistence diagram. Diagrams are compared before and after seizures.

Studying sequences of local correlation networks has also proven to be useful to study stock markets [Gid17] or detect multivariate anomalies [CRL24]. Both works use a correlation -based graph transformation on subwindows and clique filtrations to obtain a time series of persistence diagrams. Gidea [Gid17] studies the sequence of 2-Wasserstein distance between each diagram and the initial one to detect critical transitions in financial markets. Chazal et al. [CRL24] introduce an algorithm for anomaly detection in multivariate time series. They vectorize the obtained persistence diagrams using a measure quantization algorithm [RCL$^+$21] then estimate the mean and covariance matrix of the time series of vectorized diagrams. This mean and covariance matrix represent the normal behavior. An anomaly score is defined based on the Mahalanobis distance (i.e. the normalized distance from each vectorized diagram to the mean of the normal model). They give a theoretical study of several aspects of their method.

3.3.4.2 Point cloud-based methods

Multivariate time series with d channels can be seen as time-indexed point clouds in $\mathbb{R}^d$. This makes it possible to use usual persistent homology methods such as those from Section 3.3.3.

As for delay embeddings of univariate time series, 1D persistent homology can be used to detect periodic behaviors in multivariate time series. Vejdemo-Johansson et al. [VJPSK15] studied periodic motion with position and orientation time series for several body parts. They see the d-variate time series as a point cloud in $\mathbb{R}^d$, compute its Rips filtration and 1D persistence barcode, then parameterize the point cloud with circular coordinates for each persistent bar using a method from de Silva and Vejdemo-Johansson [DSVJ09]. They use a clustering technique to compute typical trajectories and interpolate between different periodic motions while preserving periodicity. Tralie and Perea [TP18] extended the delay embedding to multivariate data as a transformation from $\mathbb{R}^{W\times H}$ to $\mathbb{R}^{d\times W\times H}$ ($W \times H$ is the number of pixels), and defined (quasi-) periodicity scores based on maximal 1D and 2D persistence following ideas from [PH15, Per16, GP24]. They apply their method to (quasi-) periodicity quantification in videos (time series of images).

Gidea and Katz [GK18] studied multivariate financial time series by seeing subwindows of length w as point clouds of size w in $\mathbb{R}^d$. They use the Rips filtration and compute the

L^p norm of 1D persistence landscapes (for $p = 1, 2$) as a function of time. They studied this new univariate time series using its variance and the average spectral density at low frequencies and give empirical evidence that they increase before financial crisis.

In a methodological paper, Salch et al. [SRA$^+$21] study multivariate fMRI data as a time-varying function on voxels $f : (x, y, z, t) \mapsto f(x, y, z, t)$ (i.e. a time series for each voxel). For each time t, they compute the Čech filtration of the point cloud in $\mathbb{R}^4$ made of all points $(x, y, z, f(x, y, z, t))$. They propose to study the corresponding 1D **persistence vineyard** [CSEM06] i.e. the time series of 1D persistence diagrams, with a notion of spatial proximity between features at time t and $t+1$. They use a statistical test and visualization method to identify and study task-responsive structures. Note that this technique could also be used in higher dimensions. A drawback is that vineyards are not robust as temporal connection between features depend on their representatives [CSEM06]. The authors indicate that using random representatives or specific ones could help improve the method.

3.3.4.3 Other methods

Santoro et al. [SBPA23] generalize the notion of correlation between two time series to any number k of time series by introducing the **k-order co-fluctuation** and using to define a simplicial complex where, for all $k \geq 2$, each $(k-1)$-simplex represents a group of k channels of the multivariate time series and is associated to the corresponding k-order co-fluctuation value. Then, the complex is filtered by adding simplices by decreasing values, excluding those that do not respect the filtration condition (which are labeled as violations). The persistent homology generators and violations are used to define features (hyper-coherence and hyper-complexity) describing the persistence diagrams and homological scaffolds. They shows the relevance of their features for various types of multivariate time series (fMRI, financial, historical data of infectious diseases).

Finally, zigzag persistence has been applied to analyze multivariate time series by Corcoran and Jones [CJ17] and Myers et al. [MMKM23]. Corcoran and Jones studied swarm behavior of fish (position of several agents through time) by using zigzag persistence on a sequence of simplicial complexes computed from upper-level sets (with a fixed threshold) of a kernel density estimator. They obtain persistence landscapes on which they use K-medoids clustering with the L^2 distance. The previously described zigzag persistence-based method by Myers et al. [MMKM23] applies to time-varying graphs, so it can apply to multivariate time series with some of the graph transformations listed above.

Table 3.4: Articles from Section 3.3.4 and their characteristics.

Start of Table 3.4

Article	Year	Field	Type of time series	Transformation	Filtration	TDA tools	Method	Theoretical contribution
Petri et al. [PET+14]	2014	Biomedical	Multivariate fMRi	Weighted graph (correlations between time series)	Clique (descending)	PD and HS	Statistical features from PDs and HSs, and visualisation with HS	Introduction of HS
Vejdemo-Johansson et al. [VJPSK15]	2015	Motion capture	Position and orientation time series	Point cloud	VR	1D barcode and circular coordinates	Clustering	No
Corcoran and Jones [CJ17]	2017	Biology	Position time series	Upper-level set of kernel density estimator	Zigzag persistence	PL	K-medoid clustering (L2 distance)	No
Stolz et al. [SHP17]	2017	Biomedical	Multivariate fMRi	Weighted graph (functional connectivity)	Clique (descending)	PLs with L^2 distance	Visualization, K-means	No
Gidea [Gid17]	2017	Finance	Stock returns	Weighted graph (correlation-based distance)	Clique (ascending)	PD and 2-Wasserstein distance	Distance to initial PD	No
Gidea and Katz [GK18]	2018	Finance	Stock returns	Point cloud from subwindows	VR	L^p norm of PLs	Variance and the average spectral density	No
Tralie and Perea [TP18]	2018	Video processing	Videos	Multivariate delay embedding, SVD, normalization	VR	PD	Scores based on max 1D and 2D persistence	No
Stolz et al. [SEN+21]	2021	Biomedical	Multivariate fMRi	Weighted graph (correlation)	Clique (descending)	1D PL, persistence image	Clustering, SVM classification, statistical test	No
Salch et al. [SRA+21]	2021	Biomedical	Multivariate fMRi	Point cloud	Čech	Persistence vineyard	Visualization, statistical test	No
El-Yaagoubi et al. [EYCO23]	2023	Biomedical	Multivariate EEG	Weighted graph (dependency-based distance based on Fourier coefficients)	Clique (ascending)	PL	Permutation test (based on L^2 distance between PLs)	No
Manjunath et al. [MPS23]	2023	Biomedical	Multivariate EEG	Weighted graph (dependency-based distance based on Fourier coefficients)	Clique (ascending)	PL	Permutation test (based on L^2 distance between PLs)	No

Continuation of Table 3.4

Article	Year	Field	Type of time series	Transformation	Filtration	TDA tools	Method	Theoretical contribution
El-Yaagoubi et al. [EYJCO23]	2023	Biomedical	Multivariate EEG	Weighted graph (dependency-based distance based on Fourier coefficients)	Clique (ascending)	Spectral PLs (a PL for each frequency)	Hypothesis testing	Definition of a test statistic on spectral landscapes and associated convergence results
Santoro et al. [SBPA23]	2023	Various	Multivariate (fMRI, financial, historical data of infectious diseases)	Normalization of each channel, then simplicial complex (k-order co-fluctuation)	Decreasing k-order co-fluctuation, with violations excluded	PD and HS + violating simplices	Features (hyper-coherence and hyper-complexity)	No
El-Yaagoubi and Ombao [EYO23]	2023	Biomedical	Multivariate EEG	Weighted directed graph (partial directed coherence)	VR (distance from the anti-symmetric part of the weight matrix)	PD	Visualization	No
Chazal et al. [CRL24]	2024	Generic	Multivariate with anomalies	Graph (correlation-based distance)	Clique (ascending)	PD and vectorization (measure quantization)	Mahalanobis distance to normal model	Theoretical analysis

3.4 Discussion and perspectives

Persistent homology has proven its usefulness for several tasks in time series analysis in various fields, especially in biomedicine or dynamical systems analysis. Its usage goes from defining topological features to improve the performance of machine learning or statistical models to building robust and interpretable methods based only on persistent homology with theoretical guarantees. Many ideas from the methods reviewed here could be applied in different fields than the one they were designed for.

Theory and interpretability

Many methods reviewed here use persistent homology as a feature extractor for machine learning algorithms and are difficult to interpret, while others are based on a model and/or theoretical results such as stability theorems or bounds on persistence [KM18, EGK14, BCMR22, RMC23, BTO24b, CRL24]. These methods have the advantage of being unsupervised (unlike machine learning-based ones) and more interpretable, which can be important in various fields such as biomedicine. Working to develop this type of methods seems important for the future of the TDA research community.

Delay embedding parameters

Points in delay embeddings capture local variations of time series and persistent homology can be used to study them globally, with many possible filtrations and methods. However, as highlighted above the choice of embedding dimension and delay can have a significant impact on results and methods to find the best values are heuristic and often introduce new parameters. Finding methods that are less sensitive to these parameters is an important problem. Intrinsic distances such as the one used in [FBMG23] are a potential solution. More generally, choosing the right notion of distance to use in filtrations makes it possible to fully take advantage of TDA.

Current delay embedding methods focus on low dimensions (almost never above 1), finding ways to leverage higher dimensions could also be interesting.

Beyond correlation

In the case of multivariate data, a promising line of research would be to go beyond correlation networks by exploring tools such as k-order co-fluctuation [SBPA23], directed (causality-based) graph transforms [EYO23] or persistence vineyards [SRA$^+$21]. In particular, these tools could help better understand how the brain functions by improving EEG and fMRI analysis.

Computation time

Although we have described several ways of getting around the problem of computation time of persistent homology, it remains an important limitation which becomes a problem when the number or length of the studied time series gets too high. Efficient implementations of the persistent homology algorithm of Rips filtrations exist [Bau21, ZXW20, PHL$^+$21], as well as linear-size approximations [She12] and efficient computation of representative cycles [ČV23]. Developing similar algorithms for other filtrations and integrating them to standard TDA libraries would help researchers use persistence-based methods in real-world applications.

Persistence parameters

In some of the reviewed methods, two parameters are relevant to construct a filtered simplicial complex. For example distance and time [TMK20], density and time [CJ17] or distance and density [BTO24b]. In the two first cases, a distance/density parameter is fixed and zigzag persistence is used on a time-varying simplicial complex. In the third case, density information is integrated to the Rips filtration by the DTM filtration [ACG$^+$20], with a fixed parameter (number of neighbors). The authors [VAB11] also use several thresholds for sublevel sets but do not specify how they define the filtration, even though it is non-trivial. A way of studying the structure of data for all values of several parameters is multiparameter persistent homology [BL22]. However, it is difficult to interpret because **multiparameter persistence modules** do not have an interval-decomposition (the decomposition that gives the persistence diagram for usual persistence). There exist invariants and vectorizations for multiparameter persistent homology (usually based on 1-parameter reductions) and some are starting to be applied to time series classification [LCB24, KJ22, KJ24]. We did not include multiparameter persistence in the body this review because it would require a lot of additional background only to introduce a few applications. However, this is a subject our readers should be aware of.

Chapter 4

A method for gait signals analysis with an application to the study of multiple sclerosis

This chapter describes a method to compare pairs of gait signals, visualize the results and interpret them, based on topological data analysis techniques. Our method is non-parametric and requires no data other than gait signals acquired with inertial measurement units. We introduce tools from topological data analysis (sublevel sets, persistence barcodes) in a practical way to make it as accessible as possible in order to encourage its use by clinicians. We apply our method to study a cohort of patients suffering from progressive multiple sclerosis and healthy subjects. We show that it can help estimate the severity of the disease and also be used for longitudinal follow-up to detect an evolution of the disease or other phenomena such as asymmetry or outliers.

Contents

4.1 Introduction **69**
4.2 Persistent homology of periodic functions **70**
4.3 Protocol and data **74**
4.4 Method description **75**
4.4.1 Overview of our method and guidelines for the clinician 75
4.4.2 Mathematical description our of method 80
4.4.3 Feature extraction 84
4.5 Results **86**
4.5.1 HS/MS experiment 87
4.5.2 EDSS experiments 87
4.5.3 Longitudinal experiment 88
4.6 Analysis of the results and discussion **89**
4.6.1 Analysis of the results 89
4.7 Discussion **92**
4.8 Conclusion and perspectives **93**

4.1 Introduction

Longitudinal follow-up and inter-individual comparison of gait trials are relevant for patients suffering with many degenerative diseases [VBB+17]. One example of such a disease is progressive Multiple Sclerosis (MS), for which gait is considered the most important source of disability [HBR+08]. Throughout this chapter, we will use MS as an example to illustrate our approach. Those intra/inter-individual comparisons are usually performed using semi-quantitative clinical scales, but those have limitations. In the case of MS, several clinical scales exist such as the Expanded Disability Status Scale (EDSS) [Kur83], the Multiple Sclerosis Walking Scale-12 [NRDR+13], and the Fatigue Impact Scale [FPR+94, FRR+94]. In this study, severity of the disease was evaluated using the EDSS, which is a score from 0 to 10, ranging from normal neurological examination (0) to total impotence (9.5) or even death (10) by increments of 0.5. Those scales provide semi-quantitative or qualitative criteria for disease follow-up but they have been criticized for several reasons, including a lack of objectivity and sensitivity to clinical evolution [MMFM+14, NVWE90, Nos94, GCLBK17, LDP+13, VJQVR20]. This motivates the use of quantitative methods.

Over the past few years, gait quantification was made easier by the development of light, affordable inertial measurement units (IMUs). IMUs are portable systems integrating accelerometers, gyroscopes, and magnetometers that allow the synchronized measurement of linear accelerations and angular velocities in one single light, low-cost device [JMO+20]. Standard features such as velocity, step time or step length can be extracted from IMU signals. They have been used to discriminate healthy subjects from patients, or groups of patients with different levels of disease severity, but those studies rely on long protocols (walking for several minutes to get a high number of steps) [MHH+21, AHS+20, MSK+21] and/or gait event detection [MHH+21, AHS+20, LDCR+19, PMP+17, BGRM19, VJOM+19]. Long protocols are incompatible with patients with severely altered gait who have trouble walking a few meters, and are more difficult to include in clinical day-to-day practice. Gait event detection is either performed using expensive equipment (pressure sensitive mats, motion capture) or complex algorithms. However, those complex algorithms are difficult to apply to gait signals from pathological subjects with severely altered steps. For example, in step detection, the error on the detected start/end time of steps is typically around 10 ms for healthy subjects (HS) [DQO+20] and around 100 ms for severely affected MS patients [VJOM+20]. Moreover, the above studies do not perform comparisons between different trials of the same subjects at different dates, in which case changes can be more subtle depending on the progression of the disease. This raises the question of how to compare gait trials, especially when some of them are from pathological subjects.

The goals of this chapter are to present a method to compare pairs of gait trials, visualize the results of all the comparisons and interpret them. Our approach is based on TDA, which we use to define a notion of distance between gait signals, allowing us to compare gait trials. We then use a visualization algorithm to represent each trial as a 2D-point and compute features to study the structure of the obtained point cloud. By dividing the point cloud into different groups, our method makes it possible to perform both global studies to find differences in gait for different levels of severity of the disease, and longitudinal studies about the evolution of patients' gait in time. The main idea behind TDA is that data are a finite subset of samples of an underlying mathematical set, whose structure can bring useful information about the system under study. For instance, a gait signal is represented by a time series, i.e. a uniform sampling of a continuous 1-dimensional physiological signal. In this setting, one of the main TDA techniques, persistent homology, can

be used to study the underlying continuous signal through the finite time series.

TDA has been applied to time series in medicine and biology since the 2010s (see Chapter 3 for an full literature review). Applications include the study of cardiac arrythmia with electrocardiograms [DUC20], motor learning with fMRI data [SHP17, SRA$^+$21], gene expression time series [PDHH15], wheeze in breathing signals [EGK14], epileptic seizures with electroencephalograms [WOC18], the spread of COVID-19 [CV21] and autism spectrum disorder [MAMD20]. TDA has been applied to the study of locomotion through time series. Motion capture data has been analyzed using TDA to model bipedal walking [VAB11] or to perform action recognition (classification between dance, jump, run sit and walk) [VRT16]. It has also been used to study degenerative diseases by performing binary classification of time series of gait parameters (stride, stance, and swing time) between healthy and pathological subjects (suffering from either Parkinson's disease, Huntington's disease or Amyotrophic lateral sclerosis) [YOX$^+$20, YIMO$^+$20], multi-class classification of ground reaction force time series to assess the severity of Parkinson's disease [TZDD21], or detection of freezing-of-gait episodes [YLL$^+$22]. To the best of our knowledge, TDA was mainly used to produce features that were fed to machine learning algorithms (such as SVM, random forest, nearest neighbors or deep neural networks) [DUC20, MAMD20, VRT16, YOX$^+$20, YIMO$^+$20, TZDD21, YLL$^+$22, SDB16]. Topological features increased their performance, but they are more difficult to interpret than traditional ones (such as, for the study of locomotion, speed, step length, step time etc...) so the interpretability of the methods used in those articles is not studied. In this chapter, we propose an interpretable TDA-based method to compare gait trials. More precisely, we use objects from TDA to represent gait trials as points in a space in which a distance can be defined. This distance can be interpreted in terms of signal oscillations and used to compare gait trials. We applied our method to study a cohort of healthy and pathological subjects as a whole, and performed both inter-individual and intra-individual comparisons. In addition, the method has the advantage of working for time series measured with light, affordable IMUs during a protocol used in clinicians' day to day practice.

Section 4.2 is a theoretical study of persistent homology of periodic functions. It is the only section of this chapter that requires background in TDA from Chapter 2. It can help understand the intuition behind the method but it is not necessary to read this section to understand the rest of the chapter. In Section 4.3, we describe the protocol applied to construct our dataset. In Section 4.4, we introduce the method and its applications. We present the results of the application of our method to study MS in Section 4.5 and analyze them in Section 4.6.

4.2 Persistent homology of periodic functions

In this section, we study the persistent homology of sublevel sets of periodic functions. We start by introducing some notations and conventions before presenting our result.

Let $\mathbf{x} : I \to \mathbb{R}$ be a continuous function, where I is a closed interval. We note $\mathbf{x}_{|A}$ the restriction of $\mathbf{x}$ to A. The persistence barcode of $\mathbf{x}$ obtained using the sublevel sets filtration is made of bars (b, d) where b is a local minimum of $\mathbf{x}$ and d is either a local maximum or $+\infty$ (only one bar goes to $+\infty$). We denote this barcode by $B_{\mathbf{x}}$.

We use the following convention to deal with equal local minima: when two connected components of the filtration merge, if one started at $\mathbf{x}(t_1)$ and the other at $\mathbf{x}(t_2)$, where $t_1 < t_2$ and $\mathbf{x}(t_1) = \mathbf{x}(t_2)$, then the first one absorbs the second one. Thus, $(\mathbf{x}(t), +\infty) \in$

$B_{\mathbf{x}}$ if and only if $\mathbf{x}(t)$ is the first global minimum of $\mathbf{x}$ (recall that $\mathbf{x}$ is continuous on the compact I).

Each strict local minimum is the start of a bar. If $\mathbf{x}$ is constant on an interval $J \subset I$, and if we denote by $[a, b]$ the maximal interval such that $J \subset [a, b]$ and on which $\mathbf{x}$ is constant, then a bar starts at $\mathbf{x}(a)$ if and only if $\mathbf{x}(a)$ and $\mathbf{x}(b)$ are local minima. In that case, we choose any point $t \in [a, b]$, call $\mathbf{x}(t)$ a local minimum and ignore other points in $[a, b]$ so we can say that each bars' birth date corresponds to exactly one local minimum.

Likewise, each strict local maximum is the end of a bar. If $\mathbf{x}$ is constant on an interval $J \subset I$, and if we denote by $[a, b]$ the maximal interval such that $J \subset [a, b]$ and on which $\mathbf{x}$ is constant, then a bar ends at $\mathbf{x}(a)$ if and only if $\mathbf{x}(a)$ and $\mathbf{x}(b)$ are local maxima. In that case, we choose any point $t \in [a, b]$, call $\mathbf{x}(t)$ a local maximum and ignore other points in $[a, b]$ so we can say that each bars' death dates corresponds to exactly one local maximum or to $+\infty$ (which happens once).

Note that a connected component corresponding to a local minimum $\mathbf{x}(t)$ can only be absorbed by the previous local minimum lower than or equal to $\mathbf{x}(t)$, or by the next local minimum lower than $\mathbf{x}(t)$.

Proposition 3. *Let $\mathbf{x} : \mathbb{R} \to \mathbb{R}$ be a continuous and periodic function of period T. Let $k > 1$. The persistence barcode from sublevel sets of $\mathbf{x}$ on $[0, kT]$ can be deduced from its barcode on $[0, 2T]$. In particular, if $\mathbf{x}$ has n local minima on $[0, T]$, the barcode on $[0, kT]$ either contains n groups of k bars starting at the same local minimum among which at least $k - 1$ are identical, or it contains $n - 1$ groups of k bars starting at the same local minimum among which at least $k - 1$ are identical and one additional bar starting at the remaining local minimum.*

Remark 4. *This result is similar to a result derived in parallel work by Reise et al. ([RMC23], Lemma 2.2). They consider a 1-periodic function $\mathbf{x}$ on $[0, R]$, where $R > 1$, and show that there exists $c \in [0, 1]$ such that each bar of $B_{\mathbf{x}_{|[c,c+1]}}$ appears $\lfloor R - 1 \rfloor$ times in $B_{\mathbf{x}_{|[0,R]}}$. They also derive an upper bound on the total p-persistence of the remaining bars of $B_{\mathbf{x}_{|[0,R]}}$. The difference with our result is that we restrict to intervals $[0, k]$ but provide an explicit description of the barcode $B_{\mathbf{x}_{|[0,kT]}}$ given $B_{\mathbf{x}_{|[0,2T]}}$ (the proofs are also very different).*

Idea: we consider the restriction $\mathbf{x}_{|[0,T]}$ and successively add each period. The bars of $B_{\mathbf{x}_{|[0,kT]}}$ are not necessarily the same as those of $B_{\mathbf{x}_{|[0,T]}}$ because the additional periods may change the order in which components merge. However, knowing which changes happen when going from $B_{\mathbf{x}_{|[0,T]}}$ to $B_{\mathbf{x}_{|[0,2T]}}$ is enough to know exactly what happens when adding more periods. See Figures 4.1 and 4.2 for illustrations.

Proof. Let $(b_i = \mathbf{x}(m_i))_{1 \leq i \leq n}$ be the local minima of $\mathbf{x}_{|[0,T]}$, where $m_i < m_{i+1}$. There are four cases to consider depending on whether b_1 and b_n are equal to $\mathbf{x}(0)$ or not.

Case 1: $b_1 \neq \mathbf{x}(0)$ **and** $b_n \neq \mathbf{x}(0)$ **(see figure 4.1)**.

In that case, $\mathbf{x}_{|[0,2T]}$ has $2n$ local minima: $b_1^{(1)}, b_1^{(2)}, \ldots, b_n^{(1)}, b_n^{(2)}$, where $b_i^{(1)} = b_i^{(2)} = \mathbf{x}(m_i^{(1)}) = \mathbf{x}(m_i^{(2)})$ and $m_i^{(2)} = m_i^{(1)} + T$. Likewise, each new period adds n local minima. Thus, in this case, we can use the notation

$$B_{\mathbf{x}_{|[0,kT]}} = \{(b_i^{(j)}, d_i^{(j)}(k))\}_{1 \leq i \leq n,\ 1 \leq j \leq k}.$$

Note that the values $d_i^{(j)}$ can depend of the number of periods k, hence the notation $d_i^{(j)}(k)$. It is not the case for the values $b_i^{(j)}$.

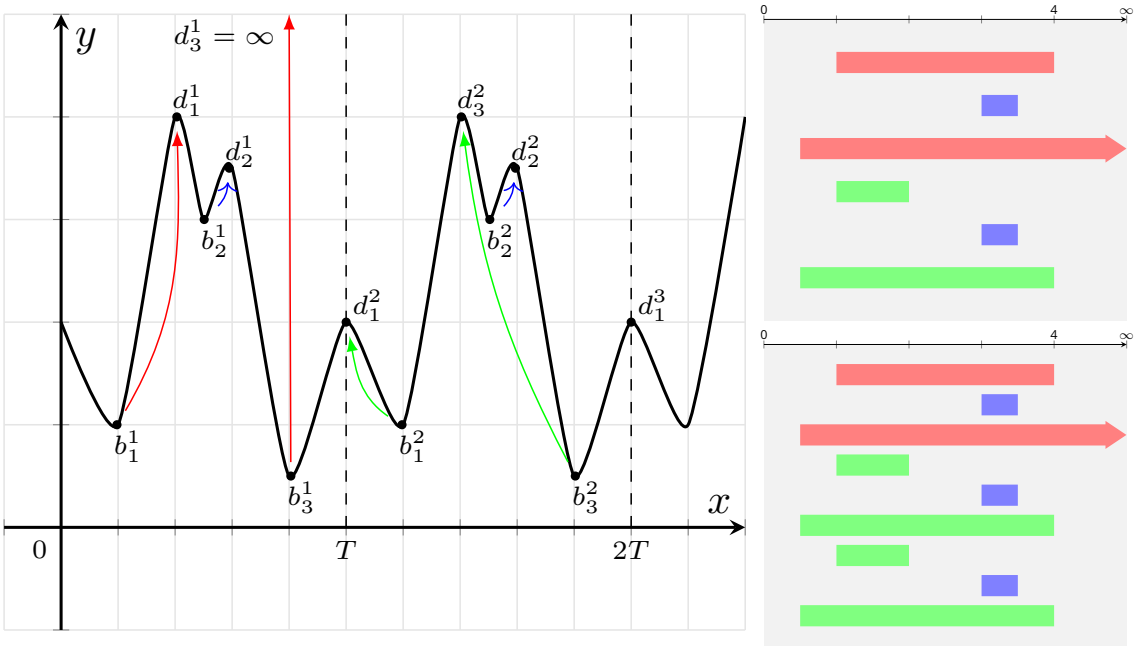


Figure 4.1: **Left:** Illustration of case 1. Each arrow shows a pair (local minimum, local maximum) that forms a bar of $B_{\mathbf{x}_{|[0,2T]}}$. Red arrows correspond to bars which appear once in $B_{\mathbf{x}_{|[0,kT]}}$. Blue arrows correspond to bars which appear k times in $B_{\mathbf{x}_{|[0,kT]}}$. Green arrows correspond to bars which appear $k-1$ times in $B_{\mathbf{x}_{|[0,kT]}}$. **Right:** barcodes $B_{\mathbf{x}_{|[0,2T]}}$ (top) and $B_{\mathbf{x}_{|[0,3T]}}$ (bottom).

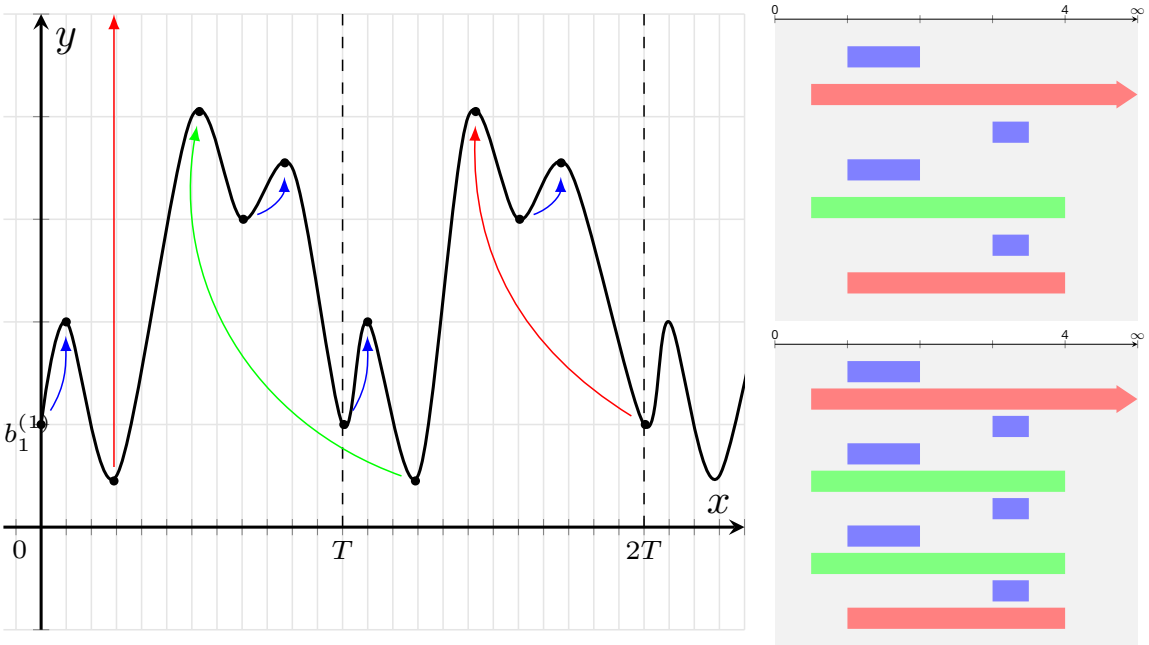


Figure 4.2: **Left:** Illustration of case 4. Each arrow shows a pair (local minimum, local maximum) that forms a bar of $B_{\mathbf{x}_{|[0,2T]}}$. Red arrows correspond to bars which appear once in $B_{\mathbf{x}_{|[0,kT]}}$. Blue arrows correspond to bars which appear k times in $B_{\mathbf{x}_{|[0,kT]}}$. Green arrows correspond to bars which appear $k-1$ times in $B_{\mathbf{x}_{|[0,kT]}}$. **Right:** barcodes $B_{\mathbf{x}_{|[0,2T]}}$ (top) and $B_{\mathbf{x}_{|[0,3T]}}$ (bottom).

Let b_{i_0} be the first global minimum of $\mathbf{x}$. Then $B_{\mathbf{x}_{|[0,2T]}}$ has one $(b_{i_0}^{(1)}, +\infty)$ bar and one $(b_{i_0}^{(2)}, \max(\mathbf{x}))$ bar. Each new period will add a $(b_{i_0}^{(j)}, \max(\mathbf{x}))$ bar.

Let $i \neq i_0$. Then each connected component starting at b_i eventually gets killed. If $(b_i^{(1)}, d_i^{(1)}) = (b_i^{(2)}, d_i^{(2)})$ in $B_{\mathbf{x}_{|[0,2T]}}$ then each new period will add an identical bar. There are two cases where this does not happen: when the connected component of $b_i^{(1)}$ is killed by a local minimum in $[T, 2T]$, and when the connected component of $b_i^{(2)}$ is killed by a local minimum in $[0, T]$.

In the first situation, the bar $(b_i^{(2)}, d_i^{(2)}(2))$ is the same as the bar $(b_i^{(1)}, d_i^{(1)}(1)) \in B_{\mathbf{x}_{|[0,T]}}$. Adding a new period turns it into a bar identical to $(b_i^{(1)}, d_i^{(1)}(2))$ and the last bar $(b_i^{(n)}, d_i^{(n)}(k))$ is equal to $(b_i^{(1)}, d_i^{(1)}(1))$.

In the second situation, the bar $(b_i^{(1)}, d_i^{(1)}(2))$ is the same as the bar $(b_i^{(1)}, d_i^{(1)}(1)) \in B_{\mathbf{x}_{|[0,T]}}$. Each new period adds a bar identical to $(b_i^{(2)}, d_i^{(2)}(2))$.

In all situations, for each $i = 1, ..., n$, $B_{\mathbf{x}_{|[0,kT]}}$ has k bars starting at b_i that already where in $B_{\mathbf{x}_{|[0,2T]}}$, and at least $k-1$ of them are identical.

Case 2: $b_1 = \mathbf{x}(0)$ **and** $b_n \neq \mathbf{x}(0)$.

In this case, barcodes are of the form

$$B_{\mathbf{x}_{|[0,lT]}} = \{(b_1^{(1)}, d_1^{(1)}(1))\} \cup \{(b_i^{(j)}, d_i^{(j)}(k))\}_{2 \leq i \leq n,\ 1 \leq j \leq k}.$$

So $B_{\mathbf{x}_{|[0,kT]}}$ has $k(n-1)+1$ bars. As in the first case, each local minimum b_i ($i > 1$) of $\mathbf{x}_{|]0,T[}$ gives k bars starting at b_i that already where in $B_{\mathbf{x}_{|[0,2T]}}$, and at least $k-1$ of them are identical. The remaining bar is $(b_1^{(1)}, d_1^{(1)}(1))$.

Case 3: $b_1 \neq \mathbf{x}(0)$ **and** $b_n = \mathbf{x}(0) = \mathbf{x}(T)$.

This case is similar to case 2. Barcodes are of the form

$$B_{\mathbf{x}_{|[0,lT]}} = \{(b_i^{(j)}, d_i^{(j)}(k))\}_{1 \leq i \leq n-1,\ 1 \leq j \leq k} \cup \{(b_n^{(1)}, d_n^{(1)}(1))\}.$$

So $B_{\mathbf{x}_{|[0,kT]}}$ has $k(n-1)+1$ bars. As in the first case, each local minimum b_i ($i < n$) of $\mathbf{x}_{|]0,T[}$ gives k bars starting at b_i that already where in $B_{\mathbf{x}_{|[0,2T]}}$, and at least $k-1$ of them are identical. The remaining bar is $(b_n^{(1)}, d_n^{(1)}(1))$.

Case 4: $b_1 = b_n = \mathbf{x}(0) = \mathbf{x}(T)$ **(see figure 4.2)**.

In this case, in $B_{\mathbf{x}_{|[0,2T]}}$, $b_n^{(1)}$ and $b_1^{(2)}$ are the same local minimum. Thus $B_{\mathbf{x}_{|[0,kT]}}$ has $k(n-1)+1$ bars. In $B_{\mathbf{x}_{|[0,2T]}}$, either $(b_n^{(1)}, d_n^{(1)}(2)) = (b_1^{(1)}, d_1^{(1)}(2)) = (b_1^{(1)}, d_1^{(1)}(1))$ or $(b_n^{(1)}, d_n^{(1)}(2)) = (b_n^{(2)}, d_n^{(2)}(2)) = (b_n^{(1)}, d_n^{(1)}(1))$.

The first situation is similar to case 3: the barcode is of the form

$$B_{\mathbf{x}_{|[0,kT]}} = \{(b_i^{(j)}, d_i^{(j)}(k))\}_{1 \leq i \leq n-1,\ 1 \leq j \leq k} \cup \{(b_n^{(1)}, d_n^{(1)}(1))\}.$$

Each local minimum b_i ($i < n$) of $\mathbf{x}_{|]0,T[}$ gives k bars starting at b_i that already where in $B_{\mathbf{x}_{|[0,2T]}}$, and at least $k-1$ of them are identical. The remaining bar is $(b_n^{(1)}, d_n^{(1)}(1))$. Note that here the k bars starting at b_1 are identical.

The second situation is similar to case 2: the barcode is of the form

$$B_{\mathbf{x}_{|[0,kT]}} = \{(b_1^{(1)}, d_1^{(1)}(1))\} \cup \{(b_i^{(j)}, d_i^{(j)}(k))\}_{2 \leq i \leq n,\ 1 \leq j \leq k}.$$

Each local minimum b_i ($i > 1$) of $\mathbf{x}_{|]0,T[}$ gives k bars starting at b_i that already where in $B_{\mathbf{x}_{|[0,2T]}}$, and at least $k-1$ of them are identical. The remaining bar is $(b_1^{(1)}, d_1^{(1)}(1))$. Note that here the k bars starting at b_n are identical.

□

Remark 5. *The infinite bar could be replaced by a bar ending at* $\max(\mathbf{x})$ *to have* k *identical long bars.*

In what follows, a gait signals can be seen as noisy periodic functions where the number k of observed periods is the number of steps made by the subject. Thus, we expect the corresponding barcode to have groups of $k-1$ or k similar bars.

4.3 Protocol and data

Our dataset is composed of gait trials from 22 MS patients and 10 young HS. The studies involving human participants were reviewed and approved by Protection des Personnes Nord Ouest III (ID RCB: 2017-A01538-45). The patients/participants provided their written informed consent to participate in this study. The protocol is a walking exercise consisting in a 12m walk with a U-turn while wearing 4 XSens® sensors (XSens® Technologies, Enschede, the Netherlands; autonomy 6 h, device dimension 47 × 30 × 13 mm, weigth 16 g, acceleration range ±160 m/s^2, angular velocity range ±2000 deg/s, dynamic accuracy roll/pitch 0.75 deg RMS, dynamic accuracy heading 1.5 deg RMS): one on the dorsal part of each foot (left foot: LF, right foot: RF), one on the lower back (T) and one on the head (H), fixed using a Velcro band designed by XSens®. Additional measurements including average walking speed were done for each trial using information from a GaitRite® mat, which detects the initial and final contacts of the feet on the ground. The experiment was conducted in two sessions, 6 months apart, that will be referred to as M0 and M6. During each session, the protocol was performed twice. The part of the signals corresponding to the U-turn was automatically removed (using the GaitRite® data, as the U-turn happened outside the mat) so that each exercise gives two signals: one for the forward path and one for the return. To sum up, for each IMU (LF, RF, T, or H) of each subject, there are 8 trials: 4 for M0 (F1: forward 1, R1: return 1, F2: forward 2, R2: return 2) and 4 for M6 (F3, R3, F4, R4).

XSens® sensors are inertial measurement units (IMUs) that measure the 3D accelerations, 3D angular velocities and 3D magnetic fields. The axes are defined on Fig 4.3: the Y-axis is parallel to the ground and orthogonal to the walking direction. The data were sampled at 100Hz. For our study, we used the angular velocity around the Y-axis (Gyr-Y) from the feet IMUs (LF, RF), which provides signals suitable for gait assessment as it corresponds to the rotation of the foot around the medio-lateral axis. [OBMM$^+$18, DQO$^+$20, Mar12, SRV$^+$04, MRG$^+$15]. In what follows, we will refer to those Gyr-Y signals as **gait signals**.

All the subjects were recruited between June and September 2018 at Percy Hospital (Clamart, France). The characteristics of the subjects are displayed in Table 4.1. Seven out of the 22 participants had an advanced disease requiring permanent walking aid (cane(s), walker and/or human help). Two patients needed human help to perform the walking test. Included participants in the MS group had an EDSS between 2 and 6.5, as disabilities greater than 7 completely impede walking.

Fig 4.4 shows an example of gait signals for both feet of a healthy subject. Gait signals can be described as a succession of gait cycles, which are composed of a support phase (when the foot touches the ground) and an oscillation phase (when it is off the ground)



Figure 4.3: XSens Sensor, with axes orientations. Source:[DQO+20]

	MS (n=22)	HS (n=10)
Sex (M/F)	9/13	4/6
Age (years)	58 (11)	26 (1)
Height (m)	1.71 (0.09)	1.72 (0.09)
Weight (kg)	71.2 (16.6)	58.2 (10.9)
BMI (kg/m2)	24.3 (5.1)	21.0 (3.0)
EDSS	5.0 [3.5-6]	-

Table 4.1: Baseline characteristics of patients with multiple sclerosis (MS) and healthy subjects (HS). For the age, height, weight and body mass index (BMI), the mean and the standard deviation (SD) are displayed. For the Expanded Diseases Status Scale (EDSS) the statistics are reported as median and interval quartile range (IQR).

[DQO+20]. The support phase starts with the **heel strike** (when the foot hits the ground) and ends with the **toe off** (when the foot leaves the ground) as shown in Fig 4.5. The plateau around 0 for angular velocity corresponds to the phase between the **foot flat** and the **heel off** events. During the oscillation phase, the angular velocity goes up, stays almost constant and then decreases until the next heel strike.

4.4 Method description

4.4.1 Overview of our method and guidelines for the clinician

Here, we give an informal description of the objects used in our method and explain how we use them, followed by guidelines on how clinicians can use our method. The mathematical concepts corresponding to terms in bold they will be described in the following sections.

4.4.1.1 Overview of our method

The goal of our method is to produce a quantitative analysis of a database of gait signals, by using comparisons based on topological properties. Given a database of gait signals

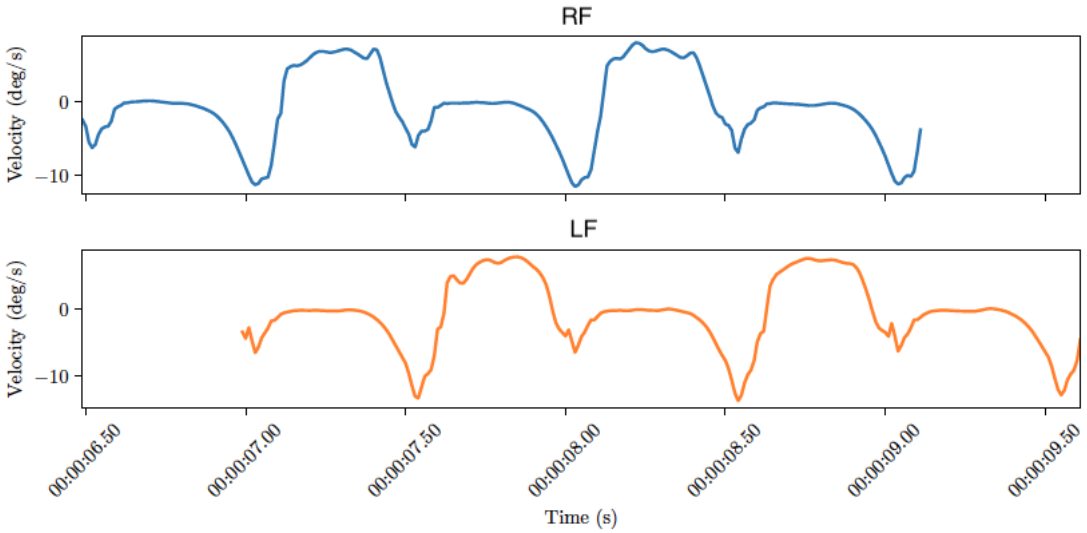


Figure 4.4: Two gait signals from a healthy subject. Top: right foot. Bottom: left foot.

and a partition of those signals into groups, it outputs a 2D point cloud and a list of features for each group of the partition.

We start by constructing a topological summary of each signal called a **persistence barcode**. A persistence barcode is a set of bars that represents the oscillations of a signal, where a long bar corresponds to a large variation. Each signal is represented by its barcode. There exists a notion of distance between barcodes called the **bottleneck distance**, that we use to compare pairs of signals via their barcodes. Before computing the bottleneck distance, we remove the longest bars from each barcode: if the barcode corresponds to a trial with k steps, we remove the k longest bars. This makes the distance less sensible to the number of steps and more sensible to the oscillations of the signal (we explain why in the mathematical description).

After having computed all the distances between pairs of barcodes, each barcode is represented as a point in the 2D Euclidean space using a **dimension reduction algorithm** called **UMAP**. This algorithm outputs a 2D point cloud whose structure is as close as possible to the structure of our set of barcodes endowed with the bottleneck distance. The obtained point cloud is a visualization of all gait signals arranged based on their topological similarity. For a given partition of the dataset, each point can be colored according to its group in order to visualize the groups on the point cloud. For each group, we compute three features: its **silhouette score** with respect to other groups (to measure their separability), and its **mean squared distance** and **diameter** (to measure its density).

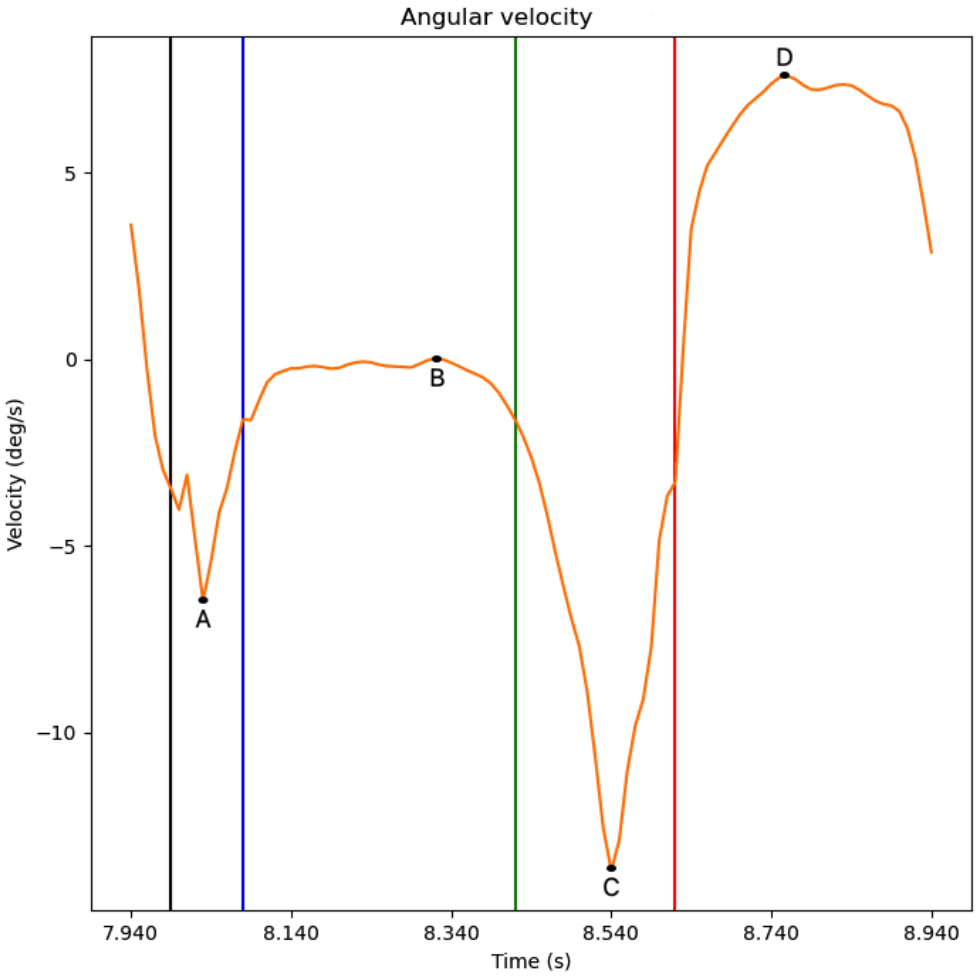


Figure 4.5: Key events of the support phase of a gait cycle. The events are represented by vertical lines. From left to right: Heel Strike (black), Foot Flat (blue), Heel Off (green) and Toe Off (red). Pairs of local extrema (A, B) and (C, D) define bars in the signal's persistence barcode.

Our method can be summarized as follows (also see Fig 4.6):

- **Input:** a database of gait signals and partitions into groups.
- Construct the persistence barcode from all the gait signals.
- For each signal, count its number of steps k and remove the k longest bars from its barcode.
- Compute the bottleneck distance between all pairs of those barcodes.
- Compute a 2D (or 3D) point cloud using UMAP.
- **Output:** the point cloud and, for each partition, the silhouette score, mean squared distance, and squared diameter for all (pairs of) groups.

4.4.1.2 Guidelines for the clinician

Here, we give guidelines on how to concretely use our method to study cohorts of patients. For users who do not want to dive into the mathematical details, all the intermediate steps of the above summary of the method can be considered to already be implemented.

- Construct a database of gait signals (preferably with information on the number of steps, if not, compute it automatically as we explain later).
- Recover the point cloud.
- Partition the cohort into groups based on additional information (clinical scales, healthy/pathological, different sessions, right/left foot etc...). Each partition is defined to study a specific aspect of the cohort.
- For each partition: color points belonging to different groups in different colors, and recover tables containing all the features (silhouette scores, mean squared distances and squared diameters, or others if needed).
- Interpret (more details below).

To interpret the results, the key idea is that the relative distances between points on the point cloud reflect the difference of structure of the oscillations of the signals.

For a given partition, if two groups are well separated on the point cloud, then the criteria that define the groups are related to the differences of gait trials. For example: if the points from the M0 session of a given patient are well separated from the M6 points, then there has been a significant change in the patient's gait and the clinician can interpret it as an evolution of the disease. On the contrary, if points of both session are not separable and form one dense group, then there is no intra/inter-session variability. If healthy subjects are well separated from patients, then the disease has an impact on gait. If points from a same session of a subject have a large mean squared distance (compared to other subjects), this means that there is a high intra-session variability.If points representing signals from the IMU placed on the left foot of a subject are well separated from those representing right foot signals, then there is an asymmetry. We followed those guidelines in our study, which we describe in the last sections of this chapter.

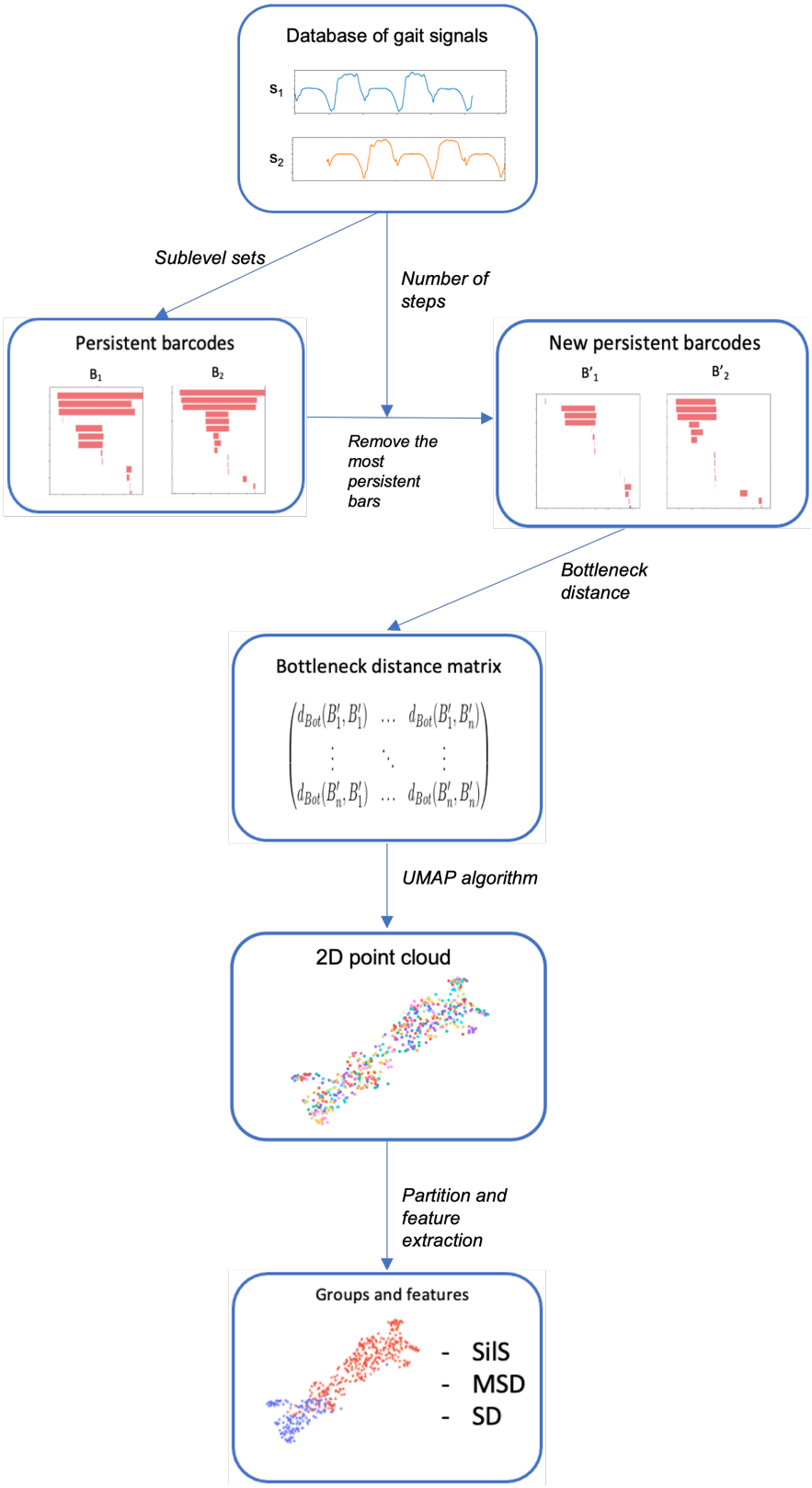


Figure 4.6: Summary of our method.

4.4.2 Mathematical description our of method

This section describes the mathematical construction of the objects from TDA used in our method, in a way that is understandable without any TDA background.

4.4.2.1 Persistence barcodes from sublevel sets

Here, we give a more detailed description of persistent homology of the sublevel sets filtration than in Chapter 2. For a given real-valued function $\mathbf{x} : t \mapsto \mathbf{x}(t)$ and threshold $\alpha \in \mathbb{R}$, the sublevel set SL_α is defined as

$$SL_\alpha = \mathbf{x}^{-1}(]-\infty, \alpha]).$$

As explained above, our goal is to study the evolution of the arrangement of data through different scales. This evolution can be summarized by a **persistence barcode**. Formally, the persistence barcode (from sublevel sets) of a signal described by a function $\mathbf{x}$ is the set of pairs (date of birth, date of death) of the connected components of the sets SL_α as α goes from $-\infty$ to $+\infty$. That is to say, for a given α, if SL_α has a connected component with no point belonging to any SL_β such that $\beta < \alpha$, we say this component was born at α. If two components from $SL_\beta,\ \beta < \alpha$ have merged in SL_α then we say that the youngest one died at α.

The persistence barcode of the sublevel sets of a time series can be constructed by pairing local minima to local maxima using the following algorithm (illustrated in Fig 4.7):

1. Mark the level on the Y-axis of all the local extrema of the signal. The first and last points can be ignored if they are local **maxima**.
2. Start drawing a vertical bar going up from the **global minimum**.
3. Each time the bars reach the level of a another **local minimum**, start another vertical bar at this minimum. Then make all the bars go up to the level of the next extrema.
4. Each time the bars reach the level of a **local maximum**, if that point has one bar at its left and one at its right, then the shortest of those two bars stops growing. Then make all the bars go up to the level of the next extrema.
5. When the bars reach the **global maximum**, stop, as the remaining bar will keep growing indefinitely.
6. The persistence barcode is made of all the pairs of (start, end) vertical coordinates of the bars obtained this way (we ignore time coordinates), where the longest bar goes up to $+\infty$. It is usually represented horizontally as in Fig 4.8. This representation is obtained by keeping only the bars and Y-axis, and rotating the graph by 90° clockwise.

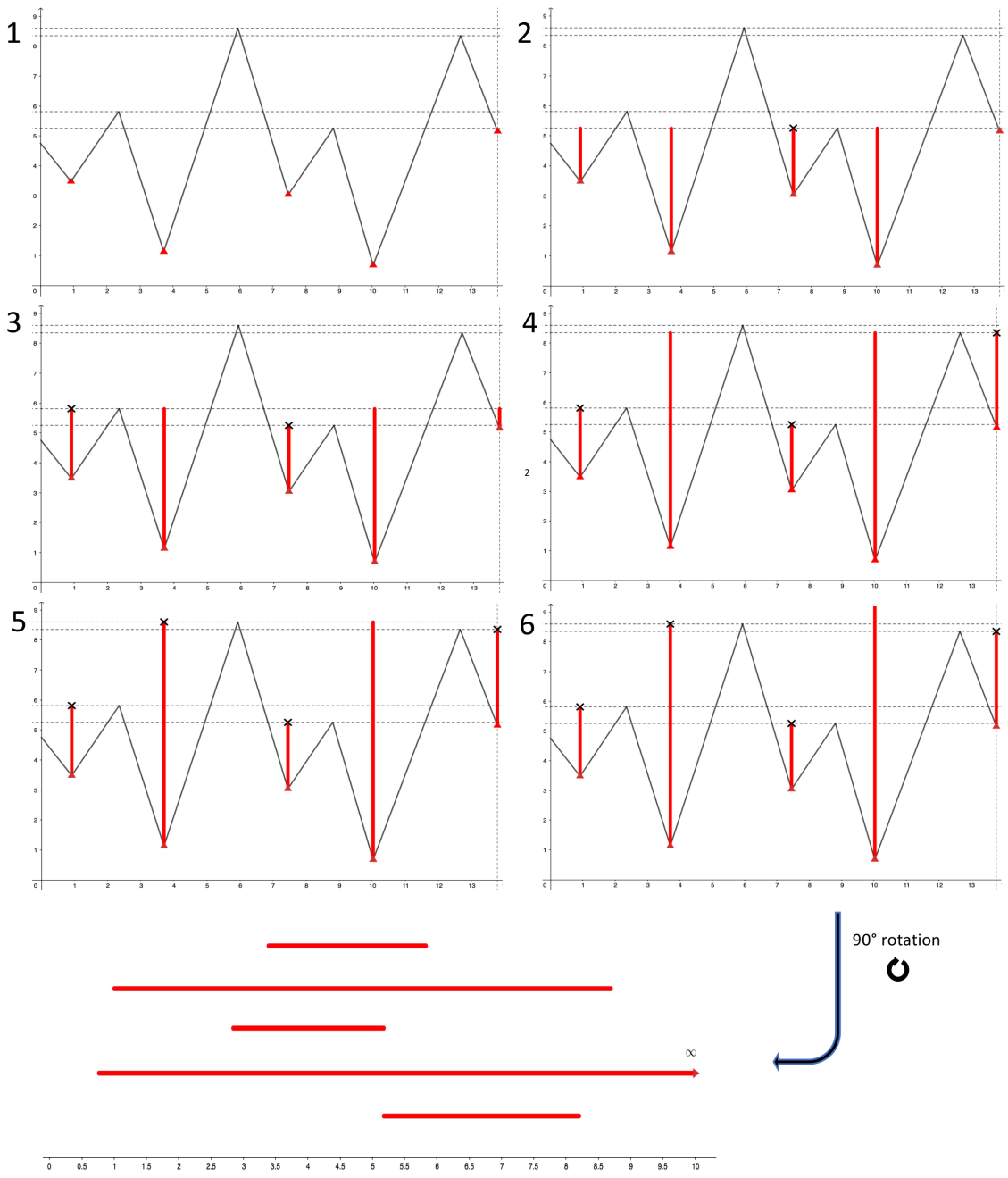


Figure 4.7: **Construction of a persistence barcode.** 1: Mark all the local extrema except for the first maximum. The minima are marked by red triangles, the maxima by horizontal lines. 2: Grow bars until the first local maximum. The third bar stops growing. 3: At the second local maximum, the first bar stops growing. 4: At the third local maximum, the fifth bar stops growing. 5: At the fourth local maximum, the second bar stops growing. 6: The fourth bar grows to infinity. Bottom: Horizontal representation of the persistence barcode.

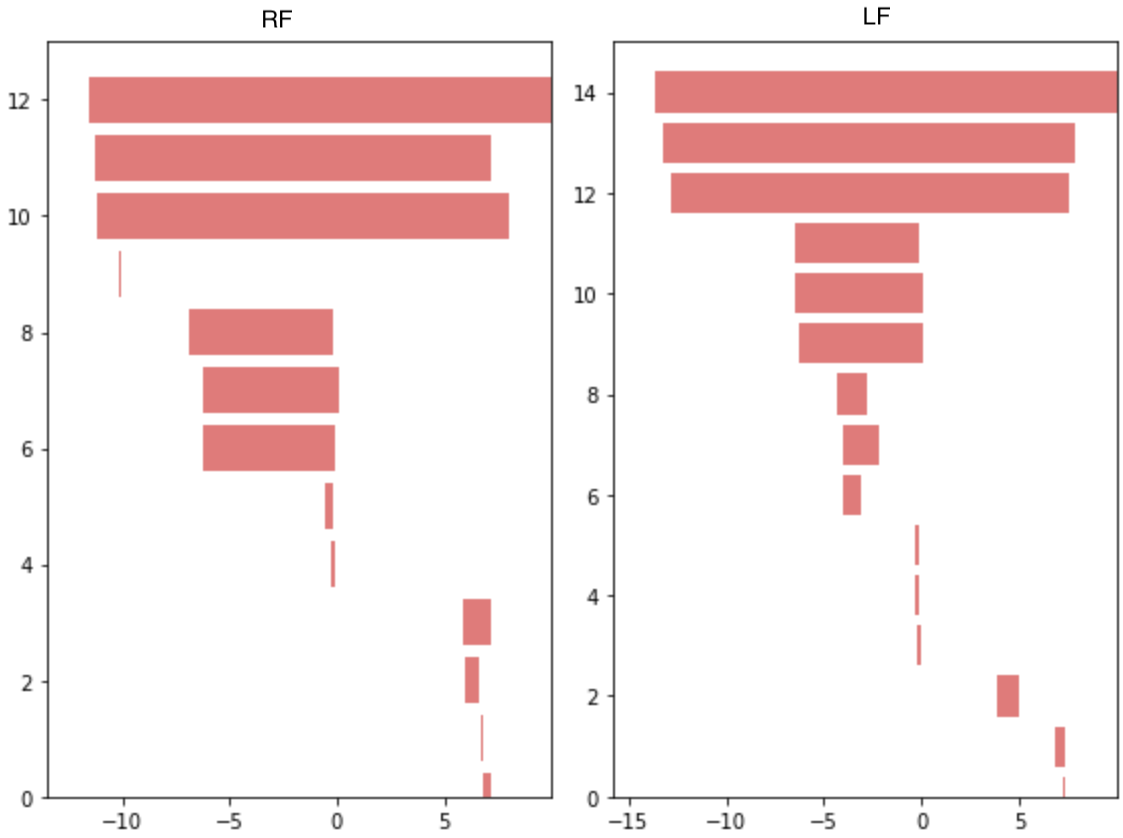


Figure 4.8: Persistence barcodes from the sublevel sets of the signals from Fig 4.4.

Let us now describe the persistence barcode corresponding to a single gait cycle. Persistence barcodes of time series can be understood in terms of pairs of local minima and maxima. On Fig 4.5, four important local extrema can be noticed:

- One minimum between the heel strike and foot flat (point A on Fig 4.5). The angular velocity keeps decreasing for some time after the heel strike before going back to zero at foot flat.
- One maximum around zero at the plateau between foot flat and heel off (point B on Fig 4.5).
- A second minimum just before toe off (point C on Fig 4.5). Heel off makes the angular velocity decrease below zero and toe off makes it go back above zero.
- A second maximum during the oscillation phase at the high plateau of the gait cycle (point D on Fig 4.5).

Let (t_P, y_P) denote the coordinates of a point P of a time series. The barcode of the signal on Fig 4.5 will have two bars that are characteristic of gait cycles: a long bar $(y_C, +\infty)$ corresponding to the pair (C, D) and a smaller bar (y_A, y_B) corresponding to (A, B). The other smaller bars are considered to be oscillations, irregular movements or noise.

Let us now consider full gait trials. Fig 4.8 shows the persistence barcodes of the two typical gait signals represented on Fig 4.4. Three gait cycles can be distinguished for both signals on Fig 4.4, each one is responsible for a long bar and a medium-sized bar on Fig 4.8. Note that only one bar goes to infinity, any other bar corresponding to a pair (P, Q) has

coordinates (y_P, y_Q). The smaller bars are oscillations. For example, on the LF barcode, the 7th, 8th and 9th bars (counting from top to bottom) correspond to the oscillation that happens just after heel strike during each gait cycle (Fig 4.5 shows where the heel strike happens on a signal). Those oscillations are less present on the RF signal.

Note that counting the longest bars is equivalent to counting the steps (including the last step, that may be incomplete): on Fig 4.8, the three long bars correspond to the three steps visible on Fig 4.4.

4.4.2.2 Distance between barcodes

Barcodes can be compared using the bottleneck distance [BCY18]. Recall that a barcode is a set of pairs (x, y) that are the start and end vertical coordinates of each bar. The same pair can be represented multiple times and y can be equal to $+\infty$ (this happens exactly once if the signal is defined on an interval). For barcodes B and B', the bottleneck distance is based on an idea from optimal transport, using bijections between the two barcodes (functions from B to B' such that each bar from B' is associated to a unique bar from B). Let $\Gamma(B, B')$ be the set of bijections from B to B'. Note that if two finite sets do not have the same number of elements there are no bijections between them, so we include to B and B' all bars (x, x) of length zero (an infinite number of times), so that there always exists a bijection between B and B'. For any $\gamma \in \Gamma(B, B')$ and any bar $b = (x, y) \in B$ such that $\gamma((x, y)) = b' = (x', y')$, the two bars can be compared using the infinite norm:

$$||b - b'||_\infty = \begin{cases} |x - x'| & \text{if } y = y' = \infty \\ \max(|x - x'|, |y - y'|) & \text{otherwise.} \end{cases} \tag{4.1}$$

For each γ, the pair of bars (b, b') such that $||b - \gamma(b)||_\infty$ is maximal gives a notion of similarity between B and B' induced by the pairing of bars defined by γ. The bottleneck distance is defined by choosing the bijection that minimizes this quantity (which means that we associate each bar of B to the one in B' that is the most similar). Formally, the bottleneck distance between B and B' is given by:

$$d_{Bot}(B, B') = \inf_{\gamma \in \Gamma(B,B')} \sup_{b \in B} ||b - \gamma(b)||_\infty. \tag{4.2}$$

Stability theorems such as those presented in Chapter 2 prove that under generic assumptions, barcodes associated with similar signals are close for the bottleneck distance.

As explained above, the number of long bars in barcodes from gait signals is the number of steps. This implies that the bottleneck distance between two barcodes corresponding to trials with a different number of steps will be high. Indeed, in that case, one of the two barcodes will have more long bars than the other so each bijection γ will pair at least one long bar b to a short bar $\gamma(b)$, resulting in a high $||b - \gamma(b)||_\infty$ and thus in a high distance.

This means that the bottleneck distance will mainly distinguish signals that have a different number of steps. However a different number of steps can be due to many factors such as experimental conditions, the subject's height, age, or the foot that does the first step (a RF and a LF signal from the same exercise can have a different number of steps if a subject starts and ends with the same foot). To reduce this step-counting effect and focus more on oscillations, we propose to count the steps on each signal and remove the k longest bars from the corresponding barcode, where k is the number of steps. The number of steps can be computed from signals using the **autocorrelation function** (ACF)

of each signal, which is defined by $ACF[i] = \rho_{\mathbf{xx}[i:]}$. The time when the second peak of the ACF is reached is the duration of the first gait cycle, and the number of steps can be deduced from this quantity and the duration of the trial. This method is heuristic and has limitations, notably with signals from patients with very deteriorated gait. For any future clinical use of our method, steps could be counted during the protocol and included in the data.

Once the barcodes from every gait signal of the database have been computed, the next step of our method is the following: for each pair of barcodes B and B' corresponding to trials with respectively k and k' steps, remove the k (resp. k') longest bars from B (resp. B') to get a new barcode $\tilde{B}$ (resp. $\tilde{B}'$) and compute $d_{Bot}(\tilde{B}, \tilde{B}')$.

4.4.2.3 Visualization algorithm

Barcodes (and the gait signals they represent) can be seen as points in a (non-Euclidean) metric space, endowed with the bottleneck distance. To be visualized, these points need to be projected onto the 2D or 3D Euclidean space. Several algorithms can compute such a projection, including the UMAP algorithm [MHM18], t-SNE [VdMH08] and multidimensional scaling (MDS) [CC08]. MDS focuses on respecting the distance matrix, while UMAP and t-SNE intend to represent the structure of the original metric space. We chose UMAP to focus on structure, and because its parameters can be chosen so that more global structure is preserved than with t-SNE.
The UMAP algorithm has been used in applications such as the study of odors and molecular structures [RJT$^+$21], physical and genetic interactions [DSQ$^+$20] or genomic data [SHK$^+$20].

The UMAP algorithm takes as input a distance matrix (here, the matrix of all bottleneck distances between all pairs of barcodes) and two parameters: n_neighbors and min_dist. It outputs a 2D (or 3D) point cloud where each point represents a gait signal, whose structure induced by the Euclidean distance is as close as possible to the structure induced by the bottleneck distance on the space of barcodes. That is to say, if two barcodes are close according to the bottleneck distance, then the corresponding points in the point cloud will be close according to the Euclidean distance.

The two parameters control the compromise between respecting the local and global structure of the data. A low n_neighbors parameter makes the UMAP algorithm focus more on the local structure around each point, whereas a high n_neighbors will make it focus on the global structure. The min_dist parameter is the minimum distance allowed between two points in the point cloud. A low min_dist allows the algorithm to represent similar barcodes as close points in the point cloud. A high min_dist will prevent it to produce very dense neighborhoods to make the global structure appear more clearly.

In what follows, we use UMAP with the metric induced by the bottleneck distance, with parameters n_neighbors $= 45$, min_dist $= 0.3$ and in 2D. See Fig 4.9 for an example.

4.4.3 Feature extraction

The goal of our method is to study a database of gait signals. As the UMAP projection preserves the structure induced by the bottleneck distance between barcodes, information can be extracted by studying the relative positions and neighborhood relations of points in the point cloud. To do this, we regroup signals that share a given characteristic (for example: both being from a healthy/pathological subject, or from the same session) and study the geometry of the groups and the relations between groups, using three fea-

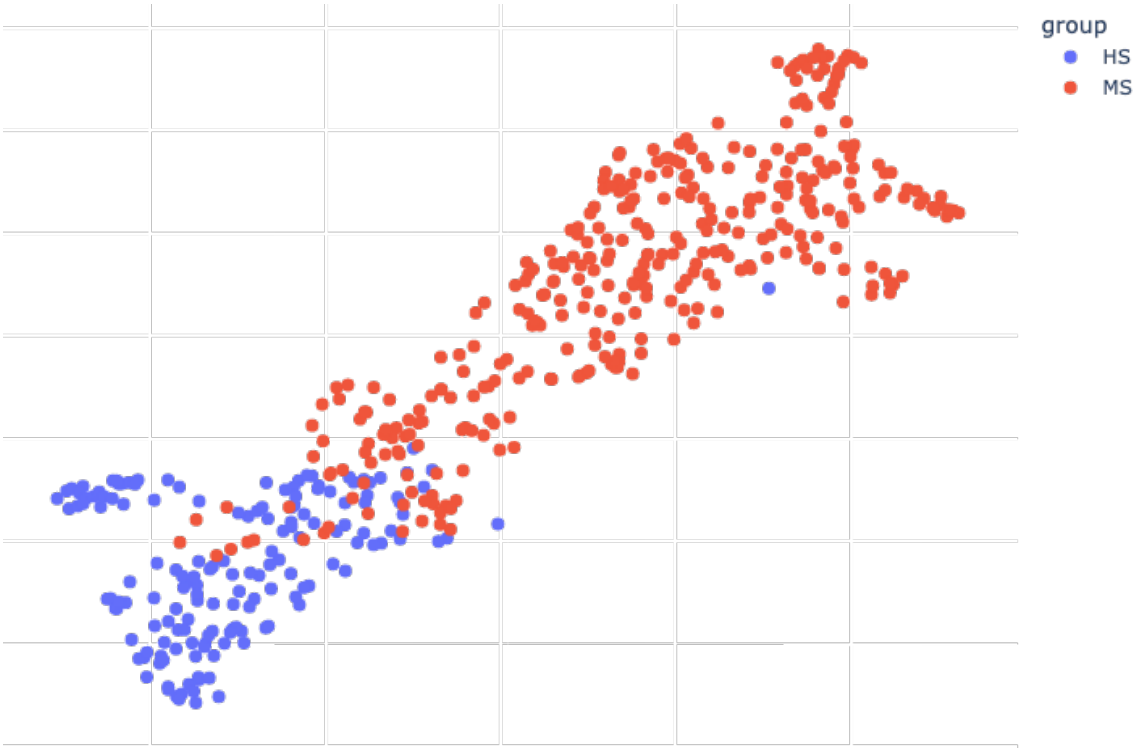


Figure 4.9: UMAP plot colored by group.

tures: the **silhouette score** (Sil), the **mean squared intra-group distance** (MSD) and the **squared diameter** (SD).

Note that, when using a UMAP projection, no information can be extracted from the absolute coordinates of the points or the absolute distance between two points. The position of a point should only be studied through its distance to other points, and distances should be studied in a relative way. For example, saying that point A and point B are closer together than they are with point C means that signal A is more similar to signal B than to signal C.

Let $S = (s_i)_{1 \leq i \leq n}$ be a database of n gait signals and $X = (x_i)_{1 \leq i \leq n}$ be a (2D or 3D) point cloud obtained with the above method, where each point x_i represents a signal s_i. Let $(C_i)_{i \in I}$ be a partition of X into groups.

4.4.3.1 Silhouette score

Let i, j be two distinct indices in I, $|C|$ denote the cardinal of set C and $||.||_2$ denote the Euclidean norm. The silhouette score of a point $x \in C_i$, with respect to group j is defined as:

$$Sil(x, C_j) = \frac{b - a}{\max(a, b)}, \tag{4.3}$$

where $a = \frac{1}{|C_i|-1} \sum_{y \in C_i, y \neq x} ||x - y||_2$ is the mean distance between x and all other points in the same group, and $b = \frac{1}{|C_j|} \sum_{y \in C_j} ||x - y||_2$ is the mean distance between x and all points in group j.

The silhouette score of group i with respect to group j is defined as the mean silhouette score of the points of group i with respect to group j :

$$Sil(C_i, C_j) = \frac{1}{|C_i|} \sum_{x \in C_i} Sil(x, C_j). \tag{4.4}$$

4.4.3.2 Mean squared distance

The MSD of group i is defined as:

$$MSD(C_i) = \frac{2}{|C_i|(|C_i|-1)} \sum_{x,y \in C_i, x \neq y} ||x-y||_2^2. \tag{4.5}$$

4.4.3.3 Squared diameter

The squared diameter of group C_i is defined as:

$$SD(C_i) = \max_{x,y \in C_i} (||x-y||_2^2). \tag{4.6}$$

Note that we have squared the distance to be consistent with the MSD.

4.4.3.4 Interpretation of the features

The silhouette score is a clustering evaluation metric that is used to determine if the groups we define can be considered as clusters in our point cloud. It takes values between -1 and 1. A value close to 1 means good clustering (the groups are well separated from one another and dense), a value close to 0 means overlapping groups, and a value close to -1 means bad clustering. To understand this, let us consider concrete examples. For a point $x \in C_1$, $Sil(x, C_2) = 0.5$ means that $b = 2a$ (using the same definition as above for a and b), i.e. that x is on average twice as far from points of C_2 than from points of C_1. Thus, $Sil(C_1, C_2) = 0.5$ means that on average a point of C_1 will be twice as far from C_2 than from C_1. On the contrary, $Sil(x, C_2) < 0$ means that x is on average closer to C_2 than to C_1, and having x in C_2 would increase the score. Negative scores are thus interpreted as bad clustering. If $Sil(x, C_2)$ is close to zero, then the difference between a and b is small compared to the size of the groups, so x can be considered to be as close to C_1 and C_2, i.e. x is "between C_1 and C_2". A small $Sil(C_1, C_2)$ then means that the two groups are overlapping. The fact that values are always between -1 and 1 is an advantage of the silhouette score compared to other clustering evaluation metrics because it can be interpreted on its own without necessarily being compared to the score of a different clustering. Note that the silhouette score is not symmetrical: $Sil(C_i, C_j)$ is not necessarily equal to $Sil(C_j, C_i)$.

The MSD and SD measure the density of the groups. They should only be interpreted relatively to other groups. The MSD measures the average (squared) distance between points of the same group, so a smaller MSD means that a group has points that are closer to one another on average. The SD measures the largest of those distances. The SD is complementary to the MSD because it focuses on the two points that are the furthest apart. For example, a group C_1 of points uniformly spread on a line and a group C_2 with one point at the beginning of the line and all the other points at the end would have the same SD but C_2 would have a significantly lower MSD as it is very dense except for one outlier. A joint analysis of the MSD and SD can thus detect outliers in a group with relatively low MSD and high SD.

4.5 Results

We applied our method to study the database of gait signals described above. The study is divided in three parts: the first one compares healthy subjects (HS) to multiple sclerosis

(MS) patients, the second one is a series of experiments that compare subjects with different EDSS scores, and the third one studies the evolution of each subject between M0 and M6. Each experiment corresponds to a different partition of the database.

4.5.1 HS/MS experiment

Here, we divide our database into two groups: HS and MS. HS is the group of points from healthy subjects and MS is the group of points from multiple sclerosis patients.

This partition can be visualized on Fig 4.9. Table 4.2 shows the Sil, MSD and SD values for the partition (HS, MS).

$Sil(HS, MS)$	$Sil(MS, HS)$	$MSD(HS)$	$MSD(MS)$	$SD(HS)$	$MSD(MS)$
0.68	**0.41**	6.8	18.6	109.5	147.2

Table 4.2: **Features for HS and MS patients.** The subjects have been divided into two groups: HS and MS patients. The silhouette scores, MSD and SD have been computed on those groups.

The values of the silhouette score are : $Sil(HS, MS) = 0.68$ and $Sil(MS, HS) = 0.41$.

4.5.2 EDSS experiments

In this section, we perform a series of experiments to study the relation between the EDSS and the relative position of points on the point cloud. For a given threshold i, we divide our database into two groups: $\{EDSS \leq i\}$ and $\{EDSS > i\}$. $\{EDSS \leq i\}$ is the group of points from subjects with EDSS lower than or equal to i and $\{EDSS > i\}$ is the group of signals from subjects with EDSS strictly higher than i. HS are given an EDSS of 0. We consider the following values of i: 0, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, and 6 as there are no patients with EDSS under 2 or above 6.5. Note that the experiment with $i = 0$ is the HS/MS experiment.

The EDSS corresponding to each point can be visualized on Fig 4.10. Table 4.3 shows the Sil, MSD and SD values for each partition $(\{EDSS \leq i\}, \{EDSS > i\})$. In this table, for the sake of clarity, we use the notation $Sil(\leq i, > i)$ instead of $Sil(\{EDSS \leq i\}, \{EDSS > i\})$.

i	$Sil(\leq i, > i)$	$Sil(> i, \leq i)$	$MSD(\{EDSS \leq i\})$	$MSD(\{EDSS > i\})$	$SD(\{EDSS \leq i\})$	$SD(\{EDSS > i\})$
0	**0.68**	**0.41**	6.8	18.6	109.5	147.2
2	**0.60**	**0.46**	11.2	16.1	136.8	137.5
2.5	**0.54**	**0.44**	13.7	16.4	136.8	137.5
3	**0.52**	**0.56**	14.0	11.2	136.8	95.3
3.5	**0.44**	**0.58**	17.6	9.9	142.0	95.3
4	**0.34**	**0.60**	22.4	8.8	168.5	57.3
4.5	**0.34**	**0.60**	22.4	8.8	168.5	57.3
5	**0.21**	**0.58**	29.9	8.5	210.3	57.3
5.5	**0.15**	**0.70**	31.2	4.8	210.3	26.0
6	**0.088**	**0.68**	34.1	4.8	210.3	26.0

Table 4.3: **Features for points with EDSS lower or equal to/ stricly higher than each threshold.** For a given threshold i, the points from each subject have been divided in two groups: points corresponding to patients with EDSS lower than or equal to i ($\{EDSS \leq i\}$), and points with EDSS higher than i ($\{EDSS > i\}$). The silhouette scores, MSD and SD have been computed on those groups for several values of i.

All silhouette scores are positive. For each i, at least one of the two silhouette scores $Sil(\{EDSS \leq i\}, \{EDSS > i\})$ and $Sil(\{EDSS > i\}, \{EDSS \leq i\})$ is above 0.5.

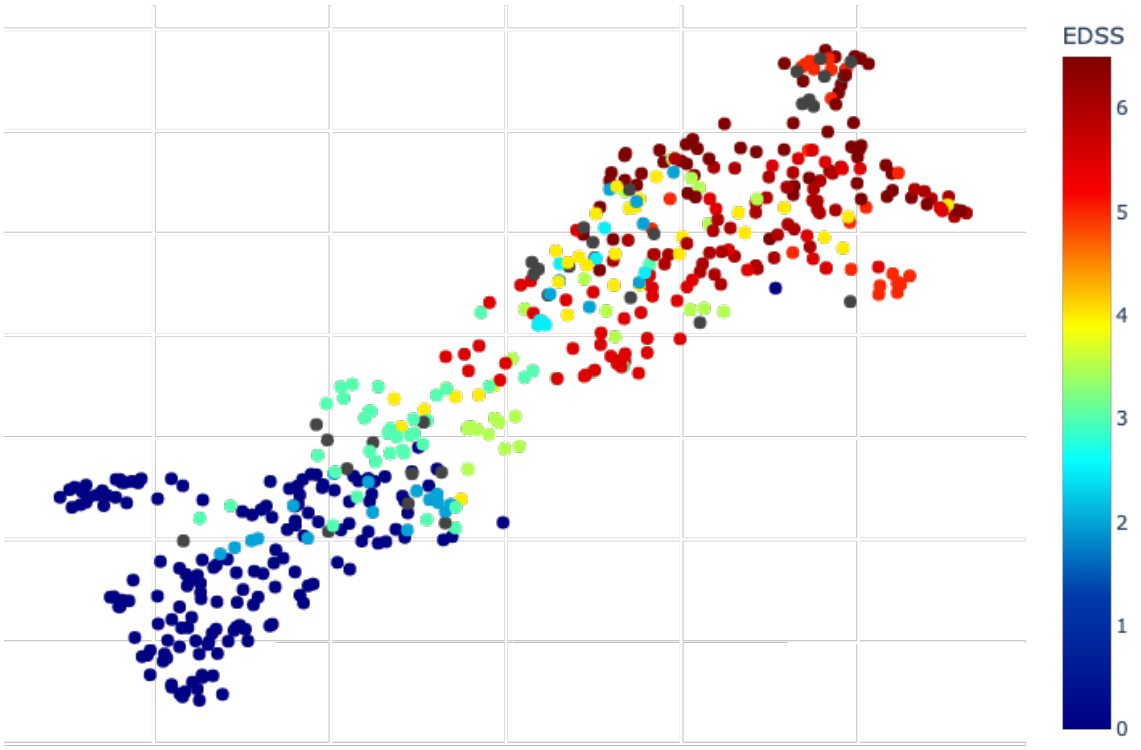


Figure 4.10: UMAP plot colored by EDSS.

4.5.3 Longitudinal experiment

In this section, for each subject, we compare the M0 session to the M6 session. We start by dividing our database into 32 groups (each corresponding to one subject): each subject is given an ID between 1 and 32, and group i corresponds to signals from subject i. Then, each group i is subdivided into two groups $M0_i$ and $M6_i$. $M0_i$ is the group of signals from the M0 session of subject i, $M6_i$ is the group of signals from their M6 session. The final partition has 64 groups: $\{(M0_i, M6_i),\ 1 \leq i \leq 32\}$. The goal of this experiment is to study the evolution of each patient, therefore silhouette scores are only computed for pairs $(M0_i, M6_i)$.

4.5.3.1 Study of subject 12

The two highest silhouette scores are from subject 12: $Sil(M0_{12}, M6_{12}) = 0.83$ and $Sil(M6_{12}, M0_{12}) = 0.96$. Subject 12 has an EDSS of 4 at M0 and 5 at M6 and is the only MS patient to have a variation of their EDSS of more than 0.5.

Fig 4.11 shows the position of points from subject 12 on the point cloud. Fig 4.12 shows gait signals from subject 12 at M0 and M6. It can be noticed that some oscillations appear at M6.

4.5.3.2 Study of subject 15

Subject 15 has silhouette scores $Sil(M0_{15}, M6_{15}) = 0.49$ and $Sil(M6_{15}, M0_{15}) = -0.21$. $M6_{15}$ has the second highest MSD and SD of all M6 sessions.

Fig 4.11 shows the position of points from subject 15 on the point cloud.

i (ID)	$Sil(M0_i, M6_i)$	$Sil(M6_i, M0_i)$	$MSD(M0_i)$	$MSD(M6_i)$	$SD(M0_i)$	$SD(M6_i)$
1	-0.17	0.19	3.1	2.1	12.1	8.9
2	-0.29	0.55	**24.9**	2.7	**99.9**	8.0
3	0.32	0.08	1.4	2.2	5.8	6.6
4	-0.03	-0.02	0.9	0.9	3.4	3.0
5	0.01	0.15	0.3	0.3	1.2	1.0
6	-0.06	0.53	1.4	0.2	3.6	0.7
7	0.28	0.40	0.9	0.7	2.7	2.7
8	-0.22	0.14	1.8	0.8	6.6	2.1
9	0.03	-0.11	2.3	3.3	5.3	12.7
10	0.31	0.22	1.9	2.9	6.2	8.7
11	0.21	-0.09	1.3	2.1	3.9	6.1
12	**0.83**	**0.96**	**1.4**	**0.1**	**5.1**	**0.3**
13	0.26	-0.14	0.2	0.4	0.6	1.0
14	0.61	0.79	1.6	0.5	6.0	1.3
15	**0.49**	**-0.21**	1.9	**9.5**	7.5	**30.6**
16	-0.07	-0.04	3.6	3.3	10.3	10.4
17	0.00	0.02	2.2	2.1	5.7	5.2
18	0.12	-0.12	2.0	3.5	4.8	9.4
19	0.42	0.21	2.3	3.3	10.8	9.1
20	0.47	0.01	0.5	2.2	1.1	8.4
21	0.38	-0.03	4.4	12.8	12.5	37.1
22	0.08	-0.09	0.9	1.1	2.9	3.8
23	0.08	0.19	2.6	1.9	7.2	5.3
24	0.62	0.30	0.6	1.4	2.2	4.7
25	-0.19	0.06	6.0	3.3	23.6	11.4
26	-0.01	0.01	2.2	2.0	8.4	7.4
27	-0.05	0.22	2.7	1.5	7.9	4.2
28	0.52	0.19	1.4	4.0	5.4	16.0
29	0.11	-0.10	2.9	3.4	7.8	14.6
30	-0.07	-0.01	3.2	3.1	7.1	8.4
31	-0.03	0.06	1.6	1.2	5.9	4.0
32	0.29	-0.30	0.5	1.8	1.7	7.2

Table 4.4: **Features for each session:** Sil, MSD and SD values for each pair $(M0_i, M6_i)$ for each subject of ID i.

4.5.3.3 Study of subject 2

The group $M0_2$ of points from the M0 session of subject 2 has the highest MSD and SD of Table 4.4: $MSD(M0_2) = 24.9$ and $SD(M0_2) = 99.9$. The second highest MSD is about 13 and the second highest SD is about 37.

On Fig 4.9, the blue point which is the furthest on the right is from the M0 session of subject 2.

4.6 Analysis of the results and discussion

4.6.1 Analysis of the results

4.6.1.1 HS/MS and EDSS experiments

The features from the HS/MS experiments (Table 4.2) quantify what could be observed on Fig 4.9: the HS and MS groups form clusters with high silhouette scores (over 0.4), but

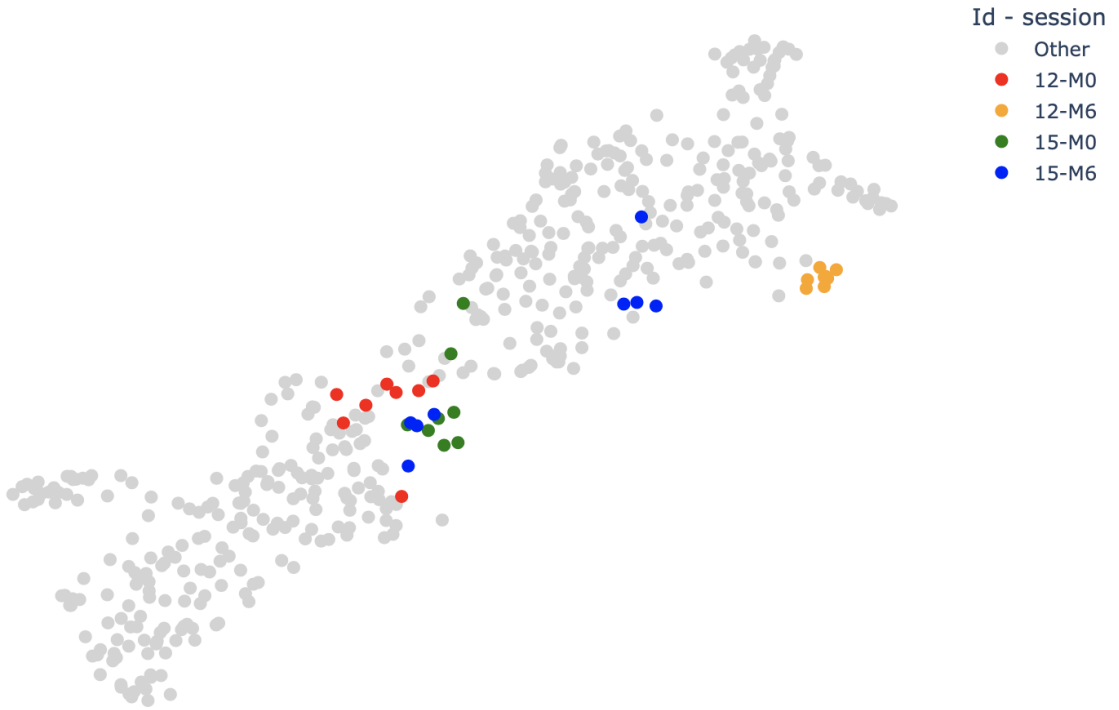


Figure 4.11: **Longitudinal study of subjects 12 and 15.** The point cloud is the same as on Fig 4.9 and Fig 4.10 but colored differently. The red (resp. orange) points are the M0 (resp. M6) points of subject 12. The green (resp. blue) points are the M0 (resp. M6) points of subject 15. Grey points correspond to other subjects.

there is some overlap. The HS group is denser that the MS group, which can be explained by the fact that the disease is more severe for some patients than others.

In Fig 4.10, a global continuity of the color of points can be observed from left (dark blue, low EDSS) to right (dark red, high EDSS). The goal of studying all partitions $(\{EDSS \leq i\}, \{EDSS > i\})$ is to quantify this left/right continuity. The silhouette scores on Table 4.3 show that $\{EDSS \leq i\}$ and $\{EDSS > i\}$ almost always form satisfactory clusters, except for $\{EDSS \leq i\}$ when i is above 5 (in that case, the group is sparse and patients with EDSS above 4 are far from the HS). This shows that our method reflects the global progression of the disease by placing MS patients with a low EDSS closer to HS than to patients with a high EDSS.

4.6.1.2 Longitudinal experiment

The objective of this experiment is to study each subject independently from the others to compare their M0 and M6 sessions. The idea behind our approach is that gait signals cannot be compared to an absolute reference but an evolution can be detected by comparing a subject at M6 to himself at M0, thus taking M0 as the reference. Analyzing the signals from subjects with significant values in Table 4.4 allowed us to highlight three different phenomena:

- A significant change in subject 12's gait between M0 and M6, which we deduce from the fact that the groups $M0_{12}$ and $M6_{12}$ are almost completely separable (see Fig 4.11). This evolution of gait can be linked to the significant evolution of the patient's disease, as their EDSS goes from 4 to 5 (and is the only one that has a variation

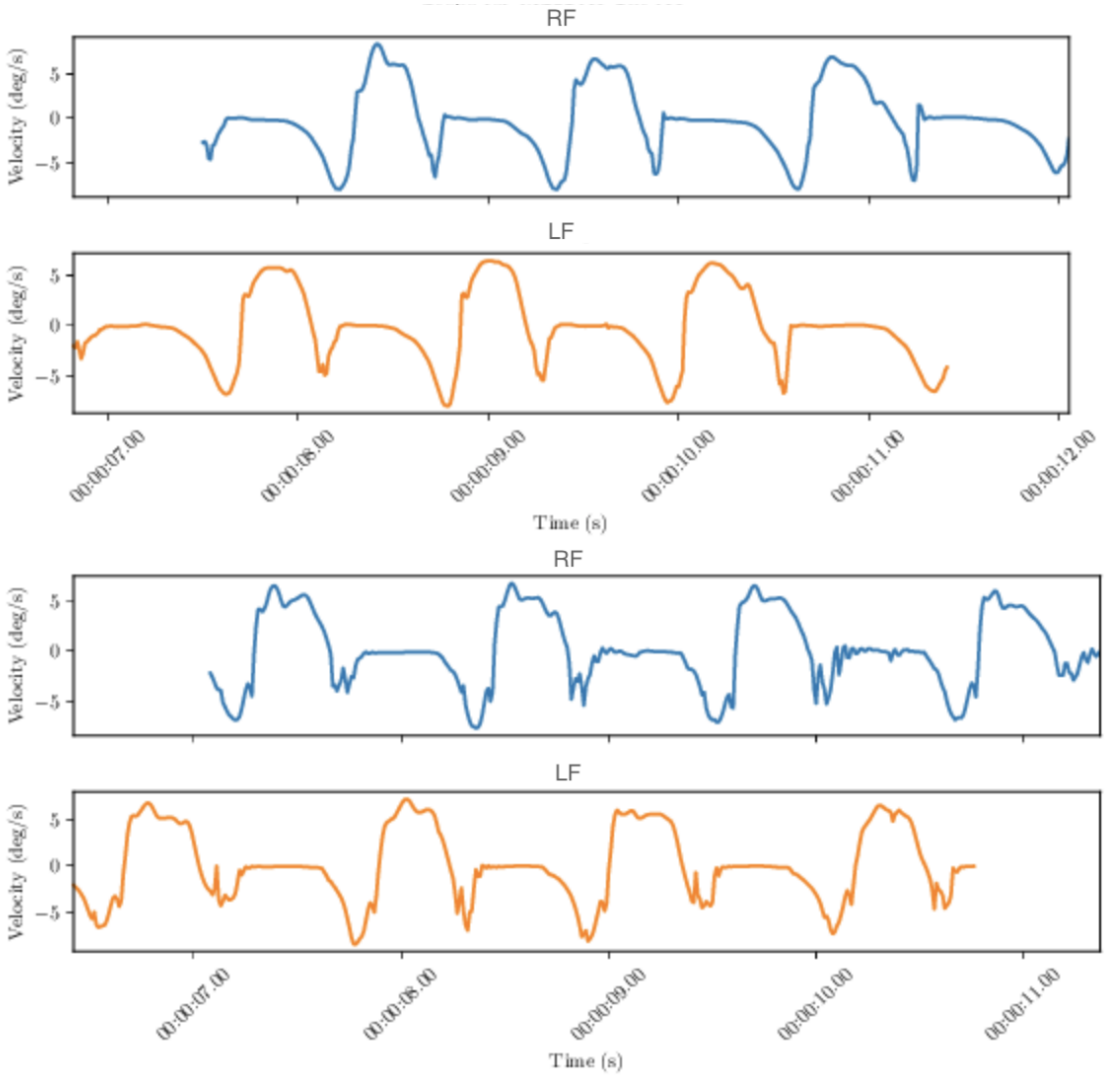


Figure 4.12: Gait signals from subject 12. **Top:** a trial at M0. **Bottom:** a trial at M6.

of more than 0.5) and can be confirmed visually by looking at the signals (see Figure 4.12).

- An asymmetrical gait at M6 for subject 15. Usually, the M0 and M6 silhouette scores of a given patient are close because if M0 is separable from M6 then M6 should be easily separable from M0. For subject 15, those scores are $Sil(M0_{12}, M6_{12}) = 0.49$ and $Sil(M6_{12}, M0_{12}) = -0.21$. This can be explained by the fact that the M6 group is sparser (its MSD and SD are among the highest of all sessions). The M6 group is split in two parts of four points each (see Fig 4.11). One part (the one closer to the M0 group) is made of the four RF signals of the M6 session, and the other one is made of the LF signals. The significant values for subject 15 on Table 4.4 can thus be explained by the apparition of an asymmetry at M6. This explanation is supported by clinical evidence.

- An outlier and a technical issue in the signal's acquisition. Subject 2's M0 group has a MSD and a SD significantly higher than every other group. It is due to the outlier of the HS group (the blue point on the right of Fig 4.9), which belongs to

$M0_2$. Visualizing this outlier allowed us, by going back to the associated gait trial, to highlight a segmentation problem during the construction of the database.

Note that the MSD detects the outlier because the remaining points of the $M0_2$ group have a low density. If it had been denser the MSD would have been lower but the SD would be similar. This justifies using the squared diameter as a complementary density measure.

Asymmetry can be detected this way in other patients such as for subject 21, and even, at a lower scale, in the HS group such as for subject 6.

Fig 4.11 illustrates the above discussion.

4.6.1.3 Comparison to other features

We compared the results of our method to the results obtained when replacing the bottleneck distance between two diagrams with the difference between two features. The considered features are:

- Standard features for the study of gait signals from MS patients [VdlVJ+23]: speed, step length, step time, double step time, variation coefficient of step time and variation coefficient of double stance time.
- Frequential features: (truncated) energy spectral density of the signal.

The experiments with frequential features was motivated by the observation of oscillations on signals from MS patients such as those presented on Figure 4.12. The intuition was that they could correspond to a change in the spectrum of the signal, probably in its high frequencies. We refer to Appendix B for details and figures. Our conclusion is that standard features used this way do not give a useful representation of the signals, and frequential features can give information about the severity of the disease but our method performs better. Our interpretation is that the energy spectral density can detect some oscillations of a signal that is caused by the disease, but our method can both detect those oscillations and more significant changes in the shape of the signal, which makes it more relevant for signal comparison.

4.7 Discussion

The method presented in this chapter has the advantage of being non-parametric, except for the two UMAP parameters, for which the default values suggested by the authors of [MHM18] seem to be appropriate. Moreover, it only relies on raw gait signals and does not need additional information such as step annotations or step detection (although, as mentioned before, knowing the number of step for each signal can be useful). Thus, it can be applied to any type of gait-affecting pathology without having to choose new parameters or previously perform step detection or manual annotations.

Our study of multiple sclerosis shows that the method can identify a global correlation between the severity of the disease represented by the EDSS and distance to HS points, and also detects changes in the patients' gait. The variations of standard gait features such as speed, step length, step time (or its variation coefficient) or double stance time (or its variation coefficient) do not appear to correlate with the distribution of EDSS scores. Frequential features have a lower correlation with the EDSS and do not detect the remarkable subjects described above.

The use of TDA with sublevel sets to create persistence barcodes and compute distances between them naturally allows to compare signals from the left foot to signals from the right foot, as those barcodes are invariant by translation along the time axis. Allowing comparison between different feet doubles the number of points for each session and thus makes the following analysis more relevant. Moreover, it provides a way to detect asymmetry in a subject's gait which can further assist clinical gait evaluation. For subjects who have a strong asymmetry, it may be necessary to separate RF and LF signals to study their evolution as the effect of an asymmetry on the distance between points may dominate the effect of any other phenomenon.

This work was limited by the size of the database, that only contains signals from 22 MS patients, and some EDSS values are not represented (4.5 or values under 2). Because of those missing values, we could not study the impact of MS on gait at its earliest stages. Indeed, our study of the $(\{EDSS \leq i\}$ and $\{EDSS > i\})$ groups would allow us to quantify how close patients with low EDSS are to healthy subjects and refine our analysis of the progression of the disease. Having more than two sessions per subject would also be beneficial for the longitudinal study.

The second limitation is the access to a ground truth. We used the EDSS as a measure of the severity of the patients' disease, but it is limited by its lack of objectivity and of sensitivity to change (and so are other clinical scores for MS) [MMFM$^+$14, NVWE90, Nos94, GCLBK17, LDP$^+$13, VJQVR20]. Indeed, in the studied cohort, EDSS scores do not vary by more than 0.5 between M0 and M6 in all cases except one, and often stays the same between two sessions whereas for several patients our method clearly separates the M0 and M6 points. A different ground truth thus seems necessary to compare the results of our method to the conclusions obtained with clinical scores.

4.8 Conclusion and perspectives

In this chapter, we described a non-parametric method to study gait signals motivated by theoretical properties of persistent homology of periodic functions, and applied it to study multiple sclerosis both globally and in a longitudinal way. Our goal was to present the method in a way that requires no background in TDA to insist on the ideas behind it and make it more easily usable by clinicians.

Future more could include testing our method on a larger cohort with multiple MS patients for each possible value of the EDSS, and possibly over a longer time to observe more significant variations of clinical scores such as the EDSS. The method can also be used to study other pathologies that have an impact on gait including some that involve a left/right asymmetry such as diseases affecting one of the brain's two hemispheres. The method can also be generalized to analyze different physiological signals or any type of time series, as the only step that is specific to the study of locomotion is the one when bars are removed from persistence barcodes according to the number of steps.

Appendix B
Analysis of gait signals with standard features

In this appendix, we give a more detailed description of the features mentioned in Section 4.6.1.3 and show the obtained UMAP plots. The standard features (introduced in [VdlVJ+23], measured using data segmentation from the GaitRite® mat) are the following:

- Speed: average speed on a trial.
- Step time: average time between two consecutive contacts of the foot with the ground.
- Variation coefficient (CV) of step time: variation coefficient (standard deviation divided by mean) of the time between two consecutive contacts of the foot with the ground.
- Step length: total length divided by the number of steps.
- Double stance time: average time with both feet in contact with the ground, divided by the total duration of the trial.
- Variation coefficient (CV) of double stance time: variation coefficient of the time with both feet in contact with the ground divided by the total duration of the trial.

Fig 4.13 shows the point clouds obtained by performing the same experiment except that the bottleneck distance was replaced by the absolute difference between features. Points are colored according to the EDSS of the corresponding subject.

Fig 4.13 shows point clouds made of dense groups of points that are completely separable from each other. Several groups are made of points with different EDSS scores, and there does not appear to be any way to correlate the distribution of points with their EDSS. This can be explained by the fact that each feature is significantly impacted by other factors than the disease and thus two subjects with different clinical conditions can have the same feature value. In particular, this means that, compared to our approach, they cannot be used to detect clinical evolution.

We also considered two frequential features: the energy spectral density, which is the integral of the squared Fourier transform over $\mathbb{R}^+$, and the truncated energy spectral density, which is the integral of the squared Fourier transform on $[f, +\infty)$, where f is a fixed frequency. Figure 4.14 show the results obtained using the energy spectral density and the truncated energy spectral density with $f = 0.35Hz$, which is the value that gave the best results in our experiments. It can be noticed that truncating with this frequency removes the three frequencies of maximal amplitude as show in Figure 4.14. In the first

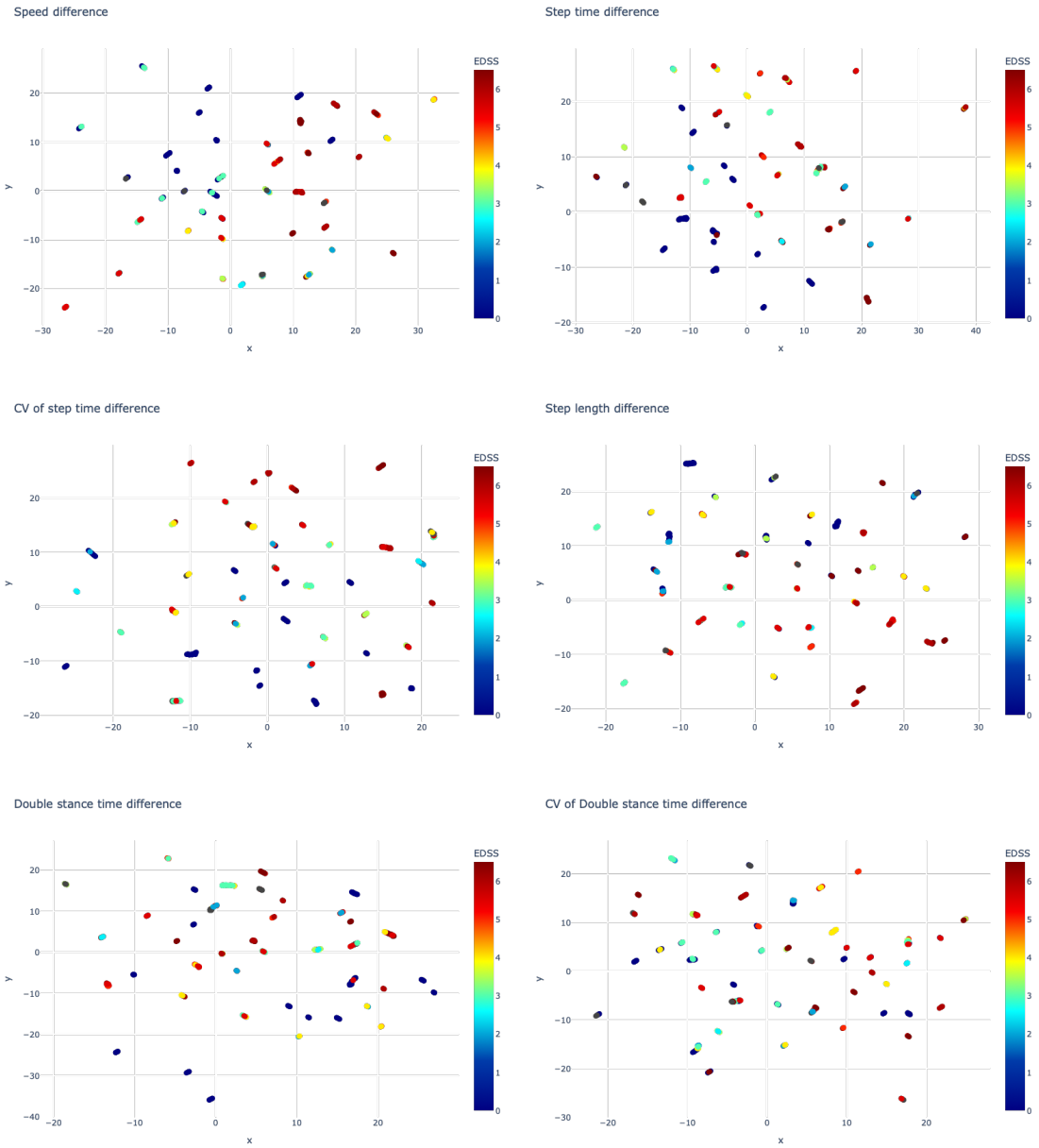


Figure 4.13: UMAP plots obtained using standard features. UMAP parameters are the same as for our method (changing them made no significant difference).

case, we obtained silhouette scores $Sil(HS, MS) = 0.47$ and $Sil(MS, HS) = 0.32$. In the second case, $Sil(HS, MS) = 0.53$ and $Sil(MS, HS) = 0.22$. Tables 4.5 and 4.6 show the results of the EDSS experiment described in Section 4.5.2.

Our conclusion is that (truncated) spectral energy can detect some evolution in the severity of the disease, but they are less efficient than the bottleneck distance between barcodes as the silhouette scores are always lower. Moreover, we found no relevant way of using these features for longitudinal follow-up.

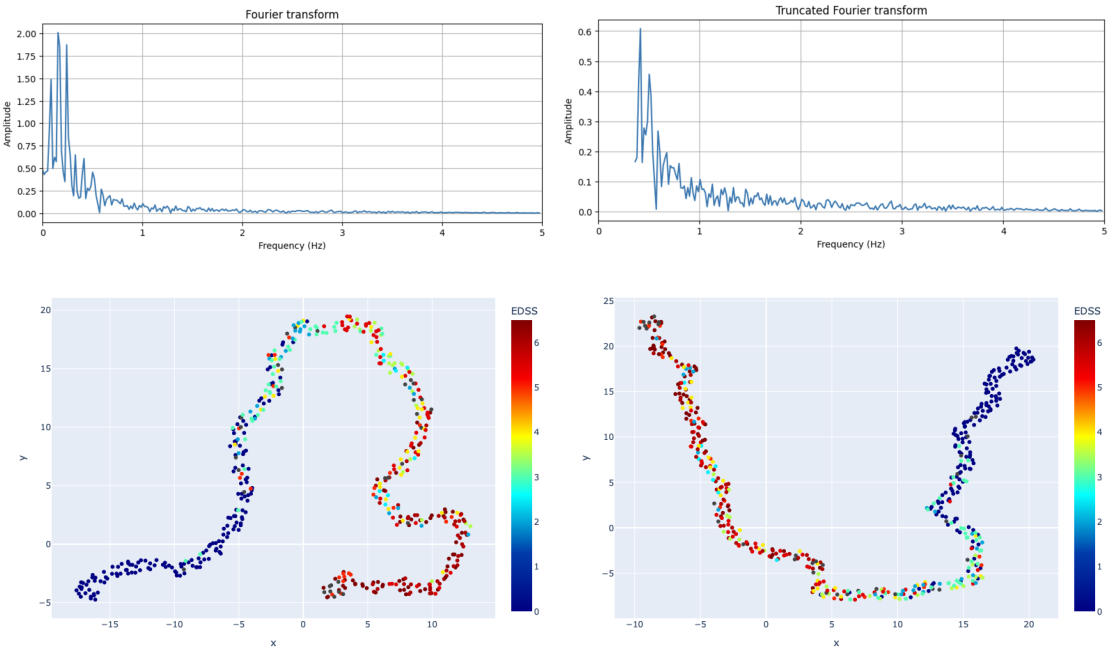


Figure 4.14: **Top:** Fourier transform (left) and truncated Fourier transform (right) of a gait signal. **Bottom:** UMAP plots obtained using the energy spectral density (left) and truncated energy spectral density with $f = 0.35Hz$ (right). UMAP parameters are set to n_neighbors $= 80$ and min_dist $= 1$.

Table 4.5: **Features for points with EDSS lower or equal to/ stritcly higher than each threshold, obtained using the energy spectral density.**

i	$Sil(\leq i, > i)$	$Sil(> i, \leq i)$	$MSD(\{EDSS \leq i\})$	$MSD(\{EDSS > i\})$	$SD(\{EDSS \leq i\})$	$SD(\{EDSS > i\})$
0.00	0.48	0.34	115.8	177.8	862.2	649.9
2.00	0.35	0.33	166.2	174.1	973.4	649.9
2.50	0.31	0.31	182.0	176.6	1,025.4	649.9
3.00	0.27	0.37	204.6	153.0	1,069.5	649.9
3.50	0.20	0.37	233.0	146.8	1,069.5	649.9
4.00	0.16	0.38	241.7	140.1	1,069.5	608.6
4.50	0.16	0.38	241.7	140.1	1,069.5	608.6
5.00	0.14	0.41	240.4	132.5	1,069.5	608.6
5.50	0.10	0.61	254.8	48.5	1,069.5	313.3
6.00	-0.0	0.57	263.7	52.3	1,069.5	313.3

For a given threshold i, the points from each subject have been divided in two groups: points corresponding to patients with EDSS lower than or equal to i ($\{EDSS \leq i\}$), and points with EDSS higher than i ($\{EDSS > i\}$). The silhouette scores, MSD and SD have been computed on those groups for several values of i.

Table 4.6: **Features for points with EDSS lower or equal to/ strictly higher than each threshold, obtained using the truncated energy spectral density.**

i	$Sil(\leq i, > i)$	$Sil(> i, \leq i)$	$MSD(\{EDSS \leq i\})$	$MSD(\{EDSS > i\})$	$SD(\{EDSS \leq i\})$	$SD(\{EDSS > i\})$
0.0	0.53	0.22	128.1	315.0	966.1	1477.9
2.0	0.41	0.22	190.2	306.3	1314.3	1471.0
2.5	0.35	0.21	212.9	311.4	1314.3	1471.0
3.0	0.37	0.28	216.5	282.4	1314.3	1471.0
3.5	0.33	0.30	236.9	268.7	1314.3	1471.0
4.0	0.25	0.29	270.2	264.2	1314.3	1471.0
4.5	0.25	0.29	270.2	264.2	1314.3	1471.0
5.0	0.19	0.31	304.0	240.9	1476.9	1456.3
5.5	0.16	0.40	314.2	189.5	1476.9	1168.2
6.0	0.16	0.49	321.2	157.8	1476.9	1037.8

For a given threshold i, the points from each subject have been divided in two groups: points corresponding to patients with EDSS lower than or equal to i ($\{EDSS \leq i\}$), and points with EDSS higher than i ($\{EDSS > i\}$). The silhouette scores, MSD and SD have been computed on those groups for several values of i.

Chapter 5

Persistence-based clustering with outlier-removing filtration

In this chapter, we describe a non-parametric hierarchical clustering algorithm with an outlier removal step, based on topological data analysis. We use a new filtration on metric spaces which is a variant of the Vietoris-Rips filtration that adds information about the points' nearest neighbor to the persistence diagram. We prove a stability theorem for this filtration, and illustrate our method on point cloud and graph datasets.

Contents

5.1 Introduction **99**
5.2 Nearest Neighbor Vietoris-Rips (NNVR) filtration and stability **99**
5.3 Outlier-removing clustering **102**
5.3.1 Algorithm 102
5.3.2 Choice of thresholds 102
5.4 Experiments **103**
5.4.1 Karate club graph 104
5.4.2 Point clouds 104
5.4.3 Synthetic graphs 105
5.5 Application to gait signals **108**
5.6 Conclusion **111**

5.1 Introduction

Agglomerative hierarchical clustering is one of the main classes of clustering algorithms [SPG+17, MC17], it consists in considering each data point as its own cluster and iteratively merging clusters. All state-of-the-art clustering algorithms require to choose parameters such as the number of clusters (K-means, spectral clustering...), some thresholds (hierarchical clustering) or a density parameter for density-based clustering (DBSCAN [EKS+96, SSE+17], ToMATo [CGOS13]). The choice of parameters can be problematic in some cases, especially when clustering is only one step in a larger algorithm.

Topological data analysis (TDA), and more specifically persistent homology [EH10] has been a popular research subject over the last decades, with many applications [CM21]. The idea is to construct a filtration from given data and to study how its structure evolves when going through the filtration, using the persistence diagram. The success of TDA is partly due to stability theorems (see Chapter 2 for examples), which state that close datasets will have close persistence diagrams, according to some distance between datasets/diagrams, ensuring in particular that TDA results are robust to noise. One of the most used filtrations is the Vietoris-Rips (or Rips, or VR) filtration, for which a stability theorem with respect to the Gromov-Hausdorff distance has been proved in [CDSO14].

Concretely, 0-dimensional persistent homology consists in linking data points until there is only one connected component left. It thus naturally induces a hierarchical clustering algorithm as the one described in [LKC+12, KHL+15], that uses the Vietoris-Rips filtration to study brain networks, or the ToMATo algorithm [CGOS13], which combines TDA with a density-based approach. Persistent homology highlights important components of the data and is thus a suitable tool for threshold choice in hierarchical clustering. In this work, we introduce a new filtration for metric spaces[1] that adds information about the points' nearest neighbor to the persistence diagram which makes it easier to identify clusters on the diagram and to remove outliers. We prove its stability, and use it to define a non-parametric algorithm to perform clustering with outlier removal [2].

In section 5.2, we introduce our filtration and prove its stability. We describe the algorithm in Section 5.3 and present some experiments on point cloud and graph datasets in Section 5.4. In section 5.5, we show a concrete application of our algorithm to pattern detection in gait signals.

5.2 Nearest Neighbor Vietoris-Rips (NNVR) filtration and stability

In this section, we introduce a new filtration: the Nearest Neighbor Vietoris-Rips (NNVR) filtration, and prove its stability.

Definition 22. *Let (X, d) be a metric space. The Nearest Neighbor Vietoris-Rips filtration $NNVR(X, \alpha)_{\alpha \geq 0}$ is defined by:*

$$\forall x \in X,\ x \in NNVR(X, \alpha) \iff \alpha \geq \inf_{y \in X, y \neq x} d(x, y)$$

and for all k-simplices σ with $k \geq 1$:

$$\sigma \in NNVR(X, \alpha) \iff \sigma \in VR(X, \alpha)$$

[1] For a review on filtrations over graphs, see [AAF19]. Note that a metric space can be seen as a non-directed weighted graph.

[2] Here, we define an outlier as an isolated point, and consider that two close points form a cluster.

This means that the NNVR filtration is the same as the VR filtration except for 0-simplices, which all enter the VR filtration at $\alpha = 0$ whereas they enter the NNVR filtration when α is equal to the distance to their nearest neighbor. It is easy to verify that NNVR is indeed a filtration. Figure 5.1 shows an example of NNVR filtration.

Remark 6. *In the case where X is a finite point cloud in $\mathbb{R}^d$ endowed with the Euclidean distance, the NNVR filtration corresponds to the DTM filtration with parameters $p = \infty$ and $q = 1$ which implies in particular that is has a stability property with respect to the Hausdorff distance. Here, we show that the NNVR filtration on any metric space is stable with respect to the Gromov-Hausdorff distance, which is stronger in the sense that, as explained in [BCY18], when considering two isometric sets P and P' and another set Q, $d_H(P, Q)$ can be much larger than $d_H(P', Q)$ while $d_b(\text{Diag}(P), \text{Diag}(Q))$ and $d_b(\text{Diag}(P'), \text{Diag}(Q))$ are equal, where* Diag *denotes the persistence diagram of the (weighted) Rips filtration. This cannot happen with the Gromov-Hausdorff distance.*

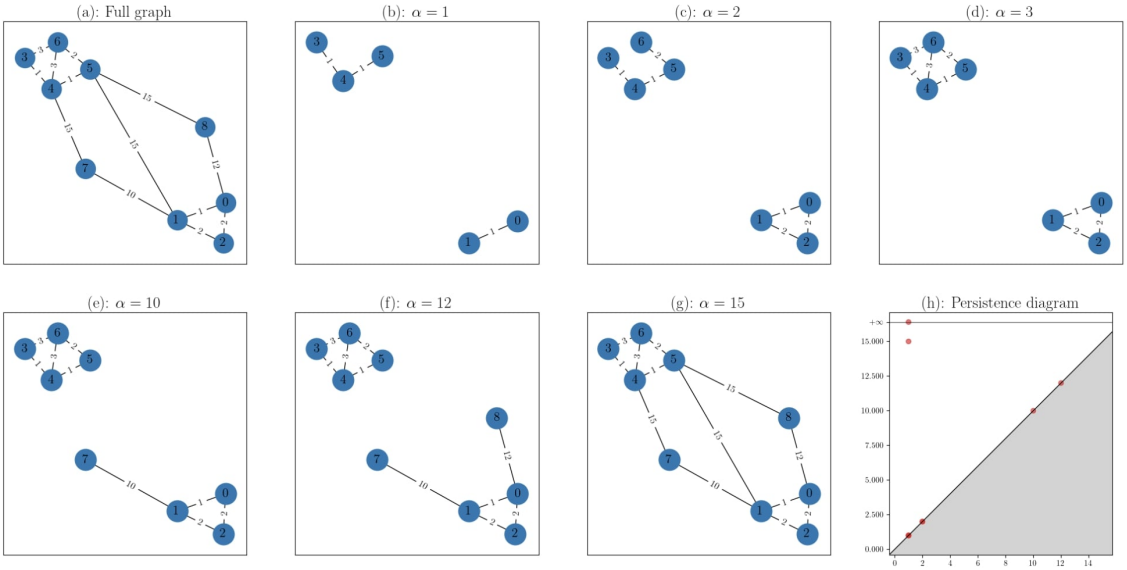


Figure 5.1: An example of NNVR filtration. The edge weights represent the distance between vertices. For each value of α, vertices and edges with filtration value equal to α are added.

The stability theorem for the VR filtration is proved in [CDSO14], we use the same framework as they did to adapt the proof to the NNVR filtration. The theorem states that the NNVR filtration is stable with respect to the **Gromov-Hausdorff** distance [BBI22], which we define below, along with the notions required to prove the theorem.

Definition 23. *Let X, Y be two sets and $C \subset X \times Y$. For any $A \subset X$ and $B \subset Y$, we note $C(A) = \{y \in Y : \exists x \in A, (x, y) \in C\}$ and $C^\top(B) = \{x \in X : \exists y \in B, (x, y) \in C\}$. C is called a correspondence between X and Y if $C(X) = Y$ and $C^\top(Y) = X$.*

Definition 24. *Let (X, d_X) and (Y, d_Y) be metric spaces and C be a correspondence between X and Y. The distortion of C is defined as*

$$\text{dis}(C) = \sup\{|d_X(x, x') - d_Y(y, y')| : (x, y), (x', y') \in C\}.$$

The Gromov-Hausdorff distance between X and Y is:

$$d_{GH}(X, Y) = \frac{1}{2}\inf\{\text{dis}(C) : C \text{ is a correspondence between } X \text{ and } Y\}.$$

We now state the stability theorem.

Theorem 3. *Let (X, d_X) and (Y, d_Y) be totally bounded metric spaces. Let D_X (resp. D_Y) be the persistence diagram of $NNVR(X)$ (resp. $NNVR(Y)$). Then:*

$$d_B(D_X, D_Y) \leq 2d_{GH}(X, Y).$$

To prove this theorem, we introduce **simplicial correspondences** and use results from [CDSO14].

Definition 25. *Let $(K^X)_{\alpha\geq 0}$ (resp. $(K^Y)_{\alpha\geq 0}$) be a filtration of a simplicial complex on a set X (resp. Y). A correspondence C is called ε-simplicial if*

- *For any $\alpha \geq 0$ and simplex $\sigma \in K^X_\alpha$, any finite subset of $C(\sigma)$ is a simplex of $K^Y_{\alpha+\varepsilon}$.*
- *For any $\alpha \geq 0$ and simplex $\sigma \in K^Y_\alpha$, any finite subset of $C^\top(\sigma)$ is a simplex of $K^X_{\alpha+\varepsilon}$.*

Proposition 4.2 from [CDSO14] states that if there is an ε-simplicial correspondence between two filtered metric spaces X and Y, then there exists a so-called ε-**interleaving** between the persistent homology groups of X and Y which is a family of linear maps between homology groups $H_k(K^X_\alpha) \to H_k(K^Y_{\alpha+\varepsilon})$ and $H_k(K^Y_\alpha) \to H_k(K^X_{\alpha+\varepsilon})$ for all α, that commute with the persistence structure. Theorem 2.3 from [CDSO14] states that if there exists such an ε-interleaving then ε is an upper bound on the bottleneck distance.[3] The following lemma synthesizes those results.

Lemma 1. *Let $(K^X)_{\alpha\geq 0}$ (resp. $(K^Y)_{\alpha\geq 0}$) be a filtration of a finite simplicial complex on a totally bounded metric space X (resp. Y), with persistence diagrams D_X and D_Y. If there exists an ε-simplicial correspondence then $d_B(D_X, D_Y) \leq \varepsilon$.*

Proof of Theorem 3. Let $\varepsilon > 2d_{GH}(X, Y)$. There exists a correspondence C such that $\text{dis}(C) \leq \varepsilon$. Let us show that C is ε-simplicial.
Let $\alpha \geq 0$, σ be a simplex of $NNVR(X, \alpha)$ and τ be a finite subset of $C(\sigma)$.

Case 1: τ only has one element: $\tau = [y]$. There exists $x \in \sigma$ such that $y \in C(x)$ and a sequence $(x_n)_{n\in\mathbb{N}} \in X^{\mathbb{N}}$ such that $0 < d_X(x, x_n)$ and $(d_X(x, x_n))_{n\in\mathbb{N}}$ converges to a value lower than or equal to α. Moreover, there exists a sequence $(y_n)_{n\in\mathbb{N}} \in Y^{\mathbb{N}}$ such that for all n, $y_n \in C(x_n)$. Then:

$$\forall n \in \mathbb{N}, \inf_{z\in Y, z\neq y} d_Y(y, z) \leq d_Y(y, y_n) \leq d_X(x, x_n) + \varepsilon$$

So: $\inf_{z\in Y, z\neq y} d_Y(y, z) \leq \alpha + \varepsilon$ and $\tau = [y] \in NNVR(Y, \alpha + \varepsilon)$.

Case 2: $\tau = [y_0, ..., y_k],\ k \geq 1$. For all i, j, there exists $x_i, x_j \in \sigma$ such that $y_i \in C(x_i)$ and $y_j \in C(x_j)$. Then :

$$d_Y(y_i, y_j) \leq d_X(x_i, x_j) + \varepsilon \leq \alpha + \varepsilon \text{ and } \tau \in NNVR(Y, \alpha + \varepsilon)$$

where the last inequality holds because $[x_i, x_j] \in NNVR(X, \alpha)$.

The same proof shows that if σ is a simplex in $NNVR(Y, \alpha)$ then all finite subsets of $C^\top(\sigma)$ are simplices of $NNVR(X, \alpha + \varepsilon)$. So C is ε-simplicial and Lemma 1 ends the proof. □

[3] This theorem requires a tameness hypothesis to be verified. Here, it is the case because Proposition 5.1 from [CDSO14] applies to the NNVR filtration with the exact same proof as for the VR filtration.

5.3 Outlier-removing clustering

5.3.1 Algorithm

Our clustering algorithm relies 0-dimensional persistent homology (defined in Chapter 2). The idea is to construct the 0D persistence diagram of the NNVR filtration of the data, then use it to chose a birth threshold (which tells us which points are outliers) and a persistence threshold (which tells us what the clusters are). The thresholds can be chosen manually or automatically. If they are chosen using a non-parametric procedure, then the whole clustering algorithm is non-parametric (see next section for more details). The clustering algorithm works as follows:

- **Input:** a finite metric space (X, d).
- Compute the filtration $NNVR(X)$ and its 0-dimensional persistence diagram.
- Choose a birth threshold and a persistence threshold.
- Mark all points whose birth dates are above the birth threshold as outliers. Let Y be the remaining points.
- Compute 0-dimensional persistent homology on $NNVR(Y)$, but do not add edges that would merge components whose persistence is above the persistence threshold.
- **Output:** a simplicial complex, where each connected component is a cluster.

The benefit of using the NNVR filtration is that, compared to the VR filtration, it separates outliers from essential components on the persistence diagram. Components corresponding to outliers indeed have late death dates in both filtrations as they are far away from other points. Thus, they give very persistent points with the VR filtration (where every birth date is 0), which can be hard to distinguish from other persistent points (as illustrated below). The NNVR filtration moves them to the right of the persistence diagram. Note that, if there are no outliers, the same algorithm can be used with the VR filtration and no birth threshold.

Seeing (X, d) as a graph with n vertices and m edges, the filtration can be computed in $O(m)$ (the filtration value of each vertex is the distance to its nearest neighbor, and the filtration value of each edge is its weight, so reading the m edge weights is enough to compute the filtration). Computing 0D persistent homology requires computing a minimal spanning tree, which can be done in $O(m\log(\max(m, n)))$ using Kruskall's algorithm. An algorithm to choose the thresholds (such as *max-jump*, described below) should work in $O(n)$ as there are n points on the persistence diagram. So the overall complexity of the method is $O(m\log(\max(m, n)))$, so $O(n^2\log(n))$ in the worst case, as $m \leq n^2$, with equality if the whole distance matrix is computed, in Euclidean spaces for example).

5.3.2 Choice of thresholds

After computing the persistence diagram D, two thresholds are needed: a **birth threshold**, which is a threshold on the list of birth dates of points on the diagram, and a **persistence threshold**, which is a threshold on the list of persistence of points on the diagram. The choice of the birth threshold determines which points are considered as outliers: if a point on the persistence diagram has a birth date above the threshold, we exclude the corresponding point in X and detect it as an outlier. The choice of the persistence threshold

determines which components are significant enough to be considered as clusters. The number of points whose persistence is above the threshold is the number of detected clusters.

A natural method to choose the birth and persistence thresholds is to apply the *max-jump* algorithm to the list of birth dates and to the list of persistence values (without the point of infinite persistence). The algorithm is defined in Algorithm 1.

Algorithm 1 Max-jump algorithm

Input: a list $L = [L(1), ..., L(n)]$.
$SL \leftarrow sorted(L)$
$J \leftarrow [SL(i+1) - SL(i)]_{1 \leq i \leq n-1}$
$i_{max} \leftarrow argmax(J)$
thr $\leftarrow (SL(i_{max}) - SL(i_{max}+1))/2$
Output: thr

The idea behind this algorithm is to separate significant values from non-significant ones by using the biggest variation. It is fully non-parametric, and will always output at least two clusters (one corresponding to the infinite persistence point, and at least one point above the persistence threshold). A drawback of this method is that it can output a persistence threshold that is too low if clusters are not at the same distance from one another (see point cloud 5 in section 5.4 for an example).

We propose two possible solutions to this problem, which both introduce the choice of a parameter. The first one is to apply a sigmoid function $s_\lambda : x \mapsto \frac{1}{1+e^{-\lambda x}}$ to the normalized list $\frac{L}{\max(L)}$ before using *max-jump*. The normalization makes the data vary from 0 to 1 which makes it easier to choose a unique parameter λ for a whole dataset, and the sigmoid function pushes significant values close to 1, which can lead to a lower threshold than the one obtained with *max-jump*. The second alternative we propose is using Otsu's method [Ots79] for gray-level threshold selection. This requires to choose a number of bins for the histograms, which we set to 1024 by default. As the persistence distribution is skewed, we apply the algorithm twice and remove values under the threshold after each iteration. The point cloud experiments on section 5.4 illustrate how those alternatives can improve clustering. For every other dataset, we will always use *max-jump* by default. Other methods can be preferred depending on the application, to make use of prior knowledge on the data (for example, knowledge about the expected number of clusters or their size).

Figure 5.2 shows the thresholded persistence diagram (with *max-jump*) and the results of our algorithm on the graph from Figure 5.1.

5.4 Experiments

We applied our algorithm to the karate club graph, which is a classical example for community detection in networks [New04], to point clouds from the scikit-learn documentation [4] to which we added four outliers (one in each corner), and to synthetic graph datasets. Results were evaluated using the Rand index [Ran71], which is the ratio of the number of correctly labeled points over the total number of points, ignoring label permutations (so perfect clustering gives a Rand index of 1 and the index is close to 0 for bad clustering).

[4] `https://scikit-learn.org/1.2/modules/clustering.html`

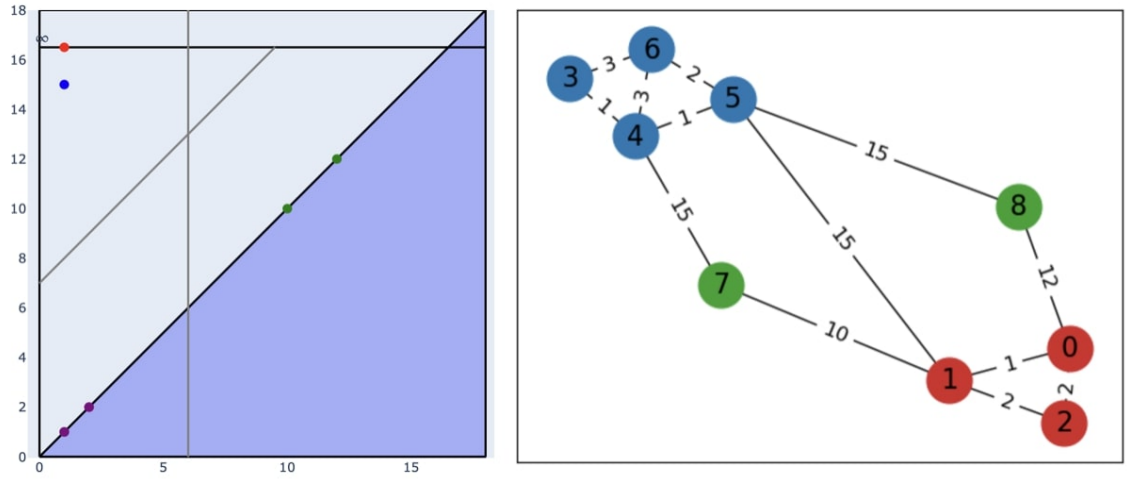


Figure 5.2: **Left :** the persistence diagram from Figure 5.1 with both thresholds chosen with max-jump. The vertical gray line represents the birth threshold, the gray line parallel to the diagonal represents the persistence threshold (as the persistence of a point is proportional to its distance to the diagonal). **Right :** clusters (blue and red) and outliers (green) obtained with our algorithm. A point of a given color on the diagram represents the component that contains the nodes of this color on the graph.

5.4.1 Karate club graph

The karate club graph shows the interactions between the 34 members of a karate club (nodes 0 to 33). Each interaction is represented by an edge with a weight between 1 and 7 (a high weight means more interaction). After a conflict between the instructor (node 0) and administrator (node 33), the club split up into two groups: one with the instructor and one with the administrator. The goal is to predict those groups based on the interaction. We added 10 outliers (nodes 34 to 43, that could represent people outside the club), each one with two edges randomly linking it to other nodes with a random weight between 0.5 and 1. We inverted the weights of the edges to get a new weighted graph and used the shortest-path distance.

The results are shown in Figure 5.3 along with persistence diagrams for the NNVR and VR filtrations, and results obtained with the Louvain Community Detection Algorithm [BGLL08], which is a modularity optimization method (we used the NetworkX [HSSC08] implementation with default parameters). The Rand indices are 0.95 for the NNVR method, 0.65 for the VR method and 0.73 for the Louvain method. Note that points from both diagrams have the same death dates, so outliers have a higher persistence than relevant points on the VR diagram, which is not the case on the NNVR diagram. The algorithm exactly recovers the true clusters for every nodes except nodes 5, 6, 16 and 10, that form a cluster. This can be explained by the fact that node 16 is not linked to node 0 and only has two links to 5 and 6, and the edges (5,6) and (5,10) are respectively heavier than (5,0) and (10,0). This point is more persistent than the blue one on Figure 5.3 because of the proximity between nodes 8 and 33.

5.4.2 Point clouds

In the case of point clouds, the metric space is a set of points in 2D equipped with the Euclidean distance. Figure 5.4 and 5.5 and table 5.1 show the results of ten clustering

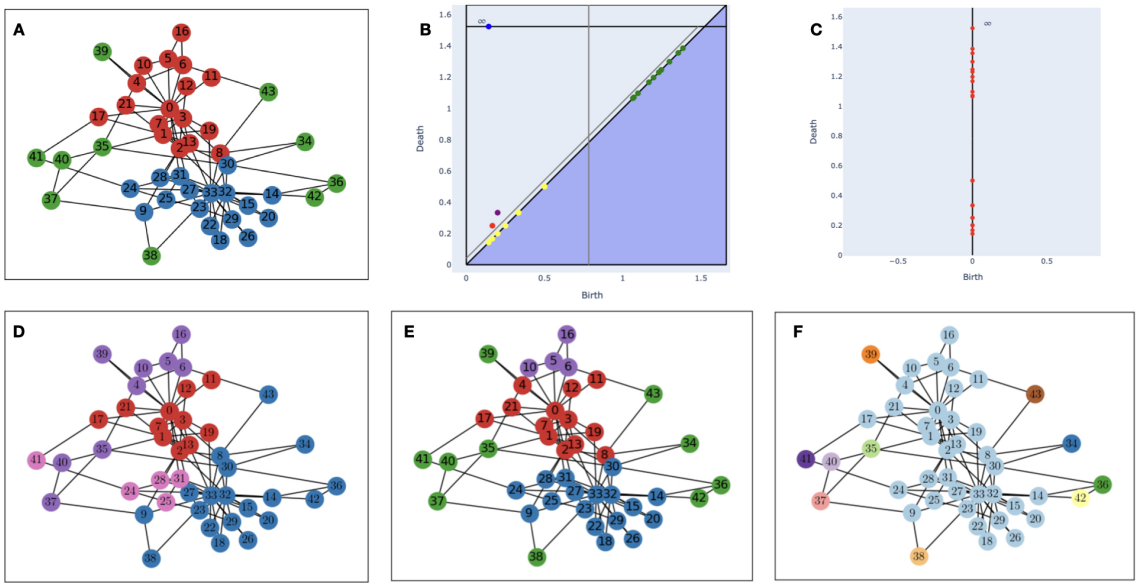


Figure 5.3: **(A)** ground truth for the karate club graph with additional outliers, **(B)** persistence diagram of the NNVR filtration, **(C)** persistence diagram of the VR filtration, **(D)** clusters obtained with the Louvain Community Detection Algorithm, **(E)** clusters obtained with the NNVR algorithm, **(F)** clusters obtained with the VR algorithm. On **(B)**, the two gray lines represent the two thresholds, the blue (resp. red, purple) point corresponds to the blue (resp. red, purple) cluster on the graph, green points correspond to outliers, and yellow points are below the persistence threshold.

algorithms [AV07, FD07, CM02, Shi03, NN16, EKS$^+$96, ABKS99, ZRL96, Moo96], three versions of our algorithm that all use *max-jump* to choose the birth threshold but different methods (described above) for the persistence threshold: one using *max-jump*, one using it after applying a sigmoid with parameter 4, and one using Otsu. We also test our algorithm with the VR filtration (and max-jump persistence threshold), on three point clouds. For all algorithms, we used the same parameters as the ones used in the scikit-learn documentation.

The NNVR algorithm with max-jump performs almost perfectly on datasets 1, 2, 4 and 6. On dataset 5, the persistence threshold is too high so the algorithm detects two clusters instead of three, which is not the case for the sigmoid and Otsu methods, which work perfectly on this dataset. The drawback of the sigmoid and Otsu methods is that they tend to choose more clusters than max-jump and can thus choose too many such as for dataset 4 and 6 (and 1 and 2 for the sigmoid). Density-based methods outperform the others on dataset 3. Dataset 6 shows that our algorithm cannot output only one cluster. The only methods capable of detecting outliers are DBSCAN, OPTICS and our NNVR method. Our method only fails to detect two of the four outliers on dataset 4. DBSCAN and OPTICS detect too many outliers on dataset 3 and 4.

5.4.3 Synthetic graphs

We used the NetworkX's [HSSC08] random_partition_graph function to generate synthetic graph datasets with a number of clusters varying from 2 to 10 and containing outliers.

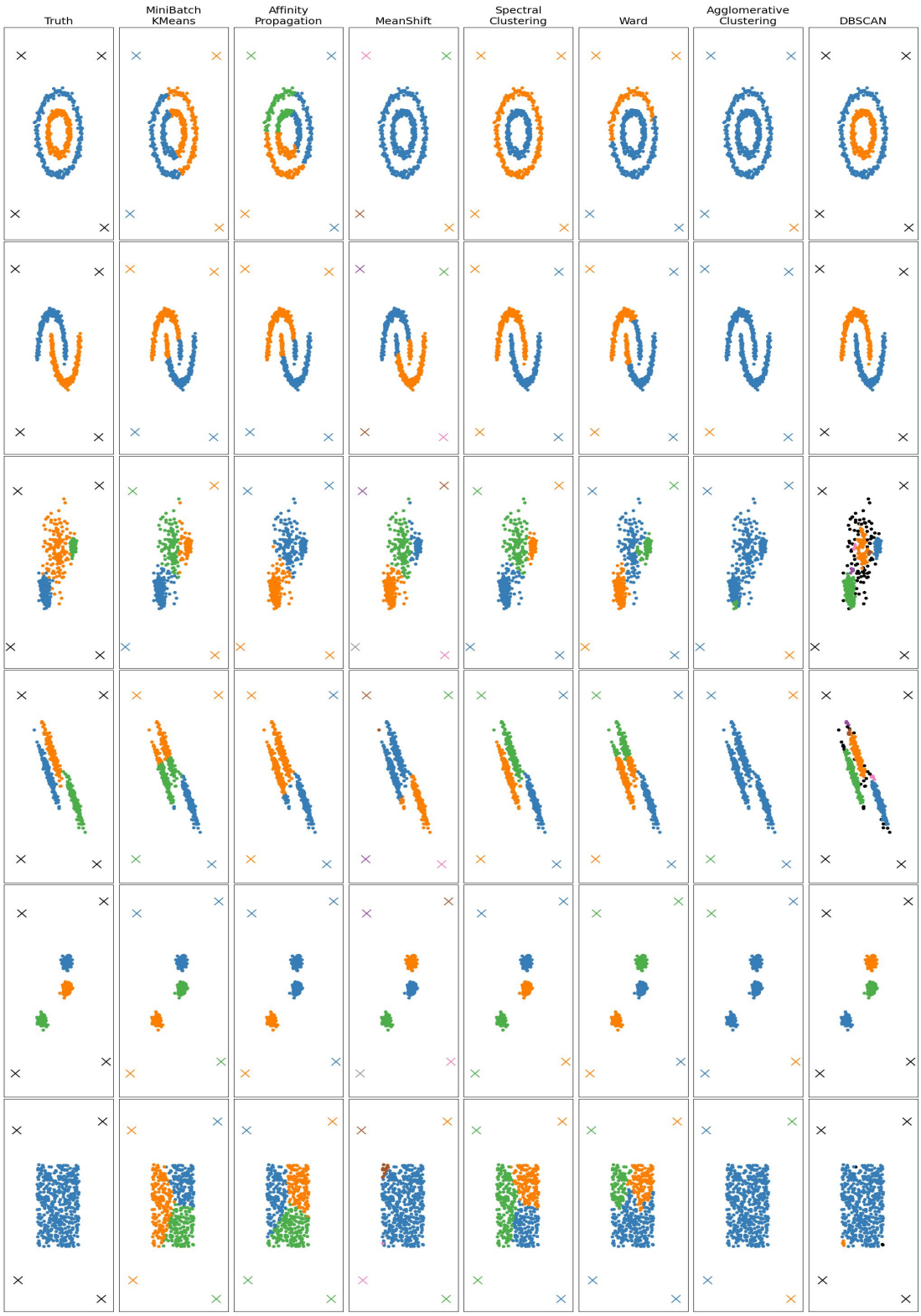


Figure 5.4: Clustering results for six point clouds obtained with several methods. Points marked with an X are outliers.

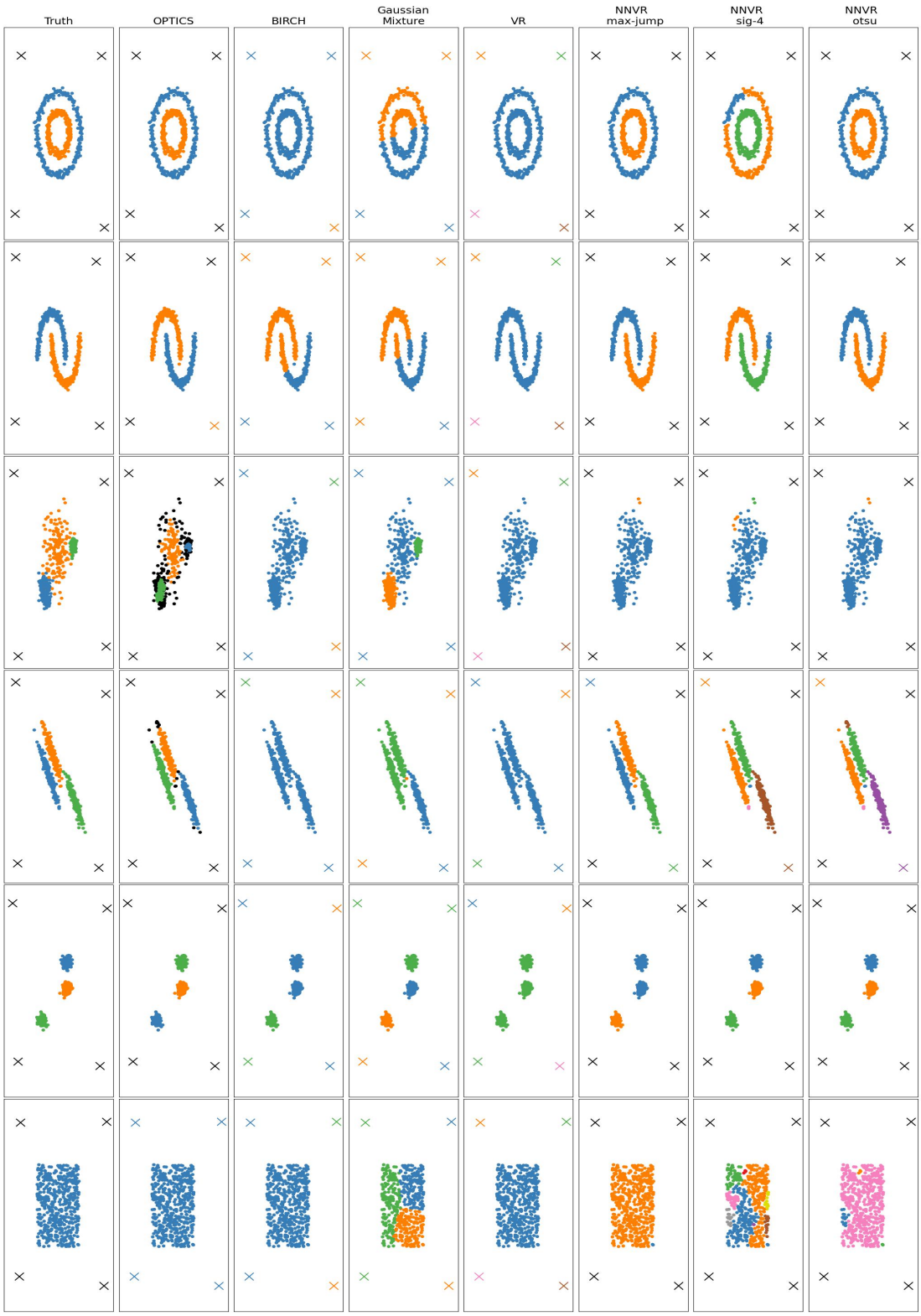


Figure 5.5: Clustering results for six point clouds obtained with several methods, including the NNVR filtration with max-jump, sigmoid of parameter 4 and Otsu, and VR filtration. Points marked with an X are outliers.

Dataset	MiniBatch KMeans	Affinity Propagation	MeanShift	Spectral Clustering	Ward	Agglomerative Clustering	DBSCAN
0	0.50	0.50	0.51	0.99	0.61	0.50	**1.00**
1	0.74	0.75	0.78	0.99	0.61	0.50	**1.00**
2	0.88	0.74	0.90	0.91	0.92	0.35	0.91
3	0.81	0.75	0.76	0.93	0.83	0.33	0.94
4	0.99	0.77	1.00	0.79	0.99	0.33	**1.00**
5	0.34	0.34	0.95	0.34	0.40	0.99	0.97

Dataset	OPTICS	BIRCH	Gaussian Mixture	VR	NNVR max-jump	NNVR sig-4	NNVR otsu
0	**1.00**	0.50	0.50	0.51	**1.00**	0.93	**1.00**
1	**1.00**	0.76	0.75	0.51	**1.00**	0.96	**1.00**
2	0.80	0.33	**0.97**	0.34	0.35	0.35	0.35
3	0.98	0.33	0.78	0.33	**1.00**	0.99	0.98
4	**1.00**	0.77	0.99	0.34	0.78	**1.00**	**1.00**
5	0.98	0.99	0.35	**1.00**	0.99	0.27	0.92

Table 5.1: Rand indices for the 14 clustering algorithms applied to the 6 datasets represented on Figure 5.4 and 5.5

.

This functions takes as input the size of each cluster, a probability p_{in} and a probability p_{out}. It outputs a random graph where each edge has a probability p_{in} of existing between two nodes from the same cluster, and p_{out} if they are from different clusters. The data was generated as follows:

- Fix a number of iterations n and a number of clusters k, $p_{in} = 0.3$ and $p_{out} = 0.01$.
- Set the size of each cluster to 5 plus a random integer between -3 and 3.
- Add 5 clusters of size 1 (the outliers).
- Generate n graphs with the above procedure, and add their adjacency matrices together. The resulting matrix is the adjacency matrix of a weighted graph $G_{n,k}$.

See Figure 5.6 for examples. Table 5.2 shows the results obtained with our algorithm (using *max-jump* for both thresholds) and with the Louvain method [BGLL08], for $n = 30, 40, 50$ and k varying from 2 to 10. Our method performs better than Louvain's on respectively 5 and 6 graphs out of 9 for n = 30 and 40, and for higher values of n, when clusters are denser, our method always detects the right clusters and outliers.

5.5 Application to gait signals

Our clustering algorithm was initially developed for the time series motif discovery algorithm proposed by Germain et al. [GTO24b], called PersistentPattern (PEPA). They use the LT-normalized distance [GTO] (distance invariant to linear trend) to construct a weighted k-nearest-neighbors graph (k-nearest non-overlapping subsequences) with additional temporal connections between successive points. Our clustering algorithm is used on this graph to exclude outliers (which can either be motifs that only occur once or subwindows where the time series is constant) and find clusters (the number of detected clusters is the number of patterns), from which each occurrence of each pattern can be reconstructed by aggregating overlapping subwindows in each cluster. We refer

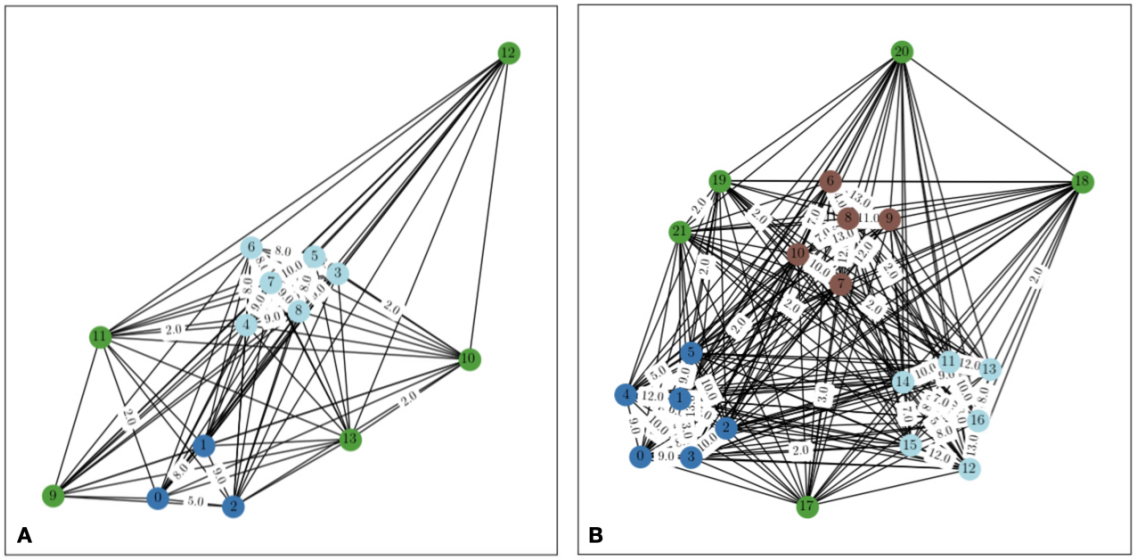


Figure 5.6: Generated graphs with $n = 30$, with respectively 2 and 3 clusters. Edges without a label have a weight of 1.

	n=30		n=40		n=50		n=60	
k	NNVR	Louvain	NNVR	Louvain	NNVR	Louvain	NNVR	Louvain
2	0.74	**0.78**	0.69	**0.78**	**1.00**	0.84	**1.00**	0.84
3	0.87	0.87	0.74	**0.79**	**1.00**	0.92	**1.00**	0.89
4	**1.00**	0.95	0.84	**0.88**	**1.00**	0.89	**1.00**	0.96
5	**1.00**	0.94	**1.00**	0.94	**1.00**	0.90	**1.00**	0.94
6	**0.97**	0.92	**1.00**	0.91	**1.00**	0.97	**1.00**	0.94
7	**0.98**	0.93	**1.00**	0.96	**1.00**	0.96	**1.00**	0.92
8	**0.98**	0.98	**1.00**	0.93	**1.00**	0.92	**1.00**	0.91
9	0.82	**0.95**	**0.98**	0.95	**1.00**	0.98	**1.00**	0.92
10	0.45	**0.94**	**1.00**	0.95	**1.00**	0.97	**1.00**	0.98

Table 5.2: Rand indices for $n = 30, 40, 50, 60$ and k varying from 2 to 10.

to Germain et al. [GTO24b] for a detailed description of the algorithm. Figure 5.7 shows the result of the PEPA algorithm [5] to gait signals from the dataset described in Chapter 1 (see Figure 1.1).

This signal has four different regimes: it stays near zero at the start and end, there are two phases of normal walk separated by the U-turn that cause a unique pattern around 9-10 seconds, and the first and last steps have smaller amplitudes than normal steps. The algorithm detects one motif that corresponds to a typical step (in green on the figure). On Figure 5.7, we can observe that the first step (the first oscillation with a small amplitude) is only partially detected, as the main variations are not shown in red. The algorithm then detects six consecutive steps, the nothing for about one second, and then six consecutive steps. This one second subwindow corresponds to the U-turn, during which steps are different so a distinct pattern appears. This pattern is considered as an outlier by the

[5]The authors made their method publicly available in an interactive application [GTO24a] than can be run on a webpage `https://persistent-pattern-discovery.onrender.com`

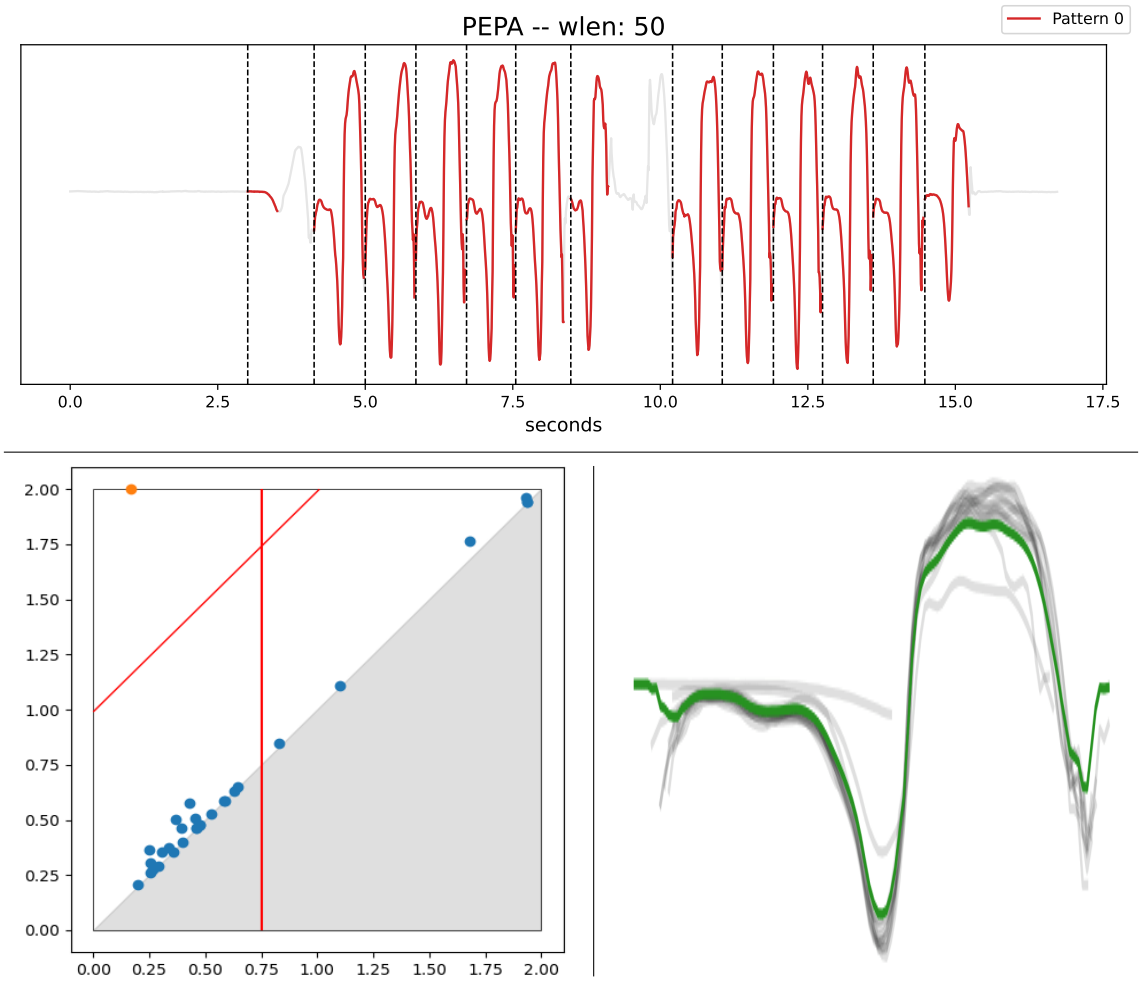


Figure 5.7: **Top:** result of PEPA with window size 50 applied to a gait signal from the [TBMM+19] dataset. Detected motifs appear in red, vertical lines correspond to their start. **Bottom left:** corresponding NNVR persistence diagram, persistence threshold and birth date threshold. **Bottom right:** normalized detected motifs (in gray) and average motif (in green). The average is computed after realigning the signals to maximize correlation.

algorithm because it only appears once, so it is not detected as a motif.

Note that using a normalized distance is important for the algorithm to exclude almost constant subwindows because in this case it normalizes the noise so the distance between two such subwindows should be high. A consequence is that the algorithm cannot distinguish the last step from normal steps (this can be desirable or not, depending on the application), or that it can detect a short subwindow with small variations of the signal as part of a motif, as in the case of the first step of our example. Another consequence is that the start and end of the signal are not detected as motifs.

This application of pattern detection to gait signals could be used to segment the data before using other algorithms designed to analyze steps individually. For example, it could be useful in a step-detection based method such as the one developed by Vienne-Jumeau et al. [VJOM+19, VJQVR20], with an application to the study of multiple sclerosis.

5.6 Conclusion

In this chapter, we have defined the NNVR filtration, proved its stability and integrated it in a non-parametric hierarchical clustering algorithm. The NNVR filtration contains more information than the VR filtration, which makes it possible to add an outlier detection step to persistence-based clustering. Our experiments show that our algorithm can perform well despite being fully non-parametric, and can thus be useful in many applications such as pattern detection. Future research could explore more advanced methods to choose thresholds on the persistence diagram, that could select both thresholds simultaneously, for example a 2D version of Otsu's method [Ots79], using transformations of the diagram to make it easier to analyze with heuristics or incorporate some knowledge depending on the application.

Chapter 6

Unsupervised anomaly detection in time series

In this chapter, we propose a new algorithm for unsupervised anomaly detection in univariate time series. It relies on delay embeddings and on the extraction of persistent cycles from the 1-dimensional persistent homology constructed from the distance to measure Rips filtration. This filtration makes it possible to identify 1-cycles corresponding to recurrent patterns by leveraging density information. Points in those cycles are considered as normal, and the algorithm can then assign an anomaly score to any point which is its distance to the normal set. We describe the algorithm, make a theoretical study, and test it on several real-world and synthetic datasets, showing that it is competitive with state-of-the-art anomaly detection methods.

Contents

6.1 Introduction **113**
6.2 Model and problem formulation **114**
6.3 Method **117**
6.3.1 Motivation 117
6.3.2 Algorithm description 118
6.4 Theoretical study **122**
6.4.1 Overview of existing stability theorems for DTM filtrations . . . 122
6.4.2 Upper bound on the interleaving distance between the real and ideal filtrations 123
6.5 Experimental setup **129**
6.5.1 Synthetic dataset 129
6.5.2 Real-world datasets and methods from TSB-UAD 130
6.6 Results **131**
6.6.1 Benchmark on real-world data 131
6.6.2 Influence of the embedding dimension and delay 133
6.6.3 Influence of the number of neighbors and points on the diagram 134
6.6.4 Computation time and influence of the subsampling parameter 134
6.6.5 Robustness to noise and benchmark on synthetic data 135
6.7 Application to gait signals **136**
6.8 Conclusion **139**

6.1 Introduction

Anomaly detection in time series is an important problem in data science, with applications in many fields such as healthcare [AFD+17] and engineering [WAAC14]. A time series is a sequence of real numbers $\mathbf{x} = (\mathbf{x}[i])_{1 \leq i \leq n}$ (n will always denote the length of the time series). In the context of anomaly detection, $\mathbf{x}$ is assumed to be composed of a normal behavior and anomalies, i.e. points or sequences of points that differ from the normal behavior. More precisely, in several application contexts such as industrial monitoring or healthcare, the time series is usually assumed to be composed of some repetitive/frequent patterns, possibly of varying lengths (think for instance of an heartbeat in ECG data) among which some occur a large number of times (the *normal* ones) and some have significantly fewer occurrences (the *abnormal* ones) : see Figure 6.1 for an illustration.

Over the past years, several unsupervised anomaly detection algorithms have been developed from different research areas [LTZ08, BKNS00, GD12, YZU+18, BLR+21, AA17, SY14, MVS+15, LMM07, MSDA18, SWS+99] (see [PKB+22] and [SWP22] for a comprehensive review). Among them, some rely on a model and use the prediction or reconstruction error as an anomaly detector and some are based on clustering or machine learning techniques applied on the subsequences in order to detect outliers. For instance, LOF [BKNS00] transforms the time series into a point cloud and studies the density of each point to assess whether or not they correspond to normal or abnormal behaviors. Some methods aim to find subsequences that represent the normal behavior and define anomalies as subsequences that differ from normality. The fact that the patterns can have different lengths, that there can be multiple normal patterns and multiple occurrences of an anomaly, and noise make it difficult to build a universal anomaly detection algorithm [PKB+22]. The main differences between the approaches actually lie in the implied definition given to the notion of normality.

Persistent homology [EH10] is a set of techniques that consists in analyzing the structure of data by constructing a filtered simplicial complex. The persistence diagram sums up when connected components, loops or higher-dimensional simplices appear and disappear when going through the filtration. It has been applied to many fields [CM21] including time series analysis. It is particularly adapted to study structured data such as time series with a periodic behavior [PH15, EGK14, BCMR22, BTM+22], and methods from TDA benefit from stability theorems that guarantee a certain robustness to noise [CDSO14, ACG+20]. The most commonly used filtrations are the Čech and the Vietoris-Rips filtrations. As those filtrations are not robust to the presence of outliers, new variants based on the notion of distance to a measure were introduced in [CCSM11, BCOS16, ACG+20], along with new stability theorems. Other relevant filtrations are defined in [DSC04, BLM+19]. The authors of [ACG+20, UIY22, FBMG23] explain how sequences of persistence diagrams can be used for change detection or anomaly detection in time series. However, this method seems more relevant for change detection than anomaly detection as it is not designed to handle typical cases such as punctual anomalies, multiple normal behaviors or repeated anomalies.

In the case of periodic functions, the importance of 1-dimensional persistent homology (the study of loops in point clouds) was theoretically studied in [PH15] and it was applied to time series in [EGK14, PDHH15] by transforming the data into a point cloud and considering the most important loop. By extension, 1-dimensional persistent homology is also relevant to study time series with repetitive patterns.

In a nutshell, our method consists in transforming the time series into a point cloud and extracting 1-cycles (i.e. loops) that are considered to correspond to normal patterns of

the time series. We will use the Vietoris-Rips filtration associated to the empirical distance to a measure as described in [ACG+20]. Those cycles are identified on the persistence diagram because density information is used to construct the filtration. Once "normal cycles" have been extracted, an anomaly score is defined for each point of the embedding as its distance to the normal cycles. The specificity of our method is that it makes a global study of the delay embedding to find loops corresponding to whole patterns and also integrates local density information through the choice of the filtration, that is used to distinguish normal and abnormal points by reading the persistence diagram.

Contributions. In this chapter, we apply methods from topological data analysis to unsupervised anomaly detection in time series. Our contributions are listed below.

- We propose a model of time series that makes it possible to formally define the anomaly detection problem. It includes the possibility of having multiple normal or abnormal behaviors, repeating anomalies and noise. We show examples of real-life time series that fit this model.
- We present a new method for anomaly detection based on persistent homology. The method uses a delay embedding and the Vietoris-Rips filtration associated to the empirical distance to a measure to find 1-cycles made of dense points and identify them to the normal behavior.
- We use a property of this filtration to deal with a subset of the point cloud while keeping information from the whole data in order to significantly decrease computation time.
- We use our model to derive an upper bound on the interleaving distance between the filtration used in our algorithm (with discrete, noisy data with anomalies) and a filtration obtained in an ideal situation with a continuous signal without noise or anomalies.
- We study the behavior of our algorithm with different parameters and the influence of noise, and show that it is competitive to state-of-the-art anomaly detection methods on different real-world datasets.

The chapter is organized as follows. in Section 6.2, we introduce our model of time series and our definition the anomaly detection problem. Section 6.3 describes our algorithm. Section 6.4 contains the theoretical study. In Section 6.5, we present the 12 state-of-the-art anomaly detection methods and real-world and synthetic datasets. In Section 6.6, we use these datasets to study the behavior of our algorithm with different parameters and the influence of noise, compare it to state-of-the-art anomaly detection methods. In Section 6.7, we show how our algorithm can be applied to the analysis of gait signals.

6.2 Model and problem formulation

In this subsection, we introduce a model for time series which makes it possible to formally define the anomaly detection problem. Notations that are used throughout the chapter are listed in Table 6.1.

Let $\mathbf{x} = (\mathbf{x}[t])_{t\in[0,1]}$ and $\varepsilon = (\varepsilon[t])_{t\in[0,1]}$ be real-valued functions defined on $[0,1]$. We call $\mathbf{x}$ the **signal** and ε is called the **noise**.

Table 6.1: Notations

Notation	Description
t	Time
n	Length of the time series.
d	Dimension of the time delay embedding.
τ	Delay of the time delay embedding.
$\mathbf{x} = (\mathbf{x}[t])_{t\in[0,1]}$	Signal: a continuous function defined on $[0,1]$.
$\hat{\mathbf{x}} = (\hat{\mathbf{x}}[i])_{1\leq i\leq n}$	Time series: a finite uniform subsampling of $\mathbf{x}$.
$\mathbf{n}_i$	Normal atom i.
$\mathbf{a}_i$	Abnormal atom i.
$\mathbf{x_n}$ (resp. $\mathbf{x_a}$)	Normal (resp. abnormal) part of the signal.
$\hat{\mathbf{x}}_\mathbf{n}$ (resp. $\hat{\mathbf{x}}_\mathbf{a}$)	Normal (resp. abnormal) part of the time series.
$M_{\mathbf{n}_i}$ (resp. $M_{\mathbf{a}_i}$)	Number of normal (resp abnormal) atoms.
$\boldsymbol{\eta}_i$ (resp. $\boldsymbol{\alpha}_i$)	Activations of $\mathbf{n}_i$ (resp. $\mathbf{a}_i$).
$k_{\mathbf{n}_i}$ (resp. $k_{\mathbf{a}_i}$)	Number of occurrences of $\mathbf{n}_i$ (resp. $\mathbf{a}_i$).
$l_{\mathbf{n}_i}$ (resp. $l_{\mathbf{a}_i}$)	Length of atom $\mathbf{n}_i$ (resp. $\mathbf{a}_i$).
$k_\mathbf{n}$	Number of normal occurrences: $\sum_{i=1}^{M_{\mathbf{n}_i}} k_{\mathbf{n}_i}$.
$k_\mathbf{a}$	Number of abnormal occurrences: $\sum_{i=1}^{M_{\mathbf{a}_i}} k_{\mathbf{a}_i}$.
$k_\mathbf{x}$	Total number of occurrences: $k_\mathbf{n} + k_\mathbf{a}$.
$V = (V_\alpha)_{\alpha\in\mathbb{R}^+}$	A filtration.
$\mathbb{V}$	Persistent homology module corresponding to the filtration V.
$\mathsf{Diag}(\mathbb{V})$	Persistence diagram corresponding to the persistence module $\mathbb{V}$.
$X_{d,\tau}$	Time delay embedding of a signal or time series $\mathbf{x}$, with dimension d and delay τ .
μ_B (where $B \subset E$)	A probability measure on E with support included in B.
$d_{\mu_B,m}$	Distance function to the measure μ_B with parameter m.
d_i	Interleaving pseudo-distance.
d_b	Bottleneck distance.
W_2	Wasserstein distance with quadratic cost.
d_H	Hausdorff distance.
$p \in \mathbb{N}$	Parameter of the weighted filtrations.
$m = \frac{q}{\mathsf{Card}(\hat{X})} \in]0,1[$ $(q \in \mathbb{N})$	Parameter m of the DTM.
$\mathsf{Cech}[X,f,p]$ $(B \subset E)$	Weighted Čech filtration with parameters (X,f,p). with function f, and parameters m,p.
$\mathsf{Rips}[X,f,p]$ $(B \subset E)$	Weighted Rips filtration with parameters (X,f,p). with function f, and parameters m,p.

Remark 7. *We will always use the term signal for continuous real-valued functions, and time series for finite sequences of real numbers.*

We use a convolutional sparse coding model [PSE17] with an additional hypothesis to describe $\mathbf{x}$ as a sequence of **atoms**, which are either normal or abnormal. Formally, we assume that $\mathbf{x}$ can be written in the following way:

$$\mathbf{x} = \sum_{i=1}^{M_\mathbf{n}} \mathbf{n}_i * \boldsymbol{\eta}_i + \sum_{i=1}^{M_\mathbf{a}} \mathbf{a}_i * \boldsymbol{\alpha}_i.$$

The $\mathbf{n}_i$ and $\mathbf{a}_i$ are respectively normal and abnormal atoms of length $l_{\mathbf{n}_i} < 1$ (resp. $l_{\mathbf{a}_i}$), i.e. continuous functions on $[0,1]$ with support $[0, l_{\mathbf{n}_i}]$ (resp. $[0, l_{\mathbf{a}_i}]$) and such that $\mathbf{n}_i(0) = \mathbf{n}_i(l_{\mathbf{n}_i}) = 0$ (resp. $\mathbf{a}_i(0) = \mathbf{a}_i(l_{\mathbf{a}_i}) = 0$). There are $M_\mathbf{n}$ normal atoms and $M_\mathbf{a}$ abnormal atoms. The $\boldsymbol{\eta}_i$ and $\boldsymbol{\alpha}_i$ are the **activations**: binary functions on $[0,1]$ that take the value 1 only a finite number of times. Thus, $\mathbf{n}_i * \boldsymbol{\eta}_i$ is a signal where an **occurrence**

of $\mathbf{n}_i$ starts at each time t where $\boldsymbol{\eta}_i[t] = 1$. For each i, the number of occurrences of atom $\mathbf{n}_i$ (resp. $\mathbf{a}_i$) is denoted by $k_{\mathbf{n}_i} = ||\boldsymbol{\eta}_i||_0$ (resp. $k_{\mathbf{a}_i} = ||\boldsymbol{\alpha}_i||_0$), where $||.||_0$ is the number of non-zero values of a function or vector.
Finally, we define the normal and abnormal parts of the signal as

$$\mathbf{x}_\mathbf{n} = \sum_{i=1}^{M_\mathbf{n}} \mathbf{n}_i * \boldsymbol{\eta}_i \text{ and } \mathbf{x}_\mathbf{a} = \sum_{i=1}^{M_\mathbf{a}} \mathbf{a}_i * \boldsymbol{\alpha}_i$$

along with the numbers $k_\mathbf{n} = \sum_{i=1}^{M_{\mathbf{n}_i}} k_{\mathbf{n}_i}$, $k_\mathbf{a} = \sum_{i=1}^{M_{\mathbf{a}_i}} k_{\mathbf{a}_i}$, and $k_\mathbf{x} = k_\mathbf{n} + k_\mathbf{a}$.

Let $n > 1 \in \mathbb{N}$. In the rest of the chapter, we will always denote by $\hat{\mathbf{s}}$ the uniform subsampling of size n of any given signal $\mathbf{s} = (\mathbf{s}[t])_{t\in[0,1]}$, that is $\hat{\mathbf{s}} = (\hat{\mathbf{s}}[j])_{1\leq j\leq n}$, where $\hat{\mathbf{s}}[j] = \mathbf{s}[\frac{j-1}{n-1}]$. All the above signals, atoms, activations and quantities can be defined the same way in the discrete case starting from

$$\hat{\mathbf{x}} = \sum_{i=1}^{M_\mathbf{n}} \hat{\mathbf{n}}_i * \hat{\boldsymbol{\eta}}_i + \sum_{i=1}^{M_\mathbf{a}} \hat{\mathbf{a}}_i * \hat{\boldsymbol{\alpha}}_i.$$

With this model, we can formally define the anomaly detection problem.

Definition 26 (Anomaly detection problem)**.** *The anomaly detection problem consists in finding the set of integers $i \in [1, n]$ such that $\hat{\mathbf{x}}_\mathbf{a}[i] \neq 0$, which means finding all the discrete abnormal activations and atom lengths.*

We now make some assumptions related to the model in the context of anomaly detection. The first assumption states that all the $\mathbf{n}_i * \boldsymbol{\eta}_i$ and $\mathbf{a}_i * \boldsymbol{\alpha}_i$ have disjoint supports, which implies that each point $\mathbf{x}[t]$ is either normal or abnormal.

Assumption 1. *if we denote by $(\boldsymbol{\zeta}_i)_{1\leq i\leq M_\mathbf{n}+M_\mathbf{a}}$ the list of all the activations $\boldsymbol{\eta}_i$ and $\boldsymbol{\alpha}_i$, the assumption is:*

$$\forall i, \forall t \in [0,1], \quad \boldsymbol{\zeta}_i[t] \neq 0 \Rightarrow \forall j \neq i, \forall s \in [t, t + l_{\boldsymbol{\zeta}_i}], \quad \boldsymbol{\zeta}_j[s] = 0.$$

Assumption 2. *The observed data $\hat{\mathbf{y}}$ (the time series) is a uniform noisy subsampling of the signal $\mathbf{x}$: $\hat{\mathbf{y}} = \widehat{\mathbf{x} + \boldsymbol{\varepsilon}}$.*

Assumption 3. *For the anomaly detection problem, we will assume that $\max_i(k_{\mathbf{a}_i})$ is significantly lower than $\min_i(k_{\mathbf{n}_i})$, which means that anomalies are rare atoms.*

Remark 8. *Note that the only assumption we will make on the noise is that $||\boldsymbol{\varepsilon}||_\infty$ is small compared to the variations of the signal. Thus, it could represent small variations at each occurrence of an atom, of noise due to the data acquisition process. Moreover, introducing $\boldsymbol{\varepsilon}$ also makes it possible to study the problem without the disjoint supports assumption as long as the overlap between supports causes a small enough change in the infinity norm of $\mathbf{x}$.*

Our model is illustrated on Figure 6.1. All the atoms start and end at zero and have disjoint supports. The green and red ones each occur ten times: they are the normal atoms. The blue one occurs twice, the orange and purple one each occur once. Note that atoms can have significantly different lengths.

Figure 6.2 show two example of time series that fit our model, from real-world datasets included in the TSB-UAD suite. The first one comes from the ECG dataset, which is a

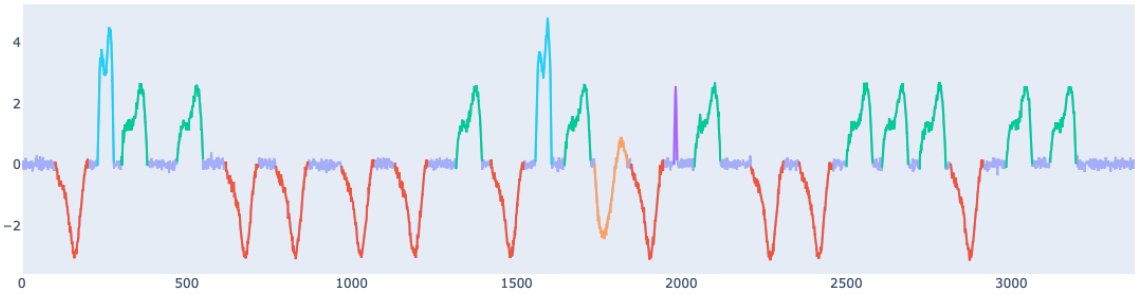


Figure 6.1: An example time series, with normal patterns in green and red (and the null parts in dark blue) and three abnormal ones (in blue, orange and purple), with Gaussian noise of amplitude 0.1.

standard electrocardiogram dataset with anomalies that represent ventricular premature contractions. This shows that our model is relevant to study structured activities that present repetitive behaviors over time where anomalies in the real-world manifest as anomalies on the signal (here: pathologies cause premature ventricular contractions that appear as anomalies on the ECG). The second one comes from the NAB dataset, which is composed of labeled real-world and artificial time series including AWS server metrics, online advertisement clicking rates, real time traffic data, and a collection of Twitter mentions of large publicly-traded companies.

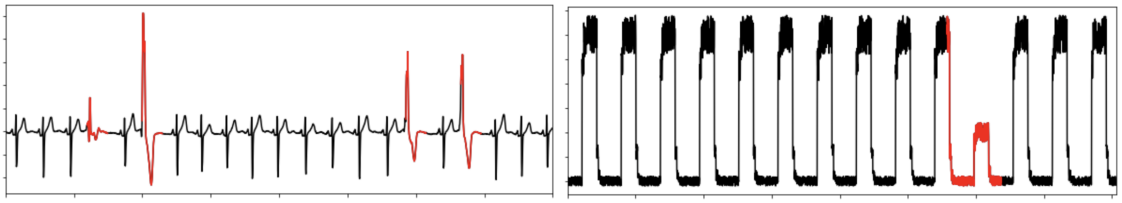


Figure 6.2: Two example of real-world time series that fit our model, with anomalies in red. **Left:** Time series from the ECG dataset. **Right:** Time series from the NAB dataset. See [PKB+22] for details on the datasets.

6.3 Method

In this section, we describe our algorithm for unsupervised anomaly detection. We start by explaining the main ideas behind the algorithm, and then describe its four steps: compute the DTM Rips filtration of a subset of the delay embedding, identify the normal 1-cycles, extract them, and compute the anomaly scores. We use the above notations for all parameters and mathematical objects. The required mathematical background was introduced in Chapter 2.

6.3.1 Motivation

The solution to the anomaly detection problem described in Section 6.2 is a binary vector of length n: each point $\hat{\mathbf{y}}[i]$ of the time series is either normal or abnormal. Our algorithm, like most anomaly detection algorithms [PKB+22, SWP22], works by giving an anomaly score to each $\hat{\mathbf{y}}[i]$ which should be high if the point is abnormal. A binary answer can then be obtained by choosing a threshold over which points are considered abnormal. We do not give a specific method to choose the threshold, as it depends on the application.

Our algorithm works by studying a delay embedding $\hat{Y}$ of $\hat{\mathbf{y}}$ to identify a set of points as normal. Then, each point of $\hat{Y}$ is given an anomaly score which is its distance to the set of normal points. Finally, we get an anomaly score for $\hat{\mathbf{y}}[i]$ by averaging the scores of all points in $\hat{Y}$ of which $\hat{\mathbf{y}}[i]$ is a coordinate.
The idea behind our algorithm is that, if we consider our model of signals from Section 6.2 and if $(d-1)\tau$ is small enough, Assumption 1 implies that the embedding of each occurrence of atom should starts and ends at $(0,\dots,0)$, so it has at least one loop. 1-dimensional persistent homology can be used to detect those loops in $\hat{Y}$ as homology classes of 1-cycles. Each point from a normal atom $\mathbf{n_i}$ will have $k_{\mathbf{n_i}}$ occurrences and each point from an abnormal atom $\mathbf{a_i}$ will have $k_{\mathbf{a_i}}$ occurrences, so if $\max(k_{\mathbf{a_i}}) < q \leq \min(k_{\mathbf{n_i}})$ (see Assumption 3), then $d_{\mu_{\hat{Y}}, \frac{q}{n-(d-1)\hat{\tau}}}$ should be close to zero for normal points and strictly higher for abnormal points, which would make it possible to discriminate the corresponding 1-cycles on the persistence diagram, as their birth date will be different. Our set of normal points is then the set of 1-cycles detected as normal.

Remark 9. *One could argue that using 1-cycles is useless because points could be studied individually using their filtration value (i.e. the birth dates on the 0-dimensional persistence diagram) which is the opposite of a measure of the density of $\hat{Y}$ around each point. This approach would then be similar to the LOF algorithm [BKNS00]. However, in practice, data do not fit our ideal model (there can be noise, differences between occurrences of the same atoms or non-disjoint supports) and there is no way to be certain that $\max(k_{\mathbf{a_i}}) < q \leq \min(k_{\mathbf{n_i}})$ so the filtration values can take a range of values and choosing a threshold would be hard. Moreover, an abnormal atom with slow variations will give dense points in $\hat{Y}$ which would make the 0D approach or LOF fail. Considering 1D persistent homology and considering only cycles with a high persistence makes the choice of normal points easier by focusing on a few cycles (and thus a few birth dates) corresponding to structured components of $\hat{Y}$. It also makes it possible to eliminate atoms with slow variations (because the corresponding cycles have low persistence).*

6.3.2 Algorithm description

The input of our algorithm is a time series $\mathbf{y}$. There are five parameters that will be described below: $d, \hat{\tau}, n_{points}, q, n_{diag}$.

6.3.2.1 Delay embedding, subsampling and DTM Rips filtration

We start by choosing a dimension d and a delay $\hat{\tau}$ and computing the delay embedding $\hat{Y} = \hat{Y}_{d,\hat{\tau}}$. Figure 6.3 shows a delay embedding of the time series from Figure 6.1. Each colored loop on Figure 6.3 corresponds to the atom of the same color on Figure 6.1. The presence of noise makes loops corresponding to normal atoms (in green and red) thicker, as these atoms have more occurrences.

As point clouds can be very large (they have $n-(d-1)\hat{\tau}$ points), computing persistent homology can be too long for the algorithm to be used in practice ($O(n^3)$ in the worst case for 1D persistent homology and cycle extraction [BCY18, ELZ02]). To solve this problem, we choose a number n_{points} and take a subsample $\tilde{Y}$ of $\hat{Y}$ made of n_{points} points. In practice, we use a greedy method: we start with a random point and, until we have n_{points} points, add the furthest one to the set of already chosen points.

We then choose an integer $q > 1$ and set the DTM parameter m to $m = \frac{q}{\mathsf{Card}(\hat{Y})} \in]0,1[$ and compute the filtration $V = \mathsf{Rips}[\tilde{Y}, d_{\mu_{\hat{Y}},m}, 1]$ with $p = 1$. We chose to always use $p = 1$ because among the three possible values for which [ACG+20] provides explicit formulas for the filtration values (1, 2, and ∞), $p = 1$ gives the strongest theoretical guarantees (see

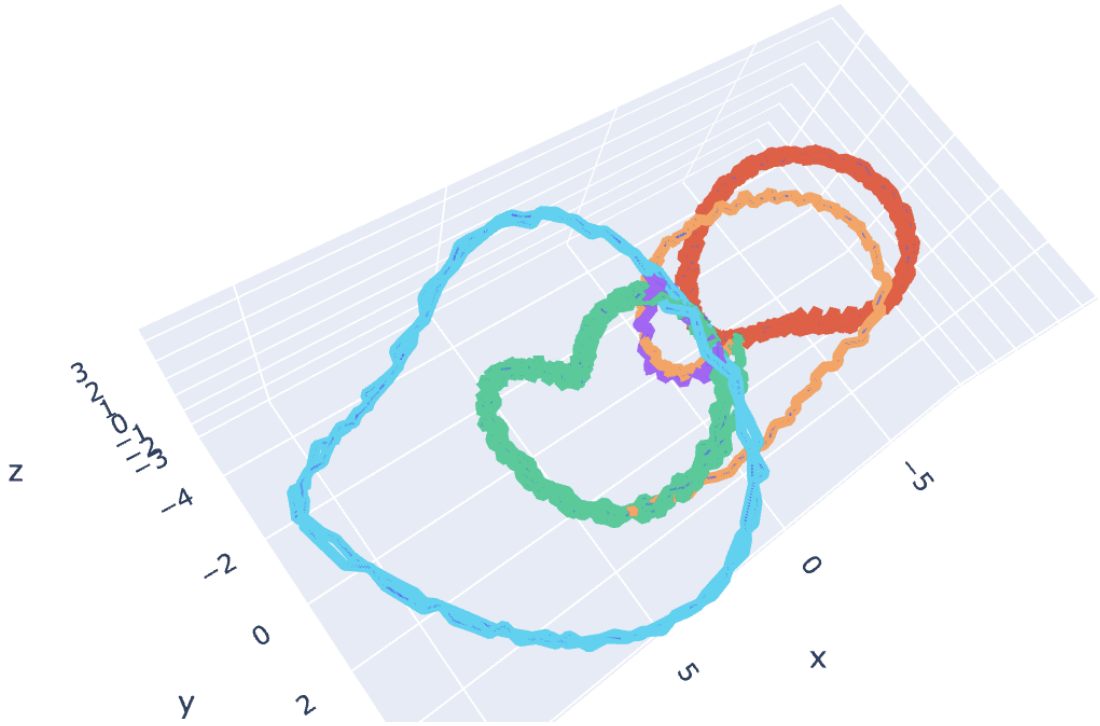


Figure 6.3: Delay embedding of the time series from Figure 6.1 (PCA in 3D), with $d = 10$ and $\hat{\tau} = 4$. Colors corrrespond to the colors of atoms on Figure 6.1.

section 6.4) and the highest value to edges, which can give more persistent cycles that are easier to identify on the diagram.

Remark 10. *Note that the filtration values of points in $\tilde{Y}$ are their original values from $d_{\mu_{\hat{Y}_{d,\hat{\tau}}},m}$ (subsampling does not change the fact that V is a filtration). This is important to keep the density information from the whole point cloud when subsampling (otherwise, the effect of the number of occurrences of normal atoms would disappear).*

6.3.2.2 Identifying normal cycles

The second step consists in identifying normal 1-cycles by reading the persistence diagram, and extracting those cycles. Let $\text{Diag}(\mathbb{V})$ be the persistence diagram corresponding to the filtration V. To identify normal cycles, we propose an algorithm that relies on the choice of two thresholds: the persistence threshold (we focus on the most persistent points, which describe important structures), and the birth date threshold (among those points, we consider those with a birth date above the cycle to be abnormal).

To choose the persistence threshold, we sort the persistence of all points by decreasing order in a list L, find the index i such that $L[i] - L[i+1]$ is maximal, and keep points corresponding to indices from 1 to $i + n_{diag}$, where n_{diag} is a chosen parameter. See Algorithm 2 for a formal description. We will use $n_{diag} \geq 2$ in practice, to keep at least three points and thus to be able to compare at least two differences in birth dates.

We choose the birth date threshold as follows: we take the points that are above the persistence threshold and the point with minimal birth date, and sort their birth dates by increasing value in a list L' (indices start at 1). Let i_{diff} be the index such that the birth date difference is maximal and i_{pers} the index of the most persistent point. If $i_{pers} \leq i_{diff}$,

Algorithm 2 PersistenceThreshold (lists start at index 1)

Require: Diag($\mathbb{V}$), n_{diag}
$L \leftarrow$ [d-b for (b,d) in Diag($\mathbb{V}$)]
Sort L in decreasing order.
$N \leftarrow$ length(L)
Diffs $\leftarrow [L[i] - L[i+1]$ for $i = 1 \dots N-1]$
$i_{thr} \leftarrow$ argmax(Diffs)
return $\frac{L[i_{thr}+n_{diag}]+L[i_{thr}+n_{diag}+1]}{2}$

then the birth date threshold is set to the birth date of $L'[i_{diff}]$ (this is the typical case where we look for the highest difference). If $i_{diff} = 1$, we consider that all the persistent points are close from one another and are all normal cycles, so we set the threshold to $+\infty$. If $1 < i_{diff} < i_{pers}$, we set the threshold to the birth date of $L[i_{pers}]$, assuming that the most persistent cycle is always normal. See Algorithm 3 for a formal description.

Algorithm 3 BirthDateThreshold (lists start at index 1)

Require: Diag($\mathbb{V}$), n_{diag}
$L \leftarrow [(b,d)$ for (b,d) in Diag($\mathbb{V}$)]
$(b_{min}, d_{b_{min}}) \leftarrow$ min$[b$ for (b,d) in $L]$ for the lexicographic order.
$p \leftarrow$ PersistenceThreshold(Diag($\mathbb{V}$), n_{diag})]
$L' \leftarrow [(b_{min}, d_{b_{min}})] + [(b,d)$ for (b,d) in L if $d - b \geq p]$
Sort L' using the lexicographic order.
$L'_{\text{birth}} \leftarrow [b$ for (b,d) in $L']$
$L'_{\text{pers}} \leftarrow [d - b$ for (b,d) in $L']$
$N' \leftarrow$ length(L')
Diffs$'_{\text{birth}} \leftarrow [L'_{\text{birth}}[i+1] - L'_{\text{birth}}[i]$ for $i = 1 \dots N'-1]$
Diffs$'_{\text{pers}} \leftarrow [L'_{\text{pers}}[i+1] - L'_{\text{pers}}[i]$ for $i = 1 \dots N'-1]$
$i_{diff} \leftarrow$ argmax(Diffs$'_{\text{birth}}$)
$i_{pers} \leftarrow$ argmax(Diffs$'_{\text{pers}}$)
if $i_{diff} == 1$ **then**
    return $+\infty$
end if
if $1 < i_{diff} < i_{pers}$ **then**
    return $\frac{L'_{\text{birth}}[i_{pers}]+L'_{\text{birth}}[i_{pers}+1]}{2}$
else
    return $\frac{L'_{\text{birth}}[i_{diff}]+L'_{\text{birth}}[i_{diff}+1]}{2}$
end if

Finally, we keep all points above the persistence threshold and whose birth dates are below the birth date threshold. If there are no 1-cycles on the diagram, we take an arbitrary "cycle" as normal. In this case, one should look for different parameters d and $\hat{\tau}$ to make the point cloud less dense.

Figure 6.4 illustrates our algorithm applied to the time series from Figure 6.1 (with delay embedding from Figure 6.3). We used $q = 10$, $n_{diag} = 2$ and $n_{points} = 200$. On this example, the birth date threshold is around 2.5, so the two points in the upper left corner are detected as normal (among the 5 most persistent ones).

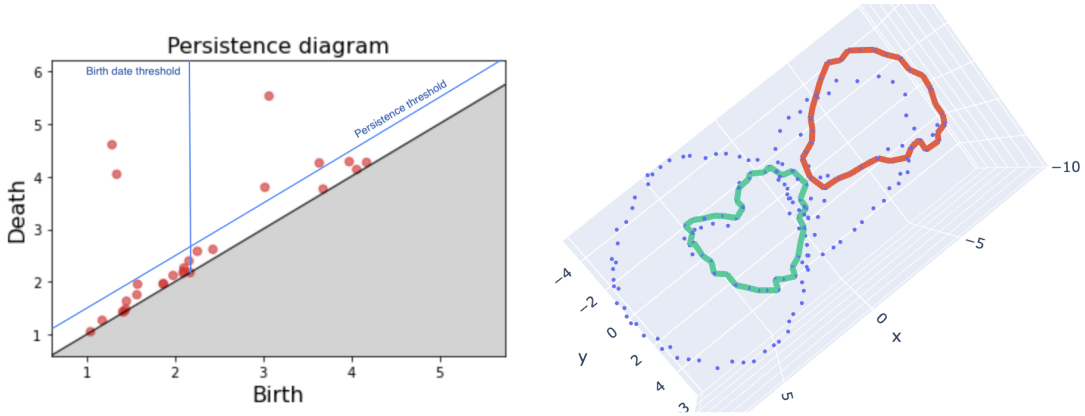


Figure 6.4: **Left:** persistence diagram of the DTM-filtration of the delay embedding from Figure 6.3, with persistence and birth date thresholds in blue.. **Right:** subsampling of the delay embedding with 200 points, and cycles detected as normal (in green and red).

6.3.2.3 Cycle extraction

The cycle extraction step consists in computing a list of cycles $L_{cycles} = (c_1, \dots, c_{n_{cycles}})$, each one representing the homology class of a cycle identified as a normal at the previous step (a cycle is stored as a list of points).

In practice, we will compute persistent homology with coefficients in $K = \mathbb{Z}/2\mathbb{Z}$. In that case, the following matrix reduction algorithm from [ELZ02] can be used.
Let us write our finite filtered simplicial complex as $V = \emptyset = V_0 \subset V_1 \subset \cdots \subset V_N$ such that $V_{i+1} = V_i \cup \sigma_{i+1}$ (where σ_{i+1} is a simplex), and let ∂ be the $N \times N$ matrix of the boundary operator ($\partial_{i,j} = 1$ if σ_i is a face of σ_j), with columns $C_1 \dots C_N$. Let low(C_j) be the greatest index i such that $C_j[i] = \partial_{i,j} \neq 0$ (or 0 if the column only has zeros). The matrix reduction algorithm is described in Algorithm 4. The representative cycle corresponding to a point $(l(j), j)$ on the persistence diagram is $\sum_{i=1}^{N} C_j[i]\sigma_i$.

Algorithm 4 MatrixReduction

Require: $\partial = (C_1 \dots C_N)$
for $i = 1, \dots, N$ **do**
  for $j = N-1, \dots, 1$ **do**
    if low$(C_i) ==$ low$(C_j) \neq 0$: **then**
      $C_j \leftarrow C_i + C_j \bmod(2)$
    end if
  end for
end for
return ∂

The green and red cycles on the point cloud $\tilde{Y}$ from Figure 6.4 are the two normal cycles extracted with this method, corresponding to the two points evoked above (notice that we found the green and red cycle from Figure 6.3).

6.3.2.4 Anomaly scores

Once the list L_{cycles} of normal cycles has been computed, an anomaly score is given to each point $x \in \hat{Y}$, which is its distance to L_{cycles}: $d(x, L_{cycles})$.

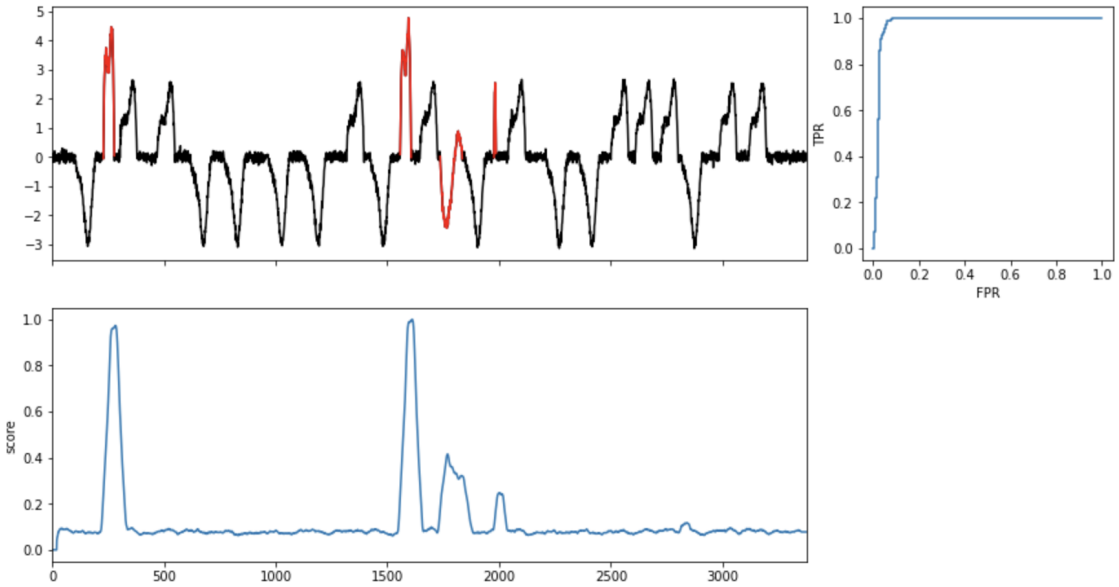


Figure 6.5: **Top left:** signal from Figure 6.1 with anomalies in red. **Bottom:** Anomaly score obtained with our algorithm (distance to the cycles from Figure 6.4, normalized). **Top right:** ROC curve (AUC = 0.98).

Finally, we get an anomaly score for $\mathbf{y}_i$ by averaging the scores of all points in $\hat{Y}$ between the first and last point that has $\mathbf{y}_i$ is a coordinate:

$$\text{Score}(\mathbf{y})_i \leftarrow \text{mean}(\{d(\hat{Y}_j, L_{cycles}) \mid \max(0, i-(d-1)\hat{\tau}) \leq j \leq i\}).$$

One can choose a threshold to the anomaly score to get a binary answer. Typically, one can chose to keep score only above a certain quantile. In Section 6.5, we will compare algorithms using the AUC-ROC curve obtained by varying the threshold from 0 to 1 in order not to be biased by an arbitrary choice of threshold. We do not give a specific method to choose the threshold, as it depends on the application (in Section 6.5, we will compare algorithms using the AUC-ROC curve of each anomaly score not to be biased by an arbitrary choice of threshold). Figure 6.5 shows the results and ROC curve of our algorithm applied to the signal from Figure 6.1 (with delay embedding from Figure 6.3 and normal cycles from Figure 6.4). The two cycles corresponding to the normal patterns have been extracted so those patterns have an anomaly score close to zero.

6.4 Theoretical study

In this section, we recall stability theorems for DTM filtrations, define a probability measure on delay embeddings of signals and use it (and some additional assumptions) to derive an upper bound on the interleaving distance between the filtration used in our algorithm (with discrete, noisy data with anomalies) and a filtration obtained in an ideal situation with a continuous signal without noise or anomalies.

6.4.1 Overview of existing stability theorems for DTM filtrations

Here, we recall stability results from [ACG+20]. In this subsection, X and Y will denote any subsets of $\mathbb{R}^d$. Let $V[X, d_{\mu,m}, p]$ be either the DTM Čech or the DTM Vietoris-Rips filtration, and let $\mathbb{V}[X, d_{\mu,m}, p]$ be the associated persistence module.

Proposition 4 ([ACG+20], Prop. 3.1)**.** *If X is bounded, $\mathbb{V}[X, d_{\mu,m}, p]$ is q-tame.*

Let W_2 denote the **Wasserstein distance with quadratic cost** between μ and ν :

$$W_2(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)} \sqrt{\int_{\mathbb{R}^d\times\mathbb{R}^d} ||u-v||^2 d\pi(u,v)}$$

where $\Pi(\mu,\nu)$ is the set of transport plans between μ and ν (see [San15] for more details).

Let d_H denote the **Hausdorff distance** between sets, defined as

$$d_H(X,Y) = \max\left(\sup_{x\in X} d(x,Y), \sup_{y\in Y} d(X,y)\right).$$

Proposition 5 ([ACG+20], Prop. 4.3)**.** *Let μ_X and μ_Y be two probability measures on $\mathbb{R}^d$ with compact supports X and Y. Then*

$$d_i(V[X, d_{\mu_X,m}, p], V[Y, d_{\mu_Y,m}, p]) \leq m^{-\frac{1}{2}} W_2(\mu_X,\mu_Y) + 2^{\frac{1}{p}} d_H(X,Y).$$

The following result is specific to the case $p = 1$ (weaker results are given for $p > 1$ in [ACG+20]).

Proposition 6 ([ACG+20], Prop. 4.4)**.** *Let μ be a probability measure on $\mathbb{R}^d$ with compact support* $\mathrm{supp}(\mu)$ *and let $c(\mu,m) = \sup_{\mathrm{supp}(\mu)} d_{\mu,m}$.*
If $\mathrm{supp}(\mu) \subset X$ *and* $\mathrm{supp}(\mu) \subset Y$*, then $V[X, d_{\mu,m}, 1]$ and $V[Y, d_{\mu,m}, 1]$ are $c(\mu,m)$-interleaved and thus*

$$d_i(V[X, d_{\mu,m}, 1], V[Y, d_{\mu,m}, 1]) \leq c(\mu,m).$$

6.4.2 Upper bound on the interleaving distance between the real and ideal filtrations

The idea of our study is to define a filtration of $X_{\mathbf{n}}$, which represents the ideal case of continuous data without any anomaly nor noise, and to show that this filtration is close in terms of interleaving distance to the filtration $V[\hat{Y}, d_{\mu_{\hat{Y}},m}]$ constructed from the observed data $\hat{Y} = \widehat{X+\mathcal{E}}$ and associated empirical measure. As explained in Section 6.2, a bound on the interleaving distance between filtration also induces a bound on the interleaving distance between the associated persistence modules and on the bottleneck distance between the associated persistence diagrams, which we use in our algorithm to identify normal cycles. So our algorithm should find cycles that approximate the true normal set $X_{\mathbf{n}}$. We start by making additional assumptions and defining a probability measure $\mu_{X_{\mathbf{n}}}$ on $X_{\mathbf{n}}$, and use the DTM filtration $V[X_{\mathbf{n}}, d_{\mu_{X_{\mathbf{n}}},m}]$ as our filtration of $X_{\mathbf{n}}$ to state our main result.

6.4.2.1 Assumption of the theorem and probability measure on curves

In this section, we add some assumption on our model from Section 6.2 and define a probability measure on $\mathbb{R}^d$ whose support is the curve $X = X_{d,\tau}$ (or $X_{\mathbf{n}}$) that is adapted to the anomaly detection problem, i.e. such that sections corresponding to anomalies have a low probability.

We first assume that the delay τ is small enough so that the curve always returns to 0 between two atoms, and at the beginning and end.

Assumption 4. *Let $(\boldsymbol{\zeta}_i)_{1\leq i\leq M_{\mathbf{n}}+M_{\mathbf{a}}}$ denote all the activations. We assume that:*

1. $\forall i, \forall t \in [0,1], \quad \boldsymbol{\zeta}_i[t] \neq 0 \Rightarrow \forall j \neq i, \forall s \in [t, t + l_{\boldsymbol{\zeta}_i} + (d-1)\tau], \quad \boldsymbol{\zeta}_j[t] = 0.$
2. $\boldsymbol{\zeta}_i[j] = 0$ *for all* i *and for all* $j \leq (d-1)\tau$ *and* $j \geq 1 - (d-1)\tau$.

Note that a small enough τ exists because of Assumption 1.

Remark 11. *This assumption ensures that each occurrence of an atom gives (at least) one loop in the delay embedding. It also ensures that each occurrence is complete, which is not mandatory but will make it simpler to compute the probability of each atom.*

Let us now define the probability measure μ_X. Here, $X^{-1}(R)$ denotes the pre-image of R under the function:

$$\begin{cases} [0, 1-(d-1)\tau] & \to \mathbb{R}^d \\ t & \mapsto X[t] = (\mathbf{x}[t], \mathbf{x}[t+\tau], \ldots, \mathbf{x}[t+(d-1)\tau]). \end{cases}$$

Definition 27. *Let λ denote the Lebesgue measure on $\mathbb{R}$. μ_X is the Borel measure on $\mathbb{R}^d$ such that for any d-rectangle $R = I_1 \times \cdots \times I_d$ where the I_i are open or closed intervals:*

$$\mu_X(R) = \frac{\lambda(X^{-1}(R))}{1-(d-1)\tau}.$$

It is clear that μ_X is a probability measure with support X.

Assumption 5. *let N_i (resp. A_i) be the delay embedding of $\mathbf{n}_i * \boldsymbol{\eta}_i$ (resp. $\mathbf{a}_i * \boldsymbol{\alpha}_i$). We assume that the intersection of any pair of sets among the $N_i \setminus \{0\}$ and $A_i \setminus \{0\}$ has null measure.*

In this case, for any i, $\mu_X(N_i \setminus \{0\}) = k_{\mathbf{n}_i} l_{\mathbf{n}_i}$ and $\mu_X(A_i \setminus \{0\}) = k_{\mathbf{a_i}} l_{\mathbf{a}_i}$.

Remark 12.
- *Assumption 5 makes sense in the context of anomaly detection as it implies that except for a set of null measure, a point $\mathbf{x}_t$ is either normal or abnormal (i.e. normal and abnormal atoms cannot coincide on an interval).*
- *If the atom lengths are equal, then the probability of each atom is proportional to its number of occurrences.*
- *If the signal is constant on an interval, there will be a point with non zero probability in X. It is the case for 0, which we consider to be part of the normal behavior.*
- *Here, it can already be noticed that the algorithm could fail if an abnormal atom was too long.*

The measure $\mu_{X_{\mathbf{n}}}$ is defined the same way starting from $X_{\mathbf{n}}$ (which is X without the anomalies).

6.4.2.2 Theorem statement

The aim of the section is to prove the following result:

Theorem 4. *With the above notations and assumptions, if $\mathbf{x_n}$ is continuously differentiable, and with $p = 1$ for DTM filtrations, we have:*

$$\begin{aligned} d_i(V[\hat{Y}, d_{\mu_{\hat{Y}},m}], V[X_{\mathbf{n}}, d_{\mu_{X_{\mathbf{n}}},m}]) \leq (b_n + ||\boldsymbol{\varepsilon}||_\infty)\sqrt{d}\left(m^{-\frac{1}{2}} + 2\right) \\ + m^{-\frac{1}{2}}\sqrt{d}||\mathbf{x_a}||_\infty \sqrt{\sum_{i=1}^{M_\mathbf{a}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}} \\ + \sqrt{\frac{d}{3}} \cdot \frac{m||\mathbf{x'_n}||_\infty}{\min\limits_i(k_{\mathbf{n}_i})} \end{aligned}$$

with $b_n \underset{n\to+\infty}{\longrightarrow} 0$.

Notice that the sampling $\sum_{i=1}^{M_\mathbf{a}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}$ is the proportion or the signal that is abnormal, so it can be assumed to be small in the context of anomaly detection. So with a high sampling frequency n, a low noise amplitude $||\boldsymbol{\varepsilon}||_\infty$, and if normal atoms occur many times, the bound should be small (after choosing m not too close to zero).

We divide the proof of this theorem in three steps:

- **Sampling step:** Proposition 7 bounds $d_i(V[\hat{X}, d_{\mu_{\hat{X}},m}, p], V[X, d_{\mu_X,m}, p])$.
- **Noise step:** Proposition 8 bounds $d_i(V[\hat{Y}, d_{\mu_{\hat{Y}},m}, p], V[\hat{X}, d_{\mu_{\hat{X}},m}, p])$.
- **Anomaly step:** Proposition 9 bounds $d_i(V[X, d_{\mu_X,m}], V[X_{\mathbf{n}}, d_{\mu_{X_{\mathbf{n}}},m}])$.

We then conclude using the triangular inequality.

Remark 13. *We now make an additional remark regarding the optional subsampling step in our algorithm. Let $\hat{Y}$ be any finite point cloud in $\mathbb{R}^d$. This steps consists in computing the filtration on $\tilde{Y}$, which is a subset of $\hat{Y}$, but we still use the DTM function $d_{\mu_{\hat{Y}},m}$ so Proposition 5 gives:*

$$d_i(V[\hat{Y}, d_{\mu_{\hat{Y}},m}], V[\tilde{Y}, d_{\mu_{\hat{Y}},m}]) \leq 2d_H(\hat{Y}, \tilde{Y}).$$

In particular, one can take only one point for each occurrence of an atom while still using the information from the number of occurrences through $d_{\mu_{\hat{Y}},m}$ (which would not be the case we used $d_{\mu_{\tilde{Y}},m}$, leading to an important change in the filtration values that would make it hard to discriminate anomalies from normal atoms). This is very useful in practice as computing persistent homology on large datasets can be long.

Remark 14. *Finally, we give a link between this section and Section 6.3.1. If $m = \frac{q}{n-(d-1)\hat{\tau}}$ with $\max(k_{\mathbf{a_i}}) < q \leq \min(k_{\mathbf{n_i}})$, then for all normal point $y \in Y$, $d_{\mu_{\hat{X}},m}(y) \leq \sqrt{d}(b_n + ||\boldsymbol{\varepsilon}||_\infty)$. So all normal points are dense, and for each abnormal point, at least $q - \max(k_{\mathbf{a_i}})$ neighbors which are considered to compute their DTM do not correspond to the same point from a distinct occurrence. This does not guarantee that abnormal point have a lower density as a single point can appear several times in one occurrence, or the variations can be very slow. So our algorithm can fail to detect an anomaly (even with a small $k_{\mathbf{a}_i}$) if its variations are slow but still forms a large enough 1-cycle to be considered significant. This could especially happen with a large atom length $l_{\mathbf{a}_i}$, which reflects in our model as we approximate a measure μ_X such that $\mu_X(A_i \setminus 0) = k_{\mathbf{a}_i} l_{\mathbf{a}_i}$.*

6.4.2.3 Sampling step

The next result states that if the signal is well-sampled (i.e. if n is large) then there should be little difference between studying a continuous signal $\mathbf{x}$ or a sampled version $\hat{\mathbf{x}}$. Recall that $\tau = \frac{\hat{\tau}}{n-1}$. For simplicity, we assume that $\frac{1}{\tau} \in \mathbb{N}$, so $\mathsf{Card}(\hat{X}) = \frac{1-(d-1)\tau}{\tau}$ (points can be counted multiple times).

Proposition 7 (Stability for sampling)**.** *Let* $b_n = \sup_{|t-s| \leq \frac{1}{n-1}} (|\mathbf{x}[t] - \mathbf{x}[s]|)$*. With the above notations and assumptions we have:*

$$d_i(V[\hat{X}, d_{\mu_{\hat{X}},m}, p], V[X, d_{\mu_X,m}, p]) \leq b_n \sqrt{d} \left(m^{-\frac{1}{2}} + 2^{\frac{1}{p}} \right).$$

Proof. From Proposition 5 we get:

$$d_i(V[\hat{X}, d_{\mu_{\hat{X}},m}, p], V[X, d_{\mu_X,m}, p]) \leq m^{-\frac{1}{2}} W_2(\mu_{\hat{X}}, \mu_X) + 2^{\frac{1}{p}} d_H(\hat{X}, X).$$

It is immediate from the definitions that $d_H(\hat{X}, X) \leq b_n \sqrt{d}$.

Let us now derive an upper bound for $W_2(\mu_{\hat{X}}, \mu_X)$. We refer to [San15, V$^+$09] for and introduction to optimal transport. Let $\lfloor . \rfloor$ denote the floor function, we define T as :

$$T : \begin{cases} X & \to \hat{X} \\ X_t & \mapsto \hat{X}\left[\lfloor (n-1)t \rfloor + 1\right] \end{cases}$$

so $T(X[t]) = \hat{X}[\lfloor (n-1)t \rfloor + 1] = X\left[\frac{\lfloor (n-1)t \rfloor}{n-1}\right]$ with $\left| t - \frac{\lfloor (n-1)t \rfloor}{n-1} \right| < \frac{1}{n-1}$.

We have for any measurable bounded function f:

$$\begin{aligned} \int_{\hat{X}} f(y) d\mu_{\hat{X}}(y) &= \sum_{y \in \hat{X}} f(y) \mu_{\hat{X}}(\{y\}) \\ &= \frac{1}{\mathsf{Card}(\hat{X})} \sum_{y \in \hat{X}} f(y) \\ &= \frac{\tau}{1-(d-1)\tau} \sum_{y \in \hat{X}} f(y) \end{aligned}$$

and

$$\begin{aligned} \int_X f(T(x)) d\mu_X(x) &= \sum_{y \in \hat{X}} \int_{\{T^{-1}(y)\}} f(y) d\mu_X(x) \\ &= \sum_{y \in \hat{X}} f(y) \mu_X(\{T^{-1}(y)\}) \\ &= \sum_{y \in \hat{X}} f(y) \frac{\tau}{1-(d-1)\tau}. \end{aligned}$$

So $\mu_{\hat{X}} = \mu_X \circ T^{-1}$ and $\pi = (\mu_X, \mu_X \circ T^{-1})$ is a (deterministic) transport plan and thus:

$$\begin{aligned} W_2(\mu_{\hat{X}}, \mu_X) &\leq \sqrt{\int_{\mathbb{R}^d \times \mathbb{R}^d} ||u-v||^2 d\pi(u,v)} \\ &\leq \sqrt{\int_{\mathbb{R}^d} ||u - T(u)||^2 d\mu_X(u)} \\ &\leq b_n \sqrt{d}. \end{aligned}$$

□

6.4.2.4 Noise step

The next result states that the construction is robust to noise.

Proposition 8 (Stability in the presence of noise)**.** *With the above notations and assumptions we have:*

$$d_i(V[\hat{Y}, d_{\mu_{\hat{Y}},m}, p], V[\hat{X}, d_{\mu_{\hat{X}},m}, p]) \leq ||\varepsilon||_\infty \sqrt{d} \left(m^{-\frac{1}{2}} + 2^{\frac{1}{p}}\right).$$

Proof. As for the previous proof, we use Proposition 5 to derive a Hausdorff distance term and a Wasserstein distance term. The fact that for all t, $||\widehat{(X+\mathcal{E})}_t - \hat{X}_t|| \leq ||\varepsilon||_\infty \sqrt{d}$ immediately implies that $d_H(\hat{X}, \hat{Y}) \leq ||\varepsilon||_\infty \sqrt{d}$. Let T be the function from $\hat{X}$ to $\hat{Y}$ defined by $T(\hat{X}[i]) = \hat{Y}[i]$ so that $\mu_{\hat{Y}} = \mu_{\hat{X}} \circ T^{-1}$ and $\pi = (\mu_{\hat{X}}, \mu_{\hat{X}} \circ T^{-1})$ is a (deterministic) transport plan and thus:

$$\begin{aligned} W_2(\mu_{\hat{X}}, \mu_{\hat{Y}}) &\leq \sqrt{\int_{\mathbb{R}^d \times \mathbb{R}^d} ||u-v||^2 d\pi(u,v)} \\ &= \frac{1}{\mathsf{Card}(\hat{X})} \sqrt{\sum_{x \in \hat{X}} ||x - T(x)||^2} \\ &\leq ||\varepsilon||_\infty \sqrt{d}. \end{aligned}$$

□

6.4.2.5 Anomaly step

We now fix $p = 1$ and use Proposition 6 to bound the change in interleaving distance between the filtrations obtained respectively with and without anomalies (and the associated DTMs). We note $V[X, d_{\mu_X,m}]$ instead of $V[X, d_{\mu_X,m}, 1]$.

Proposition 9 (Stability in the presence of anomalies)**.** *With the above notations and assumptions, if $\mathbf{x_n}$ is continuously differentiable:*

$$d_i(V[X_\mathbf{n}, d_{\mu_{X_\mathbf{n}},m}], V[X, d_{\mu_X,m}]) \leq m^{-\frac{1}{2}} \sqrt{d} ||\mathbf{x_a}||_\infty \sqrt{\sum_{i=1}^{M_\mathbf{a}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}} + \sqrt{\frac{d}{3}} \cdot \frac{m||\mathbf{x'_n}||_\infty}{\min\limits_i (k_{\mathbf{n}_i})}.$$

Proof. From Propositions 5 (which still applies in this case where $\mathsf{supp}(\mu_{X_\mathbf{n}}) = X_\mathbf{n} \subset X$ with X compact) and 6 we have:

$$\begin{aligned} d_i(V[X, d_{\mu_X,m}], V[X_\mathbf{n}, d_{\mu_{X_\mathbf{n}},m}]) &\leq d_i(V[X, d_{\mu_X,m}], V[X, d_{\mu_{X_\mathbf{n}},m}]) \\ &\quad + d_i(V[X, d_{\mu_{X_\mathbf{n}},m}], V[X, d_{\mu_{X_\mathbf{n}},m}]) \\ &\leq m^{-\frac{1}{2}} W_2(\mu_X, \mu_{X_\mathbf{n}}) + c(\mu_{X_\mathbf{n}}, m). \end{aligned}$$

Let us first give an upper bound on $W_2(\mu_X, \mu_{X_\mathbf{n}})$. Let $T : X \to X_\mathbf{n}$ be equal to the identity function on $X_\mathbf{n}$ and to the null function on $X \setminus X_\mathbf{n}$. Recall that $X = X_\mathbf{n} \cup X_\mathbf{a}$ and that by assumption $\mu_X(X_\mathbf{n} \cap (X_\mathbf{a} \setminus 0)) = 0$. Also notice that $\mu_X = \mu_{X_\mathbf{n}}$ on $X_\mathbf{n} \setminus 0$.

For any measurable bounded function f we have:

$$\begin{aligned}
\int_X f(T(x))d\mu_X(x) &= \int_{X_{\mathbf{n}}} f(T(x))d\mu_X(x) + \int_{X_{\mathbf{a}}\setminus 0} f(T(x))d\mu_X(x) \\
&= \int_{X_{\mathbf{n}}\setminus 0} f(x)d\mu_X(x) + f(0)\mu_X(0) + \int_{X_{\mathbf{a}}\setminus 0} f(0)d\mu_X(x) \\
&= \int_{X_{\mathbf{n}}\setminus 0} f(x)d\mu_{X_{\mathbf{n}}}(x) + f(0)\mu_X(0) + f(0)\mu_X(X_{\mathbf{a}}\setminus 0) \\
&= \int_{X_{\mathbf{n}}\setminus 0} f(x)d\mu_{X_{\mathbf{n}}}(x) + f(0)\left(1 - \sum_{i=1}^{M_{\mathbf{n}}} k_{\mathbf{n}_i} l_{\mathbf{n}_i} - \sum_{i=1}^{M_{\mathbf{a}}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}\right) \\
&\quad + f(0)\left(\sum_{i=1}^{M_{\mathbf{a}}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}\right) \\
&= \int_{X_{\mathbf{n}}\setminus 0} f(x)d\mu_{X_{\mathbf{n}}}(x) + f(0)\mu_{X_{\mathbf{n}}}(0) \\
&= \int_{X_{\mathbf{n}}} f(x)d\mu_{X_{\mathbf{n}}}(x).
\end{aligned}$$

The fourth equality comes from the fact that $\sum_{i=1}^{M_{\mathbf{n}}} k_{\mathbf{n}_i} l_{\mathbf{n}_i}$ and $\sum_{i=1}^{M_{\mathbf{a}}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}$ are respectively the amount of time taken by normal and abnormal atoms (the rest of the time is spent at 0), so $\mu_X(0) = 1 - \sum_{i=1}^{M_{\mathbf{n}}} k_{\mathbf{n}_i} l_{\mathbf{n}_i} - \sum_{i=1}^{M_{\mathbf{a}}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}$ and $\mu_X(X_{\mathbf{a}} \setminus 0) = \sum_{i=1}^{M_{\mathbf{a}}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}$. The fifth equality comes from the fact that when we use $\mu_{X_{\mathbf{n}}}$, we replace all the abnormal atoms with 0, so $\mu_{X_{\mathbf{n}}}(0) = 1 - \sum_{i=1}^{M_{\mathbf{n}}} k_{\mathbf{n}_i} l_{\mathbf{n}_i}$.

The above calculation implies that $\mu_{X_{\mathbf{n}}} = \mu_X \circ T^{-1}$ and $\pi = (\mu_X, \mu_X \circ T^{-1})$ is a (deterministic) transport plan and thus:

$$\begin{aligned}
W_2(\mu_X, \mu_{X_{\mathbf{n}}}) &\leq \sqrt{\int_{\mathbb{R}^d \times \mathbb{R}^d} ||u - v||^2 d\pi(u, v)} \\
&\leq \sqrt{\int_{\mathbb{R}^d} ||u - T(u)||^2 d\mu_X(u)} \\
&\leq \sqrt{\int_{X_{\mathbf{a}}} ||u||^2 d\mu_X(u)} \\
&\leq \sqrt{d||\mathbf{x}_{\mathbf{a}}||_\infty^2 \mu_X(X_{\mathbf{a}})} \\
&\leq \sqrt{d}||\mathbf{x}_{\mathbf{a}}||_\infty \sqrt{\sum_{i=1}^{M_{\mathbf{a}}} k_{\mathbf{a}_i} l_{\mathbf{a}_i}}
\end{aligned}$$

where the fourth inequality comes from the fact that for all $u = (u_1, \ldots, u_d) \in X_{\mathbf{a}}$ we have: $\forall i,\ u_i \leq ||\mathbf{x}_{\mathbf{a}}||_\infty$.

Let us now compute an upper bound on $c(\mu_{X_{\mathbf{n}}}, m)$. Let $t, s \in [0, 1 - (d-1)\tau]$. As $\mathbf{x}_{\mathbf{n}}$ is continuously differentiable we have:

$$||X_{\mathbf{n}}[t] - X_{\mathbf{n}}[s]||^2 = \sum_{i=0}^{d-1} |\mathbf{x}_{\mathbf{n}}[t + i\tau] - \mathbf{x}_{\mathbf{n}}[s + i\tau]|^2 \leq d|t - s|^2 \cdot ||\mathbf{x}'_{\mathbf{n}}||_\infty^2.$$

Let $u \in]0, 1[$. We will compute an upper bound on $\delta_{\mu_{X_{\mathbf{n}}}, u}$ (see Definition 14 for the definition). For all $r \geq 0$ we have:

$$|t - s| \leq \frac{r}{\sqrt{d}||\mathbf{x}'_{\mathbf{n}}||_\infty} \Rightarrow ||X_{\mathbf{n}}[t] - X_{\mathbf{n}}[s]|| \leq r$$

so, as there are at least $\min_i(k_{\mathbf{n}_i})$ occurrences of each atom in $\mathbf{x_n}$:

$$\mu_{X_\mathbf{n}}(\bar{B}(X_\mathbf{n}[t], r)) \geq \frac{r \cdot \min_i(k_{\mathbf{n}_i})}{\sqrt{d}||\mathbf{x'_n}||_\infty}$$

and thus:

$$r > \frac{u\sqrt{d}||\mathbf{x'_n}||_\infty}{r \cdot \min_i(k_{\mathbf{n}_i})} \quad \Rightarrow \quad \mu_{X_\mathbf{n}}(\bar{B}(X_\mathbf{n}[t], r)) > u.$$

So we have $\delta_{\mu_{X_\mathbf{n}},u}(X_\mathbf{n}[t]) \leq \frac{u\sqrt{d}||\mathbf{x'_n}||_\infty}{\min_i(k_{\mathbf{n}_i})}$ and:

$$\begin{aligned} d_{\mu_{X_\mathbf{n}},m}(X_\mathbf{n}[t]) &= \sqrt{\frac{1}{m}\int_0^m \delta^2_{\mu_{X_\mathbf{n}},u}(X_\mathbf{n}[t])du} \\ &\leq \sqrt{\frac{d}{3}} \cdot \frac{m||\mathbf{x'_n}||_\infty}{\min_i(k_{\mathbf{n}_i})} \,. \end{aligned}$$

□

Note that Proposition 6 enabled us to replace $2d_H(X, X_\mathbf{n})$ with $c(\mu_{X_\mathbf{n}}, m)$ in the upper bound (compared to using only Proposition 5). The latter can be equal to $\sqrt{d}||\mathbf{x}||_\infty$ in some cases, whereas the former has a factor $\frac{1}{\min_i(k_{\mathbf{n}_i})}$ that is assumed to be small in our anomaly detection problem.

6.5 Experimental setup

This section describes the datasets and anomaly detection methods we will use to test our algorithm in Section 6.6.

6.5.1 Synthetic dataset

We generated 120 time series following the model described in Section 6.2. Each atom is generated specifying a random length, choosing a random integer N, computing N points on a Gaussian random walk starting at zero, and interpolating them with cubic splines to get an atom of the desired length, which is then multiplied by a random amplitude factor between 0.5 and 2. Atoms length is set to 1 for punctual anomalies and can take values between 150 and 400 for normal atoms and sequential anomalies. N is set to 1 for punctual anomalies and can take values between 5 and 15 for normal atoms and sequential anomalies. For each time series, each normal atom has at most 15 occurrences, and at least 7 more occurrences than the most frequent anomaly: at least 8 when anomalies do not repeat, and 10 when they can have several occurrences (up to 3). Time series have lengths between 1964 and 13234. [1]

The dataset is divided in six equal parts, depending on the anomaly type of the time series (punctual or sequential), on the presence or absence of multiple normal atoms and on the possibility for anomalies to have several occurrences (two or three). Table 6.2 shows the characteristics of each part.

[1] The dataset can be found at `https://github.com/Alex-B-4/A-persistent-homology-based-algorithm-for-unsupervised-anomaly-detection-in-time-series`.

Indices	Anomaly type	Several normal atoms	Repeating anomalies
$1-20$	Punctual	x	x
$21-40$	Sequential	x	x
$41-60$	Punctual	✓	x
$61-80$	Sequential	✓	x
$81-100$	Sequential	✓	✓
$101-120$	Both	✓	✓

Table 6.2: Characteristics of each part of the synthetic datasets.

Name	Size	Average lengths	Data type
Dodgers	1	50400.0	Freeway traffic
ECG	53	230351.9	Electrocardiogram
IOPS	58	102119.2	Performance indicators for machines
MGAB	10	100000.0	Differential equations (used for biological models)
NAB	58	6301.7	Server metrics and internet traffic
NASA-MSL	27	2730.7	Spacecraft telemetry
NASA-SMAP	54	8066.0	Spacecraft telemetry
Sensorscope	23	27038.4	Environmental data
YAHOO	367	1561.2	Metrics on Yahoo services
KDD21	250	77415.06	Composite dataset
Daphnet	45	21760.0	Acceleration sensors on Parkison's disease patients
GHL	126	200001.0	Metrics on Gasoil reservoirs
Genesis	6	16220.0	Industrial time series
MITDB	32	650000.0	Electrocardiogram
OPP	465	31616.9	Motion sensors on humans
Occupancy	10	5725.8	Environmental data
SMD	281	25562.3	Server metrics
SVDB	115	230400.0	Electrocardiogram

Table 6.3: Characteristics of the real-world datasets from TSB-UAD.

6.5.2 Real-world datasets and methods from TSB-UAD

Our real-world datasets are the 18 public datasets provided by the TSB-UAD benchmark suite [PKB+22], for a total of 1980 univariate time series with labeled anomalies. Table 6.3 shows the names, sizes (i.e. number of time series), average lengths and data types of 18 datasets.

For each time series, we compute the area under the ROC curve (AUC-ROC) obtained by looking at all the possible thresholds on the anomaly score. The same method is applied to 12 anomaly detection algorithms [LTZ08, BKNS00, GD12, YZU+18, BLR+21, AA17, SY14, MVS+15, LMM07, MSDA18, SWS+99] in [PKB+22]. Table 6.4 provides a succinct description of the methods. Using the AUC-ROC, the evaluation does not depend on the choice of a threshold for each algorithm. The datasets and the implementations of the 12 methods can be found at `https://github.com/TheDatumOrg/TSB-UAD/tree/main`.

Method	Unsupervised	Delay embedding	Description
TDA	✓	✓	Our method: TDA to find normal 1-cycles
IForest1 [LTZ08]	✓	x	Random forest on time series to find outliers
IForest [LTZ08]	✓	✓	Random forest on delay embedding to find outliers
LOF [BKNS00]	✓	✓	Outlier score based on local density
HBOS [GD12]	✓	✓	Outlier score based on histograms
MP [YZU+18]	✓	x	Distance between closest subsequence as anomaly score
NORMA [BLR+21]	✓	x	Clustering to find normal subsequences
PCA [AA17]	✓	✓	PCA to define a normal hyperplane
AE [SY14]	x	x	Reconstruction error as anomaly score, based on deep learning
LSTM [MVS+15]	x	x	Prediction error as anomaly score, based on deep learning
POLY [LMM07]	✓	x	Prediction error as anomaly score, based on polynomial interpolation
CNN [MSDA18]	x	x	Prediction error as anomaly score, based on deep learning
OCSVM [SWS+99]	x	✓	SVM to find normal points on delay embedding

Table 6.4: Overview of the anomaly detection methods used in our benchmark. The "Delay embedding" column indicates if the methods uses a delay embedding or not.

6.6 Results

In this section, we start by showing the results our the benchmark on real-world datasets. Then, we use the synthetic dataset to study the influence of our five parameters and of noise on the performance and computation time of our algorithm.

6.6.1 Benchmark on real-world data

Here, we show the results of our algorithm on the TSB-UAD benchmark described in Section 6.5.2. The parameters of our algorithm are chosen the same way for all time series. We estimate a period L for the time series using the first maximum of the autocorrelation function (we used the find_length function from TSB-UAD, which was used for all other methods using a delay embedding). We empirically chose $\hat{\tau} = 6$ (in practice, if $\hat{\tau}$ is too small, the embedding will stay close to the line spanned by $(1, \dots, 1)$ and it will be harder to detect cycles). In [PH15] Perea and Harer show that in the case of trigonometric functions, $d\hat{\tau}$ should be a multiple of the period to maximize persistence in 1D homology. With this in mind, we empirically set $d = \max(40, \min(120, \lfloor \frac{L}{3} \rfloor))$. We chose $q = \lfloor \frac{n}{L} \rfloor$ following the intuition from Section 6.3.1, as this value would approximate the number of normal occurrences in the case where there is one normal atom. Finally, we use $n_{diag} = 2$ and $n_{points} = 400$.

The algorithm was implemented in Python. The DTM filtration was computed with the GUHDI library [MBGY14]. The cycle extraction algorithm was implemented with dionysus[2](more specifically, with a modified version of cyclonysus[3]). The code is publicly available [4] and can be used in an interactive demo [5] on the IPOL website (IPOL is a reproducible research journal). Results for the 12 other methods come from [PKB+22].

Table 6.5 shows the average AUC-ROC obtained on each dataset with our method (TDA) with the above parameters, and the results of the TSB-UAD benchmark [PKB+22]. Figure 6.6 shows the critical diagrams comparing the average rank of each method using the Friedman test followed by the Wilcoxon or Nemenyi test with $\alpha = 0.05$, as described in [Dem06].

[2] https://pypi.org/project/dionysus/

[3] https://github.com/sauln/cyclonysus

[4] https://github.com/Alex-B-4/A-persistent-homology-based-algorithm-for-unsupervised-anomaly-detection-in-time-series

[5] https://ipolcore.ipol.im/demo/clientApp/demo.html?id=77777000518&key=8B0F4F40910286727D4476F44703D7DB

	TDA	IForest	IForest1	LOF	MP	PCA	NORMA	HBOS	POLY	OCSVM	AE	CNN	LSTM
Dodgers	**0.79**	**0.79**	0.64	0.54	0.52	0.77	**0.79**	0.3	0.69	0.64	0.73	0.68	0.39
ECG	0.88	0.75	0.61	0.56	0.58	0.71	**0.95**	0.68	0.70	0.64	0.73	0.52	0.54
IOPS	**0.82**	0.54	**0.78**	0.50	0.72	0.74	0.76	0.64	0.68	0.71	0.63	0.61	0.61
MGAB	0.58	0.57	0.58	**0.96**	0.91	0.54	0.55	0.54	0.51	0.52	0.71	0.58	0.56
NAB	**0.76**	0.45	0.56	0.48	0.49	0.69	0.58	0.68	0.75	0.61	0.54	0.52	0.50
NASA-MSL	0.64	0.57	0.69	0.52	0.52	0.75	0.55	0.77	**0.81**	0.64	0.70	0.57	0.57
NASA-SMAP	**0.83**	0.72	0.68	0.68	0.62	0.74	0.80	0.77	0.80	0.65	0.77	0.68	0.64
SensorScope	0.52	0.56	0.56	0.55	0.50	0.54	0.59	0.56	**0.62**	0.51	0.52	0.52	0.53
YAHOO	0.64	0.62	0.81	0.86	0.86	0.57	0.92	0.57	0.76	0.50	0.79	**0.96**	0.94
KDD21	0.75	0.65	0.57	0.78	**0.90**	0.58	0.88	0.60	0.58	0.60	0.79	0.74	0.66
Daphnet	0.70	0.74	0.68	**0.78**	0.44	0.69	0.46	0.69	0.77	0.45	0.44	0.47	0.44
GHL	0.86	**0.94**	**0.94**	0.54	0.42	0.91	0.64	0.92	0.76	0.45	0.63	0.47	0.47
Genesis	0.84	0.78	0.66	0.68	0.35	0.85	0.6	0.59	**0.87**	0.70	0.72	0.73	0.53
MITDB	0.71	0.70	0.61	0.61	0.69	0.67	**0.86**	0.70	0.68	0.65	0.80	0.58	0.51
OPP	0.48	0.49	0.52	0.45	**0.82**	0.52	0.65	0.54	0.28	0.38	0.70	0.47	0.57
Occupancy	0.53	0.86	0.78	0.53	0.32	0.78	0.53	**0.89**	0.80	0.66	0.69	0.79	0.71
SMD	0.77	0.85	0.73	0.69	0.51	0.80	0.61	0.77	**0.87**	0.61	0.63	0.61	0.58
SVDB	0.77	0.72	0.58	0.59	0.74	0.68	**0.92**	0.71	0.67	0.68	0.79	0.58	0.55

Table 6.5: Average AUC-ROC on each dataset. Results for methods other than TDA come from [PKB+22].

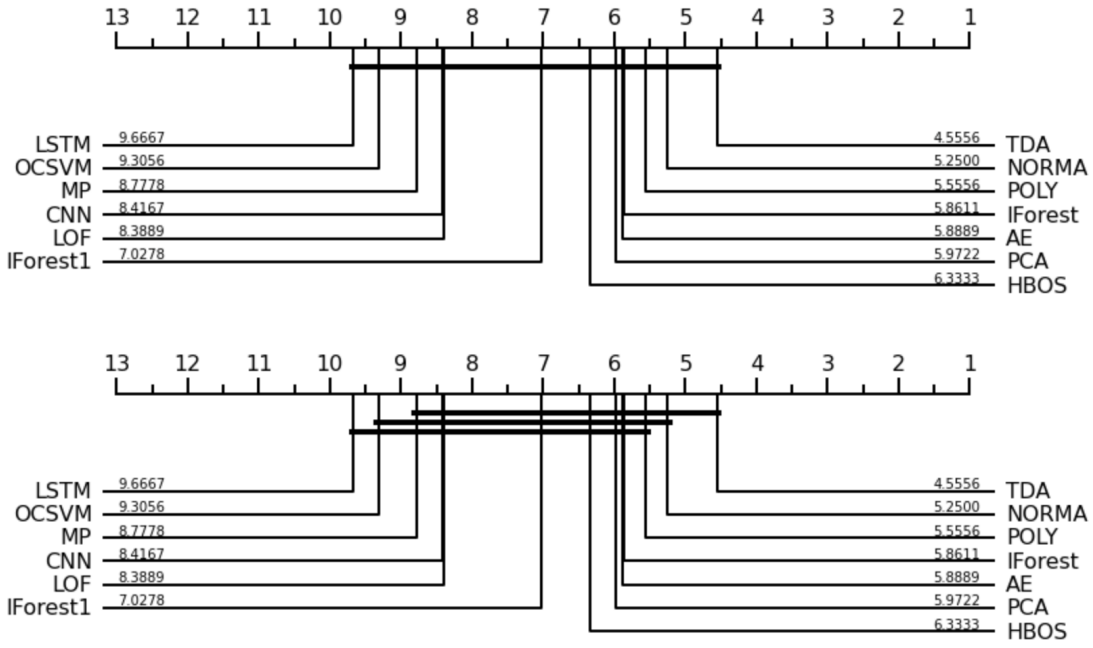


Figure 6.6: Critical diagrams for $\alpha = 0.05$. **Top:** Friedman test + Wilcoxon test. **Bottom:** Friedman test + Nemenyi test.

These results show that our method is competitive with the state-of-the-art in anomaly detection on 18 standard datasets. It has the best score on 4 of them, the best average rank (though the difference with the best methods is not significant as shown on Figure 6.6), and it is in the top 5 on 13 datasets. We found three reasons to explain what can make our algorithm perform well or not. First, our algorithm was designed for the problem described in Section 6.2, which makes it appropriate for structured data with repetitive patterns but datasets can differ from our model in several ways: the absence of a clear normal behavior, a trend, or a lack of continuity (for example binary time series) can lead to a delay embedding with no relevant normal cycles to detect. Anomalies have a nature that our algorithm cannot detect, such as a longer pause between two atoms. The second

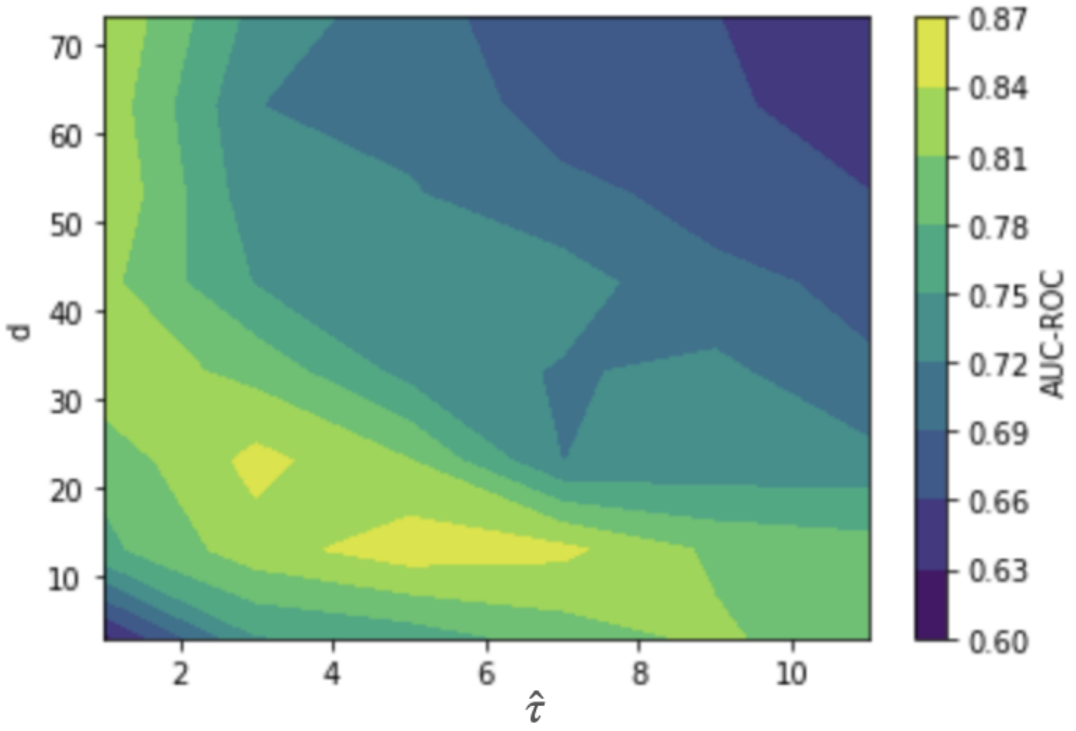


Figure 6.7: Heat map representing the average AUC-ROC on the whole dataset with different values of d and $\hat{\tau}$.

reason is the choice of parameters. Even though our heuristics to choose the parameters seem relevant (see Sections 6.6.2 and 6.6.3), it is still not optimal to have the same rule to choose parameters for datasets with different types of data, lengths, sampling rates... The delay embedding parameters d and $\hat{\tau}$ should not be too low or else the curve will not have enough space to form persistent cycles, but if they are too high points of the delay embedding will represent different atoms and the structure will be higher to detect. Moreover, if $d\hat{\tau}$ is significantly larger than the length of an anomaly, there will be false positive because the algorithm will detect too many indices as abnormal. We believe that q and n_{diag} have less influence on performance (see Sections 6.6.2 and 6.6.3). The third reason is the quality of the subsampling $\tilde{Y}$. If the time series is short, or if it has many repetitive patterns, then 400 points can be enough to represent the whole point cloud, but many time series have more than 100000 points so if they have numerous and/or long atoms then it is not enough and the distance between points will be too low for cycles to persist (this is studied in more details in Section 6.6.4.

6.6.2 Influence of the embedding dimension and delay

Here, we study the influence of the delay embedding parameters d and $\hat{\tau}$ by looking at the performance of our algorithm on our synthetic dataset with different pairs of parameters $(d, \hat{\tau})$, while the other parameters are fixed. Figure 6.7 is a heat map that shows the average AUC-ROC on the whole dataset when d varies between 3 and 70 and $\hat{\tau}$ varies between 1 and 11. We fixed $n_{points} = 100$ and $n_{diag} = 2$. We set $q = \lfloor \frac{n}{L} \rfloor$, where L is found using the autocorrelation function, as in Section 6.5.

On Figure 6.7, there is one zone where the AUC-ROC is above 0.81 and the AUC-ROC decreases as parameters go away from this zone (below or above), going to the $0.78 - 0.81$

zone then to the $0.75 - 0.78$ zone... until reaching 0.6 at the top right and bottom left corners. The shape of the zones $AUC \geq 0.81$, $AUC \geq 0.78$, and $AUC \geq 0.75$ and of their frontiers suggest that the best scores are obtained when the product $d\hat{\tau}$ is constant. This product represent the time window represented by each point of the delay embedding. The window should be large enough to capture complex patterns but the window should not cover several atoms. In our dataset, atom lengths can have random values between 50 and 400. Figure 6.7 suggests that $d\hat{\tau}$ should be approximately between 40 and 80. This empirically confirms the intuition from Section 6.5 and [PH15] that the product $d\hat{\tau}$ should be of the order of the atom lengths. Moreover, an AUC-ROC above 0.84 is never reached with d above 30 or below 10, nor with $\hat{\tau}$ above 8 or below 2. This indicates that each parameter should not take extreme values. Indeed, for a given value of the product $d\hat{\tau}$, if $\hat{\tau}$ is too small then the delay embedding will be concentrated around the line $x_1 = x_2 = \cdots = x_d$, and if it is too high then points on the embedding will not represent the local variations that constitue the shape of the atoms.

6.6.3 Influence of the number of neighbors and points on the diagram

Here, we study the parameters q, which has an influence on the filtration values and thus on the persistence diagram, and n_{diag}, which influences the way we read the persistence diagram. We look at the performance of our algorithm on our synthetic dataset with different pairs of parameters (q, n_{diag}), while the other parameters are fixed. Figure 6.8 shows the average AUC-ROC on the whole dataset when q varies between 1 and 140 and n_{diag} varies between 2 and 16 (as explained in 6.3.2.2, we always set $n_{diag} \geq 2$). We fixed $n_{points} = 100$, $d = 15$ and $\hat{\tau} = 5$.

On Figure 6.8, we can see zones of constant AUC-ROC separated by almost horizontal lines, except for $2 \leq n_{diag} \leq 6$ where the zone $AUC = 0.92$ gets thinner as n_{diag} decreases. Moreover, the AUC-ROC significantly decreases when q gets lower than 20. This suggests that the algorithm is robust to changes of q and n_{diag} except for very low values of q (the proposed heuristic makes q proportional to n to avoid this situation), and that setting n_{diag} a little higher than 2 can make it easier to reach optimal performance, but considering more cycles can make it harder to find the right birth date threshold.

6.6.4 Computation time and influence of the subsampling parameter

A nice property of our method is the possibility of computing persistent homology on a subset of the point cloud while keeping density information about the whole point cloud in the filtration. This makes the complexity of the algorithm go from $O(n^3)$ [ELZ02] to $O(n^3_{points} + n^2)$, as persistent homology is computed on a point cloud of size n_{points}, but the distance matrix of the full points cloud must be computed. Figure 6.9 shows the computation time when n_{points} varies from 50 to 600 (on average on the 20 first time series of the dataset), on a normal and logarithmic scale (to empirically confirm the theoretical complexity), along with the AUC-ROC (to study the compromise between computation time and performance). We set $d = 15$, $\hat{\tau} = 15$, $q = 50$ and $n_{diag} = 2$.

On Figure 6.9, it appears that the AUC-ROC stops increasing after $n_{points} = 200$, at a value above 0.95. Computing persistent homology without subsampling (i.e. on thousands of points) would take hours for each time series, but these examples show that performance can be almost optimal for values of n_{points} significantly smaller than n (here, of the order of 10 times smaller, which makes computation about 1000 times faster). This study, and the fact that our algorithm performed well with $n_{points} = 400$ on time series

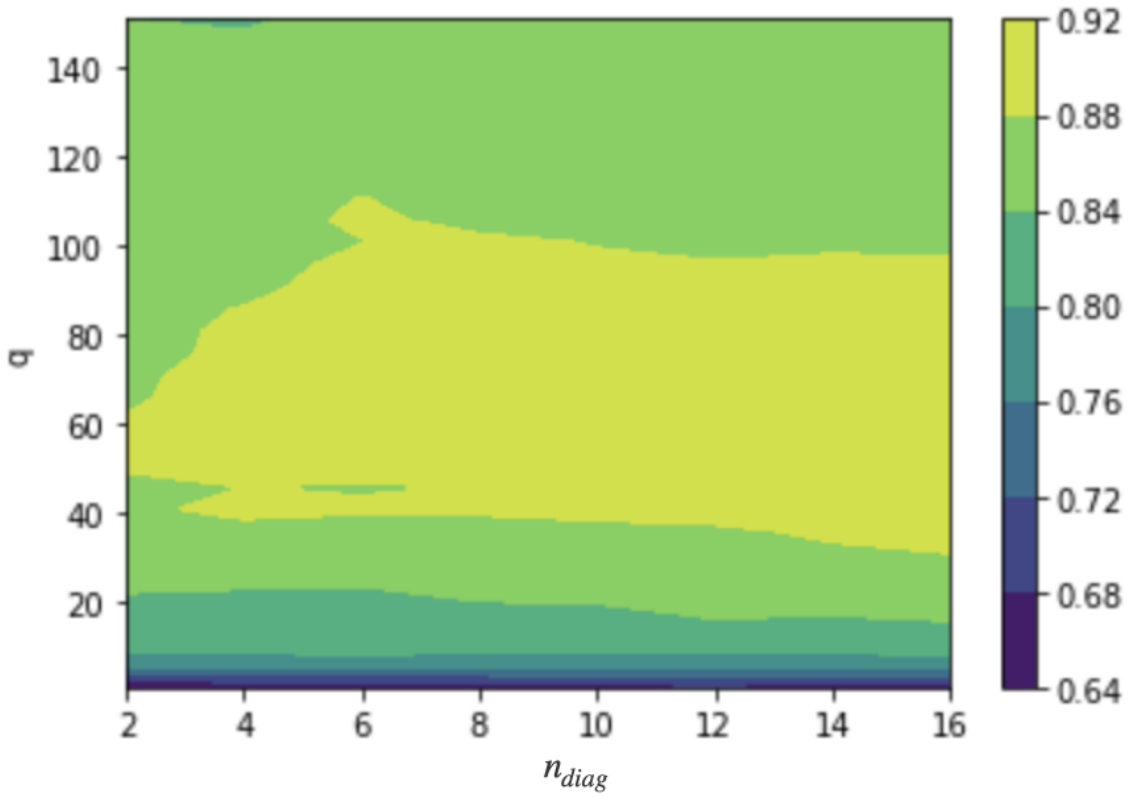


Figure 6.8: Heat map representing the average AUC-ROC on the whole dataset with different values of q and n_{diag}.

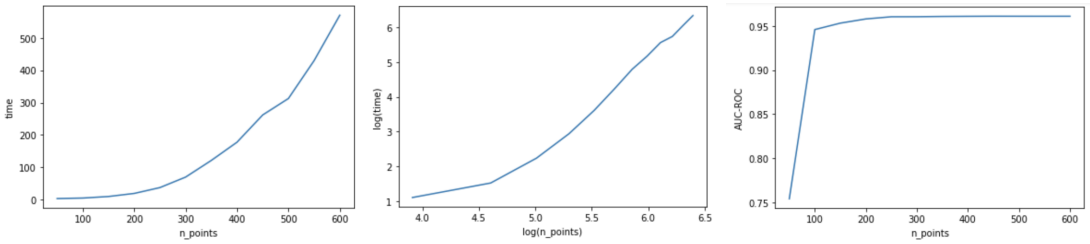


Figure 6.9: **Left:** computation time (in seconds) as a function of n_{points}. **Middle:** computation time as a function of n_{points}, logarithmic scale. **Right:** AUC-ROC as a function of n_{points}. All quantities are averaged over 20 time series.

with lengths above 100000 in our benchmark from Section 6.6.1, indicate that the algorithm has a certain scalability as with n_{points} can increase slower than n.

6.6.5 Robustness to noise and benchmark on synthetic data

Here, we study the robustness of our algorithm to noise, and compare it to the unsupervised methods from the benchmark of Section 6.5 whose code is publicly available, and DAMP [LWM+22], which is a version of the Matrix Profile designed to better handle multiple anomalies. For each signal $\mathbf{y} = \mathbf{x} + \varepsilon$, we define the **noise ratio** as $\frac{\text{Var}(\varepsilon)}{\text{Var}(\mathbf{x})}$ (the inverse of the signal-to-noise ratio). Figure 6.10 shows the average AUC-ROC of each algorithm on the whole synthetic dataset, where centered Gaussian noise is added to each time series, with a noise ratio between 0 and 1. Note that the real benchmark is in Section 6.6.1, we

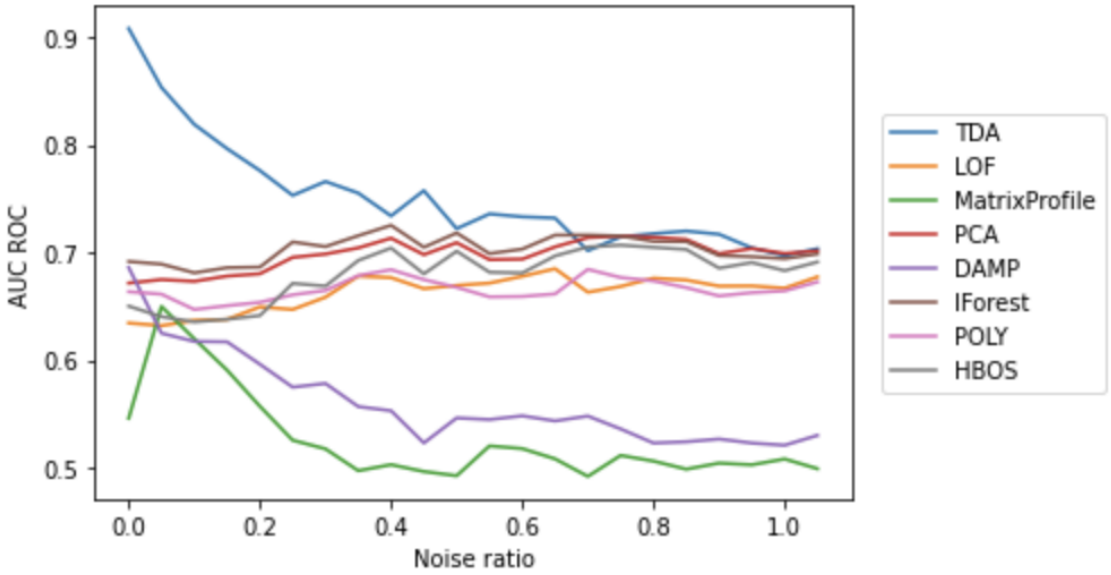


Figure 6.10: Average AUC-ROC of each algorithm on the whole synthetic dataset, when the noise ratio of the centered Gaussian noise varies between 0 and 1.

only make comparisons here to gain insights our algorithm. It should be noted that since the synthetic data are generated according to the exact model used by our method, the comparison is biased.

Figure 6.10 shows that our algorithm clearly outperforms all the other unsupervised methods up to a noise ratio of 30%, and stays in the top performers for higher noise ratios (it seems that the AUC-ROC of about 0.7 corresponds to a points where only easy anomalies are detected, which would explain the fact that it can be reached by all the algorithms, even with a lot of noise). When looking into more details at our study on synthetic data, it appears that our algorithm usually performs better than the others when there are multiple normal atoms or repeating anomalies. However, LOF can outperform it in the case of punctual anomalies, especially when the normal behavior is complex (several atoms or a lot of noise). Also note that the significant decline in performance of the Matrix Profile and DAMP in the presence of noise can be explained by the fact that they use a z-normalized Euclidean distance, which can amplify the noise when looking at sequences between atoms so those sequences are detected as anomalies and can even hide other anomalies.

Our method gets an AUC-ROC above 0.8 for noise ratios up to approximately 20%. Too much noise can fill the loops so the main cycles are less persistent and thus more difficult to identify on the persistence diagram. All the other methods except the Matrix Profile and DAMP are very robust to noise. However, it would seem more relevant to study those methods on data for which they perform very well to see how much noise makes them fail.

6.7 Application to gait signals

As in Chapter 5, we applied our algorithm to gait signals from the dataset described in Chapter 1 (see Figure 1.1). A result is shown on Figure 6.11 and 6.12 (these figures can be reproduced using the IPOL demo mentioned above). We used parameters $d = 15$,

$\hat{\tau} = 3$, $n_{points} = 250$, $q = 20$ and $n_{diag} = 2$. The threshold on anomaly scores is chosen automatically using Otsu's method [Ots79] for gray-level threshold selection.

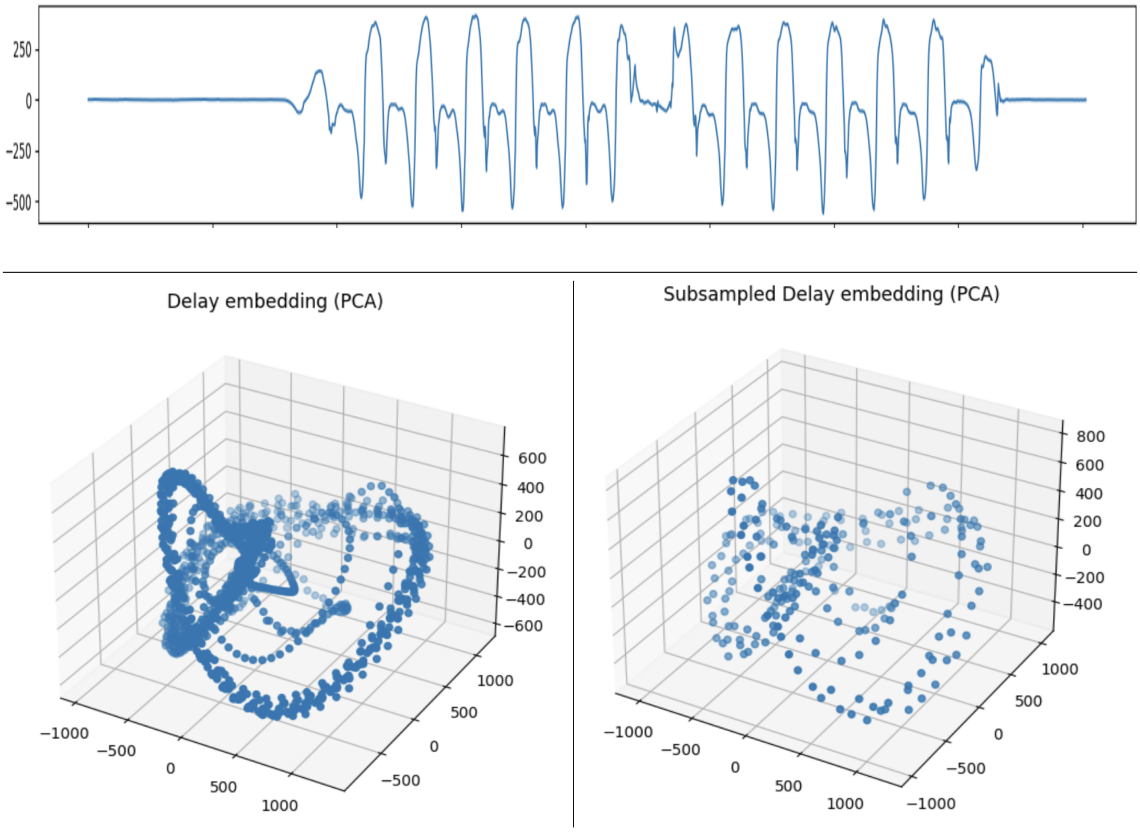


Figure 6.11: **Top:** gait signal from the [TBMM$^+$19] dataset. **Bottom left:** PCA view of the delay embedding with $d = 15$ and $\hat{\tau} = 3$. **Bottom right:** PCA view of the subsampled delay embedding with $n_{points} = 250$.

This signal has four different regimes: it stays near zero at the start and end (the subject stands still), there are two phases of normal walk separated by the U-turn, and the first and last steps have smaller amplitudes than normal steps due to lower speed at the start and end. Usually, a distinctive feature of U-turn steps is the absence or reduced amplitude of negative peaks on the signal. As explained by Dot et al. [DQO$^+$20], these peaks correspond to the heel strike and heel-off events in normal steps, which transfer weight to the ground to move forward. U-turns are lateral movements so the heel is less involved.

On the delay embedding from Figure 6.11, several loops can be distinguished one of which is denser. This dense loop corresponds to the ten normal steps. The algorithm detects it as a normal cycle (in purple on Figure 6.12), and detects four anomalies: the first step, last step and two U-turn steps. Those anomalies can be detected by a large range of thresholds. This result could be used to segment the data to provide a suitable input for our algorithm from Chapter 4. Indeed, this algorithm takes as input gait signals representing only normal steps, which is important because the method relies on the assumption that signals have some periodicity. The first step, last step and U-turns were removed using data from the GaitRite® mat during a preprocessing phase. Using anomaly detection would make it possible to perform this preprocessing without any other data than the IMU signal.

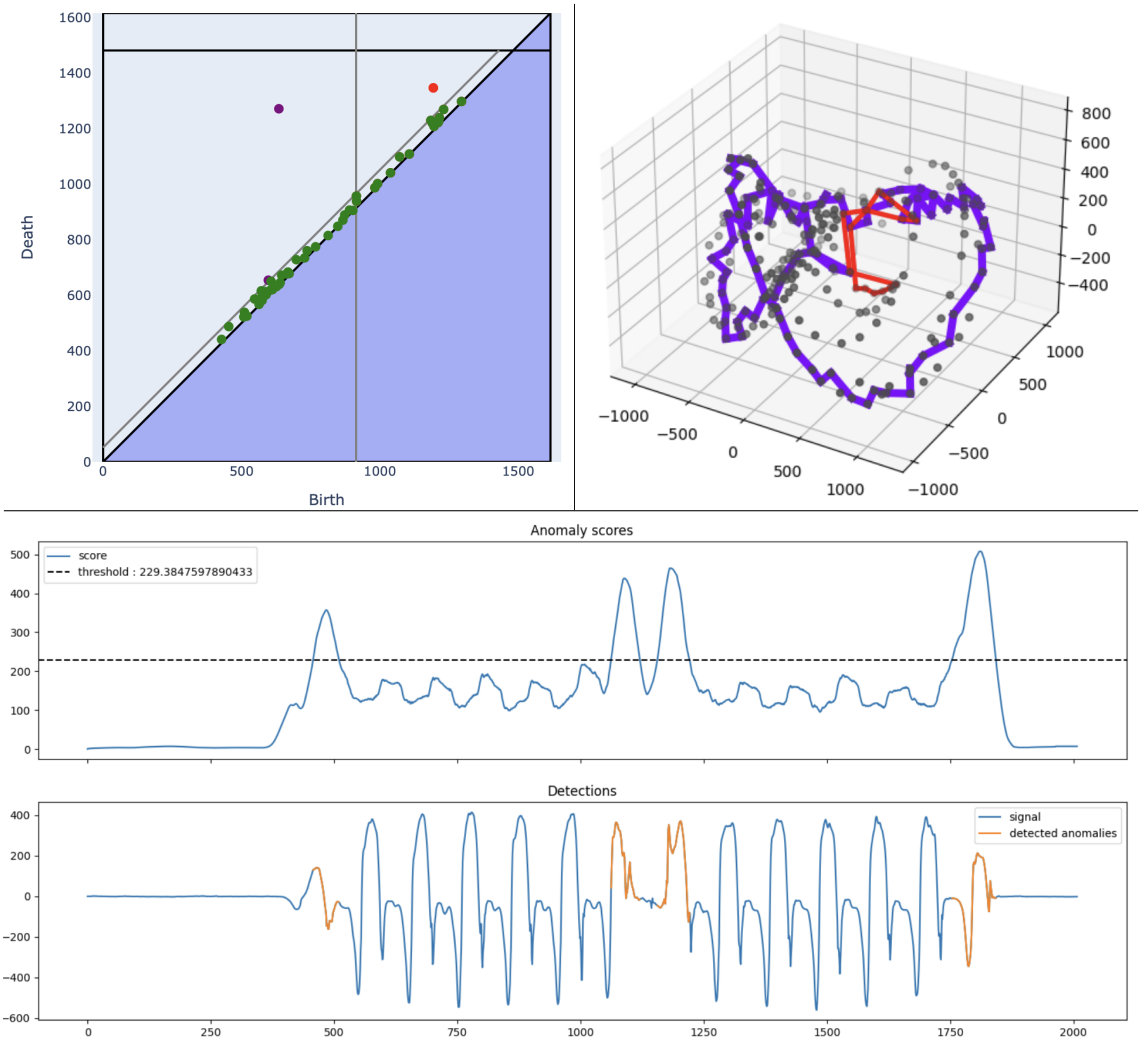


Figure 6.12: **Top left:** persistence diagram computed with $q = 20$. Gray lines represent the thresholds (chosen with $n_{diag} = 2$). **Top right:** normal cycles (in purple) and abnormal cycle (in red). **Bottom:** anomaly scores with Otsu threshold and detected anomalies (green points on the signal).

Clinicians could also examine abnormal steps, which may give information about the patient. For example, analysing U-turns via gait signals from IMUs makes it possible to identify turning strategies which can be used to evaluate the risk of fall in post-stroke patients [BRO+17]. This provides complementary information to the result obtained with the PEPA algorithm in Chapter 5: the anomaly detection method is more efficient to detect abnormal steps such as the last step. Also note that the U-turn makes the seventh step abnormal because of the peak around 9 seconds. Here the PEPA algorithm detected the first part of the step as normal ignored the peak, and the anomaly detection algorithm detected the peak. Moreover, it seems easier to choose the persistence and birth thresholds for anomaly detection, but pattern detection provides a finer segmentation by indicating the start and end of each normal step.

6.8 Conclusion

This article describes an unsupervised anomaly detection algorithm based on 1D persistent homology. We proposed a model of time series in which the anomaly detection problem is well defined, which enabled us to create our method and to derive mathematical properties of the method. The algorithm empirically proved to be competitive to state-of-the-art anomaly detection methods on different real-world datasets. The subsampling step induces a time/performance compromise that enabled us to significantly decrease computation time. Future research could improve this compromise by using faster ways of extracting cycles or finding sparse versions of the filtrations. Exploring non-Euclidean distances [FBMG23, GTO] could help make the algorithm more robust to variations of the parameters or make it possible to adapt the algorithm to more general models of time series, for example by dealing with the presence of a linear trend, or handling multivariate time series. Buchet et al. [BCOS16] generalized the construction of the distance to a measure and and the associated stability results to general metric spaces, so the construction of weighted filtrations [ACG$^+$20] can be generalized by replacing Euclidean balls with metric balls. Some technical difficulties remains: first, the tameness of the constructed persistence module depends on the fact that the *filtration value function* is continuous and proper, which is not necessarily the case in infinite spaces. Secondly, the bound from Theorem 4 is specific to the Euclidean norm, so the proof should be adapted to the new distance and we would want the bound to remain 'small' under the assumptions of the anomaly detection problem. Pseudo-metrics are often used in signal processing [Mül07, GTO] because they have invariance properties. The proposed generalization could be applied to the quotient metric space obtained by identifying indistinguishable points.

Chapter 7

Conclusion and perspectives

In this thesis, we have:

- Conducted a literature review about persistent homology applied to time series. We reviewed 71 articles, highlighted the main applications and ideas, and presented some research perspectives.
- Proposed an interpretable method for gait signals analysis based on the sublevel set filtration and the bottleneck distance between barcodes, and motivated by theoretical properties of persistent homology of periodic functions. We used this method to study a cohort of healthy subjects and patients suffering from multiple sclerosis and showed that it can help asses the severity of the disease and detect changes in the context of longitudinal follow-up.
- Defined a new filtration for metric spaces for which we proved a stability result with respect to the Gromov-Hausdorff distance. We used it to define a non-parametric clustering algorithm with an outlier removal step that can perform well on various types of data and be used to detect patterns in time series.
- Proposed an algorithm for unsupervised anomaly detection in univariate time series, based on delay embeddings and DTM filtrations. We showed that it was competitive with state-of-the-art anomaly detection methods, studied it theoretically and empirically, and illustrated its use for gait signals analysis.

Our contributions, as well as several articles reviewed in Chapter 3, show that TDA can be a powerful tool to detect structure in time series, especially in biomedical applications. The end of each chapter gives some perspectives on how to improve existing methods by making them more robust, faster, or how to generalize them to different types of signals and applications. More generally, this thesis paves the way for future research on TDA for time series following the same approach consisting in building interpretable methods based on mathematical insights such as models or stability properties. More complex models such as stochastic processes could be considered. They could be relevant in many applications because they can model complex phenomena and allow statistical studies.

The logical continuation of this thesis is to use our approach to study multivariate time series. We conclude by giving three lines of research to develop new topological methods for multivariate time series.

First, univariate methods can be extended to multivariate time series by using multivariate transformations or distances. For example, periodicity in multivariate time series

was studied in motion capture [VJPSK15] and video [TP18] using multivariate versions of the delay embedding. However, when studying more complex structures such as successions of patterns or events, generalization poses a problem because the different channels are not necessarily synchronized or events can happen in only a subset of channels. Thus, some work must be done to properly define events in the multivariate case in order to generalize methods such as our anomaly detection algorithm.

Secondly, specific methods can be developed for multivariate time series. Existing methods mostly use transformations of time series into graphs, the most common approach (often used for EEG data analysis) being to build a correlation graph whose vertices correspond to each channel of the time series [BTO24c]. An idea would be to consider more complex graphs than correlation-based ones, for example directed graphs that represent causal relations such as in El-Yaagoubi and Ombao's work [EYO23], or generalized notions of correlation [SBPA23]. Another perspective would be to make a link between graph signal processing (GSP) [OFK+18] (or its extension to simplicial complexes [BS20]) and TDA. Indeed, GSP provides relevant tool to analyze the spacial structure of a graph with notions such as graph Fourier transforms and smoothness of a graph signal, but these tools are not made to study time-varying graph signals. Symbolization methods [CMO24] can lead to promising and interpretable methods but do not quantify the regularity of temporal variations. Persistence vineyards [CSEM06, Mun13] or zigzag persistence [CDS10] are two relevant concepts for studying time-varying persistence diagrams that can be applied to multivariate time series [SRA+21, MMKM23]. It would be interesting to link them to GSP, for example by studying the stability of vineyards [Mun13] obtained from graphs or graph signals, in order to build a topological representation of time-varying graphs that is both stable with respect to temporal and spatial variations.

Thirdly, multiparameter persistence [BL22] is a promising approach to multivariate data analysis. As explained at the end of Chapter 3 two parameters are often relevant to construct a filtered simplicial complex. For example distance and time [TMK20], density and time [CJ17] or distance and density [BTO24b]. Multiparameter persistent homology studies the structure of data for all values of several parameters. There exist invariants and vectorizations for multiparameter persistent homology (usually, these vectorizations are obtained by considering 1-dimensional lines in the space of filtration parameters) and some are starting to be applied to time series classification [LSC+24, LCB24, KJ22, KJ24] but they have not clearly outperformed usual methods yet. Multiparameter invariants are more difficult to interpret than one-parameter ones because they do not have an interval-decomposition (the decomposition that gives the persistence diagram for usual persistence), so more fundamental research is needed to understand which structures are detected by multiparameter persistent homology. For example, Loiseaux et al. [LSC+24] proposed vectorizations of multiparameter persistence modules that are stable with respect to the filtration functions, but it would be interesting to study the stability of specific filtration functions with respect to metrics that reflect relevant properties of the studied data for a given task.

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